

LIFE SCIENCES SUITE

# Standard Operating Procedures Handbook

Procedures, demonstrations, good practices and diagrams for the 22 Life Sciences Suite modules and the related OCA modules — Odoo 19 Community

ENGLISH

# About this handbook

This handbook gathers the Standard Operating Procedures (SOPs) that the Life Sciences Suite can support. Each SOP rests on a workflow, a control or a record that really exists in the modules. The workflows, buttons, menus, roles and controls quoted in this book were read from the module source code and from a database in which the 22 LS modules were installed on Odoo 19.0 Community Edition.

## What this handbook is — and is not

- ✓ It is a **working basis** for your quality assurance: every sheet has blank fields (version, effective date, approval) to be completed under your own document control.
- ✓ The SOP codes (SOP-QA-001, etc.) are a **proposed numbering**, to be adapted to your document system.
- ✓ The **regulatory content** of each SOP (criteria, time limits, acceptance levels, local requirements) remains the responsibility of your QA and regulatory functions. The software helps to apply a procedure; it does not replace it and does not make an organisation compliant.
- ✓ References to regulations and standards are given at chapter or article level. Check the version in force before approval.

## How the book is organised

- **Foundations** — GMP principles, document hierarchy, the quality cycle, how the modules hand over to each other, data integrity (ALCOA+), segregation of duties, and how to write an SOP in ls\_qms.
- **Part I — Quality management system**: documents, training, deviations and CAPA, change control, complaints, audits, risk, suppliers.
- **Part II — Data integrity and compliance**: audit trail, electronic signatures, validation, calibration, environmental monitoring, recalls.
- **Part III — Industry modules**: pharmaceutical, cosmetics, medical devices, medical plastics, laboratory.
- **Part IV — Regulatory affairs**: import-export (phase 1).
- **Part V — Procedures supported by OCA modules**.
- **Part VI — Procedures needed but not covered by the suite**, with guidance on what they must contain.
- **Annexes** — SOP register, glossary, known limitations observed while preparing this book.

## Screenshots, diagrams and demonstrations

Screenshots are captures of the running software (Odoo 19.0 Community, LS modules, demonstration data). The interface is in English: the modules ship no French translation, so the French handbook quotes the English button labels and gives their meaning. Lifecycle diagrams are drawn from the states actually defined by each module; explanatory diagrams (pyramids, V-model, charts) illustrate a concept and use illustrative values, which are labelled as such. Each SOP ends with a worked demonstration describing a realistic case step by step; the people and values in these demonstrations are fictitious.

## Anatomy of a SOP sheet

| Section               | Content                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------|
| Header                | Code, title, module, menu, suggested owner, related SOPs; blank version, effective date and approval fields. |
| 1 Objective · 2 Scope | Why the procedure exists and what it covers.                                                                 |
| 3 Responsibilities    | Roles named after the security groups of the module, so that access rights and the SOP match.                |
| 4 Definitions         | Terms used in the SOP.                                                                                       |

| Section                             | Content                                                                                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 Workflow in the software          | Lifecycle diagram generated from the module states, and explanatory diagrams.                                                                              |
| 6 Procedure                         | Numbered steps. Buttons are shown as  ; menus as <b>Menu ► Sub-menu</b> . |
| 7 Controls enforced by the software | Rules the system refuses to break (it blocks the action and shows a message). The SOP does not need to police them, but users must know them.              |
| 8 Good practices · 9 Pitfalls       | Experience-based advice and frequent errors.                                                                                                               |
| 10 Records · 11 Indicators          | Evidence produced and suggested KPIs.                                                                                                                      |
| 12 References · 13 Demonstration    | Regulatory basis and a worked example.                                                                                                                     |

# Contents

|                                                                                                 |           |
|-------------------------------------------------------------------------------------------------|-----------|
| <b>About this handbook</b>                                                                      | <b>1</b>  |
| <b>Foundations</b>                                                                              | <b>7</b>  |
| Why written procedures                                                                          | 7         |
| The quality cycle                                                                               | 7         |
| The suite at a glance                                                                           | 8         |
| Data integrity — ALCOA+                                                                         | 9         |
| Segregation of duties (four eyes)                                                               | 9         |
| Roles and access rights                                                                         | 10        |
| Writing a SOP in ls_qms                                                                         | 10        |
| <b>Part I — Quality management system</b>                                                       | <b>11</b> |
| <b>Document system</b>                                                                          | <b>12</b> |
| SOP-QA-001 Drafting, review, approval and publication of quality documents                      | 14        |
| SOP-QA-002 Revision, withdrawal and archiving of documents                                      | 18        |
| SOP-QA-003 Periodic review of documents                                                         | 21        |
| SOP-QA-004 Management and retention of quality records                                          | 23        |
| SOP-QA-005 Quality objectives and measurement of their achievement                              | 27        |
| <b>Training</b>                                                                                 | <b>30</b> |
| SOP-QA-010 Creation and approval of training courses                                            | 32        |
| SOP-QA-011 Session planning and issue of certifications                                         | 35        |
| SOP-QA-012 Training linked to the publication of a new document                                 | 38        |
| SOP-QA-013 Follow-up of qualifications and requalification                                      | 40        |
| <b>Deviations and CAPA</b>                                                                      | <b>43</b> |
| SOP-QA-020 Reporting, assessment and classification of deviations                               | 45        |
| SOP-QA-021 Investigation, extension to other batches and product disposition                    | 48        |
| SOP-QA-022 CAPA management from identification to closure                                       | 51        |
| SOP-QA-023 Verification of CAPA effectiveness                                                   | 55        |
| <b>Change control</b>                                                                           | <b>57</b> |
| SOP-QA-030 Change request and impact assessment                                                 | 59        |
| SOP-QA-031 Approval, implementation and verification of changes                                 | 63        |
| <b>Complaints</b>                                                                               | <b>66</b> |
| SOP-QA-040 Reception, assessment and investigation of complaints                                | 68        |
| SOP-QA-041 Adverse events: reportability assessment and reporting deadlines                     | 71        |
| <b>Audits</b>                                                                                   | <b>74</b> |
| SOP-QA-050 Internal audit programme                                                             | 76        |
| SOP-QA-051 Auditor qualification and impartiality                                               | 79        |
| SOP-QA-052 Conducting an audit: preparation, fieldwork, findings                                | 82        |
| SOP-QA-053 Audit report and follow-up of findings                                               | 85        |
| <b>Risk management</b>                                                                          | <b>88</b> |
| SOP-QA-060 Quality risk management: matrix, assessment, control and acceptance of residual risk | 89        |
| SOP-QA-061 FMEA (failure modes and effects analysis)                                            | 93        |
| SOP-QA-062 Periodic review of risks                                                             | 96        |
| <b>Suppliers</b>                                                                                | <b>98</b> |
| SOP-QA-070 Qualification and approval of suppliers                                              | 100       |



|                                                      |                                                                                            |            |
|------------------------------------------------------|--------------------------------------------------------------------------------------------|------------|
| SOP-QA-071                                           | Supplier audits .....                                                                      | 103        |
| SOP-QA-072                                           | Supplier performance evaluation and requalification .....                                  | 106        |
| SOP-QA-073                                           | Control of purchases from non-approved suppliers .....                                     | 109        |
| <b>Part II — Data integrity and compliance</b> ..... |                                                                                            | <b>111</b> |
| <b>Audit trail</b> .....                             |                                                                                            | <b>112</b> |
| SOP-DI-001                                           | Definition of the audit trail scope (rules per model) .....                                | 113        |
| SOP-DI-002                                           | Audit trail review and integrity verification .....                                        | 116        |
| SOP-DI-003                                           | Evidence packs for inspections .....                                                       | 118        |
| SOP-DI-004                                           | Retention and controlled purge of audit data .....                                         | 120        |
| <b>Electronic signatures</b> .....                   |                                                                                            | <b>122</b> |
| SOP-DI-010                                           | Electronic signature policy: meanings, policies, signer commitment .....                   | 123        |
| SOP-DI-011                                           | Management of credentials, account lockout and security alerts .....                       | 127        |
| SOP-DI-012                                           | Verification of the signature chain .....                                                  | 130        |
| <b>Validation</b> .....                              |                                                                                            | <b>132</b> |
| SOP-VAL-001                                          | Validation master plan (VMP) .....                                                         | 134        |
| SOP-VAL-002                                          | Drafting and approval of qualification protocols: DQ, IQ, OQ, PQ .....                     | 137        |
| SOP-VAL-003                                          | Protocol execution and management of discrepancies .....                                   | 141        |
| SOP-VAL-004                                          | Summary report and validation status .....                                                 | 144        |
| SOP-VAL-005                                          | Periodic revalidation .....                                                                | 146        |
| SOP-VAL-006                                          | Validation of processes, cleaning, analytical methods and computerised systems (CSV) ..... | 148        |
| <b>Calibration</b> .....                             |                                                                                            | <b>150</b> |
| SOP-MET-001                                          | Management of the instrument inventory .....                                               | 152        |
| SOP-MET-002                                          | Planning and performing calibrations (as-found and as-left values) .....                   | 155        |
| SOP-MET-003                                          | Impact assessment of an out-of-tolerance result .....                                      | 159        |
| SOP-MET-004                                          | Calibration certificates .....                                                             | 161        |
| <b>Environmental monitoring</b> .....                |                                                                                            | <b>164</b> |
| SOP-EM-001                                           | Monitoring programme: areas, sampling points and rationale .....                           | 165        |
| SOP-EM-002                                           | Setting and approval of alert and action limits .....                                      | 168        |
| SOP-EM-003                                           | Sampling, analysis and review of results .....                                             | 171        |
| SOP-EM-004                                           | Management of excursions .....                                                             | 173        |
| SOP-EM-005                                           | Trend analysis .....                                                                       | 175        |
| <b>Recalls</b> .....                                 |                                                                                            | <b>178</b> |
| SOP-QA-080                                           | Recall plan, with coordinator and deputy .....                                             | 179        |
| SOP-QA-081                                           | Executing a recall or withdrawal: distribution tracing, notification, reconciliation ..... | 182        |
| SOP-QA-082                                           | Recall effectiveness checks .....                                                          | 185        |
| SOP-QA-083                                           | Mock recall (periodic exercise) .....                                                      | 188        |
| <b>Part III — Industry modules</b> .....             |                                                                                            | <b>190</b> |
| <b>Pharmaceutical</b> .....                          |                                                                                            | <b>191</b> |
| SOP-PRD-001                                          | Issue and execution of the batch record .....                                              | 193        |
| SOP-PRD-002                                          | Line clearance and label reconciliation .....                                              | 196        |
| SOP-QA-090                                           | Batch record review and batch release by the quality unit .....                            | 198        |
| SOP-QC-001                                           | Qualification of active substances and excipients .....                                    | 201        |
| SOP-QC-002                                           | Stability programme .....                                                                  | 204        |
| SOP-PRD-003                                          | GS1 serialisation and aggregation .....                                                    | 206        |
| SOP-RA-001                                           | Preparation of the CTD dossier (Common Technical Document) .....                           | 209        |
| <b>Cosmetics</b> .....                               |                                                                                            | <b>212</b> |

|                                                                 |                                                                                  |            |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------|------------|
| SOP-COS-001                                                     | Development and approval of formulations                                         | 214        |
| SOP-COS-002                                                     | Safety assessment (CPSR, Cosmetic Product Safety Report)                         | 216        |
| SOP-COS-003                                                     | Product information file (PIF) and its retention period                          | 218        |
| SOP-COS-004                                                     | Substantiation of claims                                                         | 220        |
| SOP-COS-005                                                     | Validation of labelling and of the INCI list                                     | 222        |
| SOP-COS-006                                                     | Maintenance of the register of regulatory restrictions on ingredients            | 224        |
| SOP-COS-007                                                     | Prior authorisation dossier in Algeria                                           | 227        |
| <b>Medical devices</b>                                          |                                                                                  | <b>230</b> |
| SOP-MD-001                                                      | Device registration and justification of its risk class                          | 231        |
| SOP-MD-002                                                      | Assignment of the UDI (Unique Device Identification)                             | 234        |
| SOP-MD-003                                                      | Risk management according to ISO 14971                                           | 236        |
| SOP-MD-004                                                      | Clinical evaluation and PMCF (Post-Market Clinical Follow-up)                    | 239        |
| SOP-MD-005                                                      | Technical documentation                                                          | 241        |
| SOP-MD-006                                                      | CE marking and relationship with the notified body                               | 243        |
| SOP-MD-007                                                      | Post-market surveillance and periodic reports                                    | 245        |
| <b>Medical plastics</b>                                         |                                                                                  | <b>248</b> |
| SOP-MP-001                                                      | Qualification and commissioning of moulds                                        | 249        |
| SOP-MP-002                                                      | Preventive maintenance of moulds (by shot count and by time) and cavity blocking | 251        |
| SOP-MP-003                                                      | Process windows: drafting, review and approval                                   | 254        |
| SOP-MP-004                                                      | Start-up, operation and closure of a moulding run                                | 256        |
| SOP-MP-005                                                      | Correction of parameter readings (without altering the original value)           | 259        |
| SOP-MP-006                                                      | Material traceability                                                            | 261        |
| <b>Laboratory</b>                                               |                                                                                  | <b>263</b> |
| SOP-QC-010                                                      | Approval of test methods and specifications                                      | 265        |
| SOP-QC-011                                                      | Receipt and registration of samples                                              | 267        |
| SOP-QC-012                                                      | Entry, review and approval of results                                            | 269        |
| SOP-QC-013                                                      | Investigation of OOS (out-of-specification) and OOT (out-of-trend) results       | 272        |
| SOP-QC-014                                                      | Stability studies in the laboratory                                              | 275        |
| SOP-QC-015                                                      | Issue of certificates of analysis                                                | 277        |
| <b>Part IV — Regulatory affairs</b>                             |                                                                                  | <b>280</b> |
| <b>Import-export compliance (phase 1)</b>                       |                                                                                  | <b>281</b> |
| SOP-RA-010                                                      | Regulatory watch: register of cited texts                                        | 282        |
| SOP-RA-011                                                      | Handling open regulatory questions                                               | 284        |
| <b>Part V — Procedures supported by OCA modules</b>             |                                                                                  | <b>287</b> |
| SOP-GEN-001                                                     | Management review                                                                | 288        |
| SOP-GEN-002                                                     | Quality indicators (KPI)                                                         | 291        |
| SOP-GEN-003                                                     | Preventive maintenance of equipment                                              | 293        |
| SOP-GEN-004                                                     | Regulatory certificates of equipment                                             | 295        |
| SOP-IT-001                                                      | Role-based access management and periodic review of access rights                | 297        |
| SOP-IT-002                                                      | Technical audit log                                                              | 299        |
| SOP-QC-020                                                      | Generic inspection controls, for example at reception                            | 301        |
| <b>Part VI — Procedures needed but not covered by the suite</b> |                                                                                  | <b>303</b> |
| <b>Annexes</b>                                                  |                                                                                  | <b>304</b> |
| Annex A — SOP register                                          |                                                                                  | 304        |
| Annex B — Glossary                                              |                                                                                  | 310        |

|                                                                          |     |
|--------------------------------------------------------------------------|-----|
| Annex C — Known limitations observed while preparing this handbook ..... | 310 |
| Annex D — How this handbook was produced .....                           | 311 |

# Foundations

## Why written procedures

Good Manufacturing Practice rests on a simple chain: **say what you do, do what you say, prove it**. A procedure says what must be done, by whom and when; the people trained on it do it; the records prove that it was done, and the quality system checks that the result is right. The suite supports each link: ls\_qms and ls\_document\_management hold the procedures, ls\_training proves the competence, each module keeps the records, and the audit, CAPA and risk modules close the loop.

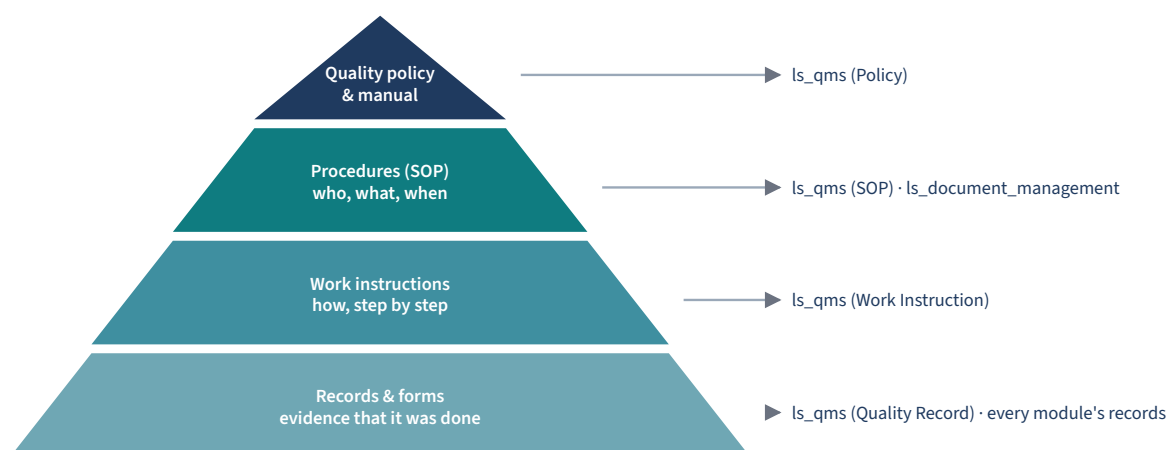


Figure 1 — Document hierarchy and where each level lives in the suite.

## The quality cycle

ISO 9001, ISO 13485 and ICH Q10 describe a quality system as a continuous improvement cycle. The modules map onto the Plan–Do–Check–Act cycle as shown below. The cycle only works if the outputs of one stage reach the next one: an audit finding that never becomes a CAPA, or a CAPA whose action is never managed as a change, breaks the loop.

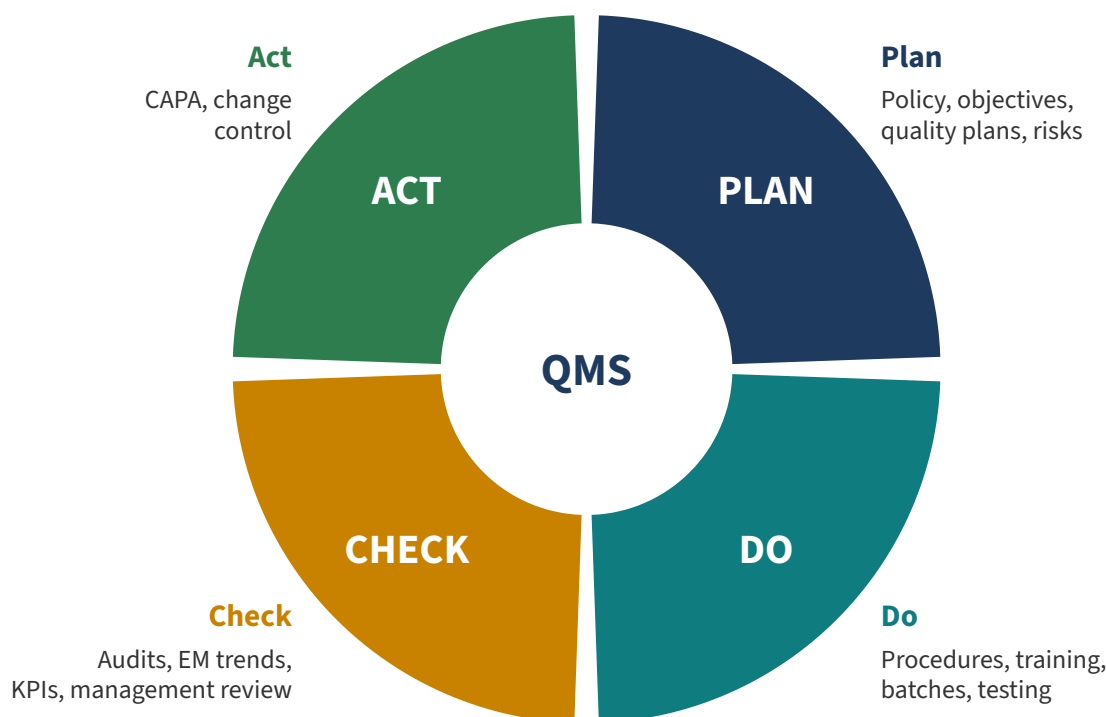
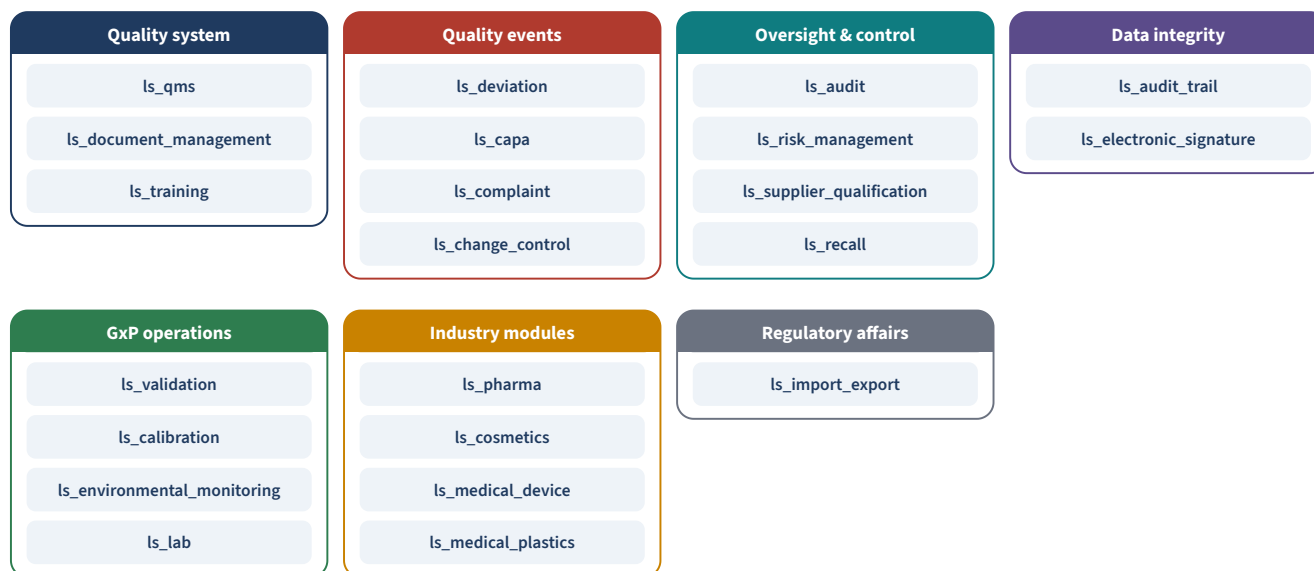


Figure 2 — Plan-Do-Check-Act cycle and the modules that support each stage.

## The suite at a glance



No LS module holds a relational link to another LS module; hand-offs (deviation → CAPA, audit finding → CAPA, excursion → deviation...) are made through a reference field typed by the user. The procedures must therefore require these references to be entered and checked.

Figure 3 — The 22 Life Sciences Suite modules grouped by domain.

Each module can be installed on its own. The price of that independence is important for the procedures: **the modules do not create records in each other**. When a deviation needs a CAPA, the deviation reaches the state *CAPA Required* and records a CAPA reference, but the CAPA itself is opened by a person in *ls\_capa*. The SOPs of this book therefore specify who opens the downstream record, within which time, and how the reference is written back.

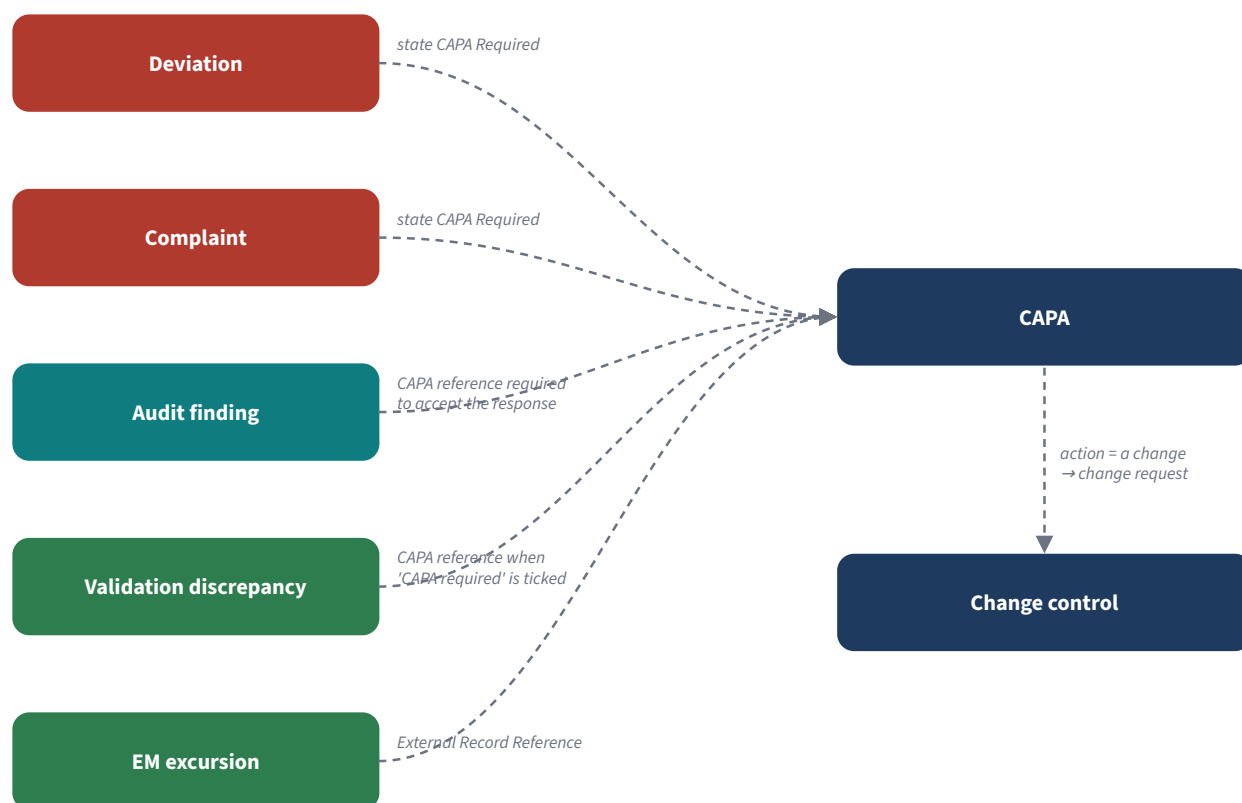


Figure 4 — Hand-offs between quality events, as implemented: by state and reference field, not by automatic link.

## Data integrity — ALCOA+

Regulators expect GxP data to be Attributable, Legible, Contemporaneous, Original and Accurate, and also Complete, Consistent, Enduring and Available. The suite contributes technical controls — immutable records after approval, append-only logs, hash chains, mandatory reasons for change — but data integrity is first a behaviour: record at the time of the activity, never share an account, never record for someone else, and correct by a traced amendment rather than by deletion.



Figure 5 — ALCOA+ principles.

## Segregation of duties (four eyes)

Most lifecycles in the suite separate the person who prepares or performs from the person who reviews and the person who approves. Where the rule is coded, the software refuses the forbidden combination and says why. Access rights must be designed so that every critical workflow has at least two eligible people, including during holidays.



The software refuses the same person in two roles on one record wherever the rule exists  
(e.g. result review, SOP approval, batch release, excursion closure, protocol approval).

Figure 6 — Segregation of duties enforced by the software.

## Roles and access rights

Each module defines its own security groups (typically Viewer, User or Operator, Reviewer or Approver, Manager). The responsibilities tables of the SOPs use these group names so that the procedure and the access matrix say the same thing. Grant the least privilege needed, review the grants periodically (see SOP-IT-001), and never give a production account a manager role "for convenience".

## Writing a SOP in ls\_qms

The *Standard Operating Procedure* model of ls\_qms has the sections Purpose, Scope, Definitions and Abbreviations, Responsibilities, Procedure and Reference Documents. Purpose, scope and procedure are mandatory before review. The sheets of this handbook follow the same structure: copy each section into the corresponding field, adapt it, then submit it through the approval workflow described in SOP-QA-001.

### Rules of good writing

- ★ One procedure, one process. Split when two teams or two triggers are involved.
- ★ Write imperative, numbered steps; one action per step; name the role, not the person.
- ★ State the time limits and the acceptance criteria as numbers, not as "promptly" or "as appropriate".
- ★ Quote the exact button and menu of the software, so that training and audit use the same words.
- ★ Say what to do when something goes wrong (the exception path), not only the nominal path.
- ★ Keep the "why" in the objective and the references; keep the steps short.

# I

## Quality management system

The procedures that organise the quality system itself: documents and records, training, quality events (deviations, CAPA, complaints, changes), audits, risk management and suppliers.

Document system — ls\_qms,  
ls\_document\_management

Training — ls\_training

Deviations and CAPA — ls\_deviation, ls\_capa

Change control — ls\_change\_control

Complaints — ls\_complaint

Audits — ls\_audit

Risk management — ls\_risk\_management

Suppliers — ls\_supplier\_qualification



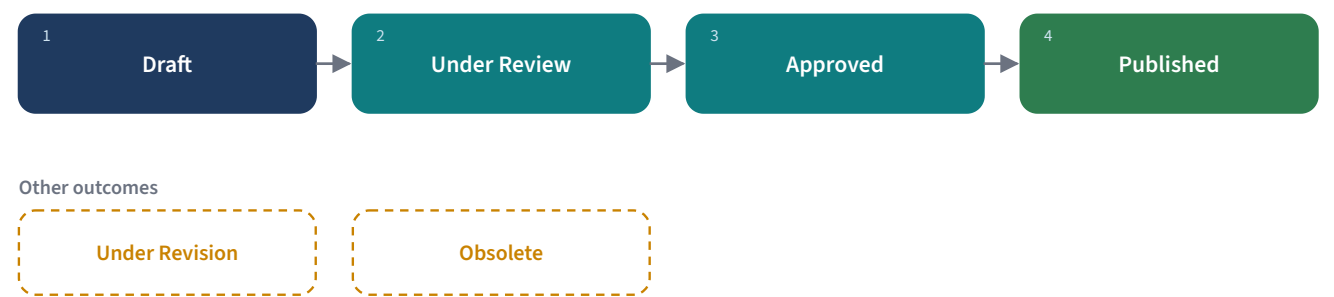
# Document system

ls\_qms · ls\_document\_management

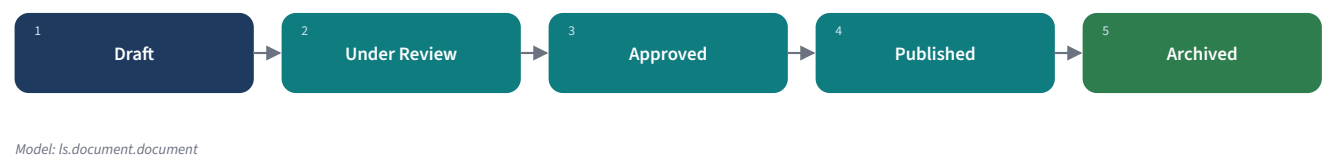
Two modules carry controlled documentation. **ls\_qms** holds the structured quality documents — quality policy, SOPs, work instructions and quality plans — whose content is typed in sections, plus quality objectives and quality records. **ls\_document\_management** holds controlled *files* (PDF, drawings, forms) organised in folders, with immutable versions protected by a SHA-256 checksum, an approval circuit per document and retention policies. Use ls\_qms for documents you write in the system and ls\_document\_management for documents you manage as files.

| Menu                                                                                    | Content                                                |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------|
| Quality Management ► Quality Policy / Procedures / Work Instructions / Quality Planning | Structured controlled documents (ls_qms)               |
| Quality Management ► Quality Objectives, Quality Records                                | Objectives with measurements; retained quality records |
| Quality Management ► Reporting ► Procedures Due for Review                              | Documents whose periodic review is due                 |
| Documents ► Documents / Versions / My Approvals                                         | Controlled files (ls_document_management)              |
| Documents ► Configuration ► Folders / Retention Policies / Tags                         | Structure and retention rules                          |

| Group                                                        | Main rights                                                                                                             |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| ls_qms: Viewer · User · Approver · Manager                   | User writes and submits; Approver approves, rejects and publishes; Manager sets obsolete, reopens, disposes of records. |
| ls_document_management: Viewer · Editor · Approver · Manager | Editor creates documents and versions; named approvers decide; only the Manager publishes and archives.                 |



Model: ls.qms.sop  
Figure 7 — Lifecycle of an ls\_qms controlled document (policy, SOP, work instruction, quality plan).



Model: ls.document.document  
Figure 8 — Lifecycle of a controlled file in ls\_document\_management.

Quality Management
Quality Policy
Quality Objectives
Procedures
Work Instructions
Quality Planning
Quality Records
Reporting
Configuration
7
14
My Company (San Francisco)

New
Standard Operating Procedures
[SOP-0001] Control of Documented Information...
Instructions 1
Attachments 0
1 / 1

New Revision
Set Obsolete
Draft
Under Review
Approved
Published
Send message
Log note
Activity

Title
Control of Documented Information
SOP-0001 - version 1

OWNERSHIP

Author ? Mitchell Admin
Owning Department Quality Assurance (demo)
Reviewers ?
Approved By Mitchell Admin
Training Required ? ☒
Company My Company (San Francisco)

LIFECYCLE

Submitted On
Approved On
Effective Date ? Jun 27
Review Period (Months) ? 24
Next Review Due ? Jun 27, 2028
Review Status ? Up To Date

Purpose
Scope
Definitions
Responsibilities
Procedure
References
Work Instructions
Revision

Define how quality documents are drafted, reviewed, approved, published, revised and withdrawn.

Today
OdooBot 10:40 AM
Standard Operating Procedure created

Figure 9 — A procedure in `ls_qms`: sections, status bar and approval buttons. (Screenshot of the running software (Odoo 19.0 Community, `LS` modules, demonstration data).)

# Drafting, review, approval and publication of quality documents

|                   |                                                                                           |                |       |
|-------------------|-------------------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_qms · ls_document_management                                                           | Version        | _____ |
| Menu              | Quality Management ► Procedures (and Quality Policy, Work Instructions, Quality Planning) | Effective date | _____ |
| Owner (suggested) | Head of Quality Assurance                                                                 | Approved by    | _____ |
| Related SOPs      | SOP-QA-002, SOP-QA-003, SOP-QA-012, SOP-QA-030                                            |                |       |

## 1 Objective

Ensure that every quality document (quality policy, SOP, work instruction, quality plan) is written to a common standard, reviewed and approved by competent people other than its author, and made effective in a controlled way, so that only approved, current instructions are used.

## 2 Scope

All quality documents managed in ls\_qms, and controlled files (forms, drawings, external documents) managed in ls\_document\_management. Excludes records, which follow SOP-QA-004.

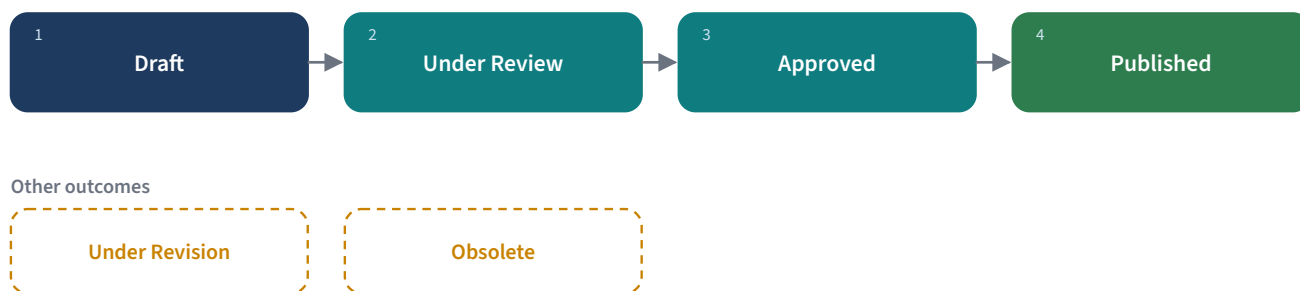
## 3 Responsibilities

| Role (software group)                            | Responsibility                                                                                        |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>QMS User (author)</b>                         | Drafts the document, completes all mandatory sections, proposes reviewers, submits for review.        |
| <b>Reviewers (named on the document)</b>         | Check technical content and consistency with other documents; return comments within the agreed time. |
| <b>QMS Approver</b>                              | Approves or rejects with a reason; publishes the approved revision on its effective date.             |
| <b>QMS Manager</b>                               | Owens the document system, numbering and templates; resolves disputes; withdraws documents.           |
| <b>Document Manager (ls_document_management)</b> | Publishes and archives controlled files once all named approvers have approved.                       |

## 4 Definitions

| Term                     | Definition                                                                                    |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Revision</b>          | A numbered version of a document. Only one revision of a document may be published at a time. |
| <b>Effective date</b>    | Date from which the published revision applies. It cannot precede the approval date.          |
| <b>Reason for change</b> | Mandatory statement explaining why a new revision is created.                                 |

## 5 Workflow in the software



Model: *ls.qms.sop*

Figure 10 — Lifecycle of a quality document in *ls\_qms*.

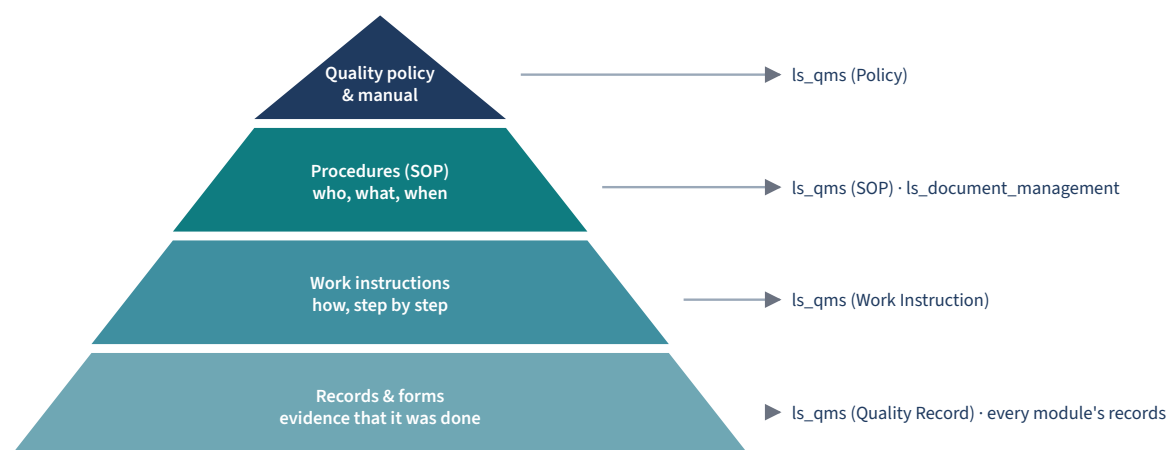


Figure 11 — Where each document type sits in the hierarchy.

## 6 Procedure

- 1 **Check the need.** Search the register (**Quality Management ► Procedures**) to avoid duplicates. For a change to an existing document, use SOP-QA-002.
- 2 **Create the draft.** Click **New**. Enter Title, Owning Department, Review Period (months; the default comes from the QMS parameter, 24 months at installation) and tick *Training Required* when users must be trained before applying the procedure. The reference and version are allocated by the system.
- 3 **Write the content.** Complete Purpose, Scope, Definitions and Abbreviations, Responsibilities, Procedure and Reference Documents. For a work instruction, select the *Parent Procedure*; for a quality plan, add at least one control line; for the policy, write the policy statement.
- 4 **Name the reviewers.** Add the reviewers who cover every affected function (production, QC, engineering, regulatory as relevant).
- 5 **Submit.** Click **Submit for Review**. The document moves to *Under Review* and its content is locked for the author.
- 6 **Review.** Reviewers record comments in the chatter. Blocking comments lead the approver to **Reject** with a written reason; the document returns to Draft.
- 7 **Approve.** A QMS Approver who is not the author clicks **Approve**. The approval date is recorded.
- 8 **Plan the training.** If *Training Required* is ticked, launch SOP-QA-012 before the effective date.
- 9 **Publish.** On the effective date the approver clicks **Publish**. The previous revision, if any, becomes *Obsolete* automatically.
- 10 **Distribute.** Inform the users (chatter, e-mail, briefing). Withdraw any paper copy of the previous revision.

- 11 Controlled files.** For a file in `ls_document_management`: create the document in its folder, upload the file with **New Version**, add at least one approver, **Submit for Review**; each approver decides in **Documents ▶ My Approvals**; the Document Manager then clicks **Publish**.

## 7. Controls enforced by the software

- ✓ Purpose, scope and procedure of an SOP must be filled before it can be submitted.
- ✓ The author of a document cannot approve it (segregation of duties).
- ✓ Only one revision of a document may be published at a time; publishing retires the previous one.
- ✓ The effective date cannot precede the approval date.
- ✓ Published or obsolete content cannot be modified — a new revision is required.
- ✓ Only draft documents can be deleted; others must be made obsolete.
- ✓ A work instruction cannot be published while its parent procedure is not published.
- ✓ `ls_document_management`: a document cannot be submitted without a version (file) and at least one approver; only a Document Manager can publish or archive; each version carries a SHA-256 checksum verified for integrity.

## 8. Good practices

- ★ Use one template per document type and keep the section headings of `ls_qms`.
- ★ Choose reviewers by function, not by availability; two reviewers are usually enough.
- ★ Set the effective date far enough ahead to complete training (SOP-QA-012).
- ★ Write the reason for change even for the first revision ("new procedure").
- ★ Quote the exact software buttons and menus in procedures that use the suite.

## 9. Pitfalls to avoid

- ✗ Publishing before the users are trained.
- ✗ Approving your own document through a second account — a data integrity breach.
- ✗ Keeping uncontrolled copies (desktop, printouts) after a new revision is published.
- ✗ Using the chatter to change content after approval: content is frozen, create a revision.

## 10 Records and evidence

- The document revision with its approval and publication dates
- Chatter history of review comments
- Rejection reasons
- Training records (SOP-QA-012)

## 11 Performance indicators

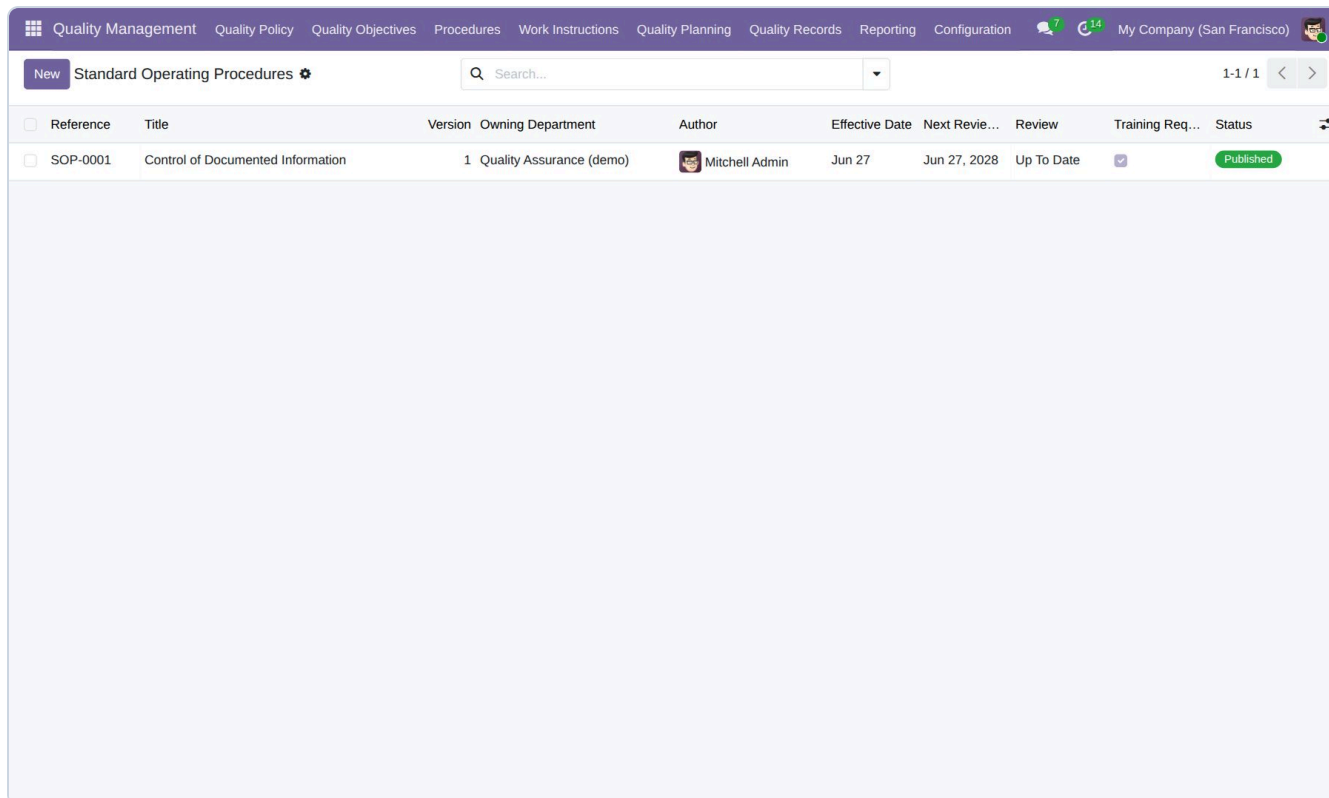
- Median time from submission to approval
- Number of rejections per document
- Documents published without training completed (target 0)

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.100 — Written procedures; deviations
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

### 13. Demonstration

**Case.** The warehouse needs a procedure for temperature mapping. The QA officer creates *SOP – Temperature mapping of storage areas*, sets a 24-month review period and ticks *Training Required*. She names the warehouse manager and the validation engineer as reviewers and clicks **Submit for Review**. The validation engineer asks for the acceptance criterion to be stated in °C; the approver clicks **Reject** with this reason. After correction the document is resubmitted and approved by the Head of QA (not the author). Training sessions are held (SOP-QA-012); on 1 March the approver clicks **Publish**: the SOP becomes effective and its next review date is computed automatically (1 March two years later).



| Quality Management   Quality Policy   Quality Objectives   Procedures   Work Instructions   Quality Planning   Quality Records   Reporting   Configuration   7   14   My Company (San Francisco) |                                   |         |                          |                |                |                |            |                                     |           |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------|--------------------------|----------------|----------------|----------------|------------|-------------------------------------|-----------|--|
| New                                                                                                                                                                                              | Standard Operating Procedures     |         | Search...                |                | 1-1 / 1        |                |            |                                     |           |  |
| Reference                                                                                                                                                                                        | Title                             | Version | Owning Department        | Author         | Effective Date | Next Review... | Review     | Training Req...                     | Status    |  |
| SOP-0001                                                                                                                                                                                         | Control of Documented Information | 1       | Quality Assurance (demo) | Mitchell Admin | Jun 27         | Jun 27, 2028   | Up To Date | <input checked="" type="checkbox"/> | Published |  |

Figure 12 — Procedure register with status and next review date. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Revision, withdrawal and archiving of documents

|                   |                                                               |                |       |
|-------------------|---------------------------------------------------------------|----------------|-------|
| Module            | ls_qms · ls_document_management                               | Version        | _____ |
| Menu              | Quality Management ► Procedures ► New Revision / Set Obsolete | Effective date | _____ |
| Owner (suggested) | QMS Manager                                                   | Approved by    | _____ |
| Related SOPs      | SOP-QA-001, SOP-QA-003, SOP-QA-004, SOP-QA-030                |                |       |

## 1 Objective

Control changes to effective documents so that every change is justified, reviewed and approved, the previous revision is withdrawn from use at the right moment and remains retrievable for the retention period.

## 2 Scope

Published documents of ls\_qms and ls\_document\_management that must be changed, replaced or withdrawn.

## 3 Responsibilities

| Role (software group)            | Responsibility                                                                            |
|----------------------------------|-------------------------------------------------------------------------------------------|
| <b>Document owner / QMS User</b> | Requests the revision, writes the reason for change and the new content.                  |
| <b>QMS Approver</b>              | Approves and publishes the new revision.                                                  |
| <b>QMS Manager</b>               | Withdraws (sets obsolete) documents that are no longer needed; checks references to them. |
| <b>Document Manager</b>          | Starts revisions and archives files in ls_document_management.                            |

## 4 Definitions

| Term                    | Definition                                                                                     |
|-------------------------|------------------------------------------------------------------------------------------------|
| <b>Obsolete</b>         | A document that is no longer applicable. It remains readable for history but must not be used. |
| <b>Archived (files)</b> | End state of a published controlled file; its retention period may start at archiving.         |

## 5 Workflow in the software



Model: ls.document.document

Figure 13 — Lifecycle of a controlled file, including archiving.



Only one revision is published at a time; published content is frozen.

Figure 14 — A revision replaces the previous one, which becomes obsolete.

## 6 Procedure

- 1 **Decide the path.** Minor editorial change or substantive change? Substantive changes affecting product, process or validated state also require a change request (SOP-QA-030).
- 2 **Create the revision.** On the published document click **New Revision**. Enter the *Reason for Change* (mandatory). A new draft revision linked to the previous one is created; the current revision moves to *Under Revision* and remains the applicable revision until the new one is published.
- 3 **Edit and route.** Edit the draft and route it through review and approval as in SOP-QA-001.
- 4 **Publish.** Publishing the new revision retires the previous one to *Obsolete* automatically; its obsolete date is recorded.
- 5 **Withdraw without replacement.** If the document is no longer needed, the QMS Manager clicks **Set Obsolete** after checking that no other published document refers to it (for a SOP, check its work instructions).
- 6 **Controlled files.** In *Is\_document\_management*, click **Start Revision** on a published document, add a new version with **New Version**, then submit it for approval. To withdraw a file, the Document Manager clicks **Archive**.
- 7 **Communicate.** Announce the change and its effective date; collect paper copies of the old revision.

### 7. Controls enforced by the software

- ✓ A reason for change is required to create a new revision.
- ✓ Only one revision may be published; publishing retires the previous one.
- ✓ Obsolete and published content cannot be edited.
- ✓ Only draft documents can be deleted; published documents are made obsolete, never deleted.
- ✓ *Is\_document\_management*: only published documents can enter revision or be archived; a published document cannot be deleted; versions are immutable.

### 8. Good practices

- ★ Summarise what changed in the reason for change ("§5.3 temperature limit changed from 25 °C to 30 °C").
- ★ Check the documents that refer to the revised one; revise them together if necessary.
- ★ Keep the effective date and the training of the new revision aligned (SOP-QA-012).

### 9. Pitfalls to avoid

- ✗ Making a substantive change without a change request.
- ✗ Setting a SOP obsolete while its work instructions are still published.
- ✗ Leaving the old revision printed at the workstation.



## 10 Records and evidence

- Chain of revisions (Previous Revision / Following Revisions)
- Reason for change
- Obsolete date
- Archived file versions with checksums

## 11 Performance indicators

- Revisions per document per year
- Documents with overdue review
- Number of obsolete documents still referenced

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.180 to 211.198 — Records and reports

### 13. Demonstration

**Case.** An audit shows that the cleaning SOP still names a detergent that is no longer purchased. The owner opens the published SOP, clicks **New Revision** and writes: "\$6.2 detergent X replaced by detergent Y (change CC-2026-014)". The new draft is reviewed and approved. On publication, revision 3 becomes *Published* and revision 2 becomes *Obsolete* with today's date, and remains available for inspection.

# Periodic review of documents

|                   |                                                            |                |       |
|-------------------|------------------------------------------------------------|----------------|-------|
| Module            | ls_qms                                                     | Version        | _____ |
| Menu              | Quality Management ► Reporting ► Procedures Due for Review | Effective date | _____ |
| Owner (suggested) | QMS Manager                                                | Approved by    | _____ |
| Related SOPs      | SOP-QA-001, SOP-QA-002, SOP-QA-004                         |                |       |

## 1 Objective

Confirm at defined intervals that every effective document is still accurate, applicable and consistent with practice, regulations and other documents.

## 2 Scope

All published ls\_qms documents (policy, SOP, work instructions, quality plans) and controlled files whose owner defines a review.

## 3 Responsibilities

| Role (software group) | Responsibility                                                                                             |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| <b>Document owner</b> | Performs the review and records its conclusion.                                                            |
| <b>QMS Manager</b>    | Monitors the list of documents due, escalates overdue reviews, reports the indicator to management review. |
| <b>QMS Approver</b>   | Approves the revision when the review concludes that a change is needed.                                   |

## 4 Definitions

| Term                   | Definition                                                                   |
|------------------------|------------------------------------------------------------------------------|
| <b>Review period</b>   | Number of months between two reviews (field <i>Review Period (Months)</i> ). |
| <b>Next review due</b> | Computed by the system: effective date + review period.                      |

## 5 Workflow in the software



Only one revision is published at a time; published content is frozen.

Figure 15 — Periodic review cycle computed from the effective date.

## 6 Procedure

- List the documents due.** Each month open [Quality Management ► Reporting ► Procedures Due for Review](#) and assign each document to its owner, with a due date.
- Review the content.** The owner checks the document against current practice, recent deviations, CAPA, changes, audit findings and regulatory updates.

- 3 **Conclude.** Either *no change needed* or *revision needed*. Record the conclusion as a quality record of type *Document Review Record* (SOP-QA-004) that refers to the document.
- 4 **Revise if needed.** Create a new revision (SOP-QA-002). The review date restarts from the effective date of the new revision.
- 5 **No change.** If no change is needed, the review period restarts only when a new revision becomes effective; to reset the next review date without content change, publish a revision whose reason for change states "*periodic review — no change*", according to your policy.
- 6 **Escalate.** Reviews overdue by more than one month are reported to the Head of QA and to management review.

### 7. Controls enforced by the software

- ✓ The next review date is computed from the effective date and the review period; the period cannot be negative.
- ✓ Published content cannot be edited in place — any change goes through a new revision.

### 8. Good practices

- ★ Plan reviews in batches by process so related documents are reviewed together.
- ★ Use the review to simplify: remove duplicated or unused documents.
- ★ Decide in your document policy how a "no change" review is evidenced (review record only, or a new revision).

### 9. Pitfalls to avoid

- ✗ Signing off a review without reading the document against current practice.
- ✗ Letting documents drift years past their review date.

## 10 Records and evidence

- Quality record *Document Review Record* per review
- New revision when applicable

## 11 Performance indicators

- % of documents reviewed on time
- Number of documents overdue by more than 3 months (target 0)

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** In January, the report lists 12 procedures due. The QMS Manager assigns them. For SOP-QA-040 the owner finds that the acknowledgement time was shortened from 5 to 2 days by the complaint category configuration; she creates a new revision to align the text. For SOP-MET-001 no change is needed; she records a *Document Review Record* stating "reviewed against CAPA-2025-031 and change CC-2025-044 — no change".

# Management and retention of quality records

|                   |                                                                                          |                |       |
|-------------------|------------------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_qms · ls_document_management                                                          | Version        | _____ |
| Menu              | Quality Management ► Quality Records · Documents<br>► Configuration ► Retention Policies | Effective date | _____ |
| Owner (suggested) | QMS Manager                                                                              | Approved by    | _____ |
| Related SOPs      | SOP-QA-001, SOP-DI-004                                                                   |                |       |

## 1 Objective

Ensure that quality records are complete, protected against unauthorised change, retained for the required period and disposed of only after that period, under control.

## 2 Scope

Quality records of ls\_qms (management review, meeting minutes, process verification, document review, objective review, customer feedback, internal communication, QMS change evaluation) and controlled files with a retention policy. Records produced by the other modules (batch records, deviations, calibrations...) are retained in their module and follow the same principles.

## 3 Responsibilities

| Role (software group)   | Responsibility                                                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>QMS User</b>         | Creates and completes the record; confirms it.                                                                |
| <b>QMS Manager</b>      | Sets retention periods, corrects confirmed records when justified (logged), archives and disposes of records. |
| <b>Document Manager</b> | Maintains retention policies and legal holds of controlled files.                                             |

## 4 Definitions

| Term                    | Definition                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------|
| <b>Retention period</b> | Minimum time a record must be kept (months). <i>Retain Until</i> is computed from the record date. |
| <b>Legal hold</b>       | Flag that suspends the end of retention of a controlled file (litigation, inspection).             |
| <b>Disposal</b>         | Controlled destruction of a record after its retention period.                                     |

## 5 Workflow in the software



Model: ls.qms.quality\_record

Figure 16 — Lifecycle of a quality record.

## 6 Procedure

- 1 **Define retention rules.** The QMS Manager lists retention periods per record type from the regulations applicable to the products (for example batch-related records, medical device records) and configures them: *Retention Period (Months)* on quality records; in **Documents ► Configuration ► Retention Policies** the duration, unit, trigger (publication or archiving date), end-of-retention action (notify or archive) and advance notice.
- 2 **Create the record.** In **Quality Management ► Quality Records** click **New**, choose the *Record Type*, date, department, description, conclusion and the related policy, procedure, plan or objective. Attach supporting files.
- 3 **Confirm.** When complete, click **Confirm**. From then on only a QMS Manager can correct it, and the correction is logged.
- 4 **Archive.** When the record is no longer active, click **Archive**.
- 5 **Monitor retention.** Use **Quality Management ► Reporting ► Records Past Retention** and the retention notices of *Is\_document\_management*.
- 6 **Dispose.** After the *Retain Until* date and checking there is no legal hold, open inspection or litigation, the QMS Manager clicks **Dispose** and records the disposal decision.

### 7. Controls enforced by the software

- ✓ Only draft quality records can be deleted; others must be retained until the end of their retention period.
- ✓ Confirmed records can only be corrected by a QMS Manager, and the correction is logged.
- ✓ Only confirmed records can be archived; only archived records can be disposed of, and not before *Retain Until*.
- ✓ Only a QMS Manager can dispose of records.
- ✓ *Is\_document\_management*: retention duration limited to 200 years; legal hold available on documents.

### 8. Good practices

- ★ Base retention periods on the longest applicable requirement (product lifetime, market, contracts).
- ★ Record the disposal decision itself as a quality record.
- ★ Test periodically that archived records can still be read (see SOP-DI-004 for audit trail data).

### 9. Pitfalls to avoid

- ✗ Disposing of records needed for an ongoing investigation or litigation.
- ✗ Keeping records in personal drives outside the controlled system.

## 10 Records and evidence

- Quality records with state history
- Disposal decisions
- Retention policy configuration

## 11 Performance indicators

- Records past retention not yet processed
- Records confirmed more than 30 days after the activity

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.180 to 211.198 — Records and reports
- ISO 13485:2016 — Medical devices — Quality management systems
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

### 13. Demonstration

**Case.** After the quarterly quality meeting, the secretary creates a quality record of type *Quality Meeting Minutes*, links it to the objective *Complaint closure time*, attaches the signed attendance list and clicks **Confirm**. The retention period is 120 months; *Retain Until* is computed automatically. Ten years later the record appears in *Records Past Retention*; the QMS Manager checks that no legal hold applies, clicks **Archive** then **Dispose**.

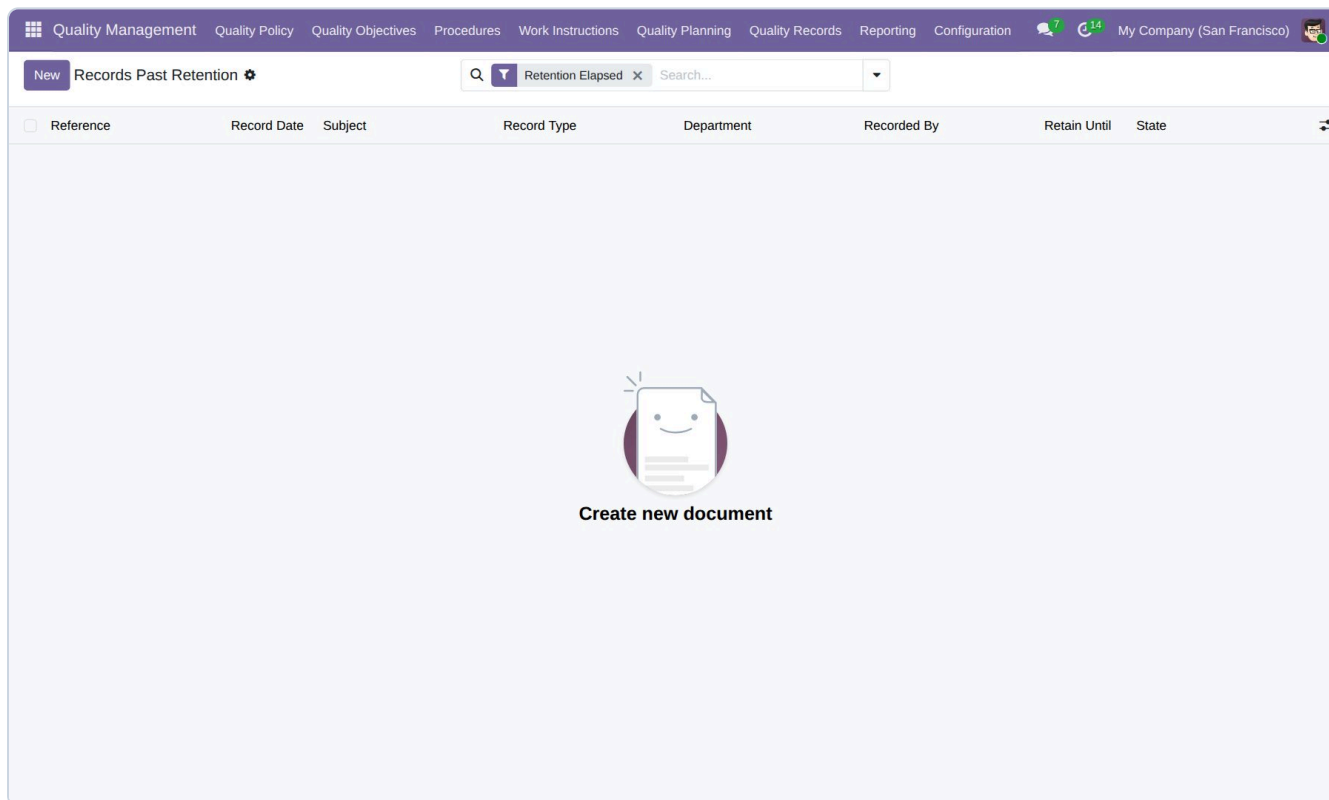


Figure 17 — Quality records with their retention date. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

| <div> <div>Documents Documents My Approvals Versions Linked Records Configuration</div> <div> <div>7</div> <div>14</div> </div> <div>My Company (San Francisco)</div> </div> |                                         |              |               |                   |                          |                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------------|---------------|-------------------|--------------------------|------------------------------------------------------------------------------------------------|
| <div> <div>New Retention Policies ⚙</div> <div>Search...</div> <div>1-1 / 1 &lt; &gt;</div> </div>                                                                           |                                         |              |               |                   |                          |                                                                                                |
| <input type="checkbox"/> Code                                                                                                                                                | Policy Name                             | Retention... | Duration Unit | Retention Trigger | End of Retention Action  | Advance...  |
| <input type="checkbox"/> GMP-5Y                                                                                                                                              | GMP records - 5 years after publication |              | 5 Years       | Publication Date  | Notify Document Managers | 90                                                                                             |
|                                                                                                                                                                              |                                         |              |               |                   |                          |                                                                                                |

Figure 18 — Retention policies of controlled files. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Quality objectives and measurement of their achievement

|                   |                                         |                |       |
|-------------------|-----------------------------------------|----------------|-------|
| Module            | ls_qms                                  | Version        | _____ |
| Menu              | Quality Management ► Quality Objectives | Effective date | _____ |
| Owner (suggested) | Head of Quality / Management            | Approved by    | _____ |
| Related SOPs      | SOP-GEN-001, SOP-GEN-002, SOP-QA-004    |                |       |

## 1 Objective

Translate the quality policy into measurable objectives, follow them with dated measurements and conclude on their achievement at the end of the period.

## 2 Scope

All quality objectives of the company and its departments recorded in ls\_qms.

## 3 Responsibilities

| Role (software group)                | Responsibility                                                                   |
|--------------------------------------|----------------------------------------------------------------------------------|
| <b>Top management</b>                | Sets the objectives consistent with the quality policy and allocates resources.  |
| <b>Objective owner (Responsible)</b> | Records measurements, explains deviations from the trajectory, proposes actions. |
| <b>QMS Manager</b>                   | Maintains the list of objectives, prepares the review for management review.     |

## 4 Definitions

| Term                                 | Definition                                                                      |
|--------------------------------------|---------------------------------------------------------------------------------|
| <b>Direction</b>                     | <i>Increase, Decrease or Maintain</i> : the expected movement of the indicator. |
| <b>Baseline / Target / Tolerance</b> | Starting value, value to reach, and accepted margin.                            |
| <b>Performance status</b>            | Computed: No Measurement, On Track, At Risk, Off Track, Target Reached.         |



## 5 Workflow in the software



Model: *ls.qms.objective*

Figure 19 — Lifecycle of a quality objective.

## 6 Procedure

- 1 Define.** In **Quality Management ► Quality Objectives** click **New**. Enter the objective, the related *Quality Policy*, department, responsible, start and target dates, direction, unit, baseline, target and tolerance.
- 2 Start.** Click **Start** at the beginning of the period.
- 3 Measure.** Add dated measurements at the planned frequency (monthly for example). The current value, achievement % and performance status are computed.
- 4 React.** When the status becomes *At Risk* or *Off Track*, the owner records the analysis in the chatter and, if the cause is systemic, raises a CAPA (SOP-QA-022).
- 5 Close.** At the target date click **Close as Achieved** or **Close as Not Achieved**. Record the conclusion as a quality record of type *Quality Objective Review*.
- 6 Review.** Present all objectives in management review (SOP-GEN-001) and set the next period's objectives.

### 7. Controls enforced by the software

- ✓ For an *Increase* objective the target must be greater than the baseline; for *Decrease* it must be lower.
- ✓ A measurement cannot be dated before the start date of the objective.
- ✓ Only draft objectives can be started or deleted; others are cancelled.
- ✓ An objective must be in progress to be closed.

### 8. Good practices

- ★ Use SMART objectives: one indicator, one number, one date, one owner.
- ★ Limit the number of objectives so that each receives attention.
- ★ Record the data source of each measurement in its note.

### 9. Pitfalls to avoid

- ✗ Changing the target during the period to make it achievable — close and create a new objective instead.
- ✗ Objectives without a link to the policy or to a real process risk.

## 10 Records and evidence

- Objective with measurements and closure
- Quality Objective Review records
- Management review minutes

## 11 Performance indicators

- % of objectives achieved
- Objectives Off Track without action for more than one month

## 12 References

- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- ICH Q10 — Pharmaceutical Quality System

### 13. Demonstration

**Case.** Objective *Reduce the median complaint closure time*: direction Decrease, unit days, baseline 48, target 30, tolerance 3, from January to December. Monthly measurements: 46, 44, 41... In June the value is 39 and the status turns *At Risk*; the owner notes that investigations wait for samples and raises CAPA-2026-022. In December the measurement is 29: the owner clicks **Close as Achieved**.

The screenshot displays the Odoo Quality Management interface. The top navigation bar includes links for Quality Management, Quality Policy, Quality Objectives, Procedures, Work Instructions, Quality Planning, Quality Records, Reporting, Configuration, and a user profile for 'My Company (San Francisco)'. The main header shows 'New Objective Analysis [OBJ-0001] Reduce the number of customer co...' with a 'Measurements' icon and '1 / 1' page indicator.

Below the header, there are buttons for 'Close as Achieved', 'Close as Not Achieved', and 'Cancel'. The 'In Progress' status is highlighted. To the right, there are buttons for 'Send message', 'Log note', and 'Activity'.

The objective details are as follows:

- Objective:** Reduce the number of customer complaints
- OBJ-0001**
- OWNERSHIP:**
  - Quality Policy: [POL-0001] Quality Policy of the Site (...)
  - Department: Quality Assurance (demo)
  - Responsible: Mitchell Admin
  - Company: My Company (San Francisco)
- PERIOD:**
  - Start Date: Mar 27
  - Target Date: Mar 27, 2027
- INDICATOR:**
  - Direction: Decrease
  - Unit: complaints per quarter
  - Baseline: 20.00
  - Target: 8.00
- PERFORMANCE:**
  - Current Value: 12.00
  - Last Measurement Date: Sep 12
  - Achievement (%): 66.67
  - Performance Status: On Track

At the bottom, there is a 'Measurements' table with the following data:

| Measureme... | Value | Recorded By    | Comment                     |
|--------------|-------|----------------|-----------------------------|
| Sep 12       | 12.00 | Mitchell Admin | Second quarter measurement. |

On the right side, there is a chat window with a message from 'OdooBot' at 10:40 AM: 'Quality Objective created'.

Figure 20 — Quality objective with baseline, target and measurements. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Training

## ls\_training

ls\_training manages a controlled course catalogue (Draft → Under Review → Approved → Obsolete), sessions with attendance and assessment scores, certifications issued automatically when a session closes, training requirements by job position, department or employee, a training matrix, and competency assessments that are distinct from attendance. Certification status (Valid, Expiring Soon, Expired, Revoked) is refreshed nightly and reminders are sent before expiry.

| Menu                                                                                                                                    | Content                             |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <a href="#">Training</a> ► <a href="#">Configuration</a> ► <a href="#">Courses / Competencies / Training Requirements</a>               | Catalogue and rules                 |
| <a href="#">Training</a> ► <a href="#">Operations</a> ► <a href="#">Sessions / Attendance / Certifications / Competency Assessments</a> | Delivery and evidence               |
| <a href="#">Training</a> ► <a href="#">Operations</a> ► <a href="#">Expiring and Expired</a>                                            | Certifications to renew             |
| <a href="#">Training</a> ► <a href="#">Reporting</a> ► <a href="#">Training Matrix</a>                                                  | Employee × course compliance matrix |

| Group            | Main rights                                                    |
|------------------|----------------------------------------------------------------|
| Learner · Viewer | Read own records / all records                                 |
| Trainer          | Runs sessions, records attendance and scores                   |
| Manager          | Approves courses, manages requirements, revokes certifications |

### Regulatory position stated by the module

The module README states that ls\_training does not implement 21 CFR Part 11 (no electronic signature, no field-level audit trail of its own). If training records are GxP records in your organisation, cover them with ls\_audit\_trail rules (SOP-DI-001) and state the approach in your validation.

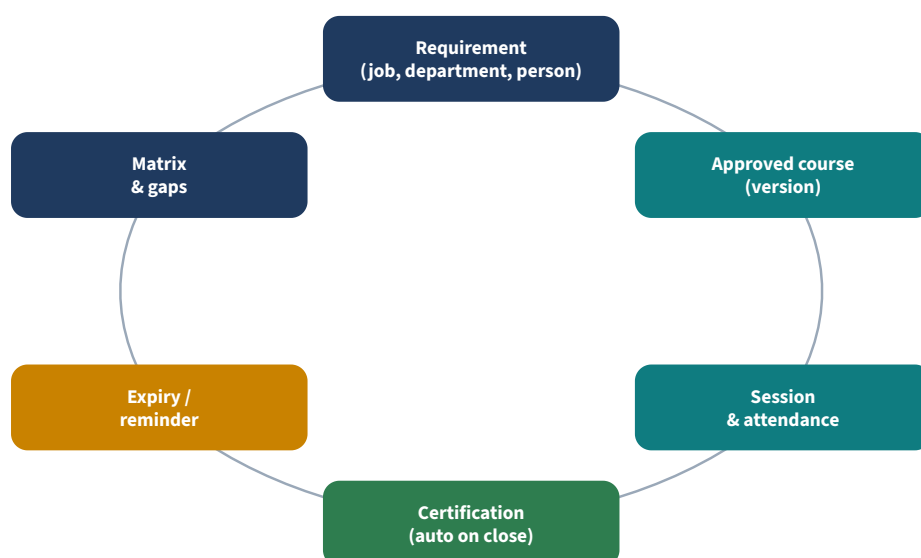
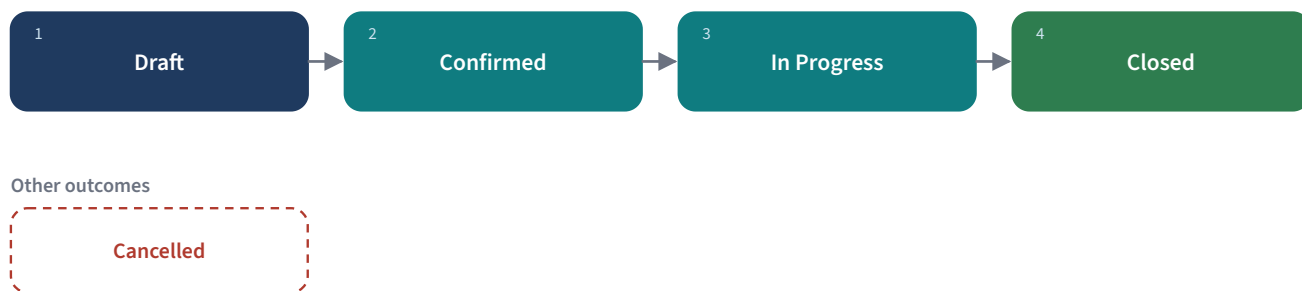


Figure 21 — The training loop.



Model: `ls.training.session`

Figure 22 — Lifecycle of a training session.

| Training Courses |                               |         |             |               |                |               |             |          |                            |
|------------------|-------------------------------|---------|-------------|---------------|----------------|---------------|-------------|----------|----------------------------|
| Course Code      | Course Title                  | Version | Course Type | Delivery Mode | Duration (H... | Certificat... | Session ... | Status   | Company                    |
| TRN/CRS/DEMO1    | GMP Fundamentals              | 2.0     | GMP / GxP   | Classroom     | 4.00           | 24            | 1           | Approved | My Company (San Francisco) |
| TRN/CRS/DEMO2    | Aseptic Gowning Qualification | 1.0     | Technical   | On-the-Job    | 2.00           | 12            | 0           | Approved | My Company (San Francisco) |
| TRN/CRS/DEMO3    | Site Induction                | 1.0     | Induction   | Self-Study    | 1.00           | 0             | 0           | Approved | My Company (San Francisco) |
|                  |                               |         |             |               | 7.00           |               |             |          |                            |

Figure 23 — Course catalogue. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Creation and approval of training courses

|                   |                                    |                |       |
|-------------------|------------------------------------|----------------|-------|
| Module            | ls_training                        | Version        | _____ |
| Menu              | Training ► Configuration ► Courses | Effective date | _____ |
| Owner (suggested) | Training Manager                   | Approved by    | _____ |
| Related SOPs      | SOP-QA-011, SOP-QA-012, SOP-QA-013 |                |       |

## 1 Objective

Ensure that every course used as evidence of qualification has approved content, learning objectives, a defined assessment rule and a validity period.

## 2 Scope

All courses of the catalogue: induction, GMP/GxP, procedure/SOP, technical, health and safety, quality system and regulatory courses, whatever the delivery mode.

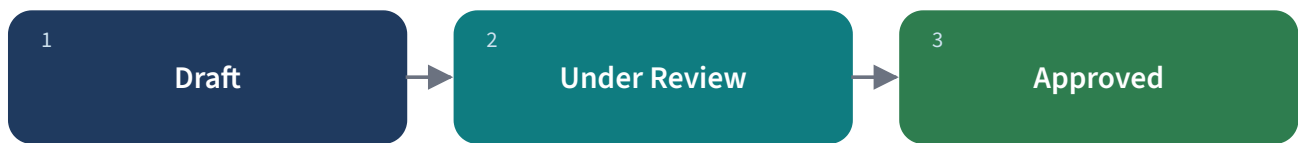
## 3 Responsibilities

| Role (software group)          | Responsibility                                                           |
|--------------------------------|--------------------------------------------------------------------------|
| <b>Course author (Trainer)</b> | Writes the content and objectives, proposes the assessment and validity. |
| <b>Subject-matter reviewer</b> | Checks technical accuracy.                                               |
| <b>Training Manager</b>        | Approves the course, makes obsolete courses that are no longer valid.    |

## 4 Definitions

| Term                          | Definition                                                                                      |
|-------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Delivery mode</b>          | Classroom, On-the-Job, Self-Study or External e-Learning (an e-Learning URL is then mandatory). |
| <b>Certification validity</b> | Months during which the certification stays valid; 0 means no expiry.                           |
| <b>Pass score</b>             | Score (%) to reach when the course requires an assessment.                                      |

## 5 Workflow in the software



Other outcomes



Model: *ls.training.course*

Figure 24 — Lifecycle of a course.

## 6 Procedure

- 1 **Create.** In **Training ► Configuration ► Courses** click **New**. Enter name, code, version, course type, delivery mode, duration (hours), learning objectives and content.
- 2 **Define assessment.** Tick *Requires Assessment* and set the pass score if knowledge must be tested (for example a written quiz for a critical SOP).
- 3 **Set validity.** Enter the certification validity in months according to the risk (for example 12 months for aseptic behaviour, 0 for a one-off induction module).
- 4 **Link competencies.** Link the competencies the course contributes to.
- 5 **Submit and approve.** Click **Submit for Review**; the Training Manager checks and clicks **Approve**. Only approved courses can be scheduled.
- 6 **Retire.** When the course is replaced, close or cancel its open sessions, then click **Make Obsolete**.

### 7. Controls enforced by the software

- ✓ Duration must be greater than zero; the pass score must lie between 0 and 100; validity cannot be negative.
- ✓ An External e-Learning course requires an e-Learning URL.
- ✓ Only a draft course can be submitted; only a course under review can be approved; only an approved course can be made obsolete.
- ✓ A course with open sessions cannot be made obsolete.
- ✓ The course version in force is frozen onto every certification issued.

### 8. Good practices

- ★ For a SOP course, use the SOP reference as course code and its revision as course version.
- ★ Write measurable learning objectives ("The operator can perform a line clearance").
- ★ Prefer on-the-job training with competency assessment for practical skills.

### 9. Pitfalls to avoid

- ✗ Using the same course for several SOP revisions without changing its version.
- ✗ A pass score set so low that the assessment proves nothing.

## 10 Records and evidence

- Approved course with version
- Course content

## 11 Performance indicators

- Courses under review for more than 30 days
- Courses without assessment for critical tasks

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

### 13. Demonstration

**Case.** For SOP-PRD-002 *Line clearance*, the trainer creates course code *SOP-PRD-002*, version 3, type *Procedure / SOP*, delivery *On-the-Job*, duration 1 h, with an assessment (pass score 100 %: the trainee must perform the clearance without error) and a 24-month validity. The Training Manager approves it.

The screenshot shows the Odoo 19.0 Community interface for the 'Training Courses' module. The top navigation bar includes 'Training', 'Operations', 'Reporting', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main header shows 'New Training Courses' with a search bar containing '[TRN/CRS/DEMO1] GMP Fundamentals (v2.0)'. Below the header, there are tabs for 'Sessions' (1) and 'Certifications' (0). The course form is displayed with the following details:

| IDENTIFICATION |                            | DELIVERY                          |           |
|----------------|----------------------------|-----------------------------------|-----------|
| Course Code ?  | TRN/CRS/DEMO1              | Delivery Mode                     | Classroom |
| Version ?      | 2.0                        | Duration (Hours)                  | 04:00     |
| Course Type    | GMP / GxP                  | Certification Validity (Months) ? | 24        |
| Company        | My Company (San Francisco) |                                   |           |

**ASSESSMENT**

Requires Assessment ? ☒

Pass Score (%) 80.00

**Content** Competencies Training Requirements

Understand the principles of Good Manufacturing Practice and their application on the shop floor.

Course content

On the right side of the form, there is a chat window with a message from 'OdooBot' at 10:40 AM: 'Training Course created'.

Figure 25 — Course form: type, delivery mode, validity, pass score. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Session planning and issue of certifications

|                   |                                  |                |       |
|-------------------|----------------------------------|----------------|-------|
| Module            | ls_training                      | Version        | _____ |
| Menu              | Training ► Operations ► Sessions | Effective date | _____ |
| Owner (suggested) | Training Manager                 | Approved by    | _____ |
| Related SOPs      | SOP-QA-010, SOP-QA-013           |                |       |

## 1 Objective

Plan and deliver training sessions so that attendance and assessment results are recorded as evidence and certifications are issued only to people who met the course rule.

## 2 Scope

All sessions of approved courses, internal or external trainer. Externally delivered training may also be recorded as a manual certification.

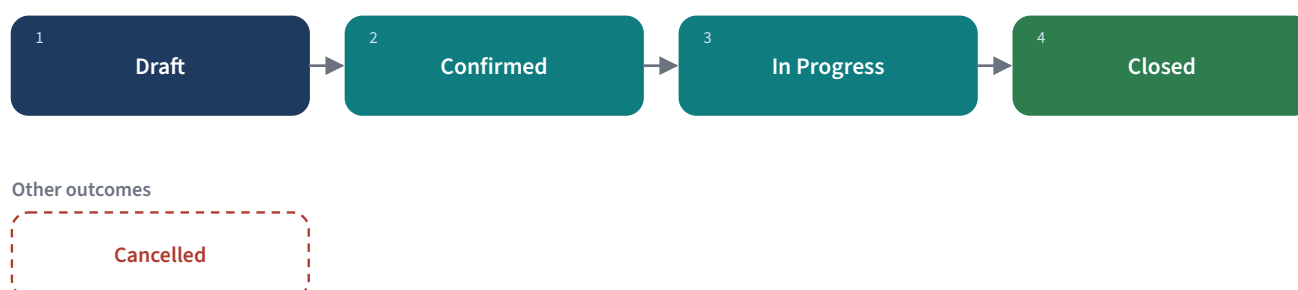
## 3 Responsibilities

| Role (software group)                 | Responsibility                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Training coordinator (Trainer)</b> | Creates sessions, registers employees, records attendance and scores, closes the session. |
| <b>Trainer</b>                        | Delivers the session and assesses attendees.                                              |
| <b>Training Manager</b>               | Supervises, revokes certifications when needed.                                           |

## 4 Definitions

| Term                 | Definition                                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Outcome</b>       | Computed from recorded evidence: presence and, when required, score $\geq$ pass mark. Absence yields <i>Pending</i> , not <i>Failed</i> . |
| <b>Certification</b> | Append-only evidence issued at session closure; renewal creates a new record.                                                             |

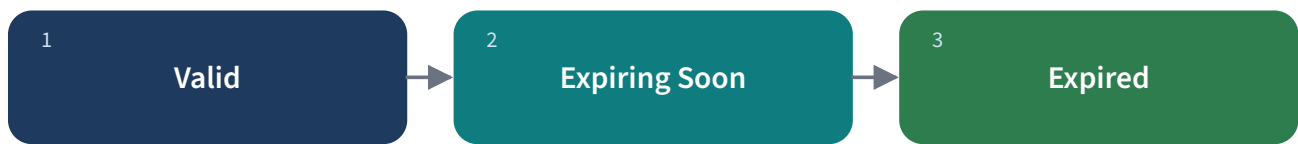
## 5 Workflow in the software



Model: ls.training.session

Figure 26 — Lifecycle of a session.





Other outcomes



Model: *ls.training.certification*

Figure 27 — Statuses of a certification.

## 6 Procedure

- 1 **Create the session.** In **Training ► Operations ► Sessions** click **New**, select the approved course, start and end, trainer (internal or external), location and maximum attendees.
- 2 **Register.** Click **Register Employees**: select employees, a department, a job position or a training requirement; optionally exclude employees who already hold a valid certification.
- 3 **Confirm.** Click **Confirm** once the list is final (at least one attendee).
- 4 **Deliver.** Click **Start** at the beginning; in the *Attendees* tab tick *Attended* for each person present and enter the assessment *Score*.
- 5 **Close.** Click **Close and Issue Certifications**. Each attendee who meets the rule receives a certification carrying the course version; validity is computed from the course.
- 6 **Handle absentees.** Absent employees remain *Pending*; register them in a later session.
- 7 **Revoke if needed.** If a certification was issued wrongly, a Manager opens it and clicks **Revoke** with a documented reason. Certifications are never deleted.

### 7. Controls enforced by the software

- ✓ A session must end after it starts and cannot exceed its maximum attendees.
- ✓ A session names either an internal or an external trainer, not both.
- ✓ Only a draft session can be confirmed and it needs at least one attendee; only a confirmed session can start; only a session in progress can be closed.
- ✓ A session cannot be closed while an attendee has no recorded outcome.
- ✓ Attendance scores lie between 0 and 100; an employee of another company cannot attend.
- ✓ Once closed, attendance evidence and the course, dates and trainer are frozen; the session can no longer be cancelled.
- ✓ Certifications cannot be deleted; revocation needs a reason.

### 8. Good practices

- ★ Keep the signed attendance sheet of classroom sessions as an attachment.
- ★ Record the score as soon as the assessment is corrected, not days later.
- ★ For external training, attach the provider's certificate to the manual certification.

## 9. Pitfalls to avoid

- ✗ Marking an employee attended because "he read the SOP" without assessment when the course requires one.
- ✗ Closing the session before scores are entered.

## 10 Records and evidence

- Session with attendance and scores
- Certifications (PDF Training Certificate)
- Individual Training Record (PDF per employee)

## 11 Performance indicators

- Sessions closed more than 5 days after the end
- Pass rate per course
- Certifications revoked

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems

## 13. Demonstration

**Case.** Session *SOP-PRD-002 v3 — 12 May* for the packaging team: 8 employees registered through the job position *Packaging operator*, 2 excluded because already certified. After the practical assessment, 7 score 100 %; one scores 80 % and fails. On **Close and Issue Certifications**, 7 certifications valid 24 months are issued; the failed employee is registered in the next session.

The screenshot displays the Odoo 19.0 Community interface for Training Sessions. The top navigation bar includes 'Training', 'Operations', 'Reporting', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main view is for a specific training session, 'TRN/SES/2026/00001 - GMP Fundamentals'. The session is currently in the 'Draft' stage, with options to 'Confirm' or 'Cancel'. The session details are as follows:

| COURSE  |                                         | SCHEDULE            |                  |
|---------|-----------------------------------------|---------------------|------------------|
| Course  | [TRN/CRS/DEMO1] GMP Fundamentals (v2... | Start               | Sep 20, 8:00 AM  |
| Company | My Company (San Francisco)              | End                 | Sep 20, 12:00 PM |
|         |                                         | Location            | Training Room 1  |
|         |                                         | Maximum Attendees ? | 20               |

| TRAINER            |  | RESULTS    |   |
|--------------------|--|------------|---|
| Internal Trainer   |  | Registered | 1 |
| External Trainer ? |  | Passed     | 1 |
|                    |  | Failed     | 0 |

Below the results, there is a section for 'Attendees' and 'Notes'. The 'Attendees' table shows the following data:

| Employee       | Attended                            | Assessment... | Result |
|----------------|-------------------------------------|---------------|--------|
| Mitchell Admin | <input checked="" type="checkbox"/> | 92.00         | Passed |

The right sidebar shows a chat window with 'OdooBot' at 10:40 AM, stating 'Training Session created'.

Figure 28 — Session with attendees, outcomes and issued certifications. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Training linked to the publication of a new document

|                   |                                                                                    |                |       |
|-------------------|------------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_qms · ls_training                                                               | Version        | _____ |
| Menu              | Quality Management ► Procedures · Training ► Configuration ► Training Requirements | Effective date | _____ |
| Owner (suggested) | QMS Manager with the Training Manager                                              | Approved by    | _____ |
| Related SOPs      | SOP-QA-001, SOP-QA-010, SOP-QA-011, SOP-QA-013                                     |                |       |

## 1 Objective

Ensure that nobody applies a new or revised procedure before being trained on it, and that the training evidence refers to the exact revision.

## 2 Scope

Every ls\_qms document with *Training Required* ticked, and every revision of such a document.

## 3 Responsibilities

| Role (software group)   | Responsibility                                                                             |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Document author</b>  | Ticks <i>Training Required</i> , identifies the target population and the training method. |
| <b>Training Manager</b> | Creates or revises the course, the requirement and the sessions.                           |
| <b>QMS Approver</b>     | Publishes only when the training completion criterion is met.                              |
| <b>Line managers</b>    | Release their staff for training; do not assign untrained staff to the task.               |

## 4 Definitions

| Term                        | Definition                                                                                                      |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Target population</b>    | Job positions, departments or named employees who apply the document.                                           |
| <b>Completion criterion</b> | Proportion of the target population to be trained before publication (for example 100 % of the people on duty). |

## 5 Procedure

- 1 Flag the need.** In the draft document, tick *Training Required* and describe in the reason for change what is new.
- 2 Create or version the course.** On approval, the Training Manager creates the course (or a new version of it) with code = document reference, version = document revision (SOP-QA-010).
- 3 Define the requirement.** In **Training ► Configuration ► Training Requirements** create the requirement: required course, job position / department / employee, *Mandatory*, and a grace period in days for newcomers.
- 4 Train.** Run sessions (SOP-QA-011). Use *Register Employees* by training requirement so nobody is forgotten.
- 5 Check completion.** Generate the matrix (**Training ► Reporting ► Training Matrix**) filtered on the course and check the gaps.
- 6 Publish.** The approver publishes the document once the completion criterion is met (SOP-QA-001) and records the training status in the chatter.

**7 Catch up.** Employees absent at publication are trained before performing the task; line managers enforce it.

### 6. Controls enforced by the software

- ✓ There is no automatic link between a published document and a training course: the SOP flag *Training Required* is informative, and the course, requirement and sessions are created by people. This SOP is the control.
- ✓ A training requirement must target an employee, a job position or a department, and cannot be duplicated.
- ✓ Only approved courses can be scheduled; the course version is frozen on each certification.

### 7. Good practices

- ★ Use "read and understand" self-study with a short quiz only for minor revisions; use classroom or on-the-job training for substantive changes.
- ★ Plan the effective date after the training capacity is known.
- ★ Keep one course per SOP, versioned, rather than a new course per revision.

### 8. Pitfalls to avoid

- ✗ Publishing on the approval date by default, before anyone is trained.
- ✗ Forgetting night shifts and temporary staff in the target population.

## 9 Records and evidence

- Course version
- Training requirement
- Certifications
- Training matrix extract attached to the publication note

## 10 Performance indicators

- % of target population trained at publication
- Tasks performed by untrained staff (from audits; target 0)

## 11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.25 — Personnel qualifications

### 12. Demonstration

**Case.** SOP-QA-020 revision 5 changes the deviation classification. The author ticks *Training Required*. After approval the Training Manager creates version 5 of course SOP-QA-020 and a requirement for the job positions *QA officer* and *Production supervisor*. Two sessions train 22 of 24 people. The matrix shows two people on leave; the approver publishes with the note "22/24 trained, 2 on leave, trained on return before any deviation assessment".

# Follow-up of qualifications and requalification

|                   |                                                                              |                |       |
|-------------------|------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_training                                                                  | Version        | _____ |
| Menu              | Training ► Reporting ► Training Matrix · Operations ► Competency Assessments | Effective date | _____ |
| Owner (suggested) | Training Manager                                                             | Approved by    | _____ |
| Related SOPs      | SOP-QA-011, SOP-QA-012, SOP-DI-011                                           |                |       |

## 1 Objective

Know at any time who is qualified for which task, renew qualifications before they expire, and assess competence where attendance alone is not enough.

## 2 Scope

All mandatory training requirements, certifications with an expiry, and competencies requiring periodic assessment.

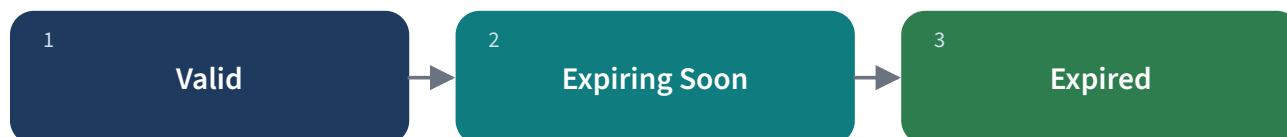
## 3 Responsibilities

| Role (software group)          | Responsibility                                                      |
|--------------------------------|---------------------------------------------------------------------|
| <b>Training Manager</b>        | Runs the matrix monthly, plans renewals, reports gaps.              |
| <b>Line manager / assessor</b> | Performs competency assessments of their staff (not of themselves). |
| <b>Employee</b>                | Follows reminders and attends renewal sessions.                     |

## 4 Definitions

| Term                         | Definition                                                                                                                                                 |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Matrix cell status</b>    | Valid, Expiring Soon, Expired, Revoked or Not Trained for each employee × course.                                                                          |
| <b>Competency assessment</b> | Evidence that a person can do the job: named assessor, demonstrated level (Not Demonstrated, Developing, Proficient, Expert), evidence, confirmation lock. |

## 5 Workflow in the software



Other outcomes



Model: ls.training.certification

Figure 29 — Certification statuses refreshed nightly.

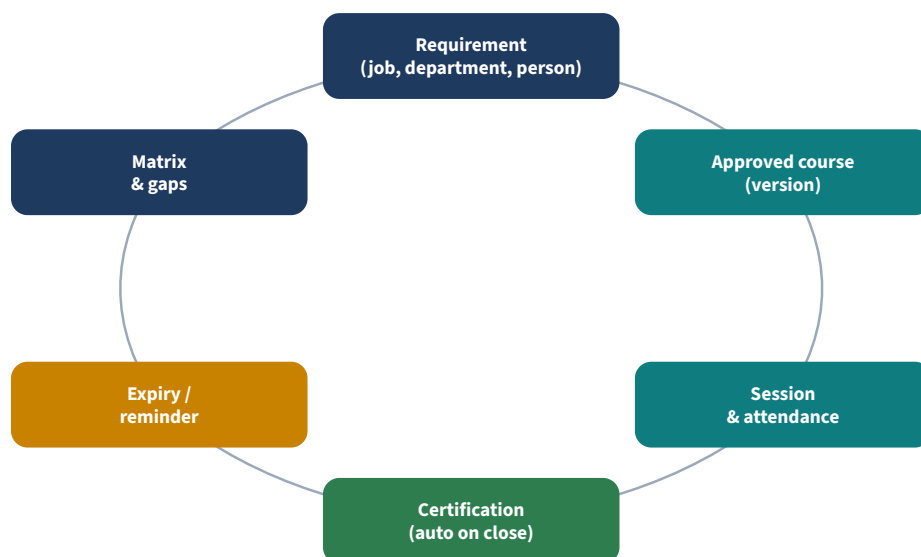


Figure 30 — Requirement, certification, expiry and matrix.

## 6 Procedure

- 1 **Generate the matrix.** Monthly, open [Training ► Reporting ► Training Matrix](#), click [Generate Matrix](#) and apply the *Gaps* filter.
- 2 **Treat expiring certifications.** Use [Training ► Operations ► Expiring and Expired](#); schedule renewal sessions before the expiry date. The system sends daily reminders within the warning window.
- 3 **Act on expired or missing training.** The line manager removes the employee from the task until retrained; record the decision.
- 4 **Assess competence.** In [Training ► Operations ► Competency Assessments](#) create the assessment: employee, competency, assessor, date, demonstrated level, evidence; set the next assessment date when the competency requires reassessment; click [Confirm](#).
- 5 **Report.** Report the compliance rate per department to management review.

### 7. Controls enforced by the software

- ✓ Certification status is computed from the expiry date and refreshed nightly; certifications are never deleted.
- ✓ An employee cannot assess their own competency.
- ✓ A competency assessment cannot be confirmed without evidence and cannot be modified or deleted once confirmed.
- ✓ The next assessment date cannot precede the assessment date.

### 8. Good practices

- ★ Link critical access rights (e.g. batch release) to a valid qualification and review them together (SOP-IT-001).
- ★ Use competency assessments for aseptic operators, analysts and auditors.
- ★ Plan renewals in the annual training plan rather than waiting for reminders.

### 9. Pitfalls to avoid

- ✗ Treating an expired certification as "administrative" and letting the employee keep working.
- ✗ Line managers assessing competencies they have not observed.

## 10 Records and evidence

- Training matrix extracts
- Certifications and renewals
- Competency assessments

## 11 Performance indicators

- Compliance rate (mandatory requirements met)
- Expired certifications of active staff (target 0)
- Competency reassessments overdue

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 2 — Personnel
- 21 CFR 211.25 — Personnel qualifications
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

### 13. Demonstration

**Case.** The March matrix shows 3 *Expired* cells for the course *Aseptic gowning* in the filling team. The Training Manager informs the supervisor, who removes the three operators from Grade A/B work. A renewal session is held 4 days later; the new certifications are valid 12 months. In parallel, the supervisor confirms a competency assessment *Aseptic behaviour — Proficient* for each operator, based on an observed media fill.

# Deviations and CAPA

ls\_deviation · ls\_capa

**ls\_deviation** records departures from approved instructions or specifications: classification (type, category, severity), impact assessment, immediate actions, investigations with a root cause analysis method, product dispositions and closure. Every state change is written in an append-only *Transition Log*. **ls\_capa** manages corrective and preventive actions from any source (audit, deviation, complaint, inspection, management review, risk assessment...): impact assessment, root cause analyses (Five Whys, Ishikawa, FMEA), actions (optionally turned into project tasks), effectiveness checks and closure.

| Menu                                                                                        | Content                                                                |
|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Quality Management ► Deviations ► All Deviations / My Deviations                            | Deviation records                                                      |
| Quality Management ► Deviations ► Immediate Actions / Investigations / Product Dispositions | Sub-records                                                            |
| Quality Management ► Reporting ► Deviation Analysis / Transition Log                        | Analysis and audit trail of transitions                                |
| Quality ► CAPA ► CAPA Records / Root Cause Analyses / Actions / Effectiveness Verification  | CAPA records                                                           |
| Settings: closure target (days) per severity                                                | Configured per company (default 30 days for minor, major and critical) |

| Group                                                 | Main rights                                                                                                                          |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Deviation: Viewer · Reporter · Investigator · Manager | Reporter declares; Investigator assesses and investigates; Manager approves dispositions, closes, and alone may edit closed records. |
| CAPA: Viewer · Investigator · Coordinator · Manager   | Investigator performs analyses and actions; Coordinator drives the record; Manager closes.                                           |

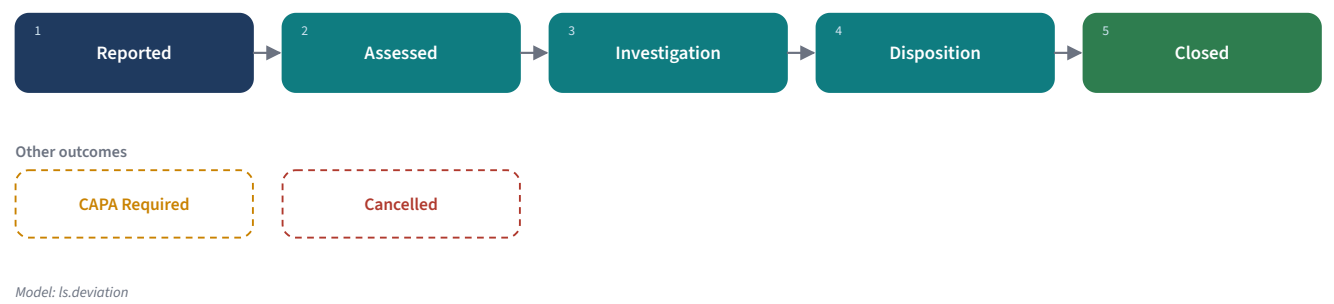


Figure 31 — Lifecycle of a deviation.

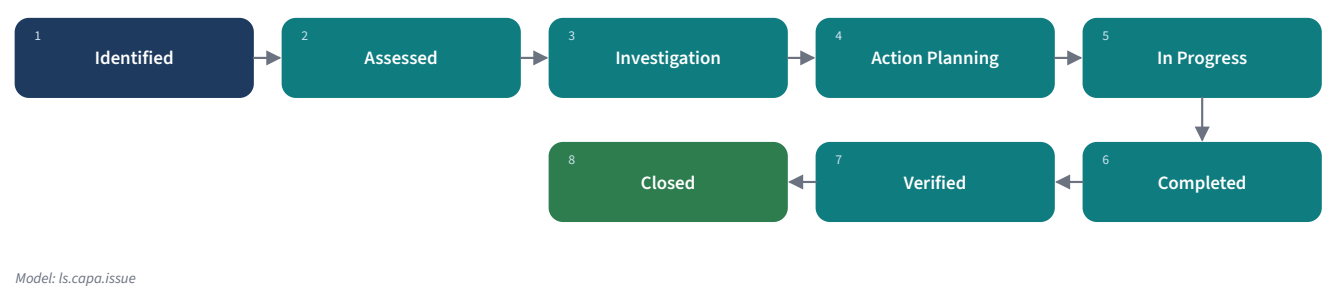


Figure 32 — Lifecycle of a CAPA record.



Quality Management
Deviations
Reporting
Configuration
7
14
My Company (San Francisco)

New
Deviations
DEV/2026/00002
1 / 1

Complete Assessment
Extend Target Date
Cancel
Reported
Assessed
Investigation
Disposition
Closed
Send message
Log note
Activity

### DEV/2026/00002

Storage temperature excursion in the finished goods quarantine are:

| CLASSIFICATION    |                            | CHRONOLOGY AND OWNERSHIP |                                     |
|-------------------|----------------------------|--------------------------|-------------------------------------|
| Deviation Type    | Environmental Excursion    | Occurrence Date          | Sep 24, 10:41 AM                    |
| Category          | Warehouse and Storage      | Detection Date           | Sep 24, 10:41 AM                    |
| Severity          | Critical                   | Report Date              | Sep 27, 10:41 AM                    |
| Planned Deviation | <input type="checkbox"/>   | Target Closure Date      | Sep 26                              |
| Tags              |                            | Is Overdue               | <input checked="" type="checkbox"/> |
| Company           | My Company (San Francisco) | Days Open                | 3                                   |
|                   |                            | Reported By              | OdooBot                             |
|                   |                            | Assigned Investigator    |                                     |
|                   |                            | QA Reviewer              |                                     |
|                   |                            | Department               |                                     |

| SUBJECT OF THE DEVIATION |                       |
|--------------------------|-----------------------|
| Product                  | Equipment             |
| Quantity Affected        | 0.00                  |
| Manufacturing Order      | Affected Lots/Serials |

Today

**OdooBot** 10:45 AM

Subject: Overdue deviation DEV/2026/00002

Deviation DEV/2026/00002 has passed its target closure date.

- Title: Storage temperature excursion in the finished goods quarantine area
- Severity: Critical
- Current state: Reported
- Target closure date: 2026-09-26
- Days open: 3

Either progress the deviation to closure or record a justified extension of the target closure date.

**OdooBot** 10:41 AM

GxP Deviation created

Figure 33 — Deviation form: classification, chronology, subject, status bar. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Reporting, assessment and classification of deviations

|                   |                                                  |                |       |
|-------------------|--------------------------------------------------|----------------|-------|
| Module            | ls_deviation                                     | Version        | _____ |
| Menu              | Quality Management ► Deviations ► All Deviations | Effective date | _____ |
| Owner (suggested) | Head of Quality Assurance                        | Approved by    | _____ |
| Related SOPs      | SOP-QA-021, SOP-QA-022, SOP-QA-030               |                |       |

## 1 Objective

Make sure that every departure from an approved procedure, specification or condition is reported without delay, contained, and classified by QA on the basis of a documented impact assessment.

## 2 Scope

Unplanned deviations and planned deviations (temporary departures approved in advance) in production, quality control, warehouse, engineering and utilities.

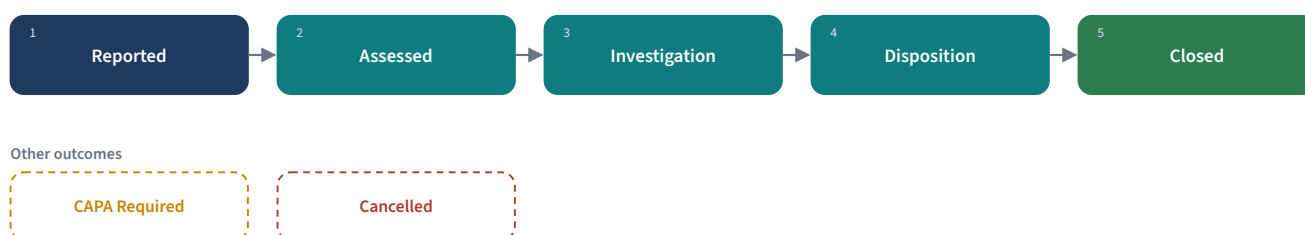
## 3 Responsibilities

| Role (software group)                    | Responsibility                                                                |
|------------------------------------------|-------------------------------------------------------------------------------|
| <b>Any employee (Deviation Reporter)</b> | Reports the event immediately and takes safe immediate actions.               |
| <b>Supervisor</b>                        | Checks the report on the same shift, completes immediate actions.             |
| <b>Deviation Investigator / QA</b>       | Documents the impact assessment, sets the severity, assigns the investigator. |
| <b>Deviation Manager (QA)</b>            | Confirms the classification; decides on planned deviations.                   |

## 4 Definitions

| Term                     | Definition                                                                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deviation</b>         | Departure from an approved instruction, specification or established standard.                                                             |
| <b>Planned deviation</b> | Departure approved before execution; it requires a justification recorded beforehand.                                                      |
| <b>Severity</b>          | Minor, Major or Critical, according to the potential impact on patient safety, product quality, regulatory compliance and validated state. |

## 5 Workflow in the software



Model: ls\_deviation

Figure 34 — Deviation lifecycle; this SOP covers Reported → Assessed.

## 6 Procedure

- 1 **Contain.** Stop or secure the activity if needed; segregate the material concerned.
- 2 **Report.** In [Quality Management ► Deviations](#) click [New](#). Enter title, description, deviation type, category, occurrence and detection dates, product, lots, manufacturing order, equipment, location and quantity affected. For a planned deviation, tick *Planned Deviation* and write the justification before execution.
- 3 **Record immediate actions.** Add each immediate action (containment, segregation, notification) with its owner; mark it done with completion evidence.
- 4 **Assess the impact.** QA completes the impact questions (patient safety, product quality, regulatory, validation, other batches) and the impact assessment text.
- 5 **Classify.** Set the severity and assign the investigator and QA reviewer. The target closure date is proposed from the company setting for that severity.
- 6 **Complete the assessment.** Click [Complete Assessment](#). The state becomes *Assessed* and the transition is logged.
- 7 **Send back if incomplete.** QA uses [Send Back](#) when the report is not usable, with an explanation in the chatter.
- 8 **Extend if necessary.** A target date can only be extended with [Extend Target Date](#) and a rationale.

### 7. Controls enforced by the software

- ✓ Occurrence date ≤ detection date ≤ recording date.
- ✓ A planned deviation requires a justification recorded before execution.
- ✓ An affected quantity requires a product.
- ✓ Assessment requires a severity, a documented impact assessment and an assigned investigator.
- ✓ An immediate action cannot be marked done without completion evidence.
- ✓ A deviation that progressed beyond *Reported* cannot be deleted — only cancelled with a reason.
- ✓ Every transition is recorded in the append-only Transition Log (cannot be edited or deleted).

### 8. Good practices

- ★ Report within the shift; a late deviation report is itself a finding.
- ★ Describe facts (what, where, when, how much), not opinions or blame.
- ★ Classify on the potential impact, not on the likely outcome of the investigation.
- ★ Check the list of open deviations at the daily production meeting.

### 9. Pitfalls to avoid

- ✗ Correcting the situation and "forgetting" the report.
- ✗ Grouping unrelated events in one deviation.
- ✗ Downgrading severity to meet the closure target.

## 10 Records and evidence

- Deviation record with transition log
- Immediate actions with evidence
- Impact assessment

## 11 Performance indicators

- Time from occurrence to report
- Deviations by type/category (trend)
- % classified within 2 working days

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.100 — Written procedures; deviations
- 21 CFR 211.192 — Production record review
- ICH Q10 — Pharmaceutical Quality System
- ICH Q9(R1) — Quality Risk Management

## 13. Demonstration

**Case.** During filling, an operator sees that the room differential pressure alarm had been active for 40 minutes. The supervisor stops filling, quarantines the 1 200 units filled in that period and reports deviation DEV/2026/00017: type *Environmental Excursion*, category *Production*, product and lot filled. Immediate actions: quarantine (done, evidence: quarantine label photo) and notification of QA (done). QA assesses a potential impact on product quality and on other batches, sets severity *Major*, assigns the investigator and clicks **Complete Assessment**.

| Quality Management   Deviations   Reporting   Configuration                                       |             |                                                              |                   |                |          |                  |                |           |                   |          |
|---------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------|-------------------|----------------|----------|------------------|----------------|-----------|-------------------|----------|
| New Deviations                                                                                    |             |                                                              |                   |                |          |                  |                |           |                   |          |
| <input type="text"/> <input type="button" value="Open"/> <input type="button" value="Search..."/> |             |                                                              |                   |                |          |                  |                |           |                   |          |
| 1-3 / 3                                                                                           |             |                                                              |                   |                |          |                  |                |           |                   |          |
|                                                                                                   | Reference   | Title                                                        | Deviation Type    | Category       | Severity | Detection Date   | Target Clos... | Days Open | Assigned Inves... | State    |
| <input type="checkbox"/>                                                                          | DEV/2026... | Storage temperature excursion in the finished goods quar...  | Environmental...  | Warehouse a... | Critical | Sep 24, 10:41 AM | Sep 26         | 3         |                   | Reported |
| <input type="checkbox"/>                                                                          | DEV/2026... | Theoretical yield below the lower limit on batch compression | Yield Discrepa... | Production     | Major    | Sep 22, 10:41 AM | Oct 22         | 5         |                   | Reported |
| <input type="checkbox"/>                                                                          | DEV/2026... | Missing second-person verification signature on a dispens... | Documentatio...   | Documentation  | Minor    | Sep 19, 10:41 AM | Oct 17         | 8         |                   | Reported |

Figure 35 — Deviation register with severity and overdue flag. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Investigation, extension to other batches and product disposition

|                   |                                                                         |                |       |
|-------------------|-------------------------------------------------------------------------|----------------|-------|
| Module            | ls_deviation                                                            | Version        | _____ |
| Menu              | Quality Management ► Deviations ► Investigations / Product Dispositions | Effective date | _____ |
| Owner (suggested) | Head of Quality Assurance                                               | Approved by    | _____ |
| Related SOPs      | SOP-QA-020, SOP-QA-022, SOP-QA-090                                      |                |       |

## 1 Objective

Find the root cause of each deviation, determine whether other batches or products are affected, decide on the fate of the affected product, and close with a written conclusion and follow-up.

## 2 Scope

All assessed deviations. Complaint and laboratory OOS investigations follow SOP-QA-040 and SOP-QC-013.

## 3 Responsibilities

| Role (software group)        | Responsibility                                                                  |
|------------------------------|---------------------------------------------------------------------------------|
| <b>Assigned investigator</b> | Leads the investigation, records findings, contributing factors and root cause. |
| <b>QA reviewer</b>           | Challenges the investigation, checks the extension to other batches.            |
| <b>Deviation Manager</b>     | Approves or rejects product dispositions; closes the deviation.                 |

## 4 Definitions

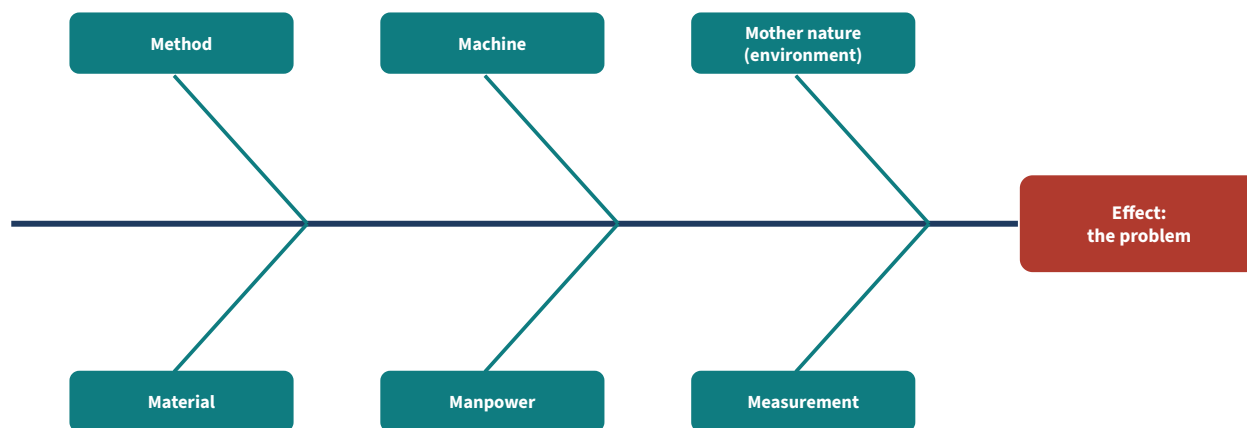
| Term                       | Definition                                                                                                        |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Extension rationale</b> | Documented reasoning on whether other batches, products or equipment are affected (required by 21 CFR 211.192).   |
| <b>Disposition</b>         | Decision on the affected quantity: Use As Is, Rework, Reprocess, Quarantine, Reject, Destroy, Return to Supplier. |

## 5 Workflow in the software



Model: ls\_deviation

Figure 36 — This SOP covers Assessed → Closed.



*Ishikawa (6M) diagram — one of the root-cause methods offered in Is\_capa (with Five Whys and FMEA).*

Figure 37 — Ishikawa diagram used during investigation.

## 6 Procedure

- 1 **Open the investigation.** Add an investigation record: subject, RCA method (from [Configuration ► Root Cause Analysis Methods](#)), investigator, start date. Click **Start**.
- 2 **Start the phase.** On the deviation click **Start Investigation** (at least one investigation record must exist).
- 3 **Investigate.** Collect facts (batch record, logbooks, EM data, interviews), apply the method (5 Whys, Ishikawa...), record findings, contributing factors and root cause; tick *Root Cause Determined* when confirmed.
- 4 **Complete.** Click **Complete** on the investigation (findings and root cause required).
- 5 **Extend.** Write the *Other Batches Evaluated* / extension rationale: which other lots, products or equipment could be affected and why they are or are not.
- 6 **Propose dispositions.** For each affected lot add a disposition: product, lot, quantity, decision and justification. Click **Proceed to Disposition**.
- 7 **Approve dispositions.** The Deviation Manager clicks **Approve** or **Reject** on each disposition.
- 8 **Decide on CAPA.** If a systemic cause exists click **Require CAPA** and open the CAPA in Is\_capa (SOP-QA-022); otherwise record the rationale for no CAPA.
- 9 **Close.** Click **Close Deviation**: enter the conclusion, the follow-up, the CAPA decision and CAPA reference, and the QA reviewer.

## 7. Controls enforced by the software

- ✓ The investigation phase needs at least one investigation record.
- ✓ An investigation cannot be completed without findings and a recorded root cause; completion date  $\geq$  start date.
- ✓ Disposition requires all investigations completed and an extension rationale; if an impact was assessed, at least one disposition is required.
- ✓ Only a Deviation Manager can approve or reject a disposition; the lot must belong to the product.
- ✓ Closure is refused while an immediate action is open or a disposition is still draft; a CAPA reference is mandatory when CAPA is required.
- ✓ A closed or cancelled deviation can only be modified by a Deviation Manager.

## 8. Good practices

- ★ Distinguish the direct cause from the root cause; ask "why" until you reach a process or system cause.
- ★ Human error is rarely a root cause: look for what made the error possible.
- ★ Base the extension on data (same equipment, same material lot, same period).
- ★ Close within the target; if not possible, extend formally with a reason.

## 9. Pitfalls to avoid

- ✗ "Operator retrained" as the only action for a systemic cause.
- ✗ Releasing product before the disposition is approved.
- ✗ Closing with "no impact" without evidence.

## 10 Records and evidence

- Investigation records
- Dispositions with approval
- Conclusion, follow-up, CAPA reference
- Transition log

## 11 Performance indicators

- Median time to closure by severity
- % closed within target
- Recurrence of the same root cause

## 12 References

- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q9(R1) — Quality Risk Management
- ICH Q10 — Pharmaceutical Quality System

## 13. Demonstration

**Case (continued from SOP-QA-020).** The investigator uses Ishikawa: the alarm was not escalated because the BMS pager had been muted after a maintenance intervention (method + machine). Root cause: no procedure to restore alarm routing after maintenance. Extension: two earlier batches filled in the same room since the maintenance are reviewed; their EM data are within limits — no impact. Disposition for the 1 200 units: *Reject*, approved by the Deviation Manager. CAPA required (restore-after-maintenance checklist); CAPA-2026-031 opened and referenced at closure.

# CAPA management from identification to closure

|                   |                                                            |                |       |
|-------------------|------------------------------------------------------------|----------------|-------|
| Module            | ls_capa                                                    | Version        | _____ |
| Menu              | Quality ► CAPA ► CAPA Records                              | Effective date | _____ |
| Owner (suggested) | CAPA Manager (QA)                                          | Approved by    | _____ |
| Related SOPs      | SOP-QA-021, SOP-QA-023, SOP-QA-030, SOP-QA-040, SOP-QA-053 |                |       |

## 1 Objective

Eliminate the causes of detected or potential nonconformities by a structured process: identify, assess, analyse root causes, plan and implement actions, verify effectiveness, close.

## 2 Scope

Corrective and preventive actions from internal and external audits, inspections, deviations, complaints, nonconformities, management review, risk assessments, environmental monitoring, suppliers and change control.

## 3 Responsibilities

| Role (software group)           | Responsibility                                                        |
|---------------------------------|-----------------------------------------------------------------------|
| <b>Initiator (any user)</b>     | Identifies the need and creates the record with its source reference. |
| <b>CAPA Owner / Coordinator</b> | Drives the CAPA through its stages and meets the due date.            |
| <b>CAPA Investigator</b>        | Performs root cause analyses and actions.                             |
| <b>CAPA Manager</b>             | Checks each gate, closes the CAPA.                                    |

## 4 Definitions

| Term                     | Definition                                                                              |
|--------------------------|-----------------------------------------------------------------------------------------|
| <b>Correction</b>        | Immediate fix of the nonconformity (recorded as <i>Immediate Action / Correction</i> ). |
| <b>Corrective action</b> | Action on the cause to prevent recurrence.                                              |
| <b>Preventive action</b> | Action on the cause of a potential nonconformity to prevent occurrence.                 |

## 5 Workflow in the software



Model: ls\_capa.issue

Figure 38 — CAPA lifecycle with its gates.



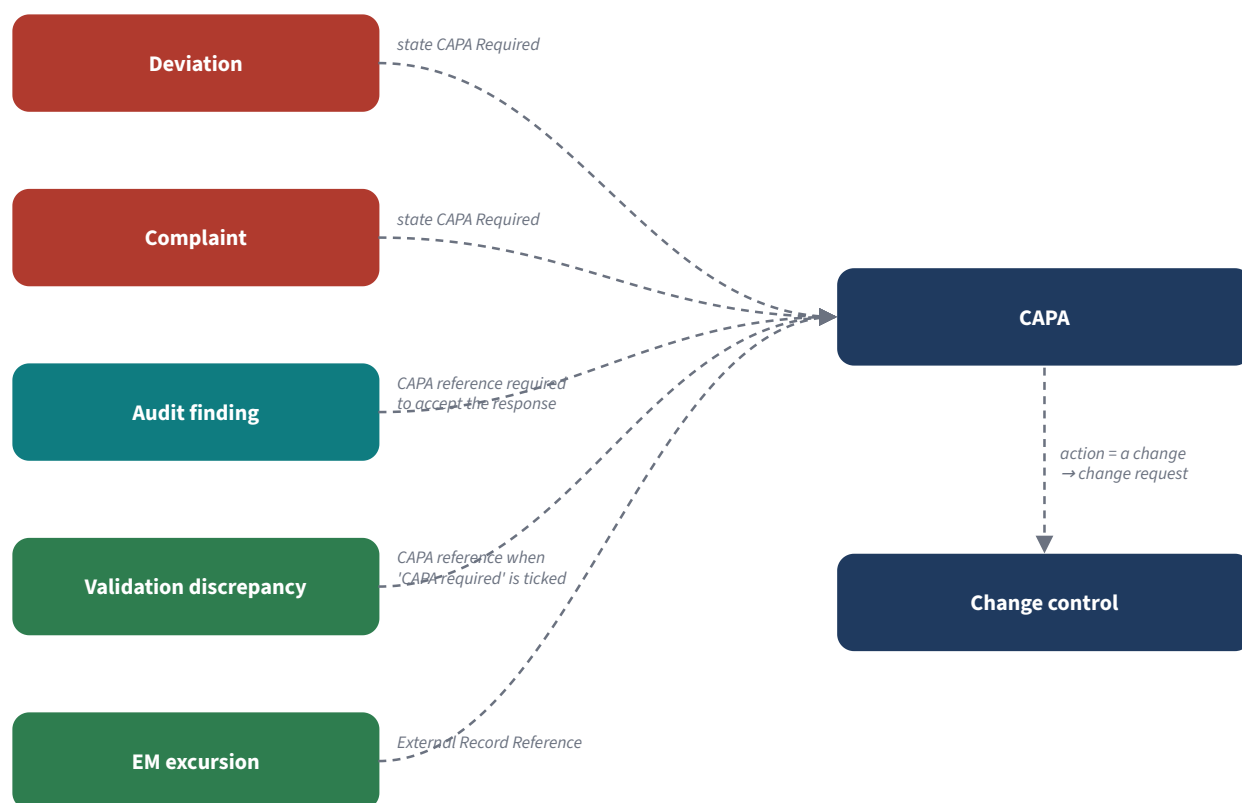


Figure 39 — Sources feeding the CAPA system.

## 6 Procedure

- 1 **Identify.** In **Quality ► CAPA ► CAPA Records** click **New** : title, description, source (e.g. *Deviation*) and *Source Reference* (e.g. DEV/2026/00017), type (corrective, preventive or both), severity, priority, category, owner and due date.
- 2 **Assess.** Document the impact assessment (product quality, regulatory, patient/user safety) and the correction already taken. Click **Assess** .
- 3 **Investigate.** Click **Start Investigation** . Add one or more root cause analyses (Five Whys, Ishikawa, FMEA or other method), each with a root cause statement; mark the primary root cause; click **Confirm** on each.
- 4 **Plan actions.** Click **Plan Actions** . Add actions: type, description, root cause addressed, responsible, planned completion date. Use **Create Task** to follow an action in a project.
- 5 **Execute.** Click **Start Execution** . Each responsible clicks **Start** then **Mark Done** with completion evidence. Cancelled actions need a reason.
- 6 **Complete.** When all actions are done or cancelled, click **Complete** .
- 7 **Verify effectiveness.** Plan and perform effectiveness checks (SOP-QA-023); when at least one concludes *Effective* and all are done, click **Verify Effectiveness** .
- 8 **Close.** Click **Close CAPA** , write the closure summary. Write the CAPA reference back into the source record (deviation, complaint, audit finding).

## 7. Controls enforced by the software

- ✓ Each transition is only allowed from the expected status.
- ✓ Assessment requires a documented impact assessment.
- ✓ Action planning requires at least one confirmed root cause analysis.
- ✓ Execution requires at least one planned action; completion requires all actions done or cancelled.
- ✓ An action cannot be completed without evidence nor cancelled without a reason; a completed action cannot be cancelled.
- ✓ Verification requires every effectiveness check completed and at least one *Effective*.
- ✓ Closure requires a closure summary; due date  $\geq$  identification date.
- ✓ A CAPA beyond *Identified* cannot be deleted — it is archived.
- ✓ Five Whys needs at least the first Why; Ishikawa needs a category; FMEA ratings are 1–10; a confirmed root cause cannot be reset once execution started.

## 8. Good practices

- ★ Size the CAPA to the risk: not every deviation needs a CAPA, but every recurring cause does.
- ★ One action, one owner, one date, one piece of evidence.
- ★ Check that the action addresses the confirmed root cause (link *Root Cause Addressed*).
- ★ Review overdue CAPAs weekly in the quality meeting.

## 9. Pitfalls to avoid

- ✗ Closing on completion of actions without effectiveness verification.
- ✗ Extending due dates repeatedly without escalation.
- ✗ Opening a CAPA without writing its reference back into the source record — the link is manual.

## 10 Records and evidence

- CAPA record with root causes, actions, evidence, effectiveness checks and closure summary

## 11 Performance indicators

- Overdue CAPAs
- Median time to closure
- % of CAPAs verified effective at first check
- Recurring root causes

## 12 References

- ICH Q10 — Pharmaceutical Quality System
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- 21 CFR Part 820 — Quality Management System Regulation (incorporates ISO 13485:2016 by reference; applicable since 2 February 2026)

### 13. Demonstration

**Case.** CAPA-2026-031, source *Deviation*, reference DEV/2026/00017, type corrective, severity major, due in 60 days. Impact assessment documented; Five Whys analysis confirmed: *no restore-after-maintenance step for alarm routing*. Actions: (1) add a restore checklist to the maintenance work order template — engineering, 30 days; (2) train maintenance technicians — 45 days; (3) weekly automatic alarm routing test — IT, 45 days. Each action is closed with evidence (new template, training certificates, test report). Effectiveness check planned at +3 months: *no alarm routing failure after 12 maintenance interventions*.

**CAPA/2026/00002**

Gowning procedure not followed in Grade C corridor

| CLASSIFICATION     |                | OWNERSHIP AND DATES   |                            |
|--------------------|----------------|-----------------------|----------------------------|
| Source ?           | Internal Audit | Identified By         | OdooBot                    |
| Source Reference ? | IA-2026-004    | CAPA Owner ?          | OdooBot                    |
| Capa Type ?        | Corrective     | Identification Date ? | Sep 21                     |
| Severity ?         | Minor          | Due Date ?            | Oct 21                     |
| Priority           | ★☆☆            | Days To Due Date ?    | 24                         |
| Category ?         | Production     | Action Progress (%) ? | 0 %                        |
|                    |                | Company               | My Company (San Francisco) |

**Issue** | Impact Assessment | Root Cause Analysis | Actions | Effectiveness

An internal audit observed two operators entering the Grade C corridor without the second pair of gloves required by SOP-CR-004.

**IMMEDIATE ACTION / CORRECTION**

Containment action already taken.

**Activity**

Today

OdooBot 10:40 AM  
CAPA Record created

Figure 40 — CAPA record form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Verification of CAPA effectiveness

|                   |                                             |                |       |
|-------------------|---------------------------------------------|----------------|-------|
| Module            | ls_capa                                     | Version        | _____ |
| Menu              | Quality ► CAPA ► Effectiveness Verification | Effective date | _____ |
| Owner (suggested) | CAPA Manager (QA)                           | Approved by    | _____ |
| Related SOPs      | SOP-QA-022, SOP-QA-031                      |                |       |

## 1 Objective

Prove, with objective data collected after implementation, that the CAPA actions eliminated or reduced the cause, and trigger a new CAPA when they did not.

## 2 Scope

Every CAPA before its verification and closure.

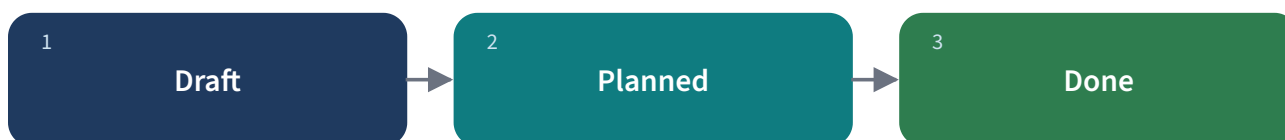
## 3 Responsibilities

| Role (software group) | Responsibility                                                                                |
|-----------------------|-----------------------------------------------------------------------------------------------|
| <b>CAPA Owner</b>     | Defines the method and acceptance criteria when planning actions.                             |
| <b>Verifier</b>       | Collects the data at the planned date and concludes; ideally independent of the action owner. |
| <b>CAPA Manager</b>   | Accepts the conclusion; raises a follow-up CAPA when not effective.                           |

## 4 Definitions

| Term                       | Definition                                                                                                     |
|----------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Verification method</b> | Data Review, Follow-up Audit, Process Monitoring, Retesting, Training Verification or Other Documented Method. |
| <b>Acceptance criteria</b> | Measurable condition that proves effectiveness (e.g. "zero recurrence in 3 months").                           |

## 5 Workflow in the software



Model: ls\_capa.effectiveness

Figure 41 — Lifecycle of an effectiveness check.

## 6 Procedure

- 1 Define.** In the CAPA, add an effectiveness check: method, acceptance criteria, planned verification date (not before the identification date), verifier.
- 2 Plan.** Click **Plan** once the actions are implemented; the check appears in **Quality ► CAPA ► Effectiveness Verification**.

- 3 **Collect data.** At the planned date, collect the data defined by the method over the whole observation period.
- 4 **Conclude.** Record the conclusion and click **Conclude Effective** or **Conclude Not Effective**.
- 5 **Not effective.** Click **Raise Follow-up CAPA**: a new CAPA is created and linked; the root cause analysis must be revisited.
- 6 **Verify the CAPA.** Return to the CAPA and click **Verify Effectiveness**, then close (SOP-QA-022).

### 7. Controls enforced by the software

- ✓ Only draft checks can be planned; a check cannot be planned before the CAPA identification date.
- ✓ A check cannot be completed while its result is *Pending* or without a conclusion.
- ✓ A follow-up CAPA can only be raised from a check concluded *Not Effective*, and only once.
- ✓ The CAPA cannot be verified until every check is completed and at least one is *Effective*.

### 8. Good practices

- ★ Set the criteria before implementation, so that they are not tailored to the results.
- ★ Choose an observation period long enough to see recurrence (number of batches, audits, interventions).
- ★ Training alone is verified by observing the task, not by the attendance list.

### 9. Pitfalls to avoid

- ✗ "Actions completed" used as effectiveness evidence.
- ✗ Verification by the person who implemented the action, without a second look.

## 10 Records and evidence

- Effectiveness checks with data, conclusion and date
- Follow-up CAPA when applicable

## 11 Performance indicators

- % effective at first verification
- Follow-up CAPAs raised
- Checks overdue

## 12 References

- ICH Q10 — Pharmaceutical Quality System
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements

### 13. Demonstration

**Case.** For CAPA-2026-031 the check is: method *Process Monitoring*, criterion "*alarm routing verified operational after 100 % of maintenance interventions over 3 months, minimum 10 interventions*". At +3 months the verifier reviews 12 work orders: all carry the completed restore checklist and the weekly routing tests passed. She clicks **Conclude Effective**; the CAPA owner clicks **Verify Effectiveness** and closes the CAPA.

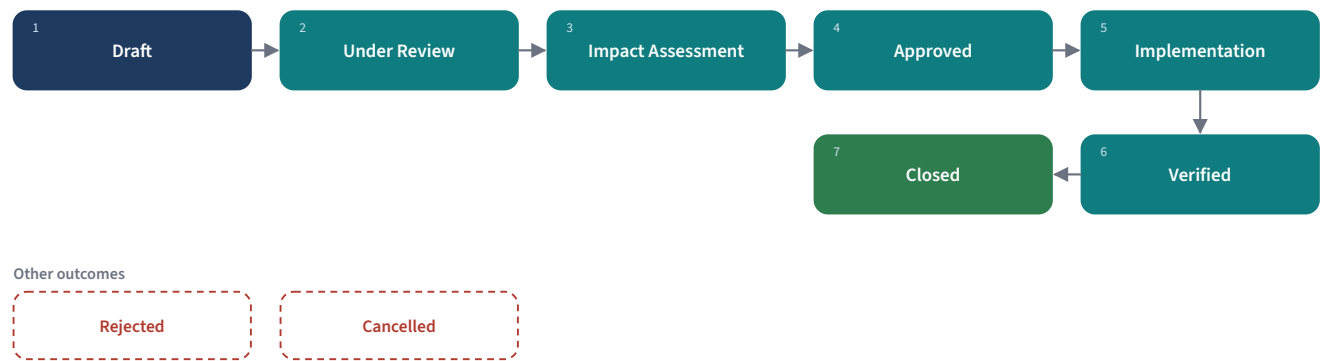
# Change control

ls\_change\_control

**ls\_change\_control** manages change requests from draft to closure: classification (minor, major, critical), permanent or temporary change, impact areas with an impact assessment per area, approvals generated from the category's approval templates, implementation actions with evidence, and effectiveness verifications. The request is approved automatically once every assessment is completed and every approval is favourable; any rejection rejects the request.

| Menu                                                                              | Content                                  |
|-----------------------------------------------------------------------------------|------------------------------------------|
| Change Control ► Change Requests ► All / My Change Requests / Impact Assessments  | Requests and assessments                 |
| Change Control ► Approvals ► My Approvals                                         | Decisions waiting for the connected user |
| Change Control ► Execution ► Implementation Actions / Effectiveness Verifications | Execution                                |
| Change Control ► Configuration ► Change Categories / Impact Areas / Settings      | Categories, approval templates, delays   |

| Group                  | Main rights                                                                         |
|------------------------|-------------------------------------------------------------------------------------|
| Viewer · Requester     | Read; create and submit requests                                                    |
| Approver               | Record approval decisions assigned to them                                          |
| Change Control Manager | Drives the workflow, starts assessment and implementation, closes, rejects, cancels |



Model: ls.change\_control.request

Figure 42 — Lifecycle of a change request.

Change Control

Change Requests

Approvals

Execution

Reporting

Configuration

7

14

My Company (San Francisco)

New

Change Requests

CC/2026/00003 - Install a new tablet press in th...

Actions Closed

0

1 / 1

Submit for Review

Cancel Request

Draft

Under Review

Impact Assessment

Approved

Implementation

...

Send message

Log note

Activity

1

Change Request

CC/2026/00003

Install a new tablet press in the solid dosage form area

CLASSIFICATION

Change Category

Equipment Change

Change Type

Permanent

Classification

Critical

Priority

☆

RESPONSIBILITIES

Requester

Mitchell Admin

Requesting Department

Change Control Manager

Company

My Company (San Francisco)

DATES

Request Date

Sep 27, 10:40 AM

Requested Target Date

Planned Implementation Date

Description

Impact

Approvals

Implementation

Verification

CURRENT SITUATION

OdooBot

10:40 AM

Change Control Request created

Figure 43 — Change request form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Change request and impact assessment

|                   |                                                 |                |       |
|-------------------|-------------------------------------------------|----------------|-------|
| Module            | ls_change_control                               | Version        | _____ |
| Menu              | Change Control ► Change Requests                | Effective date | _____ |
| Owner (suggested) | Change Control Manager                          | Approved by    | _____ |
| Related SOPs      | SOP-QA-031, SOP-QA-002, SOP-QA-060, SOP-VAL-005 |                |       |

## 1 Objective

Ensure that any change that may affect product quality, the validated state, regulatory compliance or safety is proposed, justified, assessed by every affected function and approved before implementation.

## 2 Scope

Changes to products, processes, equipment, facilities, utilities, materials and suppliers, specifications, test methods, computerised systems and quality documents with a substantive effect. Temporary changes are included.

## 3 Responsibilities

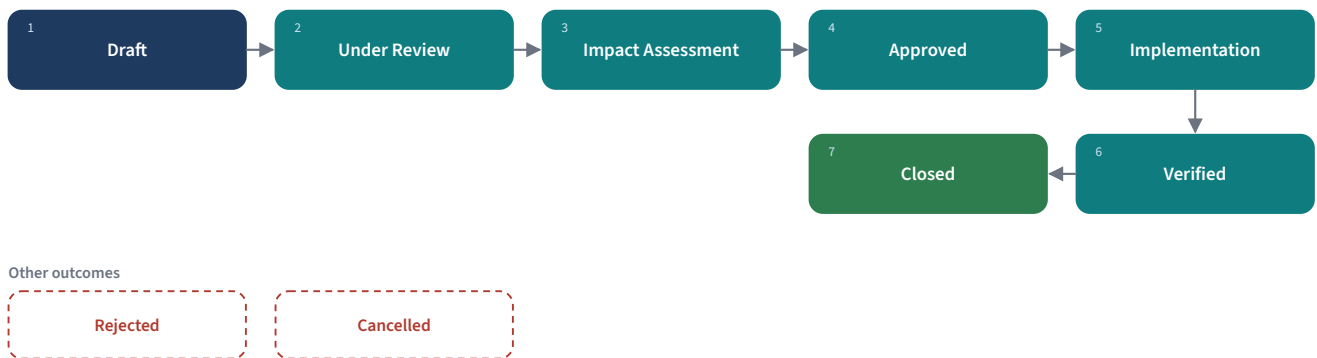
| Role (software group)              | Responsibility                                                                                     |
|------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Requester</b>                   | Describes the current situation, the proposed change and its justification; submits.               |
| <b>Change Control Manager</b>      | Screens the request, completes classification and impact areas, starts the assessment, arbitrates. |
| <b>Assessors (per impact area)</b> | Assess the impact of their area and describe the required actions.                                 |
| <b>Approvers</b>                   | Approve or reject, with a comment when rejecting.                                                  |

## 4 Definitions

| Term                     | Definition                                                                                                              |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Classification</b>    | Minor, Major or Critical.                                                                                               |
| <b>Temporary change</b>  | Change limited in time; it must declare its end date.                                                                   |
| <b>Regulatory impact</b> | No Regulatory Action, Notification to the Authority, Variation of the Marketing Authorisation, Prior Approval Required. |



## 5 Workflow in the software



Model: `ls.change_control.request`

Figure 44 — This SOP covers Draft → Approved.

## 6 Procedure

- 1 **Create.** In **Change Control ► Change Requests** click **New** : title, category, permanent/temporary (end date if temporary), classification, priority, current situation, proposed change, justification, requested target date.
- 2 **Declare impacts.** Tick the impact flags (GMP, product quality, regulatory, revalidation, training, document update, customer notification) and select the impact areas; reference the risk assessment if one exists (SOP-QA-060).
- 3 **Submit.** Click **Submit for Review** . The content is frozen for the requester.
- 4 **Screen.** The Change Control Manager checks completeness; if incomplete, **Return to Draft** with a comment.
- 5 **Start the assessment.** Click **Start Impact Assessment** : the system generates one assessment per impact area and the approvals defined by the category, and notifies the approvers.
- 6 **Assess.** Each assessor opens **Change Requests ► Impact Assessments**, sets the impact level (No Impact, Low, Medium, High), writes the assessment and, where there is an impact, the required actions; clicks **Complete Assessment** .
- 7 **Decide.** Each approver opens **Approvals ► My Approvals** and clicks **Approve** or **Reject** (a comment is required to reject).
- 8 **Approval.** When all assessments are completed and all approvals favourable, the request becomes *Approved* automatically and a planned implementation date is proposed from the category delay. A single rejection rejects the request.

### 7. Controls enforced by the software

- ✓ Once submitted, the request content is frozen; only a Change Control Manager may modify it — a change of the change itself requires a new request.
- ✓ Workflow fields (dates, states, deciders) cannot be written directly.
- ✓ A temporary change must declare an end date; a permanent change must not.
- ✓ Assessments and decisions can only be recorded while the request is in *Impact Assessment*; only the assigned assessor/approver (or the manager for assessments) may record them.
- ✓ A written assessment is required; if an impact is declared the required actions must be described.
- ✓ A rejection requires a comment; decided approvals cannot be modified or deleted.
- ✓ A submitted request cannot be deleted; requester and manager must be internal users.

## 8. Good practices

- ★ Describe the change so that a person who was not in the meeting understands it.
- ★ Involve regulatory affairs early: a variation can take months.
- ★ Use a risk assessment for major and critical changes.
- ★ Link documents to revise, validations to perform and training to deliver as implementation actions.

## 9. Pitfalls to avoid

- ✗ Implementing before approval ("it was urgent").
- ✗ Approving without reading the assessments.
- ✗ Using a temporary change that becomes permanent without a new request.

## 10 Records and evidence

- Change request with assessments, approvals and dates
- Referenced risk assessment

## 11 Performance indicators

- Time from submission to approval
- Changes implemented before approval (target 0)
- Temporary changes past their end date

## 12 References

- ICH Q10 — Pharmaceutical Quality System
- EU GMP Annex 15 (2015) — Qualification and Validation
- ISO 13485:2016 — Medical devices — Quality management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

## 13. Demonstration

**Case.** Engineering proposes to replace the tablet press punches with a new supplier's tooling. Request CC-2026-021: category *Equipment*, permanent, classification *Major*, regulatory impact *No Regulatory Action* (equivalent design), requires revalidation and document update. Impact areas: Production, Quality Control, Validation, Purchasing. The manager clicks **Start Impact Assessment**; four assessments and three approvals are generated. Validation assesses *High* impact (OQ of the press with new tooling). All approve; the request becomes *Approved*.

|                                                                                                                                                                                                                                         |               |                                                                  |                                 |                |                                                                                                  |                        |               |        |                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------|---------------------------------|----------------|--------------------------------------------------------------------------------------------------|------------------------|---------------|--------|-------------------------------------------------------------------------------------|
| <div> <div>Change Control</div> <div>Change Requests</div> <div>Approvals</div> <div>Execution</div> <div>Reporting</div> <div>Configuration</div> </div> <div> <div>7</div> <div>14</div> <div>My Company (San Francisco)</div> </div> |               |                                                                  |                                 |                |                                                                                                  |                        |               |        |                                                                                     |
| <div> <div>New</div> <div>Change Requests</div> </div> <div> <div>Q</div> <div>Open</div> <div>X</div> <div>Search...</div> </div> <div>1-3 / 3</div> <div> <div></div> <div></div> <div></div> <div></div> <div></div> </div>          |               |                                                                  |                                 |                |                                                                                                  |                        |               |        |                                                                                     |
| <input type="checkbox"/>                                                                                                                                                                                                                | Reference     | Title                                                            | Change Category                 | Classificat... | Requester                                                                                        | Change Control Mana... | Planned Im... | Status | Activities                                                                          |
| <input type="checkbox"/>                                                                                                                                                                                                                | CC/2026/00... | Install a new tablet press in the solid dosage form area         | Equipment Change                | Critical       |  Mitchell Admin |                        |               | Draft  |  |
| <input type="checkbox"/>                                                                                                                                                                                                                | CC/2026/00... | Update the blister artwork of Product B after a leaflet revision | Packaging and Labelling Chan... | Minor          |  Mitchell Admin |                        |               | Draft  |  |
| <input type="checkbox"/>                                                                                                                                                                                                                | CC/2026/00... | Replace the HPLC column of the assay method for Produc...        | Analytical Method Change        | Major          |  Mitchell Admin |                        |               | Draft  |  |

Figure 45 — Change request register. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Approval, implementation and verification of changes

|                   |                                                 |                |       |
|-------------------|-------------------------------------------------|----------------|-------|
| Module            | ls_change_control                               | Version        | _____ |
| Menu              | Change Control ► Execution                      | Effective date | _____ |
| Owner (suggested) | Change Control Manager                          | Approved by    | _____ |
| Related SOPs      | SOP-QA-030, SOP-QA-023, SOP-VAL-005, SOP-QA-012 |                |       |

## 1 Objective

Implement approved changes as planned, with evidence for each action, verify that the change achieved its purpose without adverse effect, and close the request formally.

## 2 Scope

All approved change requests.

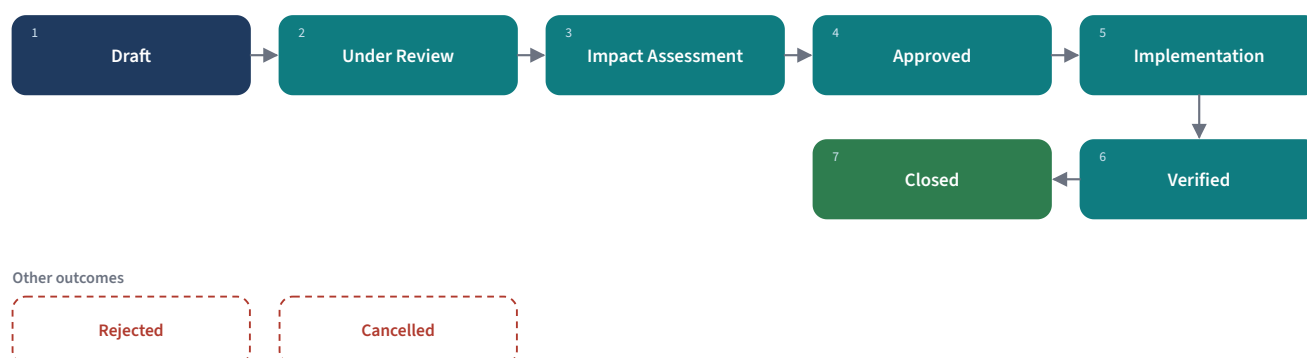
## 3 Responsibilities

| Role (software group)         | Responsibility                                                                               |
|-------------------------------|----------------------------------------------------------------------------------------------|
| <b>Action owners</b>          | Perform the implementation actions and record evidence.                                      |
| <b>Change Control Manager</b> | Starts implementation, declares the change verified, closes; cancels actions when justified. |
| <b>Verifier</b>               | Performs the effectiveness verification against the acceptance criteria.                     |

## 4 Definitions

| Term                              | Definition                                                                                                 |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Implementation action</b>      | Task with an owner and an evidence reference (e.g. document revision, validation report, training record). |
| <b>Effectiveness verification</b> | Check planned on the request with a method and acceptance criteria; required when the category demands it. |

## 5 Workflow in the software



Model: ls.change\_control.request

Figure 46 — This SOP covers Approved → Closed.

## 6 Procedure

- 1 **Plan the actions.** On the approved request add implementation actions (owner, description, due date) derived from the assessments: document revisions, validation, training, supplier notification, regulatory filing.
- 2 **Start implementation.** The manager clicks **Start Implementation** (at least one action is required).
- 3 **Execute.** Each owner clicks **Start** then **Mark as Done** with the evidence reference. Only the manager can cancel an action (**Cancel Action**), with a reason.
- 4 **Verify effectiveness.** At the planned date the verifier records the verification result and conclusion and clicks **Conclude: Effective** or **Conclude: Not Effective** (the latter requires a follow-up reference, e.g. a new change request or CAPA).
- 5 **Declare verified.** When all actions are closed and verifications complete, the manager clicks **Declare Verified**.
- 6 **Close.** Click **Close** and write the closure statement (what was changed, effective date, evidence).
- 7 **Temporary change.** At the end date, verify the return to the original state and record it before closing.

## 7. Controls enforced by the software

- ✓ Implementation actions can only be executed once the change is approved; only the owner or the manager may update an action.
- ✓ An action cannot be closed without an evidence reference; cancellation requires a reason and the manager role; a started action cannot be deleted.
- ✓ Verifications can only be completed during *Implementation*; a conclusion is required; *Not Effective* requires a follow-up.
- ✓ Verification of the request is refused while blocking items remain; closure requires a closure statement.
- ✓ A category requiring effectiveness verification must define a verification delay (on the category or the company).

## 8. Good practices

- ★ Do not declare the change effective on the day of implementation: allow a representative period.
- ★ Keep the evidence reference precise (document number and revision, report number).
- ★ Close the loop with training (SOP-QA-012) and revalidation (SOP-VAL-005).

## 9. Pitfalls to avoid

- ✗ Actions closed with "done" and no evidence.
- ✗ Leaving requests in *Implementation* for months after the change is in use.

## 10 Records and evidence

- Implementation actions with evidence
- Effectiveness verifications
- Closure statement

## 11 Performance indicators

- Actions overdue
- Requests closed within planned date
- % of changes needing a follow-up

## 12 References

- ICH Q10 — Pharmaceutical Quality System
- EU GMP Annex 15 (2015) — Qualification and Validation
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case (continued).** CC-2026-021 actions: (1) OQ of the press with new tooling — report VAL-OQ-2026-008; (2) revise SOP-PRD-014 rev 4; (3) train operators. After 10 batches the verifier checks tablet weight, hardness and friability trends: all within control limits → **Conclude: Effective**. The manager clicks **Declare Verified** then **Close**.

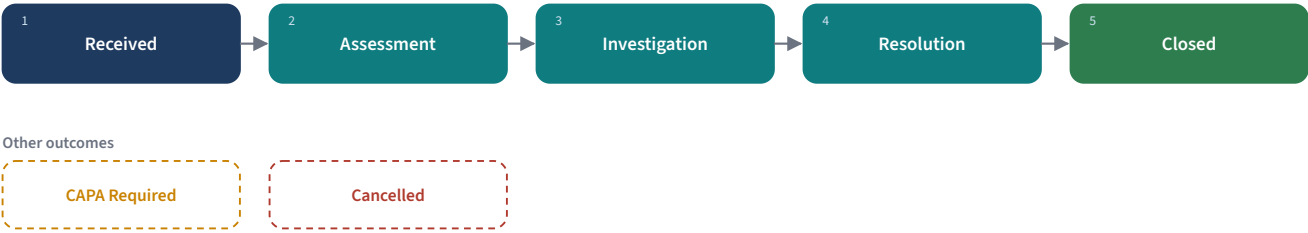
# Complaints

ls\_complaint

**ls\_complaint** records product quality complaints from any channel, with product, lot and sample, assessment, investigations (several methodologies, approval by a different user), resolutions, adverse events with reportability assessment, and closure. Complaint categories carry target times for acknowledgement, investigation and closure, and the reporting deadline for adverse events.

| Menu                                                                                 | Content                              |
|--------------------------------------------------------------------------------------|--------------------------------------|
| Complaints ► Complaint Handling ► All / My Complaints / Investigations / Resolutions | Complaint handling                   |
| Complaints ► Vigilance ► Adverse Events                                              | Adverse events and authority reports |
| Complaints ► Configuration ► Complaint Categories                                    | Target days and reporting deadline   |
| Complaints ► Reporting ► Complaint Analysis                                          | Trending                             |

| Group                                                | Main rights                                                                              |
|------------------------------------------------------|------------------------------------------------------------------------------------------|
| Complaint Viewer · Investigator · Reviewer · Manager | Investigator investigates; Reviewer approves investigations; Manager closes and cancels. |



Model: ls.complaint

Figure 47 — Lifecycle of a complaint.

Complaints
Complaint Handling
Vigilance
Reporting
Configuration
7
14
My Company (San Francisco)

New
Complaints
CMP/2026/00003 - Skin reaction reported after ...
Investigations 0
Resolutions 0
Adverse Events 1
1 / 1

Start Assessment
Cancel
Received
Assessment
Investigation
Resolution
Closed
Send message
Log note
Activity

### CMP/2026/00003

Skin reaction reported after administration

| RECEPTION       |                            | HANDLING       |                         |
|-----------------|----------------------------|----------------|-------------------------|
| Receipt Date    | Sep 27, 10:40 AM           | Responsible    |                         |
| Occurrence Date |                            | Reviewer       |                         |
| Channel         | Email                      | Priority       | ☆☆                      |
| Received By     | OdooBot                    | Category       | Product Quality Defect  |
| Company         | My Company (San Francisco) | Severity       | Critical                |
|                 |                            | Complaint Type | Safety or Adverse Event |

Complaint
Product
Assessment
Investigations
Adverse Events
CAPA
Resolutions

| COMPLAINANT      |                         | TARGETS                  |        |
|------------------|-------------------------|--------------------------|--------|
| Complainant      |                         | Acknowledgement Due Date | Sep 29 |
| Complainant Type | Healthcare Professional | Investigation Due Date   | Oct 17 |
| Contact Name     |                         | Closure Due Date         | Oct 27 |
| Contact Email    |                         |                          |        |

Today
OdooBot 10:40 AM
Product Quality Complaint created

Figure 48 — Complaint form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)



# Reception, assessment and investigation of complaints

|                   |                                    |                |       |
|-------------------|------------------------------------|----------------|-------|
| Module            | ls_complaint                       | Version        | _____ |
| Menu              | Complaints ► Complaint Handling    | Effective date | _____ |
| Owner (suggested) | Complaint Manager (QA)             | Approved by    | _____ |
| Related SOPs      | SOP-QA-041, SOP-QA-022, SOP-QA-081 |                |       |

## 1 Objective

Register every complaint, acknowledge it on time, assess its seriousness, investigate the cause when justified, respond to the complainant and feed the CAPA and recall processes.

## 2 Scope

Product quality complaints received from patients, consumers, healthcare professionals, customers, distributors, pharmacies, hospitals, authorities or internal sources, for all products of the company.

## 3 Responsibilities

| Role (software group)          | Responsibility                                                            |
|--------------------------------|---------------------------------------------------------------------------|
| <b>Receiver (any employee)</b> | Forwards the complaint to QA the same day; creates the record if trained. |
| <b>Complaint Investigator</b>  | Assesses and investigates, requests the sample, writes conclusions.       |
| <b>Complaint Reviewer</b>      | Approves or rejects investigations (not their own).                       |
| <b>Complaint Manager</b>       | Decides on waiver of investigation and CAPA, closes the complaint.        |

## 4 Definitions

| Term                           | Definition                                                                                      |
|--------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Complaint</b>               | Any written, electronic or oral communication alleging a deficiency of a product after release. |
| <b>Waiver of investigation</b> | Documented decision not to investigate (e.g. identical complaint already investigated).         |

## 5 Workflow in the software



Model: ls.complaint

Figure 49 — Complaint lifecycle.

## 6 Procedure

- 1 Register.** In **Complaints ► Complaint Handling** click **New**: receipt date, channel, received by, complainant and type, subject and description, product, lot, quantity, market, sample availability.

- 2 **Acknowledge.** Contact the complainant within the category's acknowledgement target; record *Customer Notified* and the date.
- 3 **Assess.** Click **Start Assessment** ; set category, complaint type, severity, potential safety and regulatory impact, assessment summary. If an adverse event is involved, create it (SOP-QA-041).
- 4 **Investigate or waive.** Click **Start Investigation** and add an investigation (investigator, methodology, plan). Or click **Waive Investigation** with a justification.
- 5 **Investigate.** Examine the sample, the retained samples, batch record, other complaints on the same lot; record the summary, root cause category and description, conclusion (Confirmed, Not Confirmed, Inconclusive) and the batch impact. Click **Complete** ; a reviewer clicks **Approve** .
- 6 **CAPA.** If a CAPA is needed click **CAPA Required** , open the CAPA (SOP-QA-022) and record its reference.
- 7 **Resolve.** Click **Start Resolution** ; add resolutions (reply letter, replacement, credit) and mark them done with evidence.
- 8 **Close.** Click **Close** with a closure summary. Consider a recall (SOP-QA-081) if the investigation shows a risk for other units on the market.

## 7. Controls enforced by the software

- ✓ Receipt date not in the future; occurrence date  $\leq$  receipt date; expiry date  $\geq$  manufacturing date; the lot must belong to the product.
- ✓ A product must be identified before leaving the *Received* state.
- ✓ A description is mandatory when the channel or type is *Other*.
- ✓ Investigation: completion date  $\geq$  start date; mandatory fields checked at completion; root cause description required unless *Not Determined*; the approver must differ from the investigator; a rejection needs a reason.
- ✓ Resolution requires completion evidence; cancellation requires a reason.
- ✓ Closure requires a closure summary; cancellation requires a reason; a complaint that left *Received* cannot be deleted.

## 8. Good practices

- ★ Record the complainant's words verbatim in the description.
- ★ Ask for the sample and the lot number at first contact.
- ★ Search for complaints on the same lot before concluding.
- ★ Treat suspected falsified products as urgent and inform the authority according to local rules.

## 9. Pitfalls to avoid

- ✗ Closing "not confirmed" without examining a sample or retained sample.
- ✗ Missing the reportability assessment when health is involved.

## 10 Records and evidence

- Complaint record, investigations with approval, resolutions, closure summary
- Correspondence with the complainant

## 11 Performance indicators

- Acknowledgement within target
- Median closure time
- Complaints per million units per product
- % confirmed

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR 211.198 — Complaint files
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects

### 13. Demonstration

**Case.** A pharmacy reports by phone that a blister of lot 24B117 contains one broken tablet. The receiver registers the complaint (channel *Phone*, complainant type *Pharmacy*, type *Product Quality Defect*). QA acknowledges within 2 days, assesses severity *Minor*, starts an investigation (methodology *Batch Record Review*): no event recorded, IPC hardness within limits, 2 similar complaints in 12 months across 3 lots → not systemic. Conclusion *Complaint Confirmed*, root cause *Transport or Storage* (crushed box visible on the photo). Resolution: replacement and letter. Closed in 18 days.

| Complaints |            |                  |                               |             |                                 |              |          |             |               |          |
|------------|------------|------------------|-------------------------------|-------------|---------------------------------|--------------|----------|-------------|---------------|----------|
| Complaints |            |                  |                               |             |                                 |              |          |             |               |          |
| Priority   | Reference  | Receipt Date     | Subject                       | Complainant | Product                         | Category     | Severity | Responsible | Closure Du... | Status   |
| ☆☆☆        | CMP/202... | Sep 27, 10:40 AM | Skin reaction reported aft... |             | [DEMO-TAB-10] Demo Film-Coat... | Product Q... | Critical |             | Oct 27        | Received |
| ☆☆☆        | CMP/202... | Sep 27, 10:40 AM | Leaflet missing from carton   | Demo Ph...  | [DEMO-TAB-10] Demo Film-Coat... | Packaging... | Minor    |             | Oct 27        | Received |
| ☆☆☆        | CMP/202... | Sep 27, 10:40 AM | Chipped tablets found in b... | Demo Ph...  | [DEMO-TAB-10] Demo Film-Coat... | Product Q... | Major    |             | Oct 27        | Received |

Figure 50 — Complaint register. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Adverse events: reportability assessment and reporting deadlines

|                   |                                         |                |       |
|-------------------|-----------------------------------------|----------------|-------|
| Module            | ls_complaint                            | Version        | _____ |
| Menu              | Complaints ► Vigilance ► Adverse Events | Effective date | _____ |
| Owner (suggested) | Vigilance officer / Complaint Manager   | Approved by    | _____ |
| Related SOPs      | SOP-QA-040, SOP-MD-007                  |                |       |

## 1 Objective

Identify, among complaints, the events that must be reported to a competent authority, assess reportability and causality in a documented way, and submit within the legal deadline.

## 2 Scope

Adverse events and incidents linked to complaints for medicinal products, medical devices and cosmetics. Pharmacovigilance systems for medicinal products may be managed in a dedicated tool; this SOP then covers the hand-over.

## 3 Responsibilities

| Role (software group)         | Responsibility                                                          |
|-------------------------------|-------------------------------------------------------------------------|
| <b>Complaint Investigator</b> | Creates the adverse event from the complaint and collects facts.        |
| <b>Vigilance officer</b>      | Assesses seriousness, causality and reportability; submits the report.  |
| <b>Complaint Manager</b>      | Configures the reporting deadlines per category; supervises timeliness. |

## 4 Definitions

| Term                   | Definition                                                                                 |
|------------------------|--------------------------------------------------------------------------------------------|
| <b>Awareness date</b>  | Date when the company first became aware of the event; the reporting clock starts from it. |
| <b>Report due date</b> | Computed: awareness date + reporting deadline (days) configured on the complaint category. |
| <b>Causality</b>       | Certain, Probable, Possible, Unlikely, Unrelated, Not Assessable.                          |

## 5 Workflow in the software



Model: ls.complaint.adverse\_event

Figure 51 — Lifecycle of an adverse event.

## 6 Procedure

- 1 **Configure deadlines.** For each complaint category set *Adverse Event Reporting Deadline (days)* according to the applicable rule. A single deadline per category: create separate categories when deadlines differ by seriousness (for example, for medical devices under the MDR: 2, 10 or 15 days depending on the case; for US medical device reporting: 30 calendar days, or 5 work days in defined cases; for EU pharmacovigilance: 15 days for serious and 90 days for non-serious reactions).
- 2 **Create.** From the complaint, add an adverse event: awareness date, event date, description, seriousness and criteria, outcome, subject pseudonym, age range, sex.
- 3 **Assess.** Record causality and its rationale, *Reportable to Authority*, the reportability rationale and the competent authority. Click **Assess**.
- 4 **Submit.** Prepare and send the report through the authority's channel before the due date; record the submission date, the authority acknowledgement reference and who submitted; click **Record Submission**.
- 5 **Follow up.** Record follow-up information; tick *Follow Up Required* when a follow-up report is due.
- 6 **Close.** Click **Close** with closure notes.

### 7. Controls enforced by the software

- ✓ Awareness date not in the future; event date  $\leq$  awareness date.
- ✓ Reportability rationale mandatory once assessed; causality assessment mandatory.
- ✓ A reportable event requires the competent authority; a non-reportable event cannot record a submission.
- ✓ The authority acknowledgement reference is mandatory at submission.
- ✓ Closed events cannot be modified; non-draft events cannot be deleted.

### 8. Good practices

- ★ Assess reportability on the day of awareness; when in doubt, report.
- ★ Use pseudonyms — never store the patient's name in the complaint system.
- ★ Monitor the overdue filter daily.

### 9. Pitfalls to avoid

- ✗ Starting the clock at the date QA received the complaint instead of the first awareness in the company.
- ✗ One category used for all seriousness levels with a single, inappropriate deadline.

## 10 Records and evidence

- Adverse event with assessment, submission and acknowledgement

## 11 Performance indicators

- % of reportable events submitted on time (target 100 %)
- Median time from awareness to assessment

## 12 References

- MDR Articles 87 to 92 — Vigilance
- 21 CFR Part 803 — Medical Device Reporting
- EMA Good Pharmacovigilance Practices, Module VI — Management and reporting of adverse reactions
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** A hospital reports that a catheter tip detached during use, without injury. Awareness 3 March. The vigilance officer assesses seriousness *Serious* (could have led to serious deterioration), causality *Probable*, reportable to the national competent authority. The category *Medical device — serious incident* has a 15-day deadline: the report is due 18 March. It is submitted on 12 March; the authority's reference is recorded; a follow-up report is planned after the investigation.

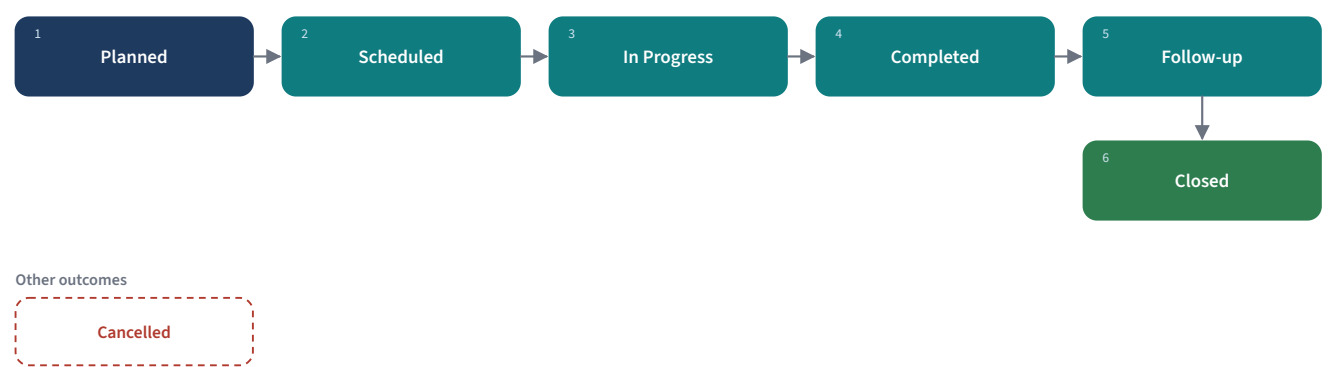
# Audits

ls\_audit

**ls\_audit** supports the internal audit process: auditable areas, audit types, versioned checklists, auditor qualifications (with lead-auditor status, qualified scope and expiry), audit programmes, individual audits with impartiality checks, checklist responses with objective evidence, findings with response, CAPA reference and verification, and audit reports reviewed and approved by people other than their author.

| Menu                                                                                                  | Content                |
|-------------------------------------------------------------------------------------------------------|------------------------|
| Quality ► Audits ► Audit Programmes / Audits / Audit Calendar / Checklists                            | Planning and execution |
| Quality ► Findings ► All Findings / My Findings                                                       | Findings               |
| Quality ► Reports ► Audit Reports / Checklist Responses                                               | Reporting              |
| Quality ► Configuration ► Audit Types / Auditable Areas / Auditor Qualifications / Finding Categories | Reference data         |

| Group                  | Main rights                                                                       |
|------------------------|-----------------------------------------------------------------------------------|
| Auditee                | Answers findings assigned to them                                                 |
| Auditor · Lead Auditor | Execute audits, raise findings, prepare reports; the lead auditor leads the audit |
| Audit Manager          | Approves programmes and checklists, manages qualifications, approves reports      |



Model: ls\_audit.schedule  
Figure 52 — Lifecycle of an audit.



Model: ls\_audit.finding  
Figure 53 — Lifecycle of an audit finding.

Quality
Audits
Findings
Reports
Configuration

7
14
My Company (San Francisco)

New
Audits
[AUD/2026/0002] Documentation Control - Raw...

Findings
0
Reports
0

1 / 1

Load Checklist
Schedule
Cancel
Planned
Scheduled
In Progress
Completed
Follow-up
Closed

Send message
Log note
Activity

AUD/2026/0002

Title
Documentation Control - Raw Material Warehou:

PLANNING

AUDIT TEAM

Audit Programme
[APG/2026/0001] Internal Audit Pr...

Lead Auditor
Amina Lead Auditor

Audit Type
Internal System Audit

Audit Team

Planned Date
Nov 11

Auditees
Sofiane Production Manager

Actual Start

Audited Organisation

Actual End

Checklist Used

Duration (hours)
00:00

Company
My Company (San Francisco)

Audited Areas
Algiers Manufacturing Site / Raw M...

Objective, Scope and Criteria
Checklist
Findings
Reports

OBJECTIVE

Determine whether goods receipt and storage records conform to the internal

Today

OdooBot
10:40 AM
Audit created

Figure 54 — Audit form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)



# Internal audit programme

|                   |                                                |                |       |
|-------------------|------------------------------------------------|----------------|-------|
| Module            | ls_audit                                       | Version        | _____ |
| Menu              | Quality ► Audits ► Audit Programmes            | Effective date | _____ |
| Owner (suggested) | Audit Manager                                  | Approved by    | _____ |
| Related SOPs      | SOP-QA-051, SOP-QA-052, SOP-QA-053, SOP-QA-071 |                |       |

## 1 Objective

Plan internal audits (self-inspections) so that every process and area of the quality system is audited at a frequency based on risk and past results, and follow the programme to completion.

## 2 Scope

All auditable areas of the company over the programme period (usually one year).

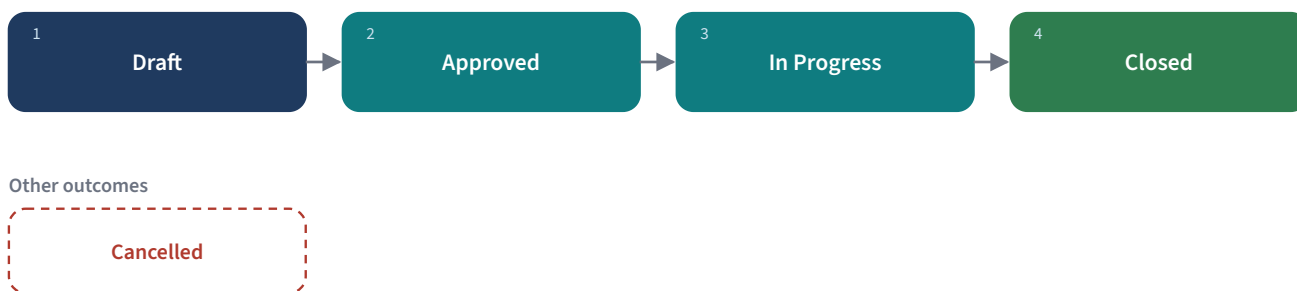
## 3 Responsibilities

| Role (software group)               | Responsibility                                                         |
|-------------------------------------|------------------------------------------------------------------------|
| <b>Audit Manager</b>                | Builds the programme, gets it approved, monitors execution, closes it. |
| <b>Head of Quality / management</b> | Approves the programme and provides resources.                         |
| <b>Lead auditors</b>                | Plan and conduct the audits assigned.                                  |

## 4 Definitions

| Term                   | Definition                                                            |
|------------------------|-----------------------------------------------------------------------|
| <b>Audit programme</b> | Set of audits planned for a period with a common objective.           |
| <b>Auditable area</b>  | Process, department or site that can be audited; areas can be nested. |

## 5 Workflow in the software



Model: ls.audit.program

Figure 55 — Lifecycle of an audit programme.

## 6 Procedure

- 1 Prepare reference data.** Maintain auditable areas, audit types and finding categories (with response deadlines) in [Quality ► Configuration](#).

- 2 **Assess risk.** Rank areas by risk: GMP criticality, results of previous audits, deviations, complaints, changes, inspection findings.
- 3 **Create the programme.** In **Audit Programmes** click **New**: title, start and end dates, owner, objective. Add the audits (title, type, areas, planned date inside the period).
- 4 **Approve.** Click **Approve** (the programme must contain at least one audit).
- 5 **Start.** Click **Start** at the beginning of the period; audits can then be scheduled (SOP-QA-052).
- 6 **Monitor.** Review the *Audit Calendar* monthly; reschedule with justification; add for-cause audits when events require it.
- 7 **Close.** At the end of the period, when every audit is closed or cancelled, click **Close** and report the programme outcome to management review.

## 7. Controls enforced by the software

- ✓ The end date cannot precede the start date; audits must be planned within the programme period.
- ✓ A programme can only be approved from Draft and must contain at least one audit; it must be approved before it starts and in progress before it closes.
- ✓ A programme cannot close while audits are neither closed nor cancelled.
- ✓ An audit cannot be scheduled while its programme is not approved.

## 8. Good practices

- ★ Audit every GMP area at least once per defined cycle and more often where risk is high.
- ★ Include the quality system itself (document control, CAPA, training) and the data integrity controls.
- ★ Keep capacity for for-cause audits.

## 9. Pitfalls to avoid

- ✗ Programmes copied from year to year without risk review.
- ✗ Postponing the same audit repeatedly.

## 10 Records and evidence

- Approved programme
- Audit calendar
- Programme closure summary

## 11 Performance indicators

- % of programme audits completed on time
- For-cause audits added

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 19011:2018 — Guidelines for auditing management systems
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements

## 13. Demonstration

**Case.** Programme *Internal audits 2027*: 14 audits covering 11 areas, with warehouse and sterile filling audited twice because of 2026 deviations. The Head of QA approves it in December; the Audit Manager clicks **Start** on 2 January.

Quality
Audits
Findings
Reports
Configuration

7
14
My Company (San Francisco)

New
Audit Programmes
[APG/2026/0001] Internal Audit Programme

Audits
2

1 / 1

Approve
Cancel

Draft
Approved
In Progress
Closed

Send message
Log note
Activity

1
2

APG/2026/0001

Title
Internal Audit Programme

PERIOD

GOVERNANCE

Start Date
Jan 1

End Date
Dec 31

Programme Owner
Nadia Quality Manager

Approved By

Approval Date

Completion Rate (%)
0%

Company
My Company (San Francisco)

Objective and Rationale
Audits
Notes

Cover every GMP-critical process at least once during the year. Frequency is set from process criticality, regulatory impact and the results of the previous programme.

Today

OdooBot
10:40 AM
Audit Programme created

Figure 56 — Audit programme with its audits. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Auditor qualification and impartiality

|                   |                                                  |                |       |
|-------------------|--------------------------------------------------|----------------|-------|
| Module            | ls_audit                                         | Version        | _____ |
| Menu              | Quality ► Configuration ► Auditor Qualifications | Effective date | _____ |
| Owner (suggested) | Audit Manager                                    | Approved by    | _____ |
| Related SOPs      | SOP-QA-050, SOP-QA-052, SOP-QA-013               |                |       |

## 1 Objective

Ensure that audits are conducted by qualified auditors, led by qualified lead auditors, within their qualified scope, and that no auditor audits their own work.

## 2 Scope

All internal auditors and lead auditors, and every audit of ls\_audit.

## 3 Responsibilities

| Role (software group) | Responsibility                                                                       |
|-----------------------|--------------------------------------------------------------------------------------|
| <b>Audit Manager</b>  | Qualifies auditors, maintains qualification records and expiry, assigns audit teams. |
| <b>Auditors</b>       | Maintain their competence, declare conflicts of interest.                            |

## 4 Definitions

| Term                   | Definition                                                         |
|------------------------|--------------------------------------------------------------------|
| <b>Qualified scope</b> | Auditable areas an auditor is qualified to audit.                  |
| <b>Impartiality</b>    | An auditor must not audit an area they own or work they performed. |

## 5 Procedure

- 1 Define criteria.** Write the qualification criteria (training in ISO 19011 principles, GMP knowledge, number of audits as observer, supervised audits) in this SOP.
- 2 Record the qualification.** In [Quality ► Configuration ► Auditor Qualifications](#) click **New**: auditor (user), employee, *Lead Auditor* flag, qualification date, expiry date, certificate reference, qualified scope (areas), notes.
- 3 Maintain.** Monitor *Qualification Status* and *Days To Expiry*; requalify before expiry (performance review, refresher training).
- 4 Assign teams.** When planning an audit, choose a lead auditor qualified as lead and team members qualified for the areas; list the auditees.
- 5 Check impartiality.** The system refuses auditors who are also auditees or who own an audited area; record other conflicts (recent job in the area) in the notes and replace the auditor.

## 6. Controls enforced by the software

- ✓ Expiry date cannot precede the qualification date; qualified scope must stay within the company.
- ✓ Only a user registered as qualified lead auditor can lead an audit.
- ✓ An audit is refused if a person appears both in the audit team and among the auditees, or if a team member owns an audited area.

## 7. Good practices

- ★ Rotate auditors across areas to keep a fresh view.
- ★ Pair new auditors with experienced lead auditors.
- ★ Link auditor training to ls\_training certifications (SOP-QA-013).

## 8. Pitfalls to avoid

- ✗ Qualification records not updated after an auditor changes department (new conflict of interest).
- ✗ Lead auditors whose qualification expired still leading audits.

## 9 Records and evidence

- Auditor qualification records
- Training and competency evidence

## 10 Performance indicators

- Qualified lead auditors available
- Qualifications expiring within 60 days

## 11 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

## 12. Demonstration

**Case.** A former production supervisor moves to QA and is qualified as auditor on 1 June, scope *Warehouse, QC laboratory*, expiry after 3 years. When the Audit Manager tries to add her to the audit of *Production — packaging*, which she owned until May, the impartiality check refuses her: she is replaced.

| <div> <div>Quality Audits Findings Reports Configuration</div> <div> <div>7</div> <div>14</div> <div>My Company (San Francisco)</div> </div> </div> |          |                                     |                 |              |                      |                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------|-----------------|--------------|----------------------|----------------------------|
| <div> <div>New Auditor Qualifications</div> <div>Search...</div> <div>1-2 / 2</div> </div>                                                          |          |                                     |                 |              |                      |                            |
| <input type="checkbox"/> Auditor                                                                                                                    | Employee | Lead Auditor                        | Qualificatio... | Expiry Date  | Qualification Status | Company                    |
| <input type="checkbox"/> Amina Lead Auditor                                                                                                         |          | <input checked="" type="checkbox"/> | Sep 27, 2025    | Sep 27, 2028 | Valid                | My Company (San Francisco) |
| <input type="checkbox"/> Karim Auditor                                                                                                              |          | <input type="checkbox"/>            | Mar 27          | Sep 27, 2027 | Valid                | My Company (San Francisco) |

Figure 57 — Auditor qualifications with status and days to expiry. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Conducting an audit: preparation, fieldwork, findings

|                   |                                    |                |       |
|-------------------|------------------------------------|----------------|-------|
| Module            | ls_audit                           | Version        | _____ |
| Menu              | Quality ► Audits ► Audits          | Effective date | _____ |
| Owner (suggested) | Lead auditor                       | Approved by    | _____ |
| Related SOPs      | SOP-QA-050, SOP-QA-051, SOP-QA-053 |                |       |

## 1 Objective

Conduct each audit in a planned, evidence-based way, and record nonconformities as findings with objective evidence and a clear requirement reference.

## 2 Scope

All internal audits of the approved programme and for-cause audits.

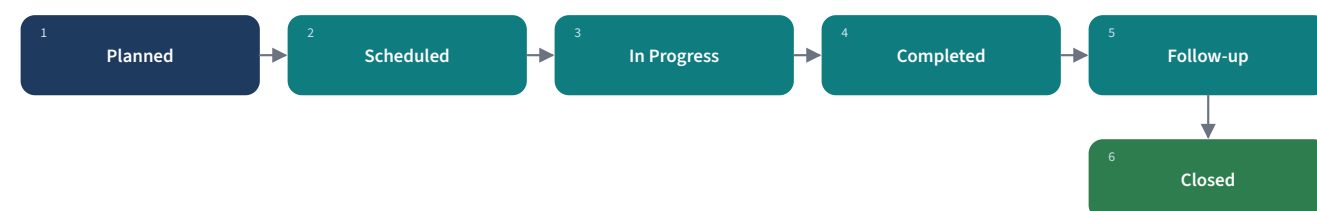
## 3 Responsibilities

| Role (software group) | Responsibility                                                                             |
|-----------------------|--------------------------------------------------------------------------------------------|
| <b>Lead auditor</b>   | Prepares, leads the opening and closing meetings, issues findings.                         |
| <b>Audit team</b>     | Performs the fieldwork, records responses with evidence.                                   |
| <b>Auditees</b>       | Provide access and information; nominate the person responsible for each finding response. |

## 4 Definitions

| Term                      | Definition                                                                                                      |
|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Checklist response</b> | Result per question: Conform, Non-conform, Observation, Not Applicable (Pending until answered).                |
| <b>Finding</b>            | Nonconformity or observation, with statement, objective evidence, requirement reference, category and severity. |

## 5 Workflow in the software



Other outcomes

Cancelled

Model: ls.audit.schedule

Figure 58 — Lifecycle of an audit.

## 6 Procedure

- 1 **Plan.** In the audit record (from the programme): objective, scope, criteria, lead auditor, team, auditees, planned date. Click [Schedule](#).
- 2 **Load the checklist.** Click [Load Checklist](#) and choose an approved checklist for the audit type; questions become responses to fill.
- 3 **Prepare.** Review previous findings, deviations and CAPAs of the area; send the agenda.
- 4 **Open fieldwork.** Click [Start Fieldwork](#) at the opening meeting.
- 5 **Collect evidence.** Answer each question with the result and the objective evidence (documents seen, records sampled, observations).
- 6 **Raise findings.** From a Non-conform or Observation response click [Raise Finding](#); complete the statement, evidence, requirement reference, category and severity, area and responsible auditee.
- 7 **Close fieldwork.** Hold the closing meeting; click [Complete Fieldwork](#).
- 8 **Issue findings.** Click [Issue to Auditee](#) on each finding; the response due date follows the finding category (SOP-QA-053).
- 9 **Checklist maintenance.** Checklists are approved by the Audit Manager; to change an approved checklist use [New Version](#).

### 7. Controls enforced by the software

- ✓ Objective evidence is required for any Non-conform or Observation response; a finding can only be raised from such a response, once per question.
- ✓ Responses can be edited only while the audit is scheduled or in progress; they can be removed only while it is scheduled.
- ✓ A finding can only be issued while the audit is in progress; it cannot be assigned to the person who raised it; its area must be in the audit scope.
- ✓ An approved checklist cannot be edited; a used checklist cannot return to draft — create a new version. A checklist without questions cannot be approved.
- ✓ Audits that left *Planned* cannot be deleted, only cancelled.

### 8. Good practices

- ★ Sample records randomly and write the identifiers in the evidence.
- ★ State the requirement breached (SOP section, GMP chapter) for each nonconformity.
- ★ Separate facts from opinions; share findings at the closing meeting — no surprises in the report.

### 9. Pitfalls to avoid

- ✗ Findings without evidence ("procedure not followed").
- ✗ Auditing only documents, never the shop floor.

## 10 Records and evidence

- Audit with checklist responses and evidence
- Findings
- Attendance of opening/closing meetings

## 11 Performance indicators

- Conformity rate per audit
- Findings per audit by severity



## 12 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

## 13. Demonstration

**Case.** Audit AUD-2027-004 *Warehouse*: the lead auditor loads checklist *WH-01 v3* (42 questions). Question 17 "Are temperature excursions investigated?": Non-conform — evidence: logger report of 14 July shows 27.8 °C for 3 h, no deviation found. She clicks **Raise Finding** : category *Major*, requirement *SOP-WH-005 §6.2*, responsible auditee: warehouse manager. After the closing meeting, **Complete Fieldwork** and **Issue to Auditee** .

The screenshot displays the Odoo 19.0 Community interface for a versioned audit checklist. The top navigation bar includes 'Quality', 'Audits', 'Findings', 'Reports', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main header shows 'New Checklists [CHK-DOC v1] Documentation Control' with a '1 / 1' indicator. Below the header, there are tabs for 'Reset to Draft', 'New Version', and 'Set Obsolete'. The checklist is currently in the 'Approved' state, with 'Draft' and 'Obsolete' also visible. On the right, there are buttons for 'Send message', 'Log note', and 'Activity'. The checklist details include: Name 'Documentation Control', CHK-DOC 'version 1', Audit Type 'Internal System Audit', Approved By 'Nadia Quality Manager', Approval Date 'Jul 27, 10:40 AM', Question Count '4', and Company 'My Company (San Francisco)'. A table lists the questions with their references and mandatory status:

| Reference       | Question                                                                                      | Mandatory                           |
|-----------------|-----------------------------------------------------------------------------------------------|-------------------------------------|
| SOP-QA-001 §4.2 | Are only current, approved versions of procedures available at the point of use?              | <input checked="" type="checkbox"/> |
| SOP-QA-001 §4.6 | Are obsolete documents removed from the point of use or clearly identified as obsolete?       | <input checked="" type="checkbox"/> |
| SOP-PR-014 §6.1 | Are batch records completed contemporaneously and signed by the operator performing the step? | <input checked="" type="checkbox"/> |
| SOP-QA-007 §3.3 | Are corrections to records made so that the original entry remains readable?                  | <input type="checkbox"/>            |

On the right side, a chat window shows a message from 'OdooBot' at 10:40 AM: 'Audit Checklist Template created'.

Figure 59 — Versioned audit checklist. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Audit report and follow-up of findings

|                   |                                              |                |       |
|-------------------|----------------------------------------------|----------------|-------|
| Module            | ls_audit                                     | Version        | _____ |
| Menu              | Quality ► Reports ► Audit Reports · Findings | Effective date | _____ |
| Owner (suggested) | Audit Manager                                | Approved by    | _____ |
| Related SOPs      | SOP-QA-052, SOP-QA-022                       |                |       |

## 1 Objective

Report the audit results to the responsible management through an independently reviewed and approved report, and follow each finding until its correction is verified.

## 2 Scope

All audits after fieldwork, and all issued findings.

## 3 Responsibilities

| Role (software group)      | Responsibility                                                                   |
|----------------------------|----------------------------------------------------------------------------------|
| <b>Lead auditor</b>        | Prepares the report; verifies finding actions.                                   |
| <b>Reviewer</b>            | Reviews the report independently.                                                |
| <b>Audit Manager</b>       | Approves and issues the report; accepts or rejects finding responses.            |
| <b>Responsible auditee</b> | Responds to findings: root cause, correction, corrective action, CAPA reference. |

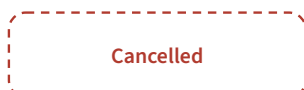
## 4 Definitions

| Term                     | Definition                                                |
|--------------------------|-----------------------------------------------------------|
| <b>Distribution list</b> | Managers who must receive the report; it cannot be empty. |
| <b>Response due</b>      | Deadline to respond, derived from the finding category.   |

## 5 Workflow in the software

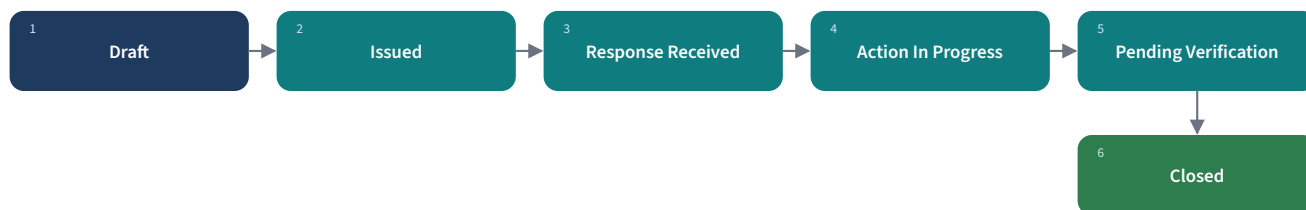


Other outcomes

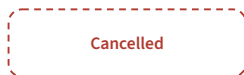


Model: ls.audit.report

Figure 60 — Lifecycle of an audit report.



Other outcomes



Model: ls.audit.finding

Figure 61 — Lifecycle of a finding.

## 6 Procedure

- 1 **Draft the report.** In **Audit Reports** click **New** : audit, date, executive summary, conclusion, positive observations, distribution list.
- 2 **Review.** Click **Submit for Review** ; a reviewer other than the preparer clicks **Record Review** .
- 3 **Approve and issue.** A user other than the preparer clicks **Approve** , then **Issue** .
- 4 **Respond.** The auditee opens **Findings ► My Findings** and clicks **Submit Response** : root cause, immediate correction, corrective action, CAPA reference when a CAPA is opened (SOP-QA-022), action due date.
- 5 **Accept or reject.** The Audit Manager clicks **Accept Response** or **Reject Response** (with reason).
- 6 **Implement.** When the actions are done, the auditee clicks **Action Completed** .
- 7 **Verify and close.** The lead auditor verifies on site or on records, documents the verification method and result, and clicks **Verify and Close** .
- 8 **Close the audit.** When all findings are closed, click **Move to Follow-up** then **Close Audit** on the audit.

### 7. Controls enforced by the software

- ✓ A report can only be submitted when the fieldwork is complete.
- ✓ The reviewer and the approver must differ from the preparer; approval requires a prior review; issue requires a distribution list.
- ✓ A response can only be accepted with a documented root cause and a CAPA reference when required.
- ✓ A finding cannot be closed without the verification method and result.
- ✓ Issued findings and reports cannot be deleted, only cancelled; final records cannot be modified.

### 8. Good practices

- ★ Issue the report within the time set in this SOP (for example 10 working days after closing meeting).
- ★ Reject responses that only correct the example found and do not address the cause.
- ★ Verify effectiveness of major findings at the next audit of the area.

### 9. Pitfalls to avoid

- ✗ Accepting "will be done" responses without dates or owners.
- ✗ Closing findings on the auditee's word without verification.

## 10 Records and evidence

- Approved and issued report
- Findings with responses and verification

## 11 Performance indicators

- Report issue time
- Findings overdue
- Repeat findings

## 12 References

- ISO 19011:2018 — Guidelines for auditing management systems
- EudraLex Vol. 4, EU GMP Part I, Chapter 9 — Self Inspection
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case (continued).** The report of AUD-2027-004 is prepared by the lead auditor, reviewed by a second auditor and approved by the Audit Manager, then issued to the site director and warehouse manager. The warehouse manager responds: root cause — logger alarms e-mailed to a former employee; correction — deviation opened; corrective action — alarm distribution list managed by role; CAPA-2027-012. Accepted; after implementation the auditor tests an alarm and clicks **Verify and Close** .

# Risk management

ls\_risk\_management

**ls\_risk\_management** follows ICH Q9(R1) and ISO 14971 concepts: configurable risk matrices (severity and probability levels, cells mapped to risk levels and acceptability), a risk register with hazard, hazardous situation and harm, successive risk assessments (initial, residual, periodic review, production/post-production information), risk control measures with the ISO 14971 option priority, residual risk acceptance, and FMEA worksheets whose failure modes can raise register entries.

| Menu                                                                                                                                         | Content                  |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <a href="#">Risk Management</a> ► <a href="#">Risk Management</a> ► <a href="#">Risk Register / Risk Assessments / Risk Control Measures</a> | Register and assessments |
| <a href="#">Risk Management</a> ► <a href="#">FMEA</a> ► <a href="#">FMEA Worksheets / Failure Modes</a>                                     | FMEA                     |
| <a href="#">Risk Management</a> ► <a href="#">Reporting</a> ► <a href="#">Reviews Due</a>                                                    | Periodic review          |
| <a href="#">Risk Management</a> ► <a href="#">Configuration</a> ► <a href="#">Risk Matrices / Risk Categories</a>                            | Matrices                 |

| Group                      | Main rights                                                            |
|----------------------------|------------------------------------------------------------------------|
| Risk Viewer · Risk Analyst | Read; perform assessments and FMEA                                     |
| Risk Manager               | Approves matrices and assessments, accepts residual risk, closes risks |

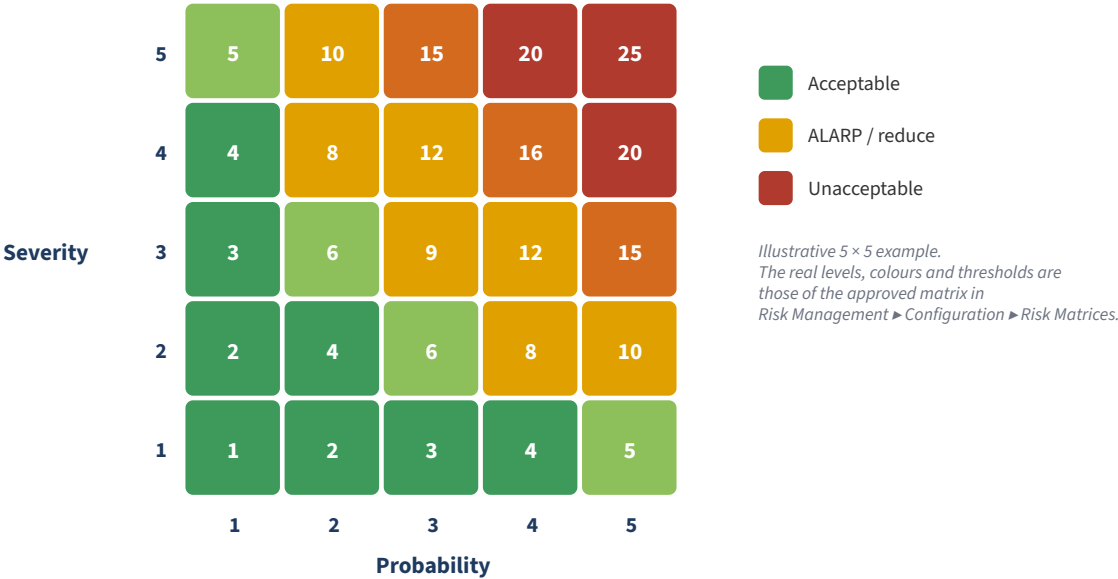
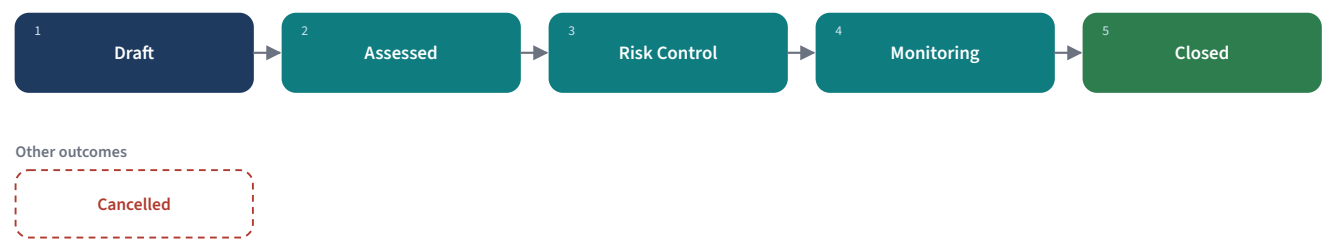


Figure 62 — Illustrative risk matrix.



Model: ls.risk.register

Figure 63 — Lifecycle of a risk register entry.

# Quality risk management: matrix, assessment, control and acceptance of residual risk

|                   |                                                                 |                |       |
|-------------------|-----------------------------------------------------------------|----------------|-------|
| Module            | ls_risk_management                                              | Version        | _____ |
| Menu              | Risk Management ► Risk Register · Configuration ► Risk Matrices | Effective date | _____ |
| Owner (suggested) | Risk Manager                                                    | Approved by    | _____ |
| Related SOPs      | SOP-QA-061, SOP-QA-062, SOP-QA-030, SOP-MD-003                  |                |       |

## 1 Objective

Identify, analyse, evaluate and control quality risks with a common, approved method, and accept residual risks formally and knowingly.

## 2 Scope

Product, process, equipment, facility, utility, material, supplier, computerised system, laboratory, packaging, regulatory and occupational risks recorded in the register. Medical device risk files follow SOP-MD-003 in addition.

## 3 Responsibilities

| Role (software group) | Responsibility                                                                                     |
|-----------------------|----------------------------------------------------------------------------------------------------|
| <b>Risk Manager</b>   | Approves the matrix and assessments, confirms control completeness, accepts residual risk, closes. |
| <b>Risk owner</b>     | Maintains the risk entry and its measures.                                                         |
| <b>Risk Analyst</b>   | Performs assessments (not the approval of their own work).                                         |
| <b>Measure owners</b> | Implement control measures and provide evidence.                                                   |

## 4 Definitions

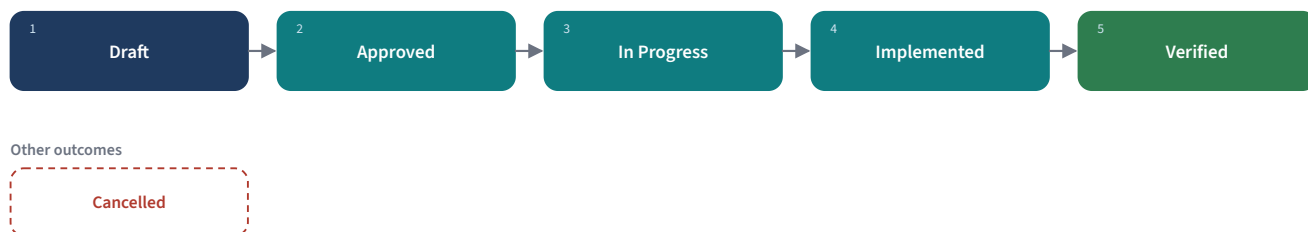
| Term                              | Definition                                                                                                                                          |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk level / acceptability</b> | Given by the matrix cell for a severity × probability pair: Negligible to Very High; Acceptable, Acceptable Only With Risk Control, Not Acceptable. |
| <b>Control option priority</b>    | Inherent safety by design, then protective measures, then information for safety (ISO 14971).                                                       |

## 5 Workflow in the software



Model: ls.risk.register

Figure 64 — Risk register entry lifecycle.



Other outcomes

Cancelled

Model: *ls.risk.mitigation*

Figure 65 — Risk control measure lifecycle.

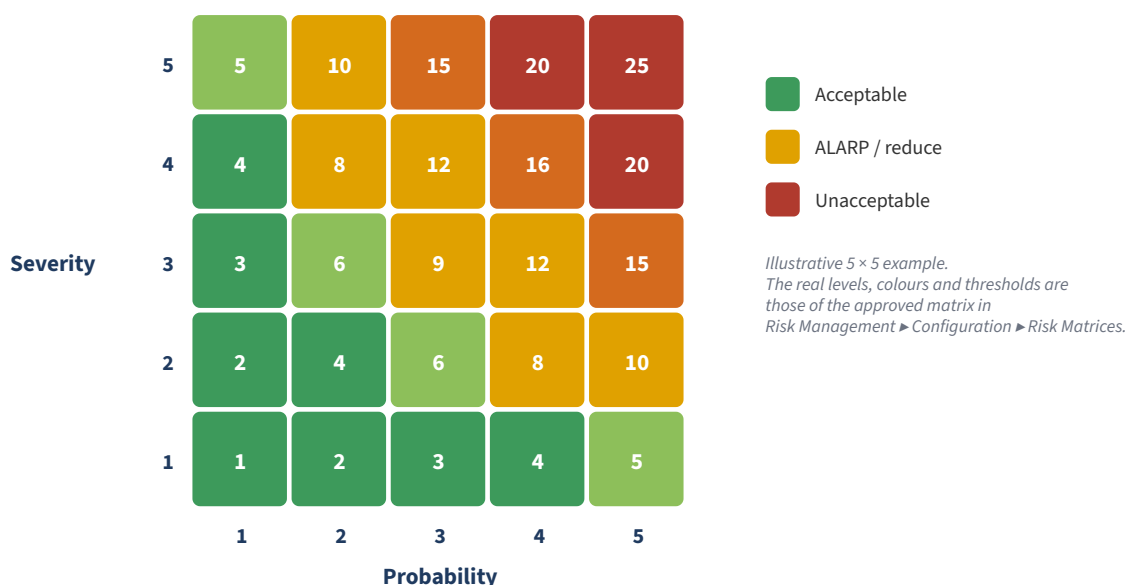


Figure 66 — Illustrative matrix: the real one is configured and approved.

## 6 Procedure

- Set up the matrix.** In **Configuration ► Risk Matrices** define severity and probability levels, click **Generate Missing Cells**, set the level and acceptability of each cell, reference the governing document, and click **Approve**. Mark one approved matrix as the company default.
- Identify.** In the *Risk Register* click **New**: title, type, category, owner, identification date, hazard, hazardous situation, sequence of events, harm, matrix, review interval.
- Assess.** Click **Assess** to create the initial assessment: severity, probability, estimation rationale. The analyst clicks **Confirm**; a Risk Manager other than the assessor clicks **Approve**. Then click **Mark Assessed**.
- Control.** Click **Start Risk Control**. Add control measures (description, option, responsible, due date); each is approved, started, marked implemented with evidence, then its effectiveness verified with evidence by a different person.
- Residual assessment.** Record a *Residual* assessment after the measures.
- New risks.** If a measure introduces a new hazard, tick *Introduces New Risk*, describe it and click **Create Risk Record for New Risk**.
- Complete and accept.** Click **Confirm Control Completeness** once every measure is verified or cancelled; if the residual risk is not plainly acceptable, click **Accept Residual Risk** with the justification and benefit-risk analysis.
- Monitor.** Click **Move to Monitoring**; periodic reviews follow SOP-QA-062; close with a reason when the risk no longer exists.

## 7. Controls enforced by the software

- ✓ The matrix must be complete (one cell per combination) and approved before assessments are approved; only one active default matrix per company.
- ✓ An assessment requires a rationale before confirmation; the assessor cannot approve it; approved assessments are frozen (cancel and record a new one).
- ✓ A risk has one initial assessment; later estimations use the other types.
- ✓ A risk cannot leave Draft without an approved initial assessment.
- ✓ Control completeness requires all measures verified or cancelled; monitoring requires completeness and, when the acceptability demands it, a residual risk acceptance decision.
- ✓ Measures: implementation evidence and effectiveness evidence are mandatory; the person who verified the implementation cannot verify the effectiveness; verified measures are frozen.
- ✓ Only a Risk Manager may perform the protected operations; assessed risks cannot be deleted.

## 8. Good practices

- ★ Write the rationale with data (failure history, detection capability), not "expert judgement" alone.
- ★ Prefer design measures over procedural ones; training is the weakest control.
- ★ Link the risk to the record it concerns (*Linked Model / Linked Record ID*) — e.g. a change request.

## 9. Pitfalls to avoid

- ✗ Lowering the probability in the residual assessment without any implemented measure.
- ✗ Accepting residual risk by default without a benefit-risk justification.

## 10 Records and evidence

- Risk register entry with assessments, measures, evidence, acceptance

## 11 Performance indicators

- Risks Not Acceptable without an active measure (target 0)
- Measures overdue
- Reviews overdue

## 12 References

- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management
- ISO 31000:2018 — Risk management — Guidelines
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System

## 13. Demonstration

**Case.** Risk *Cross-contamination on the shared granulator*: severity 4, probability 3 → level High, *Acceptable Only With Risk Control*. Measures: dedicated product-contact parts (design), cleaning validation with visual and swab limits (protective), label on the equipment status board (information). After verification, residual assessment: severity 4, probability 1 → Medium, acceptable with control. Completeness confirmed; the Risk Manager accepts the residual risk with justification; the risk moves to *Monitoring* with a 12-month review.



Figure 67 — Risk register entry. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# FMEA (failure modes and effects analysis)

|                   |                                          |                |       |
|-------------------|------------------------------------------|----------------|-------|
| Module            | ls_risk_management                       | Version        | _____ |
| Menu              | Risk Management ► FMEA ► FMEA Worksheets | Effective date | _____ |
| Owner (suggested) | Risk Manager                             | Approved by    | _____ |
| Related SOPs      | SOP-QA-060, SOP-QA-062                   |                |       |

## 1 Objective

Analyse systematically how a design, process, system or use can fail, rate severity, occurrence and detection, and drive actions where the risk priority exceeds the thresholds.

## 2 Scope

Design, process, system and use/application FMEA.

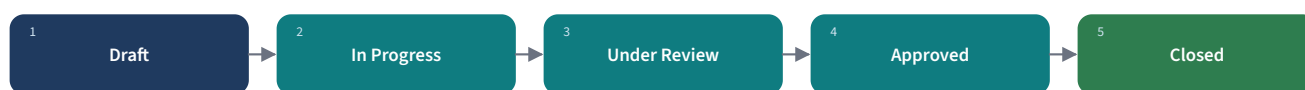
## 3 Responsibilities

| Role (software group)        | Responsibility                                                    |
|------------------------------|-------------------------------------------------------------------|
| <b>Facilitator</b>           | Leads the team, records the worksheet, submits it.                |
| <b>Team members</b>          | Contribute knowledge of the item analysed.                        |
| <b>Reviewer and approver</b> | Independent of the facilitator; review and approve the worksheet. |

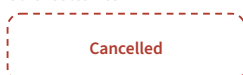
## 4 Definitions

| Term              | Definition                                                                                                 |
|-------------------|------------------------------------------------------------------------------------------------------------|
| <b>RPN</b>        | Risk Priority Number = $S \times O \times D$ , each rated 1–10.                                            |
| <b>Thresholds</b> | RPN action threshold and severity action threshold set on the worksheet: above them an action is required. |

## 5 Workflow in the software



Other outcomes



Model: ls.risk.fmea

Figure 68 — Lifecycle of an FMEA worksheet.

## 6 Procedure

- Create.** In **FMEA Worksheets** click **New**: title, FMEA type, scope, assumptions and boundaries, revision, facilitator, team, start date, RPN and severity thresholds. Click **Start**.
- Analyse.** For each item and function add failure modes: effect, severity, cause, occurrence, current controls, detection. The RPN is computed and lines above a threshold are flagged.

- 3 **Plan actions.** For flagged lines write the recommended action, responsible and target date.
- 4 **Review and approve.** Click **Submit for Review**; a reviewer other than the facilitator clicks **Record Review**, then an approver (also not the facilitator) clicks **Approve**.
- 5 **Implement and re-rate.** Record actions taken and the revised S, O and D together; the revised RPN and the reduction are computed.
- 6 **Raise risks.** Use **Create Risk** on a failure mode to open a register entry (SOP-QA-060) when the risk must be managed over time.
- 7 **Close or revise.** Click **Close** when no action remains open; use **Create Revision** for a later update.

## 7. Controls enforced by the software

- ✓ Only a draft worksheet can be started; only an in-progress worksheet with failure modes can be submitted.
- ✓ The facilitator cannot review or approve; approval requires prior review; lines above threshold must have a recommended action.
- ✓ Revised S, O and D must be given together.
- ✓ Closure requires no open action; only approved or closed worksheets can be revised.
- ✓ Raising a risk from a failure mode requires an approved default matrix; one register entry per failure mode.

## 8. Good practices

- ★ Rate severity on the effect, occurrence on the cause, detection on the current controls — never mix them.
- ★ Treat high severity even when the RPN is low (severity threshold).
- ★ Keep a cross-functional team: operators know the real failure modes.

## 9. Pitfalls to avoid

- ✗ Reducing detection scores by adding inspections instead of removing causes.
- ✗ Worksheets never revised after process changes.

## 10 Records and evidence

- Approved FMEA worksheet with revisions
- Linked register entries

## 11 Performance indicators

- Lines above threshold without action
- Average RPN reduction

## 12 References

- IEC 60812:2018 — Failure modes and effects analysis (FMEA and FMECA)
- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management

## 13. Demonstration

**Case.** Process FMEA of the blister packaging line: failure mode *missing tablet in blister*, effect *patient under-dosed* S=8, cause *feeder blockage* O=4, control *camera inspection* D=3 → RPN 96, above the RPN threshold of 80. Action: add feeder vibration sensor. After implementation O=2 → RPN 48. The worksheet is reviewed, approved, then closed when all actions are complete.

Risk Management

Risk Management

FMEA

Reporting

Configuration

7

14

My Company (San Francisco)

New

FMEA

FMEA/2026/0001 - Granulation process FMEA (rev. 1)

1 / 1

Start

Cancel

Draft

In Progress

Under Review

Approved

Closed

Send message

Log note

Activity

Q

0

The 1-10 severity, occurrence and detection scales and the Risk Priority Number are an industry convention, not a requirement of ISO 14971:2019. The action thresholds below are defined by this organisation.

FMEA/2026/0001

Granulation process FMEA

Fmea Type

Process FMEA

RPN Action Threshold

100

Revision

1

Severity Action Threshold

9

Facilitator

Mitchell Admin

Failure Mode Count

3

Start Date

Sep 27

Highest RPN

120

Company

My Company (San Francisco)

Lines Requiring Action

2

Open Actions

2

Failure Modes

Scope

Team

Review and Approval

| Item         | Function      | Failure M... | Effect of F...                           | Severity ... | Potential ...                         | Occurre... | Current C...                          | Detectio... |     |
|--------------|---------------|--------------|------------------------------------------|--------------|---------------------------------------|------------|---------------------------------------|-------------|-----|
| Granulator   | Mix powd...   | Inadequat... | Content uniformity outside specification | 9            | Mixing time below the validated range | 3          | In-process content uniformity testing | 4           | 108 |
| Binder ad... | Deliver bi... | Binder ov... | Oversized granules and clow...           | 6            | Pump calibration drift                | 4          | Quarterly pump calibration            | 5           | 120 |

OdooBot

10:39 AM

FMEA Worksheet created

Figure 69 — FMEA worksheet. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

Life Sciences Suite — SOP Handbook

95 / 311

# Periodic review of risks

|                   |                                           |                |       |
|-------------------|-------------------------------------------|----------------|-------|
| Module            | ls_risk_management                        | Version        | _____ |
| Menu              | Risk Management ► Reporting ► Reviews Due | Effective date | _____ |
| Owner (suggested) | Risk Manager                              | Approved by    | _____ |
| Related SOPs      | SOP-QA-060, SOP-QA-061, SOP-GEN-001       |                |       |

## 1 Objective

Keep risk assessments current by reviewing them at defined intervals and whenever new information (deviations, complaints, changes, post-market data) appears.

## 2 Scope

All risk register entries in *Monitoring* and approved FMEA worksheets.

## 3 Responsibilities

| Role (software group) | Responsibility                                                           |
|-----------------------|--------------------------------------------------------------------------|
| <b>Risk owner</b>     | Performs the review with the relevant data.                              |
| <b>Risk Manager</b>   | Approves the periodic assessment, decides on further control or closure. |

## 4 Definitions

| Term                              | Definition                                                                         |
|-----------------------------------|------------------------------------------------------------------------------------|
| <b>Review interval</b>            | Months between two reviews, set on each risk; <i>Next Review Date</i> is computed. |
| <b>Periodic Review assessment</b> | Assessment type used to record the re-estimation.                                  |

## 5 Procedure

- List.** Monthly open [Risk Management ► Reporting ► Reviews Due](#).
- Collect data.** Gather deviations, complaints, CAPA, changes, trends and post-market information related to the risk since the last review.
- Re-estimate.** Create an assessment of type *Periodic Review* (or *Production / Post-Production Information* when triggered by such data), confirm and have it approved.
- Decide.** If the level increased, return to risk control (new measures); if the hazard disappeared, close the risk with a reason.
- Event-driven reviews.** Any major deviation, complaint trend or change affecting a risk triggers an immediate review, not waiting for the interval.
- Report.** Summarise the risk profile in management review.

### 6. Controls enforced by the software

- ✓ Approved assessments cannot be modified; a new assessment is recorded for each review.
- ✓ The assessor cannot approve their own assessment.

## 7. Good practices

- ★ Align review intervals with risk level (shorter for High).
- ★ Record "no change" reviews as assessments too — the absence of change is information.

## 8. Pitfalls to avoid

- ✗ Reviews done on paper and never recorded.
- ✗ Ignoring post-market data in the review of product risks.

## 9 Records and evidence

- Periodic review assessments
- Decisions (new measures, closure)

## 10 Performance indicators

- Reviews overdue
- Risks whose level increased at review

## 11 References

- ICH Q9(R1) — Quality Risk Management
- ISO 14971:2019 — Medical devices — Application of risk management
- ICH Q10 — Pharmaceutical Quality System

## 12. Demonstration

**Case.** The risk *Temperature excursion during transport* is due for review. Since last year, 3 complaints and 1 deviation relate to it. The owner records a *Production / Post-Production Information* assessment: probability raised from 2 to 3, level Medium → High. The Risk Manager approves it and the risk returns to *Risk Control* with a new measure (validated shipper for summer lanes).

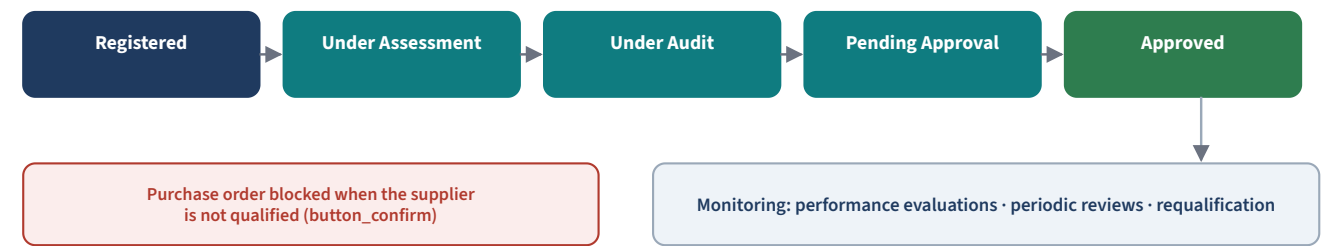
# Suppliers

## ls\_supplier\_qualification

**ls\_supplier\_qualification** manages one qualification dossier per supplier and company: scored assessments from templates, supplier audits with graded findings, the qualified scope (materials and services), performance evaluations (on-time delivery, quality acceptance, documentation, responsiveness, rated A to D), periodic reviews, and an append-only, SHA-256-chained signature log of qualification decisions. A company setting controls purchase order confirmation for suppliers that are not approved: *none*, *warn* (default) or *block*.

| Menu                                                                                                                 | Content       |
|----------------------------------------------------------------------------------------------------------------------|---------------|
| Supplier Qualification ► Qualification ► Qualification Dossiers / Qualified Scope / Expiring Approvals               | Dossiers      |
| Supplier Qualification ► Evaluation ► Assessments / Audits / Audit Findings                                          | Evaluation    |
| Supplier Qualification ► Monitoring ► Performance Evaluations / Periodic Reviews / Signature Log                     | Monitoring    |
| Supplier Qualification ► Configuration ► Supplier Categories / Assessment Templates / Criteria / Reference Standards | Configuration |

| Group                               | Main rights                                   |
|-------------------------------------|-----------------------------------------------|
| Supplier Viewer · Supplier Assessor | Read; perform assessments, audits and reviews |
| Supplier Manager                    | Approves dossiers, changes status, configures |



Other states: Conditionally Approved, Suspended, Expired, Disqualified.

Figure 70 — Supplier qualification lifecycle and purchase control.

Supplier Qualification

Qualification

Evaluation

Monitoring

Configuration

7

14

My Company (San Francisco)

New

Qualification Dossiers

SQ/2026/00001

Scope

1

Assessments

1

Audits

0

Performance

0

Reviews

0

Signatures

0

1 / 1

Start Audit

Submit for Approval

Reset to Draft

Registered

Under Assessment

Under Audit

...

Send message

Log note

Activity

0

SQ/2026/00001

Not ready for approval yet:

No completed assessment with a Pass or Conditional result is attached to the dossier.

No closed audit with an acceptable outcome is attached to the dossier.

No qualified material or service defines the approval scope.

SUPPLIER

Supplier

Demo Pharma Ingredients SARL

Supplier Category

Active Substance Manufacturer

Criticality

Critical

Responsible

OdooBot

Company

My Company (San Francisco)

VALIDITY

Registration Date

Jun 29

Approval Date

Approved By

Requalification Interval

Months

36

Approval Valid Until

Next Audit Date

Next Review Date

INDICATORS

Risk Level

Medium

Latest Assessment

Latest Assessment

Score (%)

0%

MONITORING

Latest Performance

Score

0.00

Latest Performance

Rating

Today

OdooBot

10:41 AM

Life Sciences Supplier Qualification Dossier created

Figure 71 — Qualification dossier. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)



# Qualification and approval of suppliers

|                   |                                                                    |                |       |
|-------------------|--------------------------------------------------------------------|----------------|-------|
| Module            | ls_supplier_qualification                                          | Version        | _____ |
| Menu              | Supplier Qualification ► Qualification ► Qualification<br>Dossiers | Effective date | _____ |
| Owner (suggested) | Supplier Manager (QA)                                              | Approved by    | _____ |
| Related SOPs      | SOP-QA-071, SOP-QA-072, SOP-QA-073, SOP-QC-001                     |                |       |

## 1 Objective

Approve suppliers of GMP-relevant materials and services only after an assessment proportionate to their criticality, and document the approval, its scope, its conditions and its validity.

## 2 Scope

Suppliers of active substances, excipients, packaging, components, contract services, laboratories, logistics and other GMP-relevant goods and services.

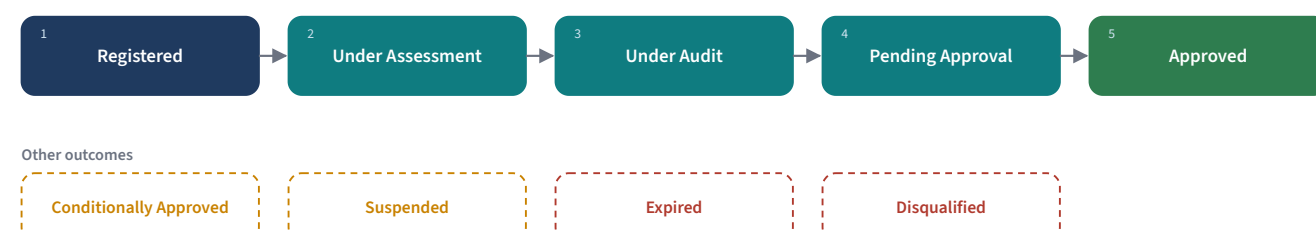
## 3 Responsibilities

| Role (software group)    | Responsibility                                                                                                      |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Supplier Assessor</b> | Performs assessments and audits; cannot approve dossiers they produced evidence for (when segregation is enforced). |
| <b>Supplier Manager</b>  | Configures categories and templates, submits and approves dossiers, sets conditions.                                |
| <b>Purchasing</b>        | Requests new suppliers and does not order before approval.                                                          |

## 4 Definitions

| Term                        | Definition                                                                                                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supplier category</b>    | Criticality, whether assessment, initial audit and periodic audit are required, audit interval, requalification and review intervals. |
| <b>Qualified scope</b>      | Materials or services for which the supplier is qualified, with validity dates.                                                       |
| <b>Conditional approval</b> | Approval with documented conditions (e.g. increased incoming testing).                                                                |

## 5 Workflow in the software



Model: ls.supplier\_qualification

Figure 72 — Lifecycle of a qualification dossier.

## 6 Procedure

- 1 **Configure.** Define supplier categories and assessment templates (criteria, weights, maximum score, minimum score for mandatory criteria, pass and conditional thresholds). In the company settings choose *Enforce Segregation of Duties* and the purchase order control level (see SOP-QA-073).
- 2 **Register.** In **Qualification Dossiers** click **New** : supplier, category, criticality, responsible, registration date, requalification interval. One active dossier per supplier and company.
- 3 **Assess.** Click **Start Assessment** ; create an assessment, choose the template, click **Load Criteria from Template** , **Start** , score each criterion (comment every criterion below the minimum), click **Complete** ; a different assessor clicks **Review** .
- 4 **Audit if required.** Click **Start Audit** when the category requires an initial audit (SOP-QA-071).
- 5 **Define the scope.** Add the qualified scope lines (product or service, qualification date, validity end) and click **Qualify** on each.
- 6 **Submit.** Click **Submit for Approval** ; the system lists any missing prerequisite.
- 7 **Approve.** The Supplier Manager clicks **Approve** and confirms approval date, validity and, for a conditional approval, the conditions. The decision is written in the signature log.
- 8 **Communicate.** Inform purchasing and the warehouse; the supplier appears as approved on purchase orders.

### 7. Controls enforced by the software

- ✓ One active dossier per supplier and company — requalify the existing one instead of creating another.
- ✓ Transitions follow the workflow: only a registered dossier moves to assessment; submission lists missing prerequisites; only a pending dossier can be approved.
- ✓ Segregation of duties (company setting, enabled by default): a person who produced assessment or audit evidence cannot approve the dossier; an assessor cannot review their own assessment.
- ✓ Assessment scoring: scores within the maximum; conditional threshold  $\leq$  pass threshold; low scores require comments; criteria are loaded only in draft.
- ✓ An approved dossier needs its mandatory information; a conditional approval needs its conditions; expiry after approval date.
- ✓ The signature log is append-only and chained; its chain can be verified ( **Verify Chain** ).

### 8. Good practices

- ★ Base the depth of qualification on criticality: questionnaire for low risk, audit for high risk.
- ★ Qualify the manufacturer, not only the distributor, for active substances.
- ★ Keep quality agreements as attachments of the dossier.

### 9. Pitfalls to avoid

- ✗ Approving under commercial pressure with incomplete evidence.
- ✗ Qualifying a supplier generically when only one material was assessed — use the qualified scope.

## 10 Records and evidence

- Dossier with assessments, audits, scope, approval and signature log

## 11 Performance indicators

- Suppliers approved per category
- Time to qualify
- Conditional approvals open

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.84 — Testing and approval or rejection of components
- ICH Q10 — Pharmaceutical Quality System

## 13. Demonstration

**Case.** New excipient supplier, category *Excipient — major*: assessment with template *Excipient GMP questionnaire* scores 82 % (pass threshold 80 %); desk review of certificates of analysis for 3 lots; no audit required for this category. Scope: *Microcrystalline cellulose PH-102*, validity 3 years. A Supplier Manager who did not perform the assessment approves the dossier; the signature log records *Approved*.

The screenshot displays the Odoo 19.0 Community LS modules interface for a supplier assessment. The top navigation bar shows 'Supplier Qualification', 'Qualification', 'Evaluation', 'Monitoring', and 'Configuration'. The left sidebar has a 'New' button and a 'Supplier Assessments' button. The main content area shows a 'Draft' status for the assessment 'SA/2026/00001'. The assessment details include:

| SUBJECT         |                              | EXECUTION   |         |
|-----------------|------------------------------|-------------|---------|
| Qualification   | SQ/2026/00001                | Date        | Jul 29  |
| Supplier        | Demo Pharma Ingredients SARL | Assessor    | OdooBot |
| Template        | Full Supplier Assessment     | Reviewer    |         |
| Assessment Type | Initial Qualification        | Review Date |         |

Below the execution details, there are two sections: 'SCORING RULES (FROZEN)' and 'RESULT'.

| SCORING RULES (FROZEN)  |       | RESULT                   |      |
|-------------------------|-------|--------------------------|------|
| Max Score Per Criterion | 5     | Total Weighted Score     | 0.00 |
| Mandatory Min Score     | 3     | Total Max Weighted Score | 0.00 |
| Pass Threshold          | 80.00 | Score (%)                | 0%   |
| Conditional Threshold   | 60.00 | Mandatory Failed Count   | 0    |
|                         |       | Finding Count            | 0    |
|                         |       | Result                   |      |

At the bottom, there is a table for 'Scored Criteria' and 'Conclusion'.

| Criterion  | Domain | Weight | Mandatory | Score | Max Sco... | Evidence | Comment | Raise Fin... |
|------------|--------|--------|-----------|-------|------------|----------|---------|--------------|
| Add a line |        |        |           |       |            |          |         |              |

On the right side of the interface, there is a chat window with a message from 'OdooBot' at 10:41 AM: 'Life Sciences Supplier Assessment created'.

Figure 73 — Scored supplier assessment. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Supplier audits

|                   |                                              |                |       |
|-------------------|----------------------------------------------|----------------|-------|
| Module            | ls_supplier_qualification                    | Version        | _____ |
| Menu              | Supplier Qualification ► Evaluation ► Audits | Effective date | _____ |
| Owner (suggested) | Supplier Manager (QA)                        | Approved by    | _____ |
| Related SOPs      | SOP-QA-070, SOP-QA-072, SOP-QA-052           |                |       |

## 1 Objective

Verify on site, remotely or on documents that a supplier's quality system and practices meet the requirements, and obtain corrective actions for the findings.

## 2 Scope

Initial, surveillance, for-cause and requalification audits of suppliers, whatever the format (on-site, remote, desktop, documentation review).

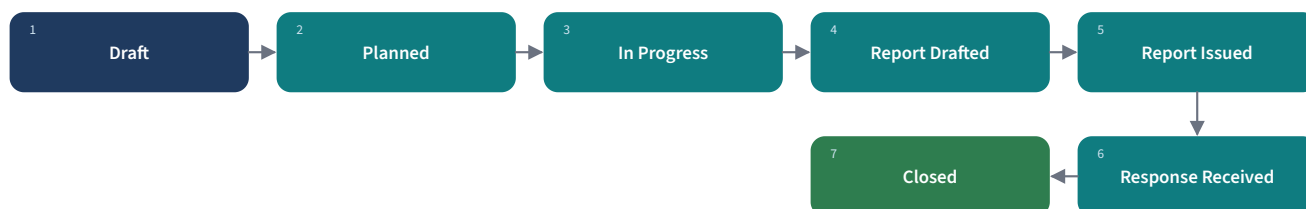
## 3 Responsibilities

| Role (software group)   | Responsibility                                     |
|-------------------------|----------------------------------------------------|
| <b>Lead auditor</b>     | Plans, conducts, reports, verifies findings.       |
| <b>Audit team</b>       | Supports the lead auditor.                         |
| <b>Supplier contact</b> | Answers findings with corrective actions.          |
| <b>Supplier Manager</b> | Takes the outcome into the qualification decision. |

## 4 Definitions

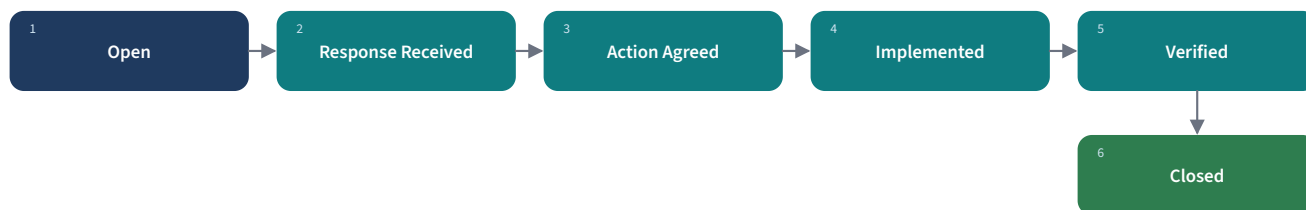
| Term                    | Definition                                           |
|-------------------------|------------------------------------------------------|
| <b>Outcome</b>          | Acceptable, Acceptable with Actions, Not Acceptable. |
| <b>Finding severity</b> | Critical, Major, Minor, Observation.                 |

## 5 Workflow in the software



Model: ls.supplier.audit

Figure 74 — Lifecycle of a supplier audit.



Other outcomes

Cancelled

Model: ls.supplier.audit.finding

Figure 75 — Lifecycle of a supplier audit finding.

## 6 Procedure

- 1 **Plan.** From the dossier click **Start Audit** or create an audit: type, scope, basis (reference standards), planned date, lead auditor, team, supplier contact. Click **Plan**.
- 2 **Conduct.** Click **Start** on the first day; record dates; add findings with severity and description.
- 3 **Report.** Click **Draft Report**; set the outcome and write the conclusion; click **Issue Report**. The response due date defaults from the company *Audit Response Lead Time* (30 days by default).
- 4 **Collect responses.** For each finding click **Register Response** with the supplier's answer; then **Agree Action** with the corrective action and due date.
- 5 **Follow up.** When the supplier reports completion click **Mark Implemented**; verify (evidence, follow-up audit) and click **Verify** with the verification description, then **Close**.
- 6 **Close the audit.** Register the overall response (**Register Response** on the audit) and click **Close Audit**; update the dossier (SOP-QA-070 / 072).

### 7. Controls enforced by the software

- ✓ Dates: end  $\geq$  start; response due  $\geq$  report date; response not before the report is issued.
- ✓ An audit with a critical finding cannot conclude *Acceptable*.
- ✓ The report cannot be issued without outcome and written conclusion; states follow Plan → Start → Draft Report → Issue Report → Response → Close.
- ✓ Findings: response before agreeing an action; the agreed action and its due date are required; verification requires a description; a finding cannot close before the report is issued; critical/major findings need a documented corrective action.
- ✓ A planned audit cannot be deleted, only cancelled.

### 8. Good practices

- ★ Send the agenda and the document request list before the audit.
- ★ Grade findings consistently with your internal audit categories.
- ★ For critical findings, involve the Supplier Manager immediately: suspension may be required.

### 9. Pitfalls to avoid

- ✗ Accepting supplier answers without evidence of implementation.
- ✗ Audits whose report is issued months later.

## 10 Records and evidence

- Audit, report, findings with responses and verification

## 11 Performance indicators

- Open findings per supplier
- Report issue time
- Audits due vs done

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 19011:2018 — Guidelines for auditing management systems
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** On-site audit of a packaging printer: 1 major finding (no line clearance record between two customers' artworks), 2 minor. Outcome *Acceptable with Actions*, report issued 5 days after the audit; response due 30 days later. The supplier introduces a line clearance form; the auditor verifies 3 completed forms sent as evidence and closes the findings; the dossier stays approved.

# Supplier performance evaluation and requalification

|                   |                                     |                |       |
|-------------------|-------------------------------------|----------------|-------|
| Module            | ls_supplier_qualification           | Version        | _____ |
| Menu              | Supplier Qualification ► Monitoring | Effective date | _____ |
| Owner (suggested) | Supplier Manager (QA)               | Approved by    | _____ |
| Related SOPs      | SOP-QA-070, SOP-QA-071, SOP-QA-073  |                |       |

## 1 Objective

Monitor approved suppliers with objective performance data and periodic reviews, and requalify, restrict, suspend or disqualify them when evidence requires it.

## 2 Scope

All approved and conditionally approved suppliers.

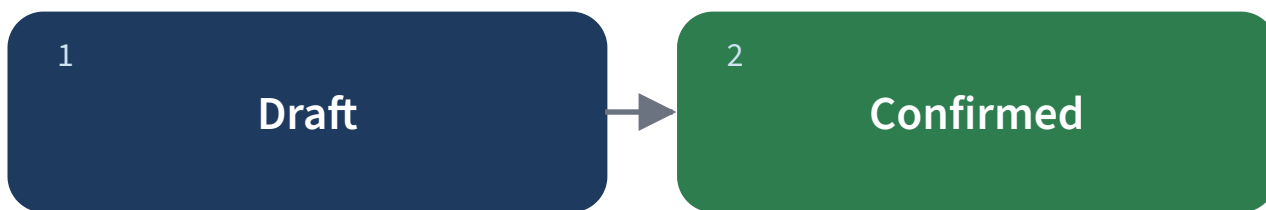
## 3 Responsibilities

| Role (software group)                | Responsibility                                |
|--------------------------------------|-----------------------------------------------|
| <b>Supplier Assessor / evaluator</b> | Performs performance evaluations and reviews. |
| <b>Supplier Manager</b>              | Takes status decisions and requalifies.       |

## 4 Definitions

| Term                     | Definition                                                                                                                                                                                                                              |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Performance score</b> | Weighted score of on-time delivery, quality acceptance, documentation compliance and responsiveness (weights set on the company, default 30/40/15/15), rated A – Preferred, B – Acceptable, C – Improvement Required, D – Unacceptable. |
| <b>Review decision</b>   | Maintain Approval, Maintain with Conditions, Requalification Required, Suspend, Disqualify.                                                                                                                                             |

## 5 Workflow in the software



### Other outcomes



*Model: ls.supplier.performance*

Figure 76 — Lifecycle of a performance evaluation.

## 6 Procedure

- 1 **Evaluate performance.** At the defined frequency (for example quarterly) create a performance evaluation for the period; click **Recompute Delivery Counters** when the purchase-stock link is installed, or enter counters; enter lots inspected/rejected, nonconformities, complaints, documentation and responsiveness scores; click **Confirm**.
- 2 **Act on ratings.** Rating C or D: open a supplier finding or CAPA, inform the supplier, consider a for-cause audit.
- 3 **Periodic review.** When due (review interval of the category), create a periodic review: period, reviewer, click **Collect Evidence** (assessments, audits, performance evaluations of the period), write the summary and the decision with justification; click **Complete Review**.
- 4 **Change status.** For suspension or disqualification the Supplier Manager clicks **Change Status** with the reason; the decision is logged.
- 5 **Requalify.** Before approval expiry (reminders are raised by the company *Expiry Reminder* lead time, 60 days by default), click **Requalify** and repeat the relevant steps of SOP-QA-070.

### 7. Controls enforced by the software

- ✓ Period end after start; counters non-negative; late deliveries  $\leq$  deliveries; rejected lots  $\leq$  inspected lots; percentages 0–100.
- ✓ Two confirmed evaluations of the same dossier cannot overlap; confirmed evaluations cannot be deleted.
- ✓ Automatic delivery counters require the purchase-stock link (module *purchase\_stock*).
- ✓ Reviews: an adverse decision must be justified; conditions must be documented when maintaining with conditions; completed reviews cannot be deleted or cancelled.
- ✓ Rating thresholds must decrease from A to C; at least one weight must be positive.



## 8. Good practices

- ★ Discuss results with key suppliers; share the scorecard.
- ★ Event-driven reviews after a recall, a critical complaint or a regulatory action at the supplier.

## 9. Pitfalls to avoid

- ✗ Suppliers left approved past their validity because nobody requalified them.
- ✗ Performance data entered by memory rather than from records.

## 10 Records and evidence

- Performance evaluations
- Periodic reviews with decisions
- Signature log entries

## 11 Performance indicators

- Suppliers rated C/D
- Approvals expired
- Reviews overdue

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 7 — Outsourced Activities
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 9001:2015 — Quality management systems — Requirements
- ICH Q10 — Pharmaceutical Quality System

## 13. Demonstration

**Case.** Q2 evaluation of an ampoule supplier: 20 deliveries, 5 late (OTD 75 %), 12 lots inspected, 2 rejected (quality 83 %); score gives rating C. The Supplier Manager raises a supplier CAPA and schedules a for-cause audit. At the annual review the reviewer collects the audit, the two evaluations and decides *Maintain with Conditions*: tightened incoming inspection until two consecutive quarters rated B.

# Control of purchases from non-approved suppliers

|                   |                                                               |                |       |
|-------------------|---------------------------------------------------------------|----------------|-------|
| Module            | ls_supplier_qualification · purchase                          | Version        | _____ |
| Menu              | Company settings ► Purchase Order Control · Purchase ► Orders | Effective date | _____ |
| Owner (suggested) | Supplier Manager with the Purchasing Manager                  | Approved by    | _____ |
| Related SOPs      | SOP-QA-070, SOP-QA-072                                        |                |       |

## 1 Objective

Prevent the purchase of GMP-relevant materials and services from suppliers that are not approved, whose approval expired, or for products outside their qualified scope.

## 2 Scope

Purchase orders in Odoo Purchase for suppliers with a qualification dossier and for GMP-relevant products.

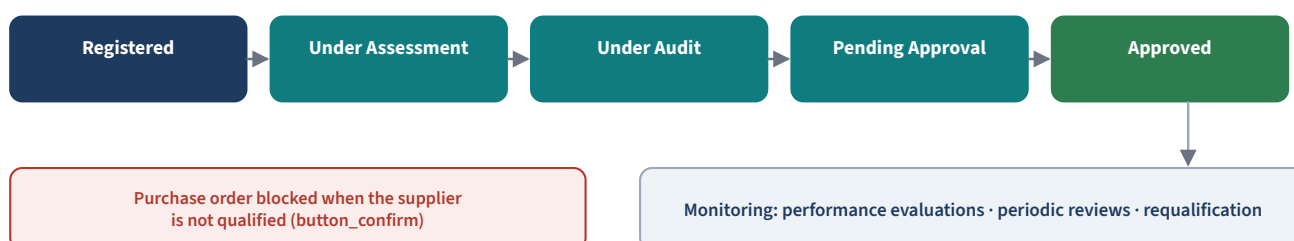
## 3 Responsibilities

| Role (software group)   | Responsibility                                                           |
|-------------------------|--------------------------------------------------------------------------|
| <b>Supplier Manager</b> | Sets the control level and keeps dossiers current.                       |
| <b>Buyers</b>           | Check the qualification warning before confirming; never bypass a block. |
| <b>QA</b>               | Handles exceptional purchases through a deviation or change request.     |

## 4 Definitions

| Term                 | Definition                                                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Control level</b> | Company setting <i>Purchase Order Control</i> : <i>none</i> (no check), <i>warn</i> (issue posted in the order chatter — default at installation), <i>block</i> (confirmation refused). |
| <b>Scope check</b>   | Optional setting <i>Check Qualified Scope on Purchase Orders</i> : each ordered product must be in the supplier's qualified scope, qualified and not expired.                           |

## 5 Workflow in the software



Other states: Conditionally Approved, Suspended, Expired, Disqualified.

Figure 77 — The purchase order control closes the loop.

## 6 Procedure

- 1 Configure.** In the company settings of Supplier Qualification set *Purchase Order Control* to **block** for GMP operations and tick *Check Qualified Scope on Purchase Orders*. Record this configuration in the validation file.

- 2 **Order.** The buyer creates the order; the *Qualification Warning* field shows any issue before confirmation.
- 3 **Confirm.** On **Confirm Order** the system checks the supplier status (approved states), the approval expiry and, if enabled, the scope. When blocked, the confirmation is refused with the list of issues.
- 4 **Resolve.** Complete the qualification (SOP-QA-070), requalify an expired supplier (SOP-QA-072), or add the product to the qualified scope.
- 5 **Exceptional purchase.** If a purchase must be made before qualification (e.g. development batch), open a planned deviation or change request, approved by QA, and quarantine the goods until the decision.
- 6 **Verify periodically.** Quarterly, list confirmed orders to suppliers not approved at the order date (report from Purchase) to check that the control works.

### 7. Controls enforced by the software

- ✓ With level *block*, confirmation is refused for a supplier that is not approved or whose approval expired, and — with the scope check — for products outside the qualified scope.
- ✓ With level *warn*, the issue is only posted in the order chatter: the order can still be confirmed.
- ✓ The status is read in the company of the order, even when the buyer has no access to qualification dossiers.

### 8. Good practices

- ★ Use *block* in production companies; *warn* is acceptable only for non-GMP entities.
- ★ Keep the product list of the qualified scope aligned with the product catalogue.

### 9. Pitfalls to avoid

- ✗ Leaving the default *warn* level in a GMP company.
- ✗ Creating a new supplier record to bypass the block — the dossier is linked to the partner.

## 10 Records and evidence

- Company setting (with its change history)
- Blocked confirmation messages
- Planned deviations for exceptional purchases

## 11 Performance indicators

- Orders confirmed to non-approved suppliers (target 0)
- Blocked confirmations per month

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** A buyer orders gelatin capsules from a supplier whose approval expired 10 days ago. On **Confirm Order** the system refuses: *"The approval of supplier X expired on ..."*. The Supplier Manager starts the requalification; meanwhile QA approves a planned deviation for a single delivery held in quarantine until requalification is complete.

# II

## Data integrity and compliance

The procedures that protect the reliability of GxP data and demonstrate that systems, instruments and environments are fit for use: audit trail, electronic signatures, validation, calibration, environmental monitoring and recalls.

|                                                 |                                                        |
|-------------------------------------------------|--------------------------------------------------------|
| Audit trail — ls_audit_trail                    | Calibration — ls_calibration                           |
| Electronic signatures — ls_electronic_signature | Environmental monitoring — ls_environmental_monitoring |
| Validation — ls_validation                      | Recalls — ls_recall                                    |

# Audit trail

## ls\_audit\_trail

**ls\_audit\_trail** records who changed what, when and from where, for the models and fields chosen in *audit rules*. Each change becomes an immutable entry of a per-company SHA-256 hash chain, so that altering, inserting or removing an entry is detectable. Integrity is verified on demand and daily by a scheduled action; *evidence packs* export a sealed ZIP (JSON, CSV and manifest) with its digest for inspectors; a controlled *retention run* removes aged entries while anchoring the chain. **Nothing is audited until a rule exists.** One-to-many and many-to-many relations and non-stored computed fields are never captured directly.

| Menu                                                                      | Content                   |
|---------------------------------------------------------------------------|---------------------------|
| <a href="#">Audit Trail ▶ Audit Log / Field Changes / Audit Dashboard</a> | Entries and field changes |
| <a href="#">Audit Trail ▶ Integrity ▶ Verifications / Verify Now</a>      | Integrity checks          |
| <a href="#">Audit Trail ▶ Evidence Packs</a>                              | Sealed exports            |
| <a href="#">Audit Trail ▶ Configuration ▶ Audit Rules / Retention Run</a> | Scope and retention       |

| Group                       | Main rights                                      |
|-----------------------------|--------------------------------------------------|
| Audit Trail / Viewer        | Read the trail of their companies                |
| Audit Trail / Auditor       | Also verify integrity and produce evidence packs |
| Audit Trail / Administrator | Also configure rules and run approved retention  |

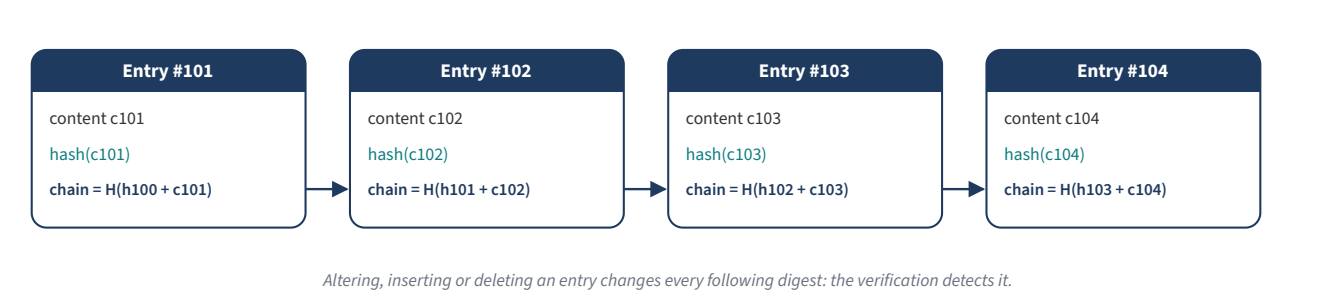


Figure 78 — How the hash chain makes tampering detectable.

## Definition of the audit trail scope (rules per model)

|                   |                                           |                |       |
|-------------------|-------------------------------------------|----------------|-------|
| Module            | ls_audit_trail                            | Version        | _____ |
| Menu              | Audit Trail ► Configuration ► Audit Rules | Effective date | _____ |
| Owner (suggested) | Data Integrity lead / System owner        | Approved by    | _____ |
| Related SOPs      | SOP-DI-002, SOP-VAL-006, SOP-IT-002       |                |       |

### 1 Objective

Define, justify and maintain the list of GxP records and fields whose creation, modification and deletion must be captured by the audit trail, based on a documented risk assessment.

### 2 Scope

All models of the installed modules that hold GxP data. The scope is part of the validated configuration of the system.

### 3 Responsibilities

| Role (software group)            | Responsibility                                                          |
|----------------------------------|-------------------------------------------------------------------------|
| <b>System owner / QA</b>         | Performs the data risk assessment and approves the scope.               |
| <b>Audit Trail Administrator</b> | Creates and maintains audit rules; cannot hide changes from auditors.   |
| <b>Validation</b>                | Verifies the rules in the computerised system validation (SOP-VAL-006). |

### 4 Definitions

| Term              | Definition                                                                                                                                                      |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Audit rule</b> | Model, operations audited (creation, modification, deletion), audited fields (empty = all stored fields except technical ones), excluded fields, justification. |
| <b>GxP record</b> | Data used to take a GMP decision or to demonstrate compliance.                                                                                                  |

### 5 Workflow in the software



Figure 79 — The audit trail supports ALCOA+ — attributable, contemporaneous, original.

## 6 Procedure

- 1 **Inventory.** List the models holding GxP data (e.g. deviations, CAPA, batch records, lab results, calibration records, stock lots) and their critical fields.
- 2 **Assess.** For each model decide which operations and fields need auditing; record the risk-based rationale.
- 3 **Create rules.** In [Audit Trail](#) ► [Configuration](#) ► [Audit Rules](#) click [New](#) : name, audited model, audit creations / modifications / deletions, audited fields or excluded fields, justification.
- 4 **Relations.** For data held in one-to-many or many-to-many relations (e.g. result lines), create a rule on the model on the other side of the relation.
- 5 **Verify.** Make a test change and check the entry in [Audit Log](#) and [Field Changes](#) before declaring the rule effective; include this test in the validation.
- 6 **Change control.** Any rule addition, change or removal goes through change control (SOP-QA-030) and is re-verified.

### 7. Controls enforced by the software

- ✓ Nothing is audited without a rule.
- ✓ Rules cannot target models of the audit engine itself, abstract models, transient models, or fields that do not belong to the model; a rule must audit at least one operation.
- ✓ Entries and field changes are immutable and cannot be deleted except through the retention run.
- ✓ Segregation of duties is enforced in the ORM: auditors cannot change the audited scope.

### 8. Good practices

- ★ Audit modifications and deletions of every GxP record type; creations where the creator/time matters.
- ★ Prefer an explicit field list for critical models, so that the scope is stable and reviewable.
- ★ Keep the rule list printed in the validation file and reconcile it at each periodic review.

### 9. Pitfalls to avoid

- ✗ Assuming that Odoo's chatter tracking is an audit trail — it is not tamper-evident and covers only tracked fields.
- ✗ Excluding fields to "reduce noise" without justification.

## 10 Records and evidence

- Audit rules with justification
- Data risk assessment
- Verification test evidence

## 11 Performance indicators

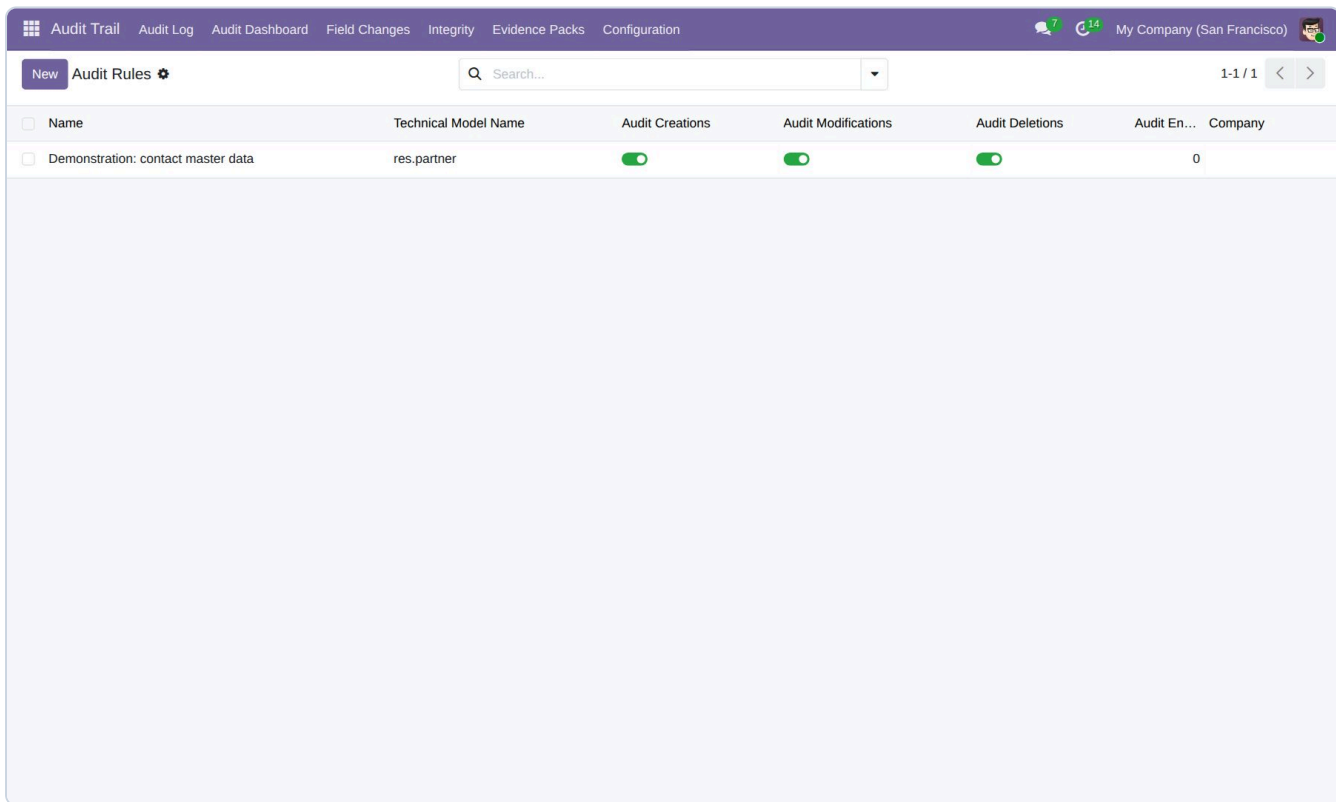
- GxP models without rule (target 0)
- Rule changes per year

## 12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)
- FDA Guidance — Data Integrity and Compliance With Drug CGMP, Questions and Answers (2018)

### 13. Demonstration

**Case.** For *ls.lab.test\_result* the rule audits modifications and deletions of all stored fields; for *stock.lot* only the fields *expiration date* and *name*. A test edit of a result value creates an entry showing old value, new value, user, date and IP address — evidence kept in the OQ.



| Audit Rules                                                 |                      |                                     |                                     |                                     |             |         |
|-------------------------------------------------------------|----------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------|---------|
| <input type="checkbox"/> Name                               | Technical Model Name | Audit Creations                     | Audit Modifications                 | Audit Deletions                     | Audit En... | Company |
| <input type="checkbox"/> Demonstration: contact master data | res.partner          | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 0           |         |

Figure 80 — Audit rules defining the audited scope. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)



# Audit trail review and integrity verification

|                   |                                                  |                |       |
|-------------------|--------------------------------------------------|----------------|-------|
| Module            | ls_audit_trail                                   | Version        | _____ |
| Menu              | Audit Trail ► Audit Log · Integrity ► Verify Now | Effective date | _____ |
| Owner (suggested) | QA / Data Integrity lead                         | Approved by    | _____ |
| Related SOPs      | SOP-DI-001, SOP-DI-003, SOP-QA-090, SOP-QC-012   |                |       |

## 1 Objective

Review audit trail data at a frequency based on risk, as part of record review, and confirm regularly that the audit trail has not been altered.

## 2 Scope

Audit trail entries of all audited models; integrity of the hash chain of each company.

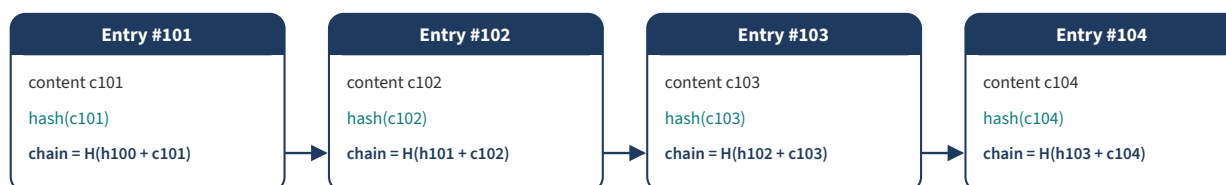
## 3 Responsibilities

| Role (software group)                       | Responsibility                                                       |
|---------------------------------------------|----------------------------------------------------------------------|
| <b>Record reviewers (QA, lab reviewers)</b> | Review the audit trail of the records they approve (batch, results). |
| <b>Audit Trail Auditor</b>                  | Runs integrity verifications, investigates failures.                 |
| <b>QA</b>                                   | Defines review frequencies and handles findings as deviations.       |

## 4 Definitions

| Term                       | Definition                                                                                                                                       |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Review by exception</b> | Focused review of changes that matter: modifications of results after entry, deletions, changes outside working hours, changes by unusual users. |
| <b>Verification scope</b>  | Recent window, explicit range or complete chain.                                                                                                 |

## 5 Workflow in the software



Altering, inserting or deleting an entry changes every following digest: the verification detects it.

Figure 81 — Integrity verification walks the chain.

## 6 Procedure

- 1 Define frequency.** Set in this SOP which audit trails are reviewed with each record (e.g. results before approval, batch record before release) and which periodically (e.g. monthly for master data).
- 2 Review.** In **Audit Trail ► Audit Log** filter by model, record, user and period; open **Field Changes** to see old and new values; use **Open Audited Record** to see the context.

- 3 **Document.** Record the review (e.g. in the batch review or as a quality record) with its scope and outcome.
- 4 **Verify integrity.** The scheduled action *Audit Trail: Verify Integrity* runs daily; in addition an Auditor runs **Integrity ► Verify Now** monthly on the complete chain and before any inspection. Results are stored in **Integrity ► Verifications**.
- 5 **React.** A failed verification or a suspicious change opens a deviation (SOP-QA-020) with data integrity classification; preserve evidence; inform QA management.

### 7. Controls enforced by the software

- ✓ Verification records are immutable and cannot be deleted.
- ✓ Entries cannot be modified; a sealed entry cannot be modified; deletion only through the retention run.

### 8. Good practices

- ★ Train reviewers on what a suspicious pattern looks like (repeated edits of a result, deletions before approval).
- ★ Review the audit trail of the system configuration too (rules, users, access rights).

### 9. Pitfalls to avoid

- ✗ Reviewing thousands of entries without focus — the review becomes a signature without content.
- ✗ Ignoring failed scheduled verifications because "it is IT".

## 10 Records and evidence

- Review records
- Verification results
- Deviations on findings

## 11 Performance indicators

- Verifications failed (target 0)
- Records approved without audit trail review (target 0)

## 12 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)

### 13. Demonstration

**Case.** Before approving sample LS-2026-0412, the reviewer filters the audit log on its results: one assay value was changed from 97.8 % to 98.4 % by the analyst 10 minutes after entry, with the comment "*transcription error, see printout p. 2*". The printout confirms 98.4 %. The reviewer documents the check in the review comment and approves.

# Evidence packs for inspections

|                   |                              |                |       |
|-------------------|------------------------------|----------------|-------|
| Module            | ls_audit_trail               | Version        | _____ |
| Menu              | Audit Trail ► Evidence Packs | Effective date | _____ |
| Owner (suggested) | Audit Trail Auditor (QA)     | Approved by    | _____ |
| Related SOPs      | SOP-DI-002, SOP-DI-004       |                |       |

## 1 Objective

Provide inspectors and auditors with complete, sealed and verifiable extracts of the audit trail on request, without giving them uncontrolled access to the system.

## 2 Scope

Requests from regulatory inspectors, customer auditors, notified bodies and internal investigations.

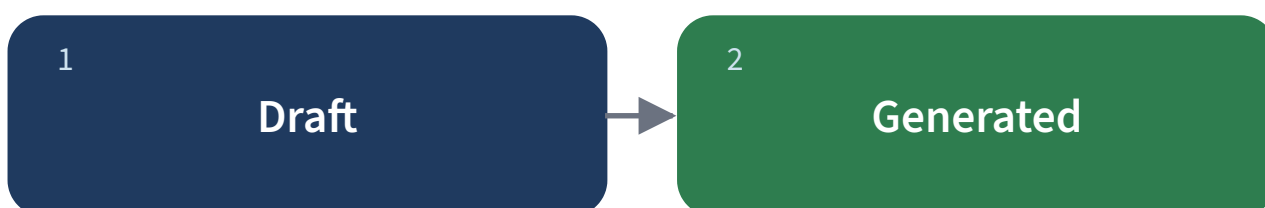
## 3 Responsibilities

| Role (software group)            | Responsibility                                                              |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Audit Trail Auditor</b>       | Prepares, generates and verifies the pack.                                  |
| <b>QA inspection coordinator</b> | Receives the request, checks the scope, hands over the pack and records it. |

## 4 Definitions

| Term                   | Definition                                                                                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Evidence pack</b>   | ZIP archive containing audit_entries.json, audit_entries.csv and manifest.json, with the SHA-256 digest of the archive stored on the record. |
| <b>Chain positions</b> | First and last chain positions exported, allowing the inspector to see the continuity.                                                       |

## 5 Workflow in the software



*Model: ls.audit\_trail.evidence\_pack*

Figure 82 — An evidence pack is sealed at generation.

## 6 Procedure

- Record the request.** Note who asked, for what purpose, which records and period.
- Create the pack.** In **Audit Trail ► Evidence Packs** click **New**: purpose, from and to dates, model filter and user filter as needed.

- 3 **Generate.** Click **Generate**. The pack becomes immutable; the number of entries and field changes, chain positions and archive digest are recorded; an integrity verification is linked.
- 4 **Check.** Click **Verify Archive Digest** to confirm the stored archive matches its digest.
- 5 **Hand over.** Click **Download Archive**; transmit it with the digest value; print the cover sheet if paper signatures are required.
- 6 **Keep.** The pack stays in the system; archive it if needed, never delete it.

### 7. Controls enforced by the software

- ✓ Only a draft pack can be generated; the criteria must match at least one entry; start  $\leq$  end.
- ✓ Generated packs are immutable and cannot be deleted (archive instead).
- ✓ A mismatch between the stored archive and its digest is reported explicitly.

### 8. Good practices

- ★ Generate packs from the production system, not from a copy, and state it in the purpose.
- ★ Give the inspector the digest separately so they can check the file themselves.

### 9. Pitfalls to avoid

- ✗ Sending screenshots or spreadsheets edited by hand instead of the sealed pack.

## 10 Records and evidence

- Evidence pack record, archive and digest
- Request log

## 11 Performance indicators

- Time to answer an inspection request

## 12 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

### 13. Demonstration

**Case.** An inspector asks for all changes made to the specifications of product P-100 in 2026. The Auditor creates a pack: purpose *"Inspection 2027-03, request #7"*, 01/01/2026–31/12/2026, model filter *ls.lab.specification*. After **Generate** and **Verify Archive Digest**, the ZIP and its digest are handed over within 30 minutes.

# Retention and controlled purge of audit data

|                   |                                             |                |       |
|-------------------|---------------------------------------------|----------------|-------|
| Module            | ls_audit_trail                              | Version        | _____ |
| Menu              | Audit Trail ► Configuration ► Retention Run | Effective date | _____ |
| Owner (suggested) | QA with the Audit Trail Administrator       | Approved by    | _____ |
| Related SOPs      | SOP-QA-004, SOP-DI-003                      |                |       |

## 1 Objective

Keep audit trail data as long as the records it relates to, and remove aged entries only under an approved procedure that preserves the verifiability of the chain.

## 2 Scope

Audit trail entries of all companies.

## 3 Responsibilities

| Role (software group)     | Responsibility                                                                                   |
|---------------------------|--------------------------------------------------------------------------------------------------|
| QA                        | Sets the retention period consistent with record retention (SOP-QA-004) and authorises each run. |
| Audit Trail Administrator | Executes the approved run.                                                                       |

## 4 Definitions

| Term             | Definition                                                                                                                                                   |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Retention anchor | Special entry written by the retention run that preserves the integrity of the remaining chain.                                                              |
| Retention period | Company setting <i>Audit Trail Retention (days)</i> ; runs also require <i>Allow Audit Trail Retention Runs</i> and a <i>Retention Procedure Reference</i> . |

## 5 Procedure

- 1 Set retention.** Set the retention period in days to at least the longest retention of the audited records. Keep *Allow Audit Trail Retention Runs* unticked until a run is authorised.
- 2 Prepare.** Before a run, generate an evidence pack of the entries to be removed (SOP-DI-003) and archive it according to SOP-QA-004.
- 3 Authorise.** QA authorises the run with a change request or quality record; set the retention procedure reference (this SOP number) and tick *Allow Audit Trail Retention Runs*.
- 4 Execute.** The Administrator opens **Configuration ► Retention Run**, confirms the number of entries, enters the justification and clicks **Execute Retention Run**.
- 5 Verify.** Run a complete-chain verification (SOP-DI-002); untick *Allow* again.

## 6. Controls enforced by the software

- ✓ Entries cannot be deleted except by the retention run.
- ✓ The retention period cannot be negative; runs require a retention period and a retention procedure reference on the company.
- ✓ The run writes a retention anchor so that the remaining chain stays verifiable.

## 7. Good practices

- ★ Plan runs rarely (e.g. yearly) and under change control.
- ★ Test the readability of archived evidence packs before deleting source entries.

## 8. Pitfalls to avoid

- ✗ Purging to improve performance without regard to record retention.

## 9 Records and evidence

- Authorisation, evidence pack of removed entries, run justification, verification result

## 10 Performance indicators

- Runs executed without authorisation (target 0)

## 11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- 21 CFR 211.180 to 211.198 — Records and reports
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

## 12. Demonstration

**Case.** Records must be kept 11 years. Retention is set to 4 018 days. In 2038, QA authorises a run under CC-2038-004; the pack of entries older than 11 years is generated and archived; the Administrator executes the run; the complete-chain verification passes.

# Electronic signatures

## ls\_electronic\_signature

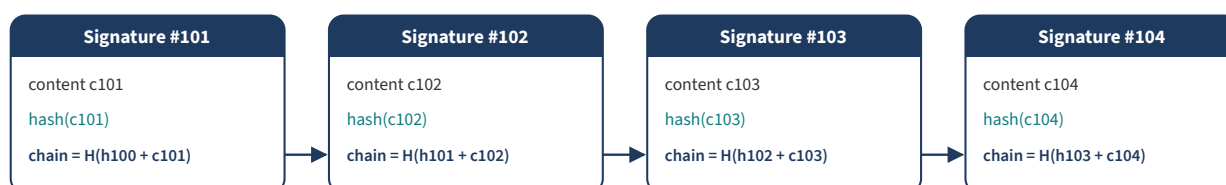
**ls\_electronic\_signature** provides electronic signatures designed to support 21 CFR Part 11 subparts B and C: printed name, date/time and meaning with every signature; identification code **and** password (or password only inside a continuous session); a binding statement; append-only signature log with a SHA-256 chain protected by a database trigger; logging of every attempt; lockout after repeated failures and immediate alert to the *Security Unit*. Signature *policies* say which model needs which signature meaning, how many signers, and whether a field transition is blocked until signed.

### Important — current coverage

- ✓ A model can carry these signatures only if it inherits *ls.signature.mixin*. In the code examined, **no production model of the 22 LS modules inherits it** (only the test fixture module does). Signature policies can therefore not be applied to LS records until a model is extended by development and validated.
- ✓ Other modules have their own mechanisms: **ls\_validation** re-authenticates the signer (login and password) and keeps immutable validation signatures; **ls\_lab** records a *signature intent* (user, time, meaning) and states it is **not** a Part 11 signature; **ls\_supplier\_qualification** keeps an append-only, chained decision log **without** re-authentication.
- ✓ Your electronic signature SOPs must state, per record type, which mechanism applies — and use handwritten signatures on printed records where none is adequate.

| Menu                                                                                                                                                             | Content                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <a href="#">Electronic Signatures</a> ► <a href="#">Signatures</a> ► <a href="#">Signature Log / My Signatures / Signature Requests</a>                          | Signatures and requests |
| <a href="#">Electronic Signatures</a> ► <a href="#">Monitoring</a> ► <a href="#">Signature Attempts / Signing Sessions / Integrity Checks / Verify Chain Now</a> | Security monitoring     |
| <a href="#">Electronic Signatures</a> ► <a href="#">Configuration</a> ► <a href="#">Signature Meanings / Signature Policies / Settings</a>                       | Configuration           |

| Group           | Main rights                                        |
|-----------------|----------------------------------------------------|
| Signer · Viewer | Sign; read signatures                              |
| Auditor         | Verify integrity, read attempts                    |
| Manager         | Configure meanings and policies                    |
| Security Unit   | Receives alerts on refused signatures and lockouts |



Altering, inserting or deleting an entry changes every following digest: the verification detects it.

Figure 83 — Each signature is bound to the previous one.

# Electronic signature policy: meanings, policies, signer commitment

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_electronic_signature               | Version        | _____ |
| Menu              | Electronic Signatures ► Configuration | Effective date | _____ |
| Owner (suggested) | QA with the System Owner              | Approved by    | _____ |
| Related SOPs      | SOP-DI-011, SOP-DI-012, SOP-VAL-002   |                |       |

## 1 Objective

Define where electronic signatures are used, what each signature means, who may sign, and obtain each signer's commitment that the electronic signature is the legally binding equivalent of their handwritten signature.

## 2 Scope

All electronic signatures executed in the system: ls\_electronic\_signature, validation signatures, lab signature intents and supplier decision log, as applicable.

## 3 Responsibilities

| Role (software group) | Responsibility                                                   |
|-----------------------|------------------------------------------------------------------|
| QA                    | Owens the policy; approves meanings and policies.                |
| Signature Manager     | Configures meanings and policies after approval.                 |
| Each signer           | Signs the commitment before first use; never shares credentials. |
| HR / QA               | Keeps the signed commitments.                                    |

## 4 Definitions

| Term              | Definition                                                                                                                                                                                 |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Meaning           | Stated purpose of a signature (e.g. Authored, Reviewed, Approved, Released); code of 2–32 upper-case characters; optionally reason mandatory and restricted to authorised groups.          |
| Policy            | Rule on a model: meaning, trigger (manual or field transition from/to value), number of signatures, distinct signers, authorised signers, applicability domain, full credentials required. |
| Binding statement | Text shown to the signer at each signature (Settings).                                                                                                                                     |



## 5 Workflow in the software



Model: *ls.signature.request*

Figure 84 — Lifecycle of a signature request.

## 6 Procedure

- 1 Map signature points.** List, per record type, the signature required by your procedures and the mechanism available (see the coverage box of this chapter). Where no adequate mechanism exists, keep a handwritten signature on the printed record.
- 2 Define meanings.** In **Configuration ► Signature Meanings** create each meaning with a code, description, *Reason Mandatory* when needed and the authorised groups.
- 3 Define policies.** For models inheriting the signature capability, create policies in **Configuration ► Signature Policies**: model, meaning, trigger, signatures required, distinct signers, authorised signers, domain, require full credentials.
- 4 Set the binding statement.** In **Configuration ► Settings** enter the binding statement approved by legal/QA.
- 5 Collect commitments.** Each signer signs a paper or electronic certification of their commitment before receiving the Signer role; for the US market, the organisation's certification to the FDA required by 21 CFR 11.100(c) is handled by Regulatory Affairs.
- 6 Request signatures.** When needed, a user creates a signature request (record, meaning, expected signers, deadline) and clicks **Send Request**; signers use **Sign** or **Decline** with a reason.
- 7 Review.** Review meanings and policies at each periodic system review (SOP-VAL-005).

### 7. Controls enforced by the software

- ✓ Meaning codes must follow the pattern (2–32 characters, upper case, digits, underscores).
- ✓ A policy can only target a model present in the registry that inherits the signature capability; a field-transition policy needs both the field and the target value; the domain must be a valid list.
- ✓ Signature records cannot be modified or deleted; a correction is a new signature with a reason.
- ✓ An operation governed by a policy is refused until the required number of valid signatures with the right meaning is present.
- ✓ Requests: only a draft request can be sent; only expected signers may sign; only pending requests can be declined; overdue requests expire by scheduled action.

## 8. Good practices

- ★ Few, clear meanings; avoid synonyms ("Checked" vs "Verified").
- ★ Require distinct signers when a policy needs two signatures.
- ★ Show the signature manifestation (name, date/time, meaning) on printed reports.

## 9. Pitfalls to avoid

- ✗ Assuming that clicking an *Approve* button is an electronic signature — it is not unless the mechanism re-authenticates and records meaning.
- ✗ Declaring the LS modules Part 11 compliant without validating the actual signature points.

## 10 Records and evidence

- Approved meanings and policies (configuration)
- Signer commitments
- Signature log

## 11 Performance indicators

- Signers without commitment on file (target 0)
- Signature requests expired

## 12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

## 13. Demonstration

**Case.** QA maps signature points: validation protocols and reports → ls\_validation signatures (re-authentication); lab result approval → lab signature intent, complemented by the printed CoA signed by hand until a Part 11 mechanism is validated; supplier approval → decision log plus signed approval form. Meanings AUTHORED, REVIEWED, APPROVED, RELEASED are created in ls\_electronic\_signature for future use.

Electronic Signatures

Signatures

Monitoring

Configuration

7

14

My Company (San Francisco)

John

New

Signature Meanings

Search...

1-7 / 7

<

>

| <div><input type="checkbox"/></div> | Code                                                    | Meaning                    | Reason Man...                                | Company                    |
|-------------------------------------|---------------------------------------------------------|----------------------------|----------------------------------------------|----------------------------|
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>AUTHORED</div></div>    | Authored by                | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>REVIEWED</div></div>    | Reviewed by                | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>APPROVED</div></div>    | Approved by                | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>RESPONSIBLE</div></div> | Responsibility accepted by | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>PERFORMED</div></div>   | Performed by               | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>WITNESSED</div></div>   | Witnessed by               | <div><input type="checkbox"/></div>          | My Company (San Francisco) |
| <div><input type="checkbox"/></div> | <div><div></div><div></div><div>REJECTED</div></div>    | Rejected by                | <div><div></div><div></div><div></div></div> | My Company (San Francisco) |

Figure 85 — Signature meanings. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Management of credentials, account lockout and security alerts

|                   |                                                                                    |                |       |
|-------------------|------------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_electronic_signature                                                            | Version        | _____ |
| Menu              | Electronic Signatures ► Configuration ► Settings · Monitoring ► Signature Attempts | Effective date | _____ |
| Owner (suggested) | IT security with QA                                                                | Approved by    | _____ |
| Related SOPs      | SOP-DI-010, SOP-IT-001, SOP-QA-013                                                 |                |       |

## 1 Objective

Ensure that each user identity is unique, credentials remain under the sole control of their owner, repeated failed signature attempts lock the identity, and security events are investigated.

## 2 Scope

User accounts of the system and the signature controls of ls\_electronic\_signature (and ls\_validation re-authentication).

## 3 Responsibilities

| Role (software group) | Responsibility                                                                              |
|-----------------------|---------------------------------------------------------------------------------------------|
| IT administrator      | Creates, disables and never reassigns accounts; resets passwords only after identity check. |
| Security Unit (group) | Receives alerts, investigates refused signatures and lockouts.                              |
| Users                 | Keep passwords secret; report suspected compromise at once.                                 |

## 4 Definitions

| Term               | Definition                                                                                                             |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| Continuous session | Period during which a signer may sign with password only; closed after the idle timeout.                               |
| Lockout            | Refusal of signatures for an identification code after the configured number of failures, for the configured duration. |

## 5 Procedure

- 1 Configure.** In [Electronic Signatures ► Configuration ► Settings](#) set *Continuous Session Timeout (minutes)* (default 15), *Failures Before Lockout* (default 3) and *Lockout Duration (minutes)* (default 15); check the *Database Immutability Trigger* state and the password verification API mode.
- 2 Assign the Security Unit.** Add the security officers to the group *Security Unit*; they receive an e-mail on each refused signature.
- 3 Manage accounts.** One account per person; never reuse or rename an account for another person; disable accounts of leavers the same day.
- 4 Monitor.** The Security Unit reviews [Monitoring ► Signature Attempts](#) weekly (results: incorrect password, identification code mismatch, not authorised, blocked by lockout, reason missing, precondition not met, system error).

- 5 **Investigate.** Each lockout or identification mismatch is checked with the user; suspected misuse becomes a deviation (data integrity).
- 6 **Close idle sessions.** The scheduled action closes idle signing sessions every 15 minutes; administrators can close a session manually.

### 6. Controls enforced by the software

- ✓ A signature requires identification code and password, or password only within a continuous session; both are required again after the session lapses.
- ✓ A signature under another person's identification code is refused.
- ✓ Every attempt is logged in an immutable security log; refusals trigger an alert to the Security Unit.
- ✓ An identification code is locked out after repeated failures.
- ✓ `ls_validation`: the login entered must match the connected user and the password must be verified, otherwise the signature is not recorded.

### 7. Good practices

- ★ Align password rules with IT policy (length, rotation where required, no reuse).
- ★ Never lower the failure threshold to zero or disable alerts "temporarily".

### 8. Pitfalls to avoid

- ✗ Shared "team" accounts in the laboratory or on the shop floor.
- ✗ Administrators resetting a password over the phone without identity verification.

## 9 Records and evidence

- Settings configuration
- Signature attempt log
- Account creation/disable records
- Investigation records

## 10 Performance indicators

- Lockouts per month
- Accounts of leavers not disabled within 24 h (target 0)

## 11 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

### 12. Demonstration

**Case.** On Monday the Security Unit receives three alerts "*Identification code does not match the session*" from the same workstation in the packaging area. Investigation: an operator tried to sign with a colleague's login left open on the terminal. A deviation is opened; the session timeout is confirmed at 15 minutes; the team is retrained on account sharing.

[illegible]

Figure 86 — Log of signature attempts. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Verification of the signature chain

|                   |                                                                          |                |       |
|-------------------|--------------------------------------------------------------------------|----------------|-------|
| Module            | ls_electronic_signature                                                  | Version        | _____ |
| Menu              | Electronic Signatures ► Monitoring ► Integrity Checks / Verify Chain Now | Effective date | _____ |
| Owner (suggested) | Signature Auditor (QA)                                                   | Approved by    | _____ |
| Related SOPs      | SOP-DI-010, SOP-DI-002                                                   |                |       |

## 1 Objective

Confirm regularly that no electronic signature has been altered, inserted or removed, and that each signature remains bound to the content that was signed.

## 2 Scope

The signature log of ls\_electronic\_signature, the validation signature records, and the supplier decision log chain.

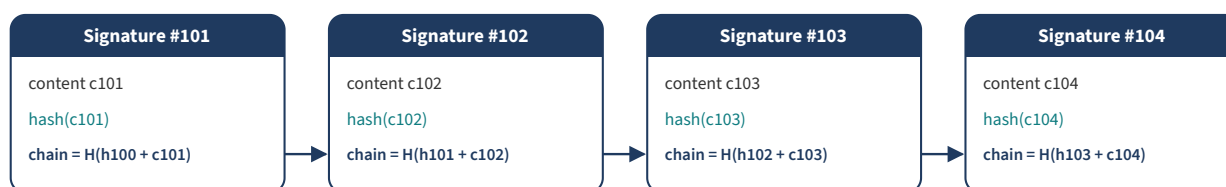
## 3 Responsibilities

| Role (software group)    | Responsibility                                          |
|--------------------------|---------------------------------------------------------|
| <b>Signature Auditor</b> | Runs and reviews verifications, investigates failures.  |
| <b>QA</b>                | Handles failures as critical data integrity deviations. |

## 4 Definitions

| Term                            | Definition                                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Record hash / chain hash</b> | SHA-256 of the signed payload, and digest binding it to the previous signature of the same company. |
| <b>Integrity check</b>          | Result Passed or Failed of a verification, manual or scheduled.                                     |

## 5 Workflow in the software



*Altering, inserting or deleting an entry changes every following digest: the verification detects it.*

Figure 87 — Signature chain verification.

## 6 Procedure

- Scheduled verification.** The scheduled action *Electronic Signatures: verify hash chain* runs daily; results appear in **Monitoring ► Integrity Checks**.
- Manual verification.** Monthly and before inspections, run **Monitoring ► Verify Chain Now** and record the result.

- 3 **Single signature.** To check one signature, open it in the signature log and click **Verify Integrity** ; **Open Signed Record** shows the record.
- 4 **Supplier decisions.** In **Supplier Qualification ► Monitoring ► Signature Log** use **Verify Chain** .
- 5 **Failure.** Stop signing on the affected scope, open a critical deviation, preserve database backups, involve IT and QA management.

### 7. Controls enforced by the software

- ✓ Signature records are append-only (ORM and database trigger refuse update and delete).
- ✓ A signature whose record no longer exists or whose payload no longer matches is reported by the verification.

### 8. Good practices

- ★ Include the chain verification in the periodic review of the system (SOP-VAL-005).
- ★ Keep the verification results with the validation evidence.

### 9. Pitfalls to avoid

- ✗ Restoring a database backup without re-verifying the chain afterwards.

## 10 Records and evidence

- Integrity check results
- Deviations on failures

## 11 Performance indicators

- Failed checks (target 0)
- Days since last successful verification

## 12 References

- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

### 13. Demonstration

**Case.** The monthly manual verification of March passes for 1 482 signatures. In April, after a database restore on the test server, a verification on production is run again as a precaution: passed; the result is attached to the change record of the restore.



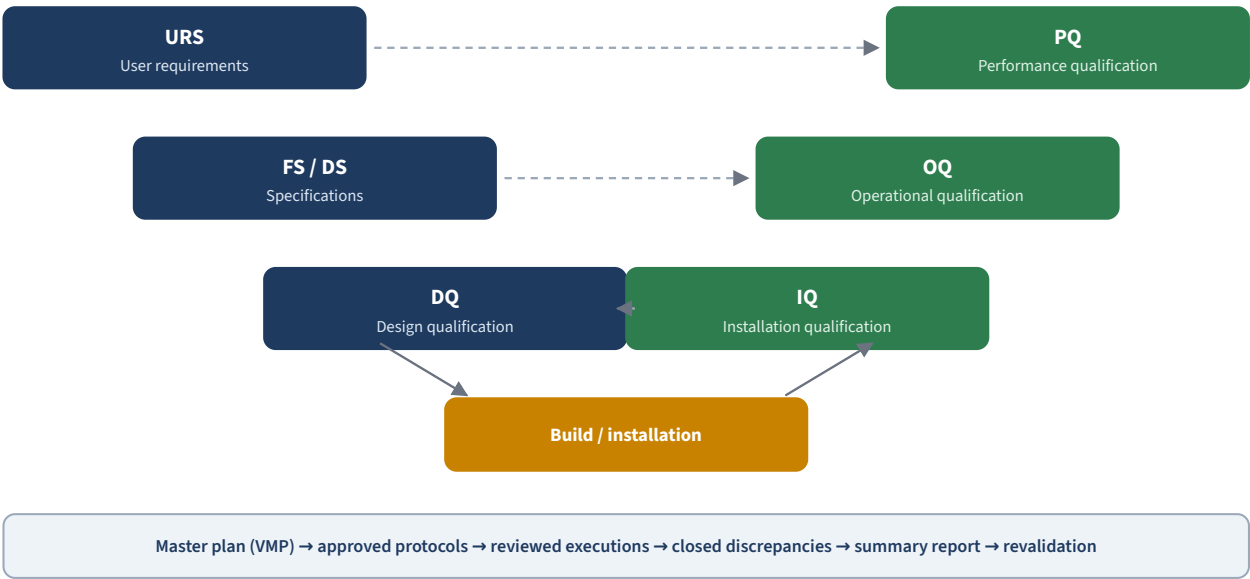
# Validation

## ls\_validation

**ls\_validation** manages validation master plans, validation items (equipment, utilities, facilities, processes, cleaning procedures, computerised systems, analytical methods, shipping lanes) with their validation status and validity, protocols of eight types (DQ, IQ, OQ, PQ, PV, CV, CSV, MV) with test cases and acceptance criteria, execution records with per-test verdicts, discrepancies, and summary reports. Key steps are signed with re-authentication (login and password) and the signatures are immutable.

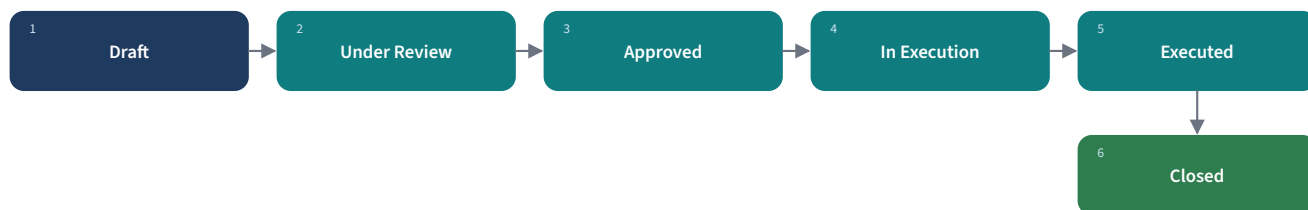
| Menu                                                                                                                          | Content                |
|-------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <a href="#">Validation</a> ► <a href="#">Planning</a> ► <a href="#">Master Plans / Validation Items / Expiring or Expired</a> | Planning               |
| <a href="#">Validation</a> ► <a href="#">Execution</a> ► <a href="#">Protocols / Execution Records / Discrepancies</a>        | Execution              |
| <a href="#">Validation</a> ► <a href="#">Reporting</a> ► <a href="#">Summary Reports / Electronic Signatures</a>              | Reports and signatures |

| Group                         | Main rights                           |
|-------------------------------|---------------------------------------|
| Validation: Viewer · Engineer | Read; author protocols, execute tests |
| Validation: Approver          | Review and approve with signature     |
| Validation: Manager           | Manage plans, items, revalidation     |

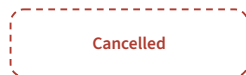


*Dashed arrows link each specification level to the test that verifies it.*

Figure 88 — The V-model of qualification.



Other outcomes



Model: ls.validation.protocol

Figure 89 — Lifecycle of a protocol.

# Validation master plan (VMP)

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_validation                         | Version        | _____ |
| Menu              | Validation ► Planning ► Master Plans  | Effective date | _____ |
| Owner (suggested) | Validation Manager                    | Approved by    | _____ |
| Related SOPs      | SOP-VAL-002, SOP-VAL-005, SOP-VAL-006 |                |       |

## 1 Objective

Describe the validation policy, strategy, responsibilities and the list of items to validate, and keep it approved, active and current.

## 2 Scope

All GxP equipment, utilities, facilities, processes, cleaning procedures, computerised systems, analytical methods and transport lanes of the site.

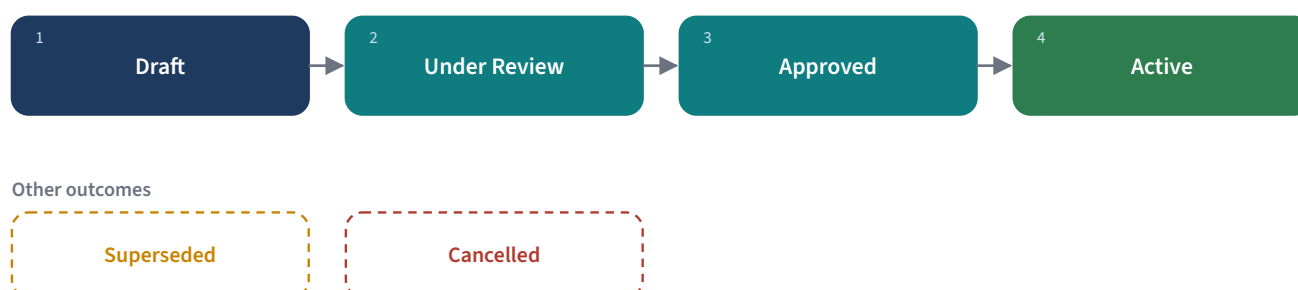
## 3 Responsibilities

| Role (software group)     | Responsibility                                  |
|---------------------------|-------------------------------------------------|
| <b>Validation Manager</b> | Writes and maintains the VMP and the item list. |
| <b>Reviewer</b>           | Reviews the VMP (QA, engineering).              |
| <b>Approver</b>           | Approves with electronic signature.             |

## 4 Definitions

| Term                     | Definition                                                                                                                               |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Validation item</b>   | Object to validate, with type, GxP impact (none, indirect, direct), criticality (low, medium, high), owner and periodic review interval. |
| <b>Validation status</b> | Not Validated, In Validation, Validated, Expiring Soon, Expired, Retired — derived from the approved reports.                            |

## 5 Workflow in the software



Model: ls.validation.master.plan

Figure 90 — Lifecycle of a validation master plan.

## 6 Procedure

- 1 **Register items.** In **Validation Items** create each item: code, designation, type, GxP impact, criticality, manufacturer, model, serial number, location, owner, periodic review interval.
- 2 **Write the VMP.** In **Master Plans** click **New** : title, scope, validation strategy (risk-based approach, V-model, deliverables per item type), responsibilities, review interval; add the items in scope.
- 3 **Review.** Click **Submit for Review** ; the reviewer comments; **Back to Draft** if changes are needed.
- 4 **Approve.** The approver clicks **Approve and Sign** (login and password), then **Activate** on the effective date.
- 5 **Maintain.** When scope or strategy changes, click **New Version** on the approved or active plan; the new version supersedes the old one when activated.
- 6 **Review periodically.** Review the VMP at its *Next Review Date* and report progress in management review.

### 7. Controls enforced by the software

- ✓ The scope must be documented before review; only a draft plan can be submitted; only a plan under review can be approved or returned to draft.
- ✓ Only an approved plan can be activated; a new version only from an approved or active plan; approved/active content is frozen.
- ✓ An active or superseded plan cannot be cancelled; plans that are not draft or cancelled cannot be deleted.
- ✓ Signatures require re-authentication and are immutable.

### 8. Good practices

- ★ Scale the deliverables to GxP impact and criticality (GAMP 5 categories for systems).
- ★ Include cleaning, transport and computerised systems — they are often forgotten.

### 9. Pitfalls to avoid

- ✗ A VMP that lists items but no strategy.
- ✗ Items validated long ago but never reviewed.

## 10 Records and evidence

- Approved VMP with signatures
- Validation item register

## 11 Performance indicators

- % of items validated
- Items expired
- VMP review on time

## 12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- FDA Guidance — Process Validation: General Principles and Practices (2011)
- ICH Q10 — Pharmaceutical Quality System

### 13. Demonstration

**Case.** VMP 2027 lists 64 items: 28 equipment, 6 utilities, 9 areas, 7 processes, 5 cleaning procedures, 6 computerised systems (including the Odoo LS suite), 3 transport lanes. The strategy states: direct-impact equipment → IQ/OQ/PQ; indirect → IQ/OQ; GAMP category 4 systems → CSV with risk-based testing. Approved and activated on 15 January.

Validation Planning Execution Reporting
7 14 My Company (San Francisco)

New Validation Master Plans
VMP/2026/0001 v1 - Site Validation Master Plan
Protocols 1 Signatures 0
1 / 1

Submit for Review Cancel Draft Under Review Approved Active
Send message Log note Activity

### VMP/2026/0001

#### Site Validation Master Plan

| DOCUMENT CONTROL |         | VALIDITY                 |                            |
|------------------|---------|--------------------------|----------------------------|
| Version          | 1       | Effective Date           |                            |
| Author           | OdooBot | Review Interval (months) | 24                         |
| Reviewer         |         | Next Review Date         |                            |
|                  |         | Company                  | My Company (San Francisco) |

Scope Strategy Responsibilities Items in Scope Protocols Electronic Signatures

This demonstration plan covers the qualification of the production and quality control equipment of the site, and the validation of the cleaning procedures of production line A.

Today

OdooBot 10:39 AM  
Validation Master Plan created

Figure 91 — Validation master plan. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Drafting and approval of qualification protocols: DQ, IQ, OQ, PQ

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_validation                         | Version        | _____ |
| Menu              | Validation ► Execution ► Protocols    | Effective date | _____ |
| Owner (suggested) | Validation Manager                    | Approved by    | _____ |
| Related SOPs      | SOP-VAL-001, SOP-VAL-003, SOP-VAL-006 |                |       |

## 1 Objective

Write qualification protocols whose test cases each have a clear procedure and a pre-defined acceptance criterion, and approve them before any execution.

## 2 Scope

Design, installation, operational and performance qualification protocols (DQ, IQ, OQ, PQ). Process, cleaning, method and computerised system validation protocols follow the same steps (SOP-VAL-006).

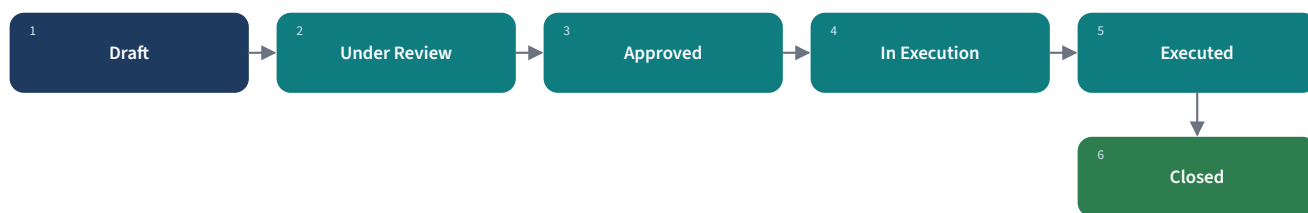
## 3 Responsibilities

| Role (software group)               | Responsibility                                                |
|-------------------------------------|---------------------------------------------------------------|
| <b>Validation Engineer (author)</b> | Writes the protocol and test cases.                           |
| <b>Reviewer</b>                     | Checks traceability to requirements and adequacy of criteria. |
| <b>Approver (QA)</b>                | Approves with signature before execution.                     |

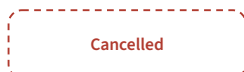
## 4 Definitions

| Term                 | Definition                                                                                                                   |
|----------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>DQ</b>            | Documented verification that the proposed design is suitable for the intended purpose.                                       |
| <b>IQ</b>            | Documented verification that the equipment is installed according to the approved design and manufacturer's recommendations. |
| <b>OQ</b>            | Documented verification that it operates as intended throughout the anticipated operating ranges.                            |
| <b>PQ</b>            | Documented verification that it performs effectively and reproducibly in routine conditions.                                 |
| <b>Critical test</b> | Test case whose failure prevents the conclusion <i>Validated</i> .                                                           |

## 5 Workflow in the software

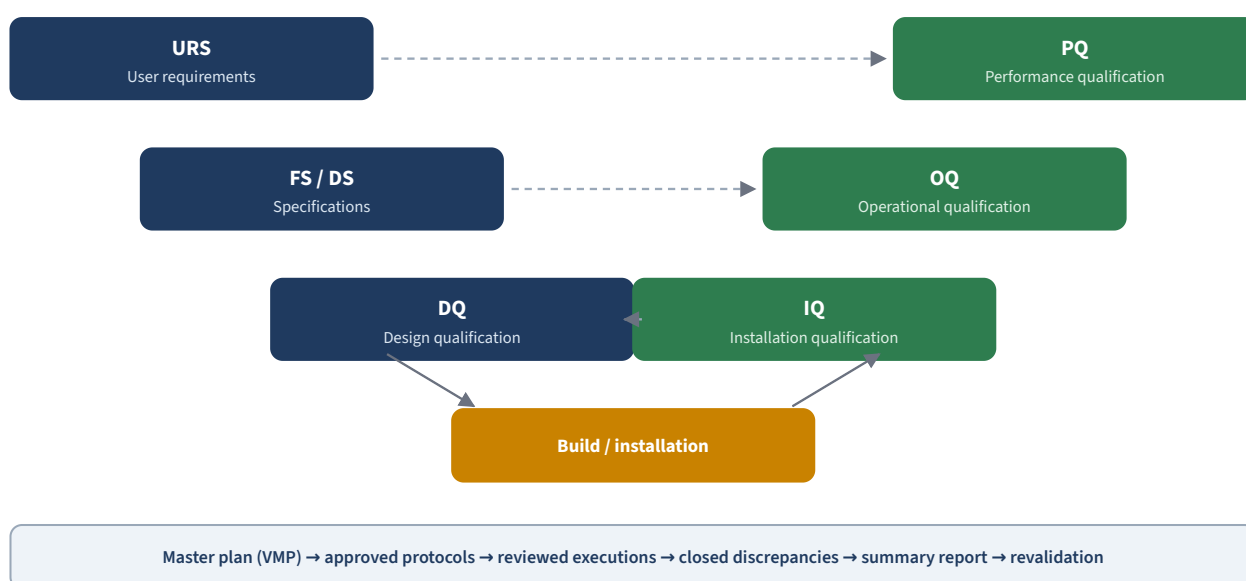


Other outcomes



Model: ls.validation.protocol

Figure 92 — Protocol lifecycle.



Dashed arrows link each specification level to the test that verifies it.

Figure 93 — Each qualification stage answers a specification level.

## 6 Procedure

- 1 **Create.** In **Protocols** click **New** : title, protocol type, validation item, master plan, objective, scope, prerequisites, reference documents, planned start and end.
- 2 **Write test cases.** Add test cases: number, title, objective, procedure, acceptance criterion (mandatory), expected evidence; tick *Critical Test* where applicable.
- 3 **Review.** Click **Submit for Review** ; use **Back to Draft** for corrections.
- 4 **Approve.** The approver clicks **Approve and Sign** and re-enters login and password.
- 5 **Change after approval.** If the protocol must change, click **New Version** ; never modify an approved protocol.

## 7. Controls enforced by the software

- ✓ Only a draft protocol can be submitted; a protocol without test cases cannot be approved; every test case needs an acceptance criterion.
- ✓ The item must be in the scope of the referenced master plan; planned end  $\geq$  planned start.
- ✓ Test cases cannot be changed after approval — a new protocol version is required.
- ✓ Executions can only be created from an approved protocol.
- ✓ Protocols with execution records cannot be cancelled; approved or executed protocols are frozen.

## 8. Good practices

- ★ Trace each test to a user requirement or risk (reference in the objective).
- ★ Write criteria as measurable values with units and tolerances.
- ★ Mark as critical the tests linked to product quality or patient safety.

## 9. Pitfalls to avoid

- ✗ Criteria such as "works correctly".
- ✗ Starting tests before approval "to save time" — the results are not valid.

## 10 Records and evidence

- Approved protocol with signatures

## 11 Performance indicators

- Protocols approved before execution (target 100 %)
- Review cycles per protocol

## 12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- ISO 13485:2016 — Medical devices — Quality management systems

## 13. Demonstration

**Case.** OQ of autoclave AC-02: 18 test cases, e.g. *OQ-07 Chamber temperature uniformity, empty chamber, 121 °C/15 min: all probes 121.0–124.0 °C during the plateau* (critical). The protocol is reviewed by QA and approved with signature before the tests are scheduled.



Validation
Planning
Execution
Reporting

7
14
My Company (San Francisco)

New
Protocols
PROT/2026/0001 v1 - Operational Qualification ...
Executions
0
Signatures
0
1 / 1

Submit for Review
Cancel
Draft
Under Review
Approved
In Execution
Executed
Closed

Send message
Log note
Activity

## PROT/2026/0001

### Operational Qualification - Steam Sterilizer 500 L

SUBJECT

DOCUMENT CONTROL

|                 |                                          |               |         |
|-----------------|------------------------------------------|---------------|---------|
| Protocol Type   | OQ - Operational Qualification           | Version       | 1       |
| Validation Item | [EQ-0001] Steam Sterilizer 500 L         | Author        | OdooBot |
| Master Plan     | VMP/2026/0001 v1 - Site Validation Ma... | Reviewer      |         |
| Company         | My Company (San Francisco)               | Planned Start |         |
|                 |                                          | Planned End   |         |

Test Cases
Objective
Scope
Prerequisites
Reference Documents
Executions
Electronic Signatures

| Test Case | Title                      | Acceptance Criteria                                                                                                     | Critical T...                       |  |
|-----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--|
| TC-001    | Chamber temperature con... | The chamber temperature remains within 121.0 degrees Celsius and 124.0 degrees Celsius during the whole exposure phase. | <input checked="" type="checkbox"/> |  |
| TC-002    | Chamber pressure control   | The chamber pressure remains within the range declared in the operating manual during the whole exposure phase.         | <input checked="" type="checkbox"/> |  |
| TC-003    | Door interlock alarm       | The cycle is aborted and the alarm is recorded in the equipment log.                                                    | <input type="checkbox"/>            |  |

OdooBot 10:39 AM

Validation Protocol created

Figure 94 — Protocol with test cases. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Protocol execution and management of discrepancies

|                   |                                                            |                |       |
|-------------------|------------------------------------------------------------|----------------|-------|
| Module            | ls_validation                                              | Version        | _____ |
| Menu              | Validation ► Execution ► Execution Records / Discrepancies | Effective date | _____ |
| Owner (suggested) | Validation Manager                                         | Approved by    | _____ |
| Related SOPs      | SOP-VAL-002, SOP-VAL-004, SOP-QA-022                       |                |       |

## 1 Objective

Execute approved protocols contemporaneously, record the observed result and verdict of each test, and handle every failure as a discrepancy investigated before approval.

## 2 Scope

All execution rounds of approved protocols.

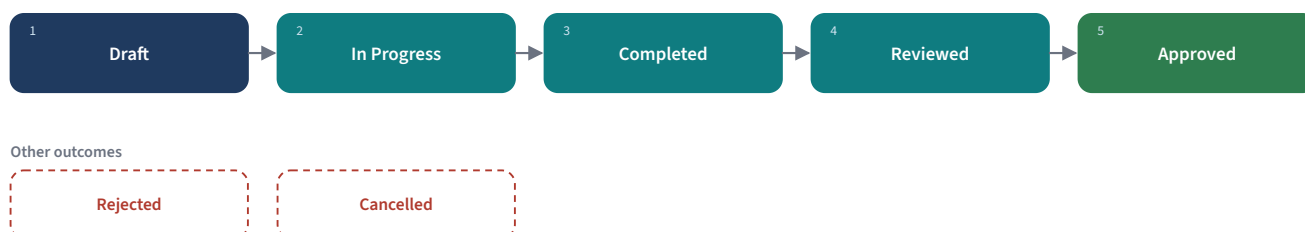
## 3 Responsibilities

| Role (software group)                 | Responsibility                                              |
|---------------------------------------|-------------------------------------------------------------|
| <b>Executor (Validation Engineer)</b> | Performs the tests and records results and evidence.        |
| <b>Reviewer</b>                       | Reviews the execution; must be different from the executor. |
| <b>Approver</b>                       | Approves the execution with signature.                      |
| <b>Discrepancy owner</b>              | Investigates and resolves discrepancies.                    |

## 4 Definitions

| Term                   | Definition                                                                           |
|------------------------|--------------------------------------------------------------------------------------|
| <b>Execution round</b> | Numbered execution of a protocol; a new round is created after a rejected execution. |
| <b>Verdict</b>         | Pending, Pass, Fail or Not Applicable per test case.                                 |
| <b>Discrepancy</b>     | Failure or unexpected event, classified Minor, Major or Critical.                    |

## 5 Workflow in the software



Model: ls.validation.execution

Figure 95 — Lifecycle of an execution record.



Model: *ls.validation.discrepancy*

Figure 96 — Lifecycle of a validation discrepancy.

## 6 Procedure

- 1 **Create the execution.** From the approved protocol click **New Execution**; result lines are generated from the test cases ( **Refresh Test Cases** while draft or in progress).
- 2 **Start.** Click **Start** when testing begins.
- 3 **Record results.** For each test record the observed result and verdict at the time of the test; attach raw data (printouts, logger files).
- 4 **Raise discrepancies.** For each failed test click the warning icon *Create a discrepancy* on the result line (or create one in **Discrepancies**): classification, description, immediate action.
- 5 **Investigate.** The owner clicks **Start Investigation**, documents root cause, corrective action and impact assessment, ticks *CAPA Required* and enters the CAPA reference if needed, clicks **Resolve**, then **Close and Sign**.
- 6 **Complete.** When all lines have a verdict and observed result, click **Complete**.
- 7 **Review and approve.** A reviewer (not the executor) clicks **Review and Sign**; the approver clicks **Approve and Sign**. If the execution is not acceptable, **Reject** with a reason and start a new round.
- 8 **Close the protocol.** When executions are approved, close the protocol ( **Close** ).

### 7. Controls enforced by the software

- ✓ Executions only from approved protocols; only a draft execution starts; only an execution in progress completes.
- ✓ Completion requires every line with a verdict and an observed result, and every failed test linked to a discrepancy.
- ✓ The reviewer must differ from the executor; approval only after review and with no open discrepancy.
- ✓ Discrepancies: investigation only from open; resolution requires root cause, corrective action and impact assessment; a CAPA reference is required when flagged; only resolved discrepancies close; closed ones are frozen and cannot be cancelled.
- ✓ Results of a completed execution cannot be modified — record a new round; an approved execution cannot be cancelled.

### 8. Good practices

- ★ Record results as the test is done (contemporaneous), never from memory at the end of the day.
- ★ Use "Not Applicable" only with a justification in the observed result.

### 9. Pitfalls to avoid

- ✗ Editing the acceptance criterion to make a result pass.
- ✗ Closing discrepancies with "retest passed" without root cause.

### 10 Records and evidence

- Execution with results, evidence and signatures
- Discrepancies with investigation

### 11 Performance indicators

- Discrepancies per protocol
- Executions rejected
- Time to close discrepancies

### 12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- 21 CFR Part 11 — Electronic Records; Electronic Signatures

### 13. Demonstration

**Case (continued).** During OQ-07 one probe reads 124.6 °C for 40 s. The executor sets the verdict *Fail* and creates discrepancy DISC-2027-005 (Major). Investigation: the probe was touching the chamber wall (setup error); impact: none on the equipment; corrective action: probe fixture added to the procedure; retest in round 2 passes. The discrepancy is resolved and closed with signature; round 2 is reviewed and approved.

## Summary report and validation status

|                   |                                          |                |       |
|-------------------|------------------------------------------|----------------|-------|
| Module            | ls_validation                            | Version        | _____ |
| Menu              | Validation ► Reporting ► Summary Reports | Effective date | _____ |
| Owner (suggested) | Validation Manager                       | Approved by    | _____ |
| Related SOPs      | SOP-VAL-003, SOP-VAL-005                 |                |       |

### 1 Objective

Conclude on the validation of an item on the basis of approved executions, state any restriction, and set the validity that drives the item's validation status.

### 2 Scope

All validation items after execution of their protocols.

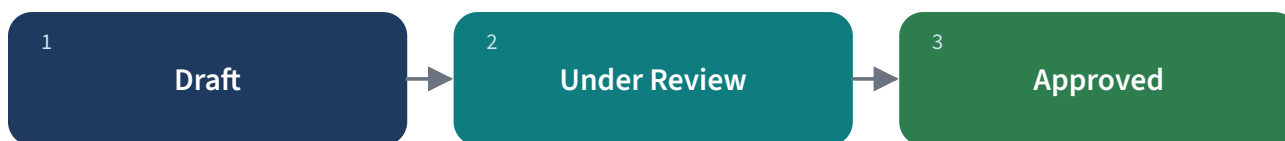
### 3 Responsibilities

| Role (software group)      | Responsibility                                              |
|----------------------------|-------------------------------------------------------------|
| <b>Author</b>              | Writes the report summarising executions and discrepancies. |
| <b>Reviewer / Approver</b> | Review and approve with signature.                          |

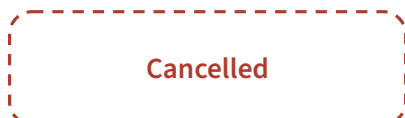
### 4 Definitions

| Term              | Definition                                                                                                    |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Conclusion</b> | Validated, Validated with Restrictions (restrictions mandatory), Not Validated.                               |
| <b>Validity</b>   | Valid from date and validity in months give <i>Valid Until</i> ; the item becomes Expiring Soon then Expired. |

### 5 Workflow in the software



Other outcomes



Model: ls.validation.report

Figure 97 — Lifecycle of a summary report.

## 6 Procedure

- 1 **Create.** In **Summary Reports** click **New** ; title, item, master plan, execution records (approved), report date, summary, conclusion, restrictions, valid from and validity (months).
- 2 **Check.** All referenced executions must be approved and concern the item; no linked discrepancy may be open; a failed critical test forbids *Validated*.
- 3 **Review and approve.** Click **Submit for Review** ; the approver clicks **Approve and Sign** .
- 4 **Release for use.** The item's validation status becomes *Validated* for the validity period; inform the owner and production.
- 5 **Restrictions.** If *Validated with Restrictions*, translate each restriction into an operating instruction or label on the equipment.

### 7. Controls enforced by the software

- ✓ The report must reference at least one approved execution of the same item and have a summary.
- ✓ No linked discrepancy may remain open; failed critical tests forbid the conclusion *Validated*; restrictions are mandatory for *Validated with Restrictions*.
- ✓ Validity cannot start before the report date; approved reports are frozen and cannot be cancelled.

### 8. Good practices

- ★ Summarise deviations from the protocol and their impact honestly.
- ★ Set validity consistent with the revalidation strategy (SOP-VAL-005).

### 9. Pitfalls to avoid

- ✗ Using an item before its report is approved.

## 10 Records and evidence

- Approved summary report with signatures

## 11 Performance indicators

- Time from last execution to report approval
- Items used while not validated (target 0)

## 12 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems

### 13. Demonstration

**Case.** Report for autoclave AC-02: IQ, OQ (round 2) and PQ executions approved, one discrepancy closed, no critical failure. Conclusion *Validated*, valid from 1 April for 36 months. The item status becomes *Validated* until 31 March three years later.

# Periodic revalidation

|                   |                                             |                |       |
|-------------------|---------------------------------------------|----------------|-------|
| Module            | ls_validation                               | Version        | _____ |
| Menu              | Validation ► Planning ► Expiring or Expired | Effective date | _____ |
| Owner (suggested) | Validation Manager                          | Approved by    | _____ |
| Related SOPs      | SOP-VAL-004, SOP-QA-030, SOP-QA-031         |                |       |

## 1 Objective

Keep items in a validated state: review them periodically, revalidate before their validity expires or after significant changes, and retire items that are no longer used.

## 2 Scope

All validation items with a validity or a periodic review interval.

## 3 Responsibilities

| Role (software group) | Responsibility                                      |
|-----------------------|-----------------------------------------------------|
| Item owner            | Provides change, maintenance and deviation history. |
| Validation Manager    | Decides on revalidation scope; plans protocols.     |

## 4 Definitions

| Term            | Definition                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Periodic review | Documented evaluation of changes, deviations, maintenance and performance since the last validation, concluding whether revalidation is needed. |
| Retirement      | End of use of an item with a date and a reason.                                                                                                 |

## 5 Procedure

- 1 Monitor.** Monthly open **Planning ► Expiring or Expired**.
- 2 Review.** For each item due, review changes (SOP-QA-030), deviations, CAPA, calibration and maintenance history, and performance data; record the conclusion.
- 3 Plan revalidation.** On the item click **Plan Revalidation**: the wizard creates the protocol to write (type, scope reduced to what the review requires).
- 4 Execute and report.** Follow SOP-VAL-002 to SOP-VAL-004; the new report renews the validity.
- 5 Change-triggered.** An approved change with *Requires Revalidation* ticked triggers revalidation before use.
- 6 Retire.** For an item out of use, click **Retire** with the retirement date and reason; **Cancel Retirement** if the decision is reversed.

### 6. Controls enforced by the software

- ✓ A retirement date is required to retire an item; an item cannot be retired twice.
- ✓ The validation status is derived from the approved reports and their validity.

## 7. Good practices

- ★ Start revalidation early enough (for example 3 months before expiry).
- ★ Scale revalidation to the review: a full PQ is not always needed.

## 8. Pitfalls to avoid

- ✗ Items used after *Expired* because nobody watched the list.

## 9 Records and evidence

- Periodic review conclusions
- Revalidation protocols and reports
- Retirement records

## 10 Performance indicators

- Items expired while in use (target 0)
- Periodic reviews on time

## 11 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

## 12. Demonstration

**Case.** HPLC system HPLC-04 shows *Expiring Soon*. The review finds a pump replaced (change CC-2027-033) and no deviation. The manager clicks **Plan Revalidation** and chooses an OQ limited to pump flow accuracy and gradient composition; the report renews validity for 24 months.



# Validation of processes, cleaning, analytical methods and computerised systems (CSV)

|                   |                                                      |                |       |
|-------------------|------------------------------------------------------|----------------|-------|
| Module            | ls_validation                                        | Version        | _____ |
| Menu              | Validation ► Execution ► Protocols (PV, CV, MV, CSV) | Effective date | _____ |
| Owner (suggested) | Validation Manager                                   | Approved by    | _____ |
| Related SOPs      | SOP-VAL-002, SOP-VAL-003, SOP-DI-001, SOP-QC-010     |                |       |

## 1 Objective

Apply the protocol–execution–report cycle to process validation (PV), cleaning validation (CV), analytical method validation (MV) and computerised system validation (CSV), with the specific content each requires.

## 2 Scope

Manufacturing processes, cleaning procedures, analytical methods and computerised systems registered as validation items — including this Odoo installation and the LS modules used for GxP decisions.

## 3 Responsibilities

| Role (software group)                             | Responsibility                                            |
|---------------------------------------------------|-----------------------------------------------------------|
| <b>Process / cleaning / method / system owner</b> | Provides requirements and risk assessment.                |
| <b>Validation Engineer</b>                        | Writes and executes protocols.                            |
| <b>QA</b>                                         | Approves protocols and reports.                           |
| <b>IT</b>                                         | Supports CSV (installation, configuration, backup tests). |

## 4 Definitions

| Term       | Definition                                                                                                                                                                |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PV</b>  | Process validation — demonstration that the process consistently produces product meeting its quality attributes (process design, qualification, continued verification). |
| <b>CV</b>  | Cleaning validation — demonstration that cleaning removes residues to below justified limits.                                                                             |
| <b>MV</b>  | Analytical method validation — accuracy, precision, specificity, linearity, range, detection/quantitation limits as applicable.                                           |
| <b>CSV</b> | Computerised system validation — risk-based verification that the configured system meets user requirements and data integrity controls work.                             |

## 5 Procedure

- 1 Register the item.** Item type Manufacturing Process, Cleaning Procedure, Analytical Method or Computerised System; set GxP impact and criticality.
- 2 Risk assessment.** Identify critical parameters/attributes (PV), worst-case products and locations (CV), method performance characteristics (MV), GxP functions and data flows (CSV); record it in ls\_risk\_management.

- 3 **Protocol.** Choose protocol type PV, CV, MV or CSV and write test cases with criteria (SOP-VAL-002).
- 4 **CSV specifics for this suite.** Test the workflows and controls you rely on (for example the state gates, segregation of duties, audit trail rules, signature mechanisms, purchase order block), the backup and restore, and the user access matrix. Record the Odoo build and module versions tested. Note: at the time of writing, four OCA modules of the repository failed to install on a fresh Odoo 19.0 database (see the annex *Known limitations*) — include installation checks in the IQ.
- 5 **Execute and report.** SOP-VAL-003 and SOP-VAL-004.
- 6 **Maintain.** Continued process verification for PV; periodic review and revalidation after changes (SOP-VAL-005); for CSV, re-test after each module upgrade under change control.

## 6. Controls enforced by the software

- ✓ Same controls as SOP-VAL-002 to SOP-VAL-004 (approval before execution, criteria mandatory, discrepancies closed before approval, signatures with re-authentication).

## 7. Good practices

- ★ For CSV, test what the software claims to enforce: each "control" box of this handbook is a candidate test case.
- ★ For CV, justify limits (e.g. health-based exposure limits) and sampling recovery.
- ★ For MV, follow ICH Q2(R2) characteristics appropriate to the method category.

## 8. Pitfalls to avoid

- ✗ Validating the software "as delivered" instead of the configuration actually used.
- ✗ Skipping the negative tests (trying the forbidden action and checking it is refused).

## 9 Records and evidence

- Protocols, executions, discrepancies, reports per item

## 10 Performance indicators

- GxP systems without current CSV report (target 0)

## 11 References

- EU GMP Annex 15 (2015) — Qualification and Validation
- FDA Guidance — Process Validation: General Principles and Practices (2011)
- ISPE GAMP 5 Guide, Second Edition (2022) — A Risk-Based Approach to Compliant GxP Computerized Systems
- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- ICH Q2(R2) — Validation of Analytical Procedures
- 21 CFR Part 11 — Electronic Records; Electronic Signatures

## 12. Demonstration

**Case.** CSV of Odoo LS — *deviation, CAPA and change control*: 46 test cases, including negative tests such as "attempt to approve a change request assessment as a non-assigned user → refused with the message 'Only ... may complete assessment'", "close a deviation with an open immediate action → refused". Backup/restore test and audit trail rule verification are included. Report *Validated with Restrictions*: the electronic signature policies are not used (no LS model inherits the signature capability) — approvals are confirmed on printed, hand-signed records.

# Calibration

ls\_calibration

**ls\_calibration** holds the instrument register (criticality, GxP impact, range, tolerance, alert lead time, calibration status Valid / Due Soon / Overdue), calibration plans with test points and interval, calibration records with as-found and as-left readings per point, reference standards, out-of-tolerance impact assessment, review and approval by a Calibration Manager, and internal or external certificates. It depends on the standard Maintenance application (instruments can be linked to equipment).

| Menu                                                                              | Content                      |
|-----------------------------------------------------------------------------------|------------------------------|
| Calibration ► Instruments                                                         | Instrument register          |
| Calibration ► Calibration Plans / Calibration Records / Certificates              | Plans, records, certificates |
| Calibration ► Calibration Status ► Due and Overdue / Generate Calibration Records | Scheduling                   |
| Calibration ► Reporting ► Calibration Analysis                                    | Analysis                     |

| Group                             | Main rights                                     |
|-----------------------------------|-------------------------------------------------|
| Calibration / Viewer · Technician | Read; perform calibrations and submit records   |
| Calibration / Manager             | Approves records, manages plans and instruments |

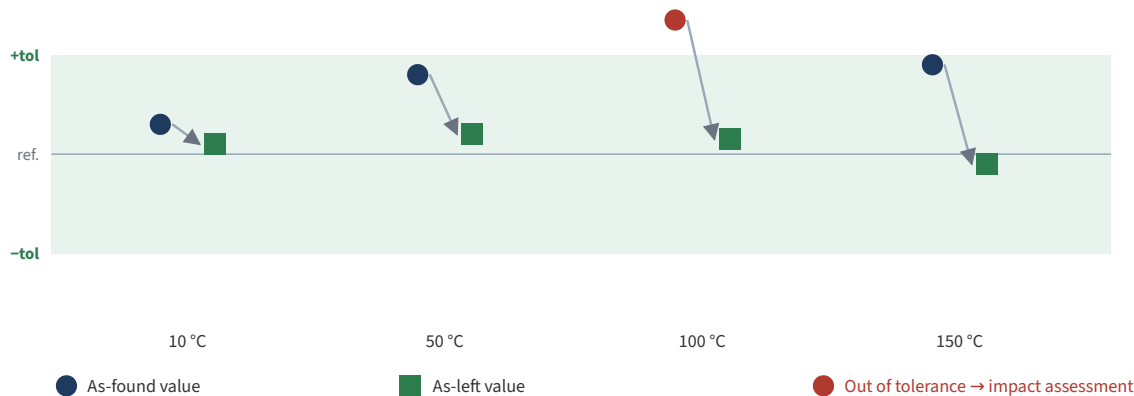
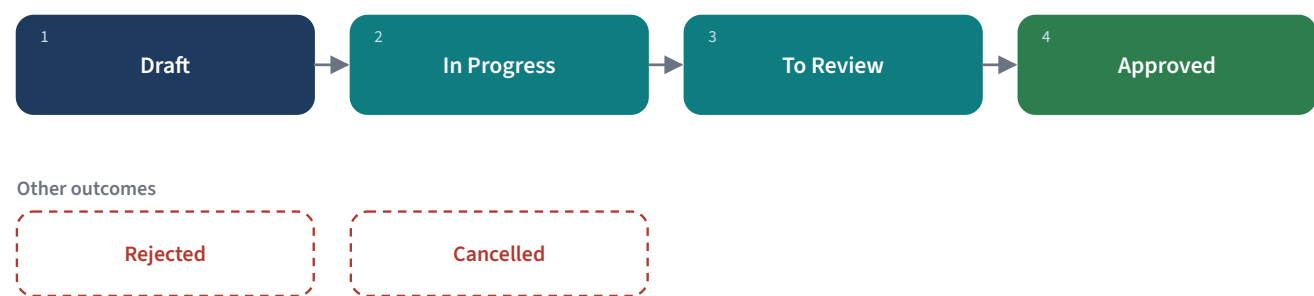


Figure 98 — As-found and as-left readings against the tolerance band (illustrative values).



Model: ls\_calibration.record

Figure 99 — Lifecycle of a calibration record.

Calibration

Instruments

Calibration Plans

Calibration Records

Certificates

Calibration Status

Reporting

7

14

My Company (San Francisco)

New

Instruments

[INS/00001] Analytical balance QC laboratory

Plans

Records

Certificates

1 / 1

Set Out of Service

Retire

Draft

In Service

Out of Service

Send message

Log note

Activity

Instrument Name

Analytical balance QC laboratory

Instrument Reference ?

INS/00001

IDENTIFICATION

Instrument Category

Manufacturer

Manufacturer Model

Serial Number

Related Equipment ?

Physical Location ?

Company

CLASSIFICATION

Criticality ?

Gxp Impact ?

Responsible ?

Alert Lead Time (Days) ?

Calibration Status ?

Last Calibration Date ?

Next Calibration Date ?

Metrological Data

Calibration Plans

Calibration Records

Certificates

Notes

MEASURING RANGE

Range Minimum

Range Maximum

0.000000

220.000000

DEFAULT ACCEPTANCE CRITERIA

Tolerance Type ?

Maximum Permissible Error ?

Absolute

0.001000

Today

OdooBot

10:40 AM

Calibration Instrument created

Figure 100 — Instrument form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Management of the instrument inventory

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_calibration                        | Version        | _____ |
| Menu              | Calibration ► Instruments             | Effective date | _____ |
| Owner (suggested) | Calibration Manager (metrology)       | Approved by    | _____ |
| Related SOPs      | SOP-MET-002, SOP-MET-003, SOP-GEN-003 |                |       |

## 1 Objective

Keep a complete register of measuring instruments that affect GxP data or product quality, with their criticality, tolerance and status, so that only calibrated instruments are used.

## 2 Scope

All measuring and test equipment used in production, QC, warehouse, utilities and validation: balances, thermometers, loggers, pressure gauges, pipettes, sensors, reference standards.

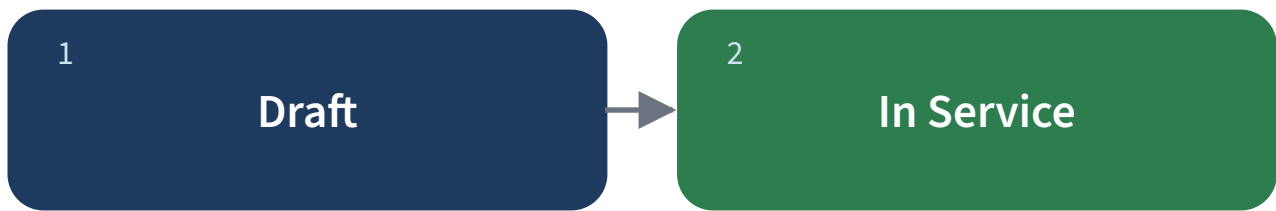
## 3 Responsibilities

| Role (software group)                 | Responsibility                                                                      |
|---------------------------------------|-------------------------------------------------------------------------------------|
| <b>Calibration Manager</b>            | Maintains the register, classifies instruments, sets tolerances.                    |
| <b>Instrument owner (Responsible)</b> | Uses the instrument correctly, reports damage, removes it from service when needed. |
| <b>Technician</b>                     | Labels instruments and updates locations.                                           |

## 4 Definitions

| Term                                         | Definition                                                                            |
|----------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Maximum permissible error / tolerance</b> | Accepted deviation, absolute or percentage of reading.                                |
| <b>Calibration status</b>                    | Not Applicable, Not Scheduled, Valid, Due Soon (within the alert lead time), Overdue. |

## 5 Workflow in the software



### Other outcomes



*Model: ls.calibration.instrument*

Figure 101 — States of an instrument.

## 6 Procedure

- 1 **Register.** In **Calibration ► Instruments** click **New** : name, reference (unique tag), related equipment, category, manufacturer, model, serial number, location, responsible, criticality, GxP impact, range, unit, tolerance type and maximum permissible error, alert lead time.
- 2 **Classify.** Set GxP impact (direct, indirect, none) and criticality; instruments with no GxP impact can be recorded for inventory only.
- 3 **Put in service.** After the initial calibration (SOP-MET-002), click **Set In Service** ; label the instrument with its reference and next due date.
- 4 **Monitor status.** Use **Calibration Status ► Due and Overdue** weekly.
- 5 **Out of service.** When damaged, overdue or failed, click **Set Out of Service** and physically segregate it ("do not use" label).
- 6 **Retire.** Click **Retire** when the instrument is disposed of; history is kept.

### 7. Controls enforced by the software

- ✓ Only draft or out-of-service instruments can be set in service; only in-service instruments can be set out of service; a retired instrument can only be reset to draft.
- ✓ Status is computed from the next calibration date and the alert lead time.

### 8. Good practices

- ★ One unique physical tag per instrument, matching the reference in the system.
- ★ Include reference standards and data loggers — they are often forgotten.

## 9. Pitfalls to avoid

- ✗ Instruments moved between rooms without updating the location.
- ✗ Personal instruments (pocket thermometers) used without registration.

## 10 Records and evidence

- Instrument register with history

## 11 Performance indicators

- Instruments overdue (target 0)
- Instruments without GxP classification

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 10012:2003 — Measurement management systems

## 13. Demonstration

**Case.** A new analytical balance BAL-017 (0.01 mg readability) is registered: criticality *Critical*, GxP *Direct Impact*, range 0–220 g, tolerance absolute 0.1 mg, alert lead time 14 days. After installation calibration it is set in service and labelled.

| Calibration Instruments Calibration Plans Calibration Records Certificates Calibration Status Reporting |                                         |                     |                                   |             |             |                 |                |                    |            |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------|-----------------------------------|-------------|-------------|-----------------|----------------|--------------------|------------|
| New Instruments                                                                                         |                                         |                     |                                   |             |             |                 |                |                    |            |
| Search...                                                                                               |                                         |                     |                                   |             |             |                 |                |                    |            |
| 1-3 / 3                                                                                                 |                                         |                     |                                   |             |             |                 |                |                    |            |
| Instrument Refere...                                                                                    | Instrument Name                         | Instrument Categ... | Physical Location                 | Criticality | Responsible | Last Calibra... | Next Calibr... | Calibration Sta... | State      |
| <input type="checkbox"/> INS/00001                                                                      | Analytical balance QC laboratory        |                     | QC laboratory - weighing room     | Critical    | OdooBot     | Mar 13          | Mar 13, 2027   | Valid              | In Service |
| <input type="checkbox"/> INS/00002                                                                      | Reference thermometer stability cham... |                     | Stability room 1                  | Major       | OdooBot     |                 | Oct 17         | Due Soon           | In Service |
| <input type="checkbox"/> INS/00003                                                                      | Mass standard set E2                    |                     | QC laboratory - metrology cabi... | Critical    | OdooBot     |                 |                | Not Scheduled      | In Service |

Figure 102 — Instrument register with calibration status. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Planning and performing calibrations (as-found and as-left values)

|                   |                                                       |                |       |
|-------------------|-------------------------------------------------------|----------------|-------|
| Module            | ls_calibration                                        | Version        | _____ |
| Menu              | Calibration ► Calibration Plans / Calibration Records | Effective date | _____ |
| Owner (suggested) | Calibration Manager                                   | Approved by    | _____ |
| Related SOPs      | SOP-MET-001, SOP-MET-003, SOP-MET-004                 |                |       |

## 1 Objective

Calibrate each instrument at a defined interval, against traceable reference standards, recording the value found before any adjustment and the value after adjustment, and have each record approved by someone other than the performer.

## 2 Scope

All instruments in service with a calibration plan, calibrated internally or by an external provider.

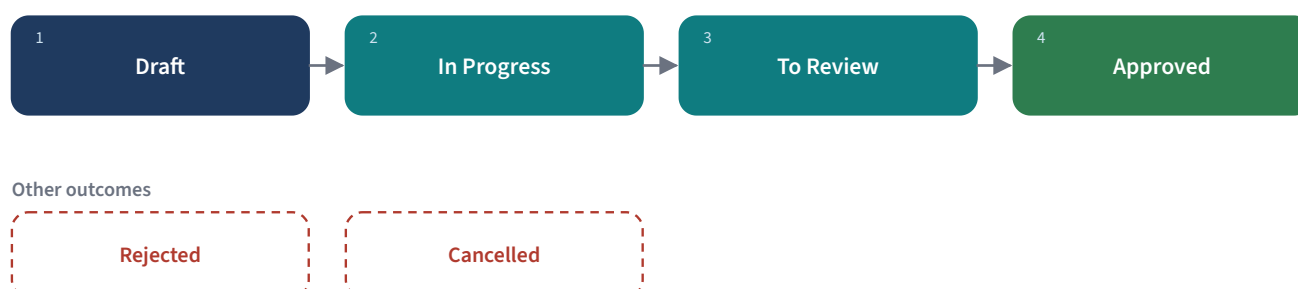
## 3 Responsibilities

| Role (software group)         | Responsibility                                                     |
|-------------------------------|--------------------------------------------------------------------|
| <b>Calibration Technician</b> | Performs the calibration and records readings; submits for review. |
| <b>Calibration Manager</b>    | Approves or rejects records; cannot approve own calibrations.      |
| <b>External provider</b>      | Performs external calibrations and issues certificates.            |

## 4 Definitions

| Term                      | Definition                                                                                                  |
|---------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>As-found</b>           | Reading before any adjustment — shows whether the instrument was within tolerance during its period of use. |
| <b>As-left</b>            | Reading after adjustment — shows the state in which it returns to service.                                  |
| <b>Reference standard</b> | Calibrated instrument used to calibrate another; it must itself be in calibration.                          |

## 5 Workflow in the software



Model: ls\_calibration.record

Figure 103 — Lifecycle of a calibration record.



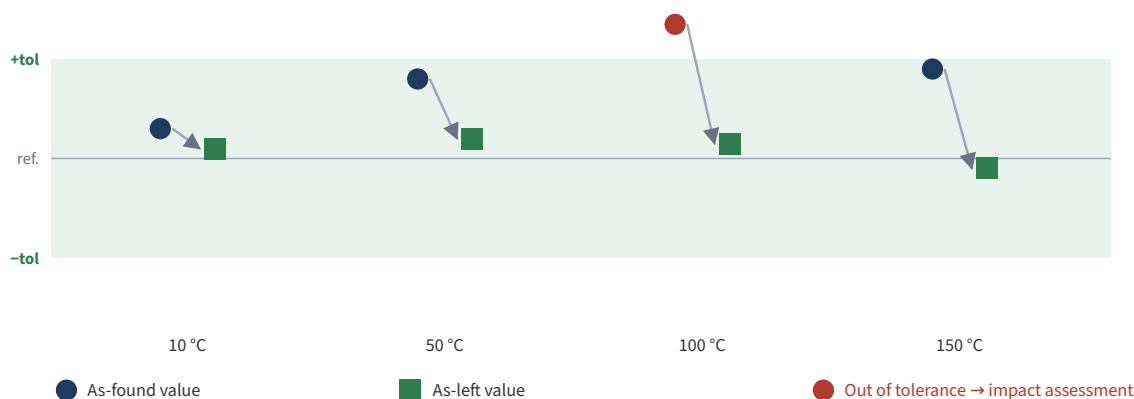


Figure 104 — As-found and as-left per test point (illustrative).

## 6 Procedure

- 1 **Create the plan.** In **Calibration Plans** click **New** : instrument, description, procedure reference, method, interval and unit, start date, responsible, external provider if any; add test points (nominal value, tolerance type and tolerance). Click **Activate** .
- 2 **Schedule.** Use **Calibration Status ▶ Generate Calibration Records** or **Create Calibration Record** on the plan for instruments due.
- 3 **Prepare.** Check that the reference standards are in calibration; record ambient temperature and humidity when relevant.
- 4 **Perform.** Click **Start** ; for each test point enter the as-found reading; adjust if needed and enter the as-left reading. The deviations and in-tolerance flags are computed.
- 5 **Conclude.** Record the calibration date, performer, reference standards (or external standard reference), result (Pass, Pass After Adjustment, Fail), adjustment performed, conclusion; if as-found is out of tolerance, complete SOP-MET-003.
- 6 **Submit.** Click **Submit to Review** .
- 7 **Approve.** The Calibration Manager (not the performer) clicks **Approve** or **Reject** with a reason. The next due date is updated.
- 8 **Certificate.** Issue or attach the certificate (SOP-MET-004).

### 7. Controls enforced by the software

- ✓ A plan needs at least one test point to be activated; records can only be created from active plans; the plan must belong to the instrument.
- ✓ Review requires a calibration date, the performer, at least one test point and at least one reference standard or external standard reference.
- ✓ An instrument cannot be its own reference; an overdue reference standard cannot be used.
- ✓ If as-found readings are out of tolerance, an impact assessment is required.
- ✓ Only a Calibration Manager can approve; the performer cannot approve; rejection requires a reason.
- ✓ Test points can only be edited in draft or in progress; approved or cancelled records are frozen; non-draft records cannot be deleted.

## 8. Good practices

- ★ Always record as-found before touching the instrument.
- ★ Choose test points covering the range of use, including the points used for release decisions.
- ★ Shorten the interval of instruments often found out of tolerance; extend it only with history.

## 9. Pitfalls to avoid

- ✗ Adjusting first and recording only the final value.
- ✗ Using an external certificate without checking the points and uncertainties against your tolerance.

## 10 Records and evidence

- Calibration records with readings, standards and approval
- Certificates

## 11 Performance indicators

- % calibrations performed before due date
- % as-found out of tolerance per instrument type

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories
- ISO 10012:2003 — Measurement management systems

## 13. Demonstration

**Case.** Temperature probe TT-104 (tolerance  $\pm 0.5$  °C) is calibrated at 20, 60 and 121 °C against reference RT-001 (in calibration). As-found at 121 °C: 121.7 °C → out of tolerance. The technician adjusts: as-left 121.1 °C. Result *Pass After Adjustment*; SOP-MET-003 is triggered for the as-found failure; the manager approves after the impact assessment.

Calibration

Instruments

Calibration Plans

Calibration Records

Certificates

Calibration Status

Reporting

7

14

My Company (San Francisco)

New

Calibration Plans

CP/00002 - [INS/00002] Reference thermomete...

Records

0

1 / 1

Create Calibration Record

Suspend

Set Obsolete

Draft

Active

Obsolete

Send message

Log note

Activity

Q

🔗

👤

Plan Reference

CP/00002

SCOPE

SCHEDULE

Instrument

[INS/00002] Reference thermometer ...

Interval

6

Months

Description ?

Half-yearly calibration of the reference

Start Date ?

Oct 17

Procedure Reference ?

SOP-MET-004

Last Calibration Date ?

Responsible

OdooBot

Next Due Date ?

Oct 17

Company

My Company (San Francisco)

Externally Calibrated ?

☐

Test Points

Method

| Point             | Nominal Value | Unit of Measurement | Tolerance Type | Tolerance | Lower Limit | Upper Limit |
|-------------------|---------------|---------------------|----------------|-----------|-------------|-------------|
| Point 1 - 5 degC  | 5.000000      | degC                | Absolute       | 0.200000  | 4.800000    | 5.200000    |
| Point 2 - 25 degC | 25.000000     | degC                | Absolute       | 0.200000  | 24.800000   | 25.200000   |
| Add a line        |               |                     |                |           |             |             |

Today

OdooBot

10:40 AM

Calibration Plan created

Figure 105 — Calibration plan with test points. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Impact assessment of an out-of-tolerance result

|                   |                                                                        |                |       |
|-------------------|------------------------------------------------------------------------|----------------|-------|
| Module            | ls_calibration                                                         | Version        | _____ |
| Menu              | Calibration ► Calibration Records (Out-of-Tolerance Impact Assessment) | Effective date | _____ |
| Owner (suggested) | Calibration Manager with QA                                            | Approved by    | _____ |
| Related SOPs      | SOP-MET-002, SOP-QA-020, SOP-QA-021                                    |                |       |

## 1 Objective

When an instrument is found out of tolerance, determine which measurements, batches and decisions made since the last good calibration may be affected, and act on them.

## 2 Scope

Every calibration record with an as-found reading out of tolerance, and every instrument found damaged or unstable in use.

## 3 Responsibilities

| Role (software group)           | Responsibility                                              |
|---------------------------------|-------------------------------------------------------------|
| <b>Calibration Manager</b>      | Documents the metrological impact and the period concerned. |
| <b>Instrument owner / users</b> | List the uses of the instrument during the period.          |
| <b>QA</b>                       | Decides on product impact through a deviation.              |

## 4 Definitions

| Term                                     | Definition                                                                             |
|------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Period at risk</b>                    | Time between the last calibration found in tolerance and the current as-found failure. |
| <b>Out-of-tolerance action reference</b> | Reference of the deviation or CAPA opened for the impact.                              |

## 5 Procedure

- 1 Stop use.** Set the instrument out of service if the as-left is not in tolerance (SOP-MET-001).
- 2 Quantify.** Compare the as-found error with the process or specification tolerance: an instrument error of 0.7 °C may be irrelevant for a 2–8 °C storage alarm set at 7 °C, or critical for a sterilisation cycle.
- 3 Trace uses.** List measurements taken with the instrument during the period at risk (batches, analyses, EM samples, validation tests).
- 4 Assess.** Write the *Out-of-Tolerance Impact Assessment* on the calibration record: error, direction, affected uses, conclusion.
- 5 Escalate.** If product or data may be affected, open a deviation (SOP-QA-020) and record its reference in *Out-of-Tolerance Action Reference*; QA decides on product disposition.
- 6 Close.** Submit the record for approval (SOP-MET-002).

## 6. Controls enforced by the software

- ✓ A calibration record whose as-found readings are out of tolerance cannot be submitted without an impact assessment.

## 7. Good practices

- ★ Think about the direction of the error: a thermometer reading low hides over-temperatures.
- ★ Use this assessment to adjust the calibration interval.

## 8. Pitfalls to avoid

- ✗ Concluding "no impact" without looking at what the instrument was used for.
- ✗ Limiting the period at risk to the last week.

## 9 Records and evidence

- Impact assessment on the record
- Deviation and product decisions

## 10 Performance indicators

- Out-of-tolerance events with product impact
- Median time from finding to impact decision

## 11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- 21 CFR 211.192 — Production record review
- ISO 10012:2003 — Measurement management systems

## 12. Demonstration

**Case (continued).** TT-104 read 0.7 °C high at 121 °C. It controls autoclave AC-02, whose cycles were therefore actually 0.7 °C cooler than displayed over the last 6 months. The validation data show a minimum F0 margin that covers 0.7 °C; QA opens deviation DEV/2026/00041 to document the evaluation of the 38 cycles; conclusion: sterility assurance maintained, no batch impact.

# Calibration certificates

|                   |                            |                |       |
|-------------------|----------------------------|----------------|-------|
| Module            | ls_calibration             | Version        | _____ |
| Menu              | Calibration ► Certificates | Effective date | _____ |
| Owner (suggested) | Calibration Manager        | Approved by    | _____ |
| Related SOPs      | SOP-MET-002                |                |       |

## 1 Objective

Issue internal certificates and register external certificates so that each calibration is supported by a controlled, retrievable certificate linked to the instrument and the calibration record.

## 2 Scope

Internal certificates generated after approved calibrations and certificates issued by external laboratories.

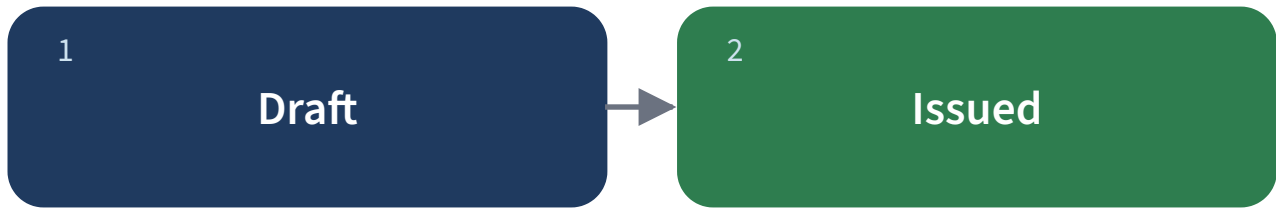
## 3 Responsibilities

| Role (software group)      | Responsibility                                                  |
|----------------------------|-----------------------------------------------------------------|
| <b>Calibration Manager</b> | Reviews and issues certificates; supersedes wrong certificates. |
| <b>Technician</b>          | Attaches external certificate files.                            |

## 4 Definitions

| Term               | Definition                                                                |
|--------------------|---------------------------------------------------------------------------|
| <b>Issuer type</b> | Internal or External (issuing laboratory, accreditation reference).       |
| <b>Superseded</b>  | Certificate replaced by a corrected one; the old one remains for history. |

## 5 Workflow in the software



### Other outcomes



*Model: ls.calibration.certificate*

Figure 106 — Lifecycle of a calibration certificate.

## 6 Procedure

- 1 **Create.** From the approved record or in **Certificates** click **New**: certificate number, instrument, calibration record, issuer type, issuing laboratory and accreditation reference for external, issue date, valid until.
- 2 **External review.** For an external certificate, check before issuing it in the system: instrument identification, points, results, uncertainty compared with your tolerance, traceability statement, accreditation. Attach the PDF.
- 3 **Issue.** Click **Issue**. The content is frozen.
- 4 **Correct.** If an issued certificate is wrong, create the corrected one and click **Supersede** on the old one.

### 7. Controls enforced by the software

- ✓ A certificate must refer to a calibration record of the same instrument; the validity end cannot precede the issue date.
- ✓ Only a draft certificate can be issued; an external certificate needs its document attached.
- ✓ Issued certificates cannot be modified or deleted; only an issued certificate can be superseded.

### 8. Good practices

- ★ Prefer accredited laboratories (ISO/IEC 17025) for reference standards.
- ★ Keep the certificate number of the provider, not a new internal number, for external certificates.

### 9. Pitfalls to avoid

- ✗ Accepting certificates that give results without uncertainty or without as-found values.

## 10 Records and evidence

- Issued certificates with files

## 11 Performance indicators

- External certificates rejected at review

## 12 References

- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories
- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- ISO 10012:2003 — Measurement management systems

### 13. Demonstration

**Case.** The external laboratory returns the certificate of reference standard RT-001 with an expanded uncertainty of 0.05 °C, accredited. The Calibration Manager checks it, attaches the PDF, enters the accreditation reference and clicks **Issue** .



# Environmental monitoring

ls\_environmental\_monitoring

**ls\_environmental\_monitoring** manages areas (nested, with cleanroom grade), sampling points (position, selection rationale, critical location), parameters (viable air, settle plates, contact plates, personnel, non-viable particles, temperature, humidity, differential pressure, air velocity, air change rate), methods, approved monitoring plans with frequencies, approved alert/action/specification limits per point, parameter and occupancy state, samples with results evaluated automatically, excursions raised automatically for action and specification breaches (and alert breaches when configured), and trend analyses.

| Menu                                                                                           | Content        |
|------------------------------------------------------------------------------------------------|----------------|
| Environmental Monitoring ► Programme ► Monitoring Plans / Sampling Points / Limits             | Programme      |
| Environmental Monitoring ► Operations ► Samples Due / All Samples / Generate Scheduled Samples | Sampling       |
| Environmental Monitoring ► Results ► Results / Threshold Breaches / Excursions                 | Results        |
| Environmental Monitoring ► Trending ► Run Trend Analysis / Trend Analyses                      | Trending       |
| Environmental Monitoring ► Configuration ► Areas / Cleanroom Grades / Methods / Parameters     | Reference data |

| Group                    | Main rights                                                                |
|--------------------------|----------------------------------------------------------------------------|
| EM / Viewer · Technician | Read; collect samples, enter results                                       |
| EM / Manager             | Approves plans and limits, reviews and approves samples, closes excursions |

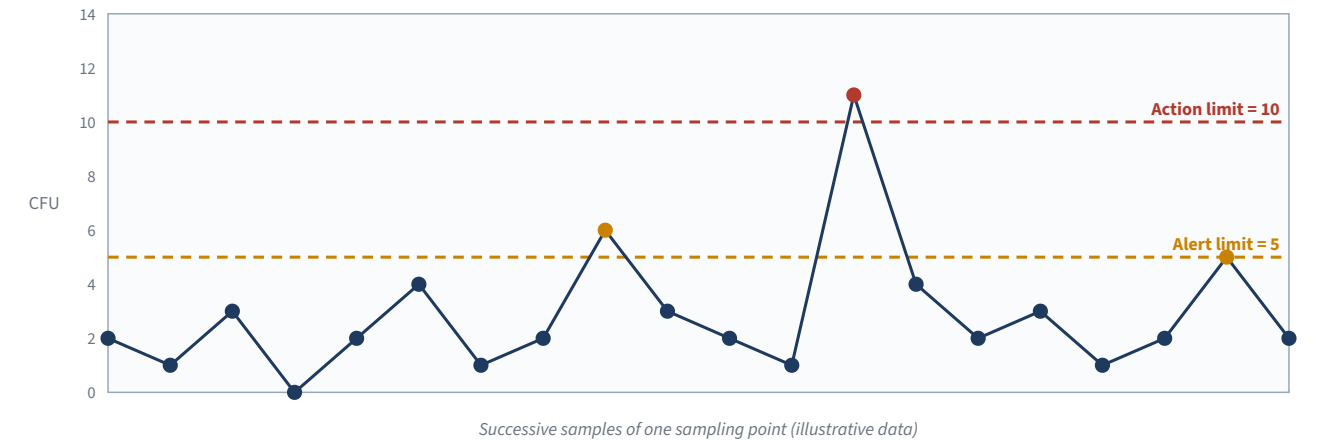


Figure 107 — Alert and action limits on a trend (illustrative values).

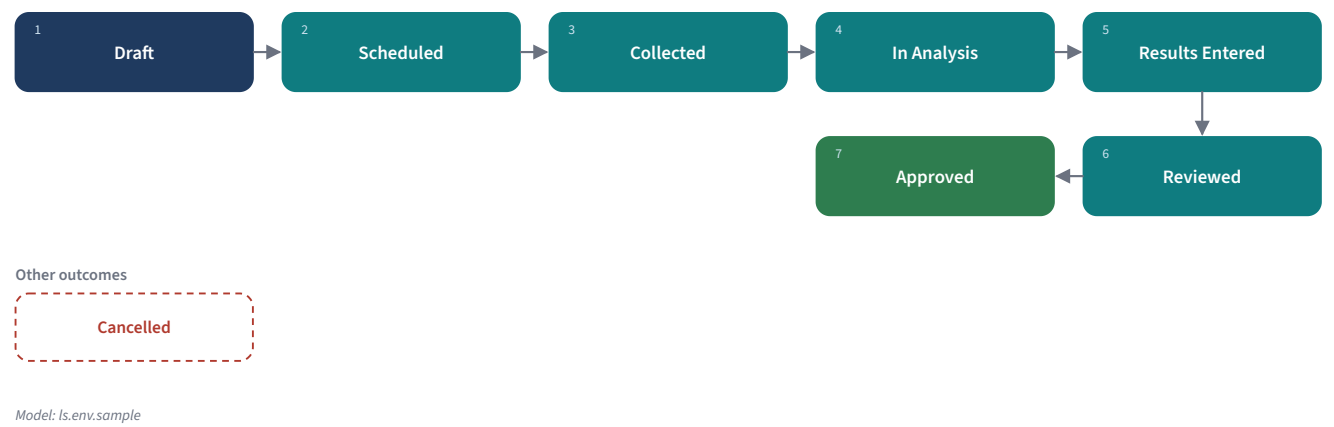


Figure 108 — Lifecycle of an EM sample.

# Monitoring programme: areas, sampling points and rationale

|                   |                                                                           |                |       |
|-------------------|---------------------------------------------------------------------------|----------------|-------|
| Module            | ls_environmental_monitoring                                               | Version        | _____ |
| Menu              | Environmental Monitoring ► Programme ► Monitoring Plans / Sampling Points | Effective date | _____ |
| Owner (suggested) | EM Manager (QA / microbiology)                                            | Approved by    | _____ |
| Related SOPs      | SOP-EM-002, SOP-EM-003, SOP-QA-060                                        |                |       |

## 1 Objective

Define, on the basis of a documented risk assessment, which areas and points are monitored, for which parameters, in which occupancy state and how often, and keep this programme approved.

## 2 Scope

Cleanrooms and controlled areas of all grades, utilities with environmental parameters, and personnel monitoring.

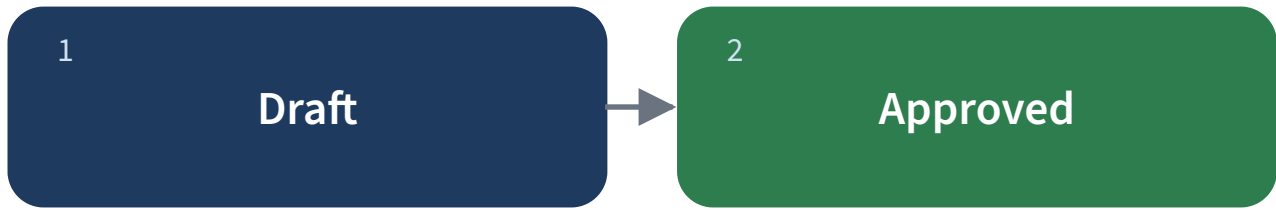
## 3 Responsibilities

| Role (software group)    | Responsibility                                                   |
|--------------------------|------------------------------------------------------------------|
| <b>EM Manager</b>        | Designs and approves (with a second person) the programme.       |
| <b>Microbiology / QC</b> | Provides method capabilities.                                    |
| <b>Production</b>        | Provides process knowledge for the choice of critical locations. |

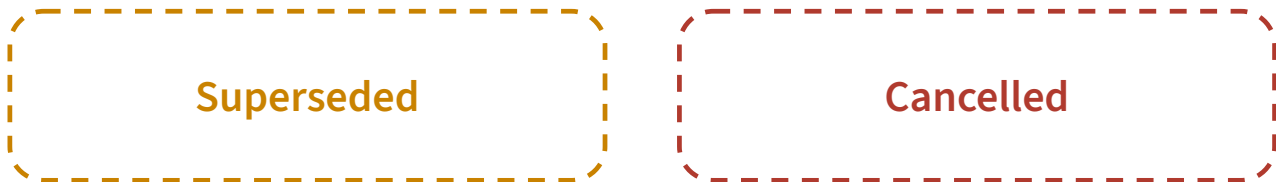
## 4 Definitions

| Term                     | Definition                                                                          |
|--------------------------|-------------------------------------------------------------------------------------|
| <b>Critical location</b> | Point where product or sterile surfaces are exposed; flagged on the sampling point. |
| <b>Occupancy state</b>   | At Rest or In Operation (limits and plans distinguish them).                        |

## 5 Workflow in the software



### Other outcomes



*Model: ls.env.plan*

Figure 109 — Lifecycle of a monitoring plan.

## 6 Procedure

- 1 Areas.** In **Configuration ► Areas** create the area tree with grades and responsible.
- 2 Sampling points.** In **Programme ► Sampling Points** create each point: code, area, grade, position, *Selection Rationale* (from the risk assessment, airflow studies, process), *Critical Location*.
- 3 Parameters and methods.** Check parameters (type, quantitative or pass/fail) and methods (which parameter each measures).
- 4 Plan.** In **Monitoring Plans** click **New**: code, version, primary area, effective from, rationale; add plan lines (point, parameter, method, occupancy state, every N days/weeks/months/years, start date, responsible).
- 5 Approve.** A user other than the author clicks **Approve**.
- 6 Revise.** To change an approved plan click **Create Revision**; the new version supersedes the old at approval.
- 7 Generate samples.** Use **Operations ► Generate Scheduled Samples** to create the due samples from approved plans.

### 7. Controls enforced by the software

- ✓ A plan needs at least one line and must be approved by a user other than its creator; approved plans cannot be cancelled or deleted — they are superseded by revision.
- ✓ The method of a plan line must measure its parameter; point, parameter and plan must belong to the same company; supported frequency units only.

### 8. Good practices

- ★ Write the selection rationale for every point: an inspector will ask why this point and not another.
- ★ Include worst-case locations (near doors, operator hands, filling needles).

### 9. Pitfalls to avoid

- ✗ Programmes copied from another site without local risk assessment.
- ✗ Monitoring only At Rest.

### 10 Records and evidence

- Approved plans with versions
- Sampling point rationale
- Risk assessment

### 11 Performance indicators

- Points without rationale (target 0)
- Plan review on time

### 12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration
- ICH Q9(R1) — Quality Risk Management

### 13. Demonstration

**Case.** Grade A filling zone: 6 points — 2 critical (needle area, stopper bowl) for continuous particle counting and active air each shift In Operation, 4 for settle plates. The rationale refers to the smoke studies of 2026 and to the process FMEA. The plan is approved by the QA microbiologist (not the author).

# Setting and approval of alert and action limits

|                   |                                               |                |       |
|-------------------|-----------------------------------------------|----------------|-------|
| Module            | ls_environmental_monitoring                   | Version        | _____ |
| Menu              | Environmental Monitoring ► Programme ► Limits | Effective date | _____ |
| Owner (suggested) | EM Manager                                    | Approved by    | _____ |
| Related SOPs      | SOP-EM-001, SOP-EM-004, SOP-EM-005            |                |       |

## 1 Objective

Set, justify and approve alert, action and, where applicable, specification limits for each sampling point, parameter and occupancy state, and change them only through a revision.

## 2 Scope

All quantitative and pass/fail parameters of the monitoring plans.

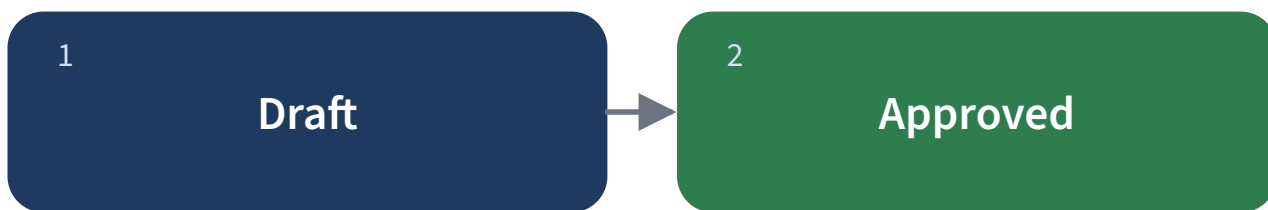
## 3 Responsibilities

| Role (software group)             | Responsibility                                                           |
|-----------------------------------|--------------------------------------------------------------------------|
| EM Manager                        | Proposes limits with justification (regulatory maxima, historical data). |
| Approver (second EM Manager / QA) | Approves limits; cannot be the creator.                                  |

## 4 Definitions

| Term                | Definition                                                                            |
|---------------------|---------------------------------------------------------------------------------------|
| Alert limit         | Level that signals a potential drift; investigation according to procedure.           |
| Action limit        | Level that requires investigation and corrective action; always opens an excursion.   |
| Specification limit | Regulatory or product maximum; always opens an excursion.                             |
| Direction           | Upper bound (breach above) or lower bound (breach below, e.g. differential pressure). |

## 5 Workflow in the software



### Other outcomes



Model: *ls.env.limit*

Figure 110 — Lifecycle of a limit.

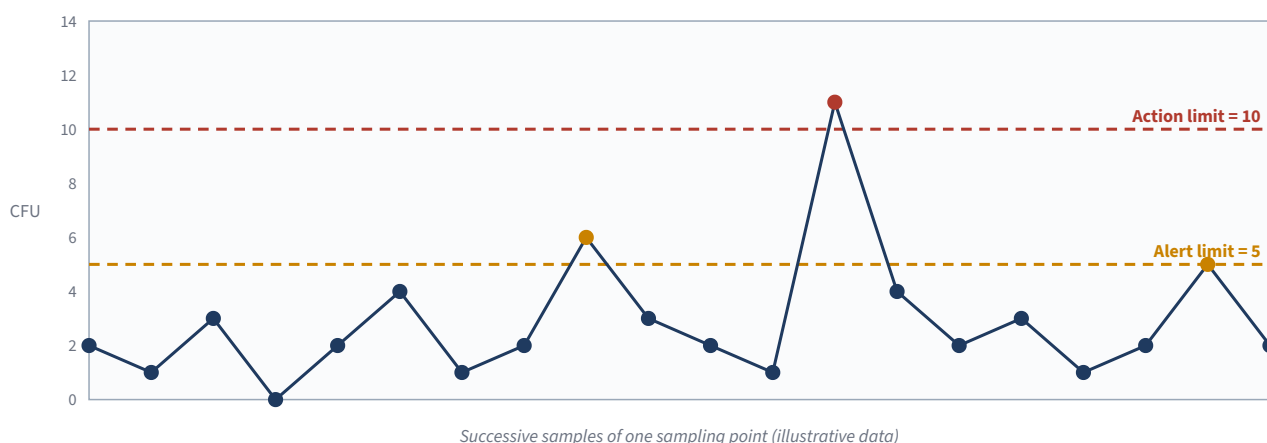


Figure 111 — Alert and action limits (illustrative values).

## 6 Procedure

- 1 **Collect basis.** Regulatory grade limits (e.g. EU GMP Annex 1 tables) and historical data of the point.
- 2 **Create.** In [Programme ► Limits](#) click **New** : sampling point, parameter, occupancy state, direction, alert, action and specification limits, *Raise Excursion on Alert* if alerts must be formally investigated, justification, controlling document, effective from.
- 3 **Approve.** A user other than the creator clicks **Approve** .
- 4 **Revise.** To change thresholds click **Create Revision** ; approve the revision (the previous one is superseded).
- 5 **Review.** Review limits yearly with trend data (SOP-EM-005).

### 7. Controls enforced by the software

- ✓ At least one threshold is required; the order of thresholds must match the direction (for an upper bound alert  $\leq$  action  $\leq$  specification).
- ✓ Only one approved limit per point, parameter, occupancy state and direction — supersede before approving a replacement.
- ✓ A justification is required; the approver must differ from the creator; approved thresholds are frozen; only draft limits can be deleted.

### 8. Good practices

- ★ Never set alert limits above regulatory action levels.
- ★ Tighten alert limits on the basis of historical performance (e.g. a statistical percentile).

### 9. Pitfalls to avoid

- ✗ Samples evaluated as *No Approved Limit* — they never raise an excursion. Check that every plan line has an approved limit.

## 10 Records and evidence

- Approved limits with justification and versions

## 11 Performance indicators

- Plan lines without approved limit (target 0)

## 12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration

### 13. Demonstration

**Case.** Point A-01, active air, In Operation, upper bound: action limit 1 CFU/m<sup>3</sup> (grade A), no alert limit; specification not used. Point C-12, settle plate, In Operation: alert 25, action 50 CFU/4 h, *Raise Excursion on Alert* ticked because the room had two alerts last quarter. Both approved by the QA microbiologist.

# Sampling, analysis and review of results

|                   |                                                 |                |       |
|-------------------|-------------------------------------------------|----------------|-------|
| Module            | ls_environmental_monitoring                     | Version        | _____ |
| Menu              | Environmental Monitoring ► Operations ► Samples | Effective date | _____ |
| Owner (suggested) | EM Manager                                      | Approved by    | _____ |
| Related SOPs      | SOP-EM-001, SOP-EM-002, SOP-EM-004              |                |       |

## 1 Objective

Collect scheduled samples on time, analyse them, record results traceably and have them reviewed and approved by people other than the analyst, so that breaches are detected and escalated.

## 2 Scope

All scheduled and unscheduled EM samples.

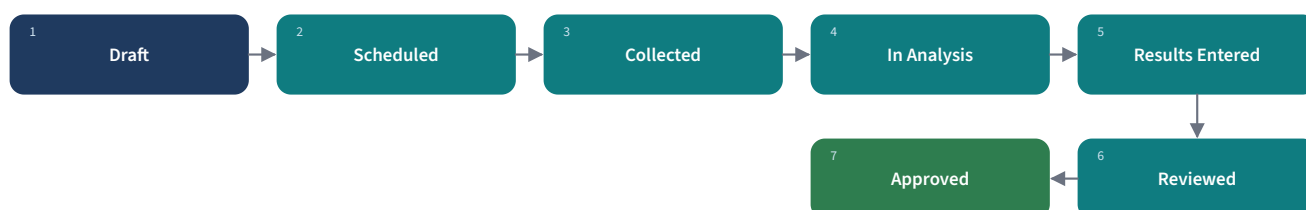
## 3 Responsibilities

| Role (software group) | Responsibility                                                            |
|-----------------------|---------------------------------------------------------------------------|
| <b>EM Technician</b>  | Collects, analyses, enters results.                                       |
| <b>EM Manager</b>     | Reviews and approves (different from the person who entered the results). |

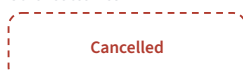
## 4 Definitions

| Term              | Definition                                                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Evaluation</b> | Automatic comparison with the approved limit: Within Limits, Alert/Action/Specification Limit Exceeded, No Approved Limit. |
| <b>Amendment</b>  | Traced correction of a result after approval, with a reason, once only.                                                    |

## 5 Workflow in the software



Other outcomes



Model: ls.env.sample

Figure 112 — Sample lifecycle.

## 6 Procedure

**1 Prepare.** Open **Operations ► Samples Due**; click **Schedule** for the day's samples; print labels.



- 2 **Collect.** Take the sample at the point in the stated occupancy state; click **Record Collection** with the actual date/time and collector; record the batch or campaign reference when relevant.
- 3 **Analyse.** Click **Start Analysis**; incubate/read according to method.
- 4 **Enter results.** Enter the value (or pass/fail) of each result; click **Submit Results**. The evaluation is computed.
- 5 **Review.** An EM Manager other than the result entrant clicks **Review**; if a value is doubtful, **Return to Analysis**.
- 6 **Approve.** An EM Manager other than the result entrant clicks **Approve**; excursions are raised automatically for action and specification breaches (and alerts when configured).
- 7 **Correct after approval.** Use **Amend Result** on the result with a reason; further corrections go through an investigation.
- 8 **Cancel.** A sample not taken is cancelled with a reason (**Cancel**) — never deleted.

### 7. Controls enforced by the software

- ✓ Collection date/time cannot be in the future; a sample needs at least one parameter before collection; every result needs a value before submission.
- ✓ Review and approval are reserved to the EM Manager role and must be done by users other than the one who entered results.
- ✓ Approved samples are frozen; results on approved samples change only by amendment (once, with reason).
- ✓ Only draft samples can be deleted; cancellation requires a reason; counts cannot be negative.

### 8. Good practices

- ★ Enter results the day they are read; record colony identification requests in the notes.
- ★ Watch *Threshold Breaches* daily, not only at approval.

### 9. Pitfalls to avoid

- ✗ Samples collected but recorded days later.
- ✗ Returning a sample to analysis to replace an inconvenient value without justification.

## 10 Records and evidence

- Samples with results, reviewer and approver
- Amendments

## 11 Performance indicators

- % samples collected on schedule
- Samples awaiting approval > 3 days

## 12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ISO 14644-1:2015 — Cleanrooms — Classification of air cleanliness by particle concentration
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

### 13. Demonstration

**Case.** Settle plate C-12, 4 h In Operation, read after incubation: 31 CFU. The technician enters 31 and submits; evaluation *Alert Limit Exceeded*. The EM Manager reviews and approves; because *Raise Excursion on Alert* is ticked on the limit, excursion EXC-2026-018 is opened automatically (SOP-EM-004).

# Management of excursions

|                   |                                                 |                |       |
|-------------------|-------------------------------------------------|----------------|-------|
| Module            | ls_environmental_monitoring                     | Version        | _____ |
| Menu              | Environmental Monitoring ► Results ► Excursions | Effective date | _____ |
| Owner (suggested) | EM Manager with QA                              | Approved by    | _____ |
| Related SOPs      | SOP-EM-003, SOP-QA-020, SOP-QA-022              |                |       |

## 1 Objective

Assess, investigate and close every environmental excursion with a documented product impact decision, and escalate to deviation and CAPA when required.

## 2 Scope

Excursions raised automatically at sample approval and excursions created manually.

## 3 Responsibilities

| Role (software group)           | Responsibility                                                                                  |
|---------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Excursion owner</b>          | Records immediate actions, impact assessment, root cause, corrective actions; proposes closure. |
| <b>EM Manager / QA (closer)</b> | Closes the excursion; must differ from the owner.                                               |
| <b>QA</b>                       | Decides on batch impact through the deviation process.                                          |

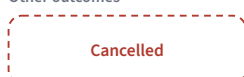
## 4 Definitions

| Term                  | Definition                                                            |
|-----------------------|-----------------------------------------------------------------------|
| <b>Breach type</b>    | Alert, Action or Specification Limit Exceeded.                        |
| <b>Product impact</b> | No Product Impact Identified, Potential, Confirmed, Not Yet Assessed. |

## 5 Workflow in the software



Other outcomes



Model: ls.env.excursion

Figure 113 — Lifecycle of an excursion.

## 6 Procedure

- Immediate actions.** Record immediate actions (resampling, cleaning, room check, identification of isolates).
- Assess.** Click **Start Assessment**; write the impact assessment: batches in the room during the period (*Batch / Campaign Reference*), criticality of the point, isolate identity, other results.

- 3 **Investigate.** When required, click **Open Investigation**, determine the root cause and corrective actions. The module cannot create a CAPA or deviation itself: open them in `ls_deviation` / `ls_capa` and write the reference in *External Record Reference*.
- 4 **Propose closure.** Set the product impact and click **Propose Closure**.
- 5 **Close.** A user other than the owner clicks **Close** and writes the closure justification; **Reopen Investigation** if the proposal is insufficient.

### 7. Controls enforced by the software

- ✓ Excursions cannot be deleted, only cancelled; closed excursions are frozen.
- ✓ An impact assessment is required before investigation and before proposing closure; the product impact must be assessed; a root cause is required when an investigation was required.
- ✓ Closure requires a justification and a user other than the owner.
- ✓ The *create external record* action explains that no corrective action module is linked: the reference must be recorded manually.

### 8. Good practices

- ★ Identify isolates to species level for grade A/B action breaches.
- ★ Link the excursion to the batch record review: QA must see it before release.

### 9. Pitfalls to avoid

- ✗ Closing with *No Product Impact Identified* before the batch has been evaluated.
- ✗ Treating repeated alerts individually instead of looking at the trend.

## 10 Records and evidence

- Excursion with assessment, investigation, closure
- Deviation / CAPA references

## 11 Performance indicators

- Open excursions > 30 days
- Excursions per area per quarter

## 12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- ICH Q9(R1) — Quality Risk Management

### 13. Demonstration

**Case (continued).** EXC-2026-018 (alert, grade C corridor). Immediate action: resample and check door interlock — found faulty. Impact assessment: grade C corridor, no open product; product impact *No Product Impact Identified*. Root cause: door interlock sensor failure; corrective action: sensor replaced (work order WO-2231). The owner proposes closure; the QA microbiologist closes it with justification.

# Trend analysis

|                   |                                     |                |       |
|-------------------|-------------------------------------|----------------|-------|
| Module            | ls_environmental_monitoring         | Version        | _____ |
| Menu              | Environmental Monitoring ► Trending | Effective date | _____ |
| Owner (suggested) | EM Manager                          | Approved by    | _____ |
| Related SOPs      | SOP-EM-002, SOP-EM-004, SOP-GEN-001 |                |       |

## 1 Objective

Detect adverse trends in environmental data before limits are breached, review them formally, and feed limit reviews and management review.

## 2 Scope

All EM results, per area, point and parameter.

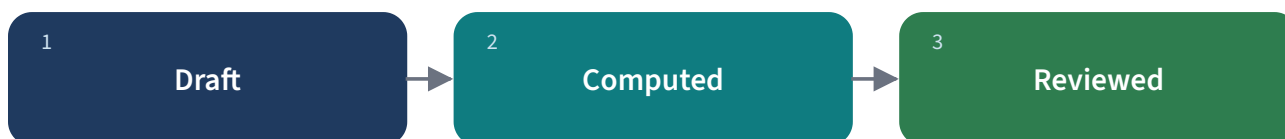
## 3 Responsibilities

| Role (software group) | Responsibility                                  |
|-----------------------|-------------------------------------------------|
| EM Manager            | Runs and reviews the analyses.                  |
| QA                    | Receives the conclusions for management review. |

## 4 Definitions

| Term                | Definition                                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| Trend line          | Per point and parameter: counts, exceedances, direction (Increasing, Stable, Decreasing, Insufficient Data). |
| Direction threshold | Change threshold used to call a trend increasing or decreasing.                                              |

## 5 Workflow in the software



Model: ls.env.trend

Figure 114 — Lifecycle of a trend analysis.

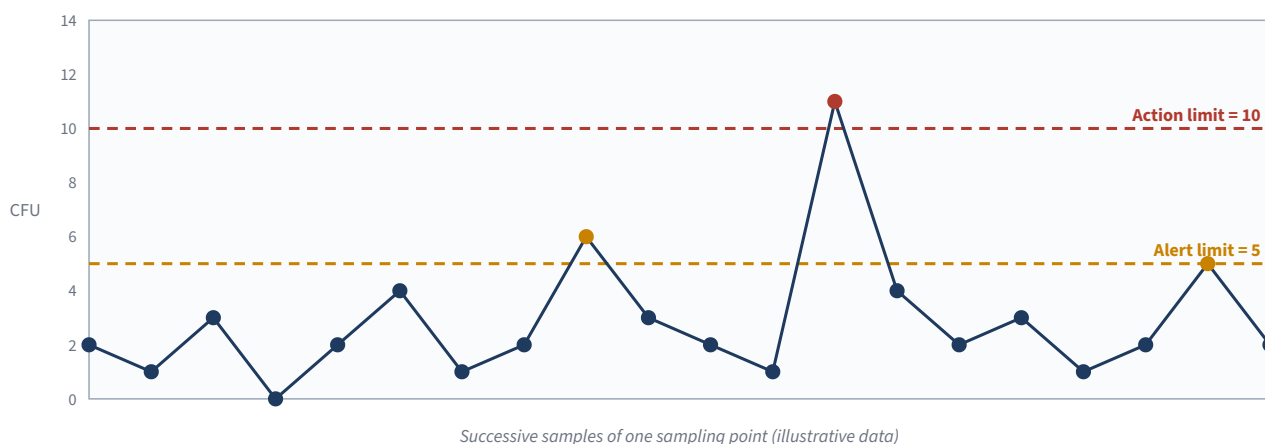


Figure 115 — A trend reveals drifts that single results hide (illustrative).

## 6 Procedure

- 1 **Run.** Monthly (and quarterly for the report), open **Trending ► Run Trend Analysis**: period, areas, points, parameters, direction threshold; click **Run Analysis** (or create an analysis and click **Compute**).
- 2 **Interpret.** Look at increasing directions, repeated alerts, new isolates, seasonal effects.
- 3 **Conclude.** Write the conclusion and click **Mark as Reviewed**; a reviewed analysis cannot be recomputed — create a new one if needed.
- 4 **Act.** Open a CAPA for adverse trends; revise limits (SOP-EM-002) or the plan (SOP-EM-001) when justified.
- 5 **Report.** Include the quarterly conclusions in management review.

### 7. Controls enforced by the software

- ✓ The end of the period must not precede its start; only a computed analysis can be reviewed; a conclusion is required; reviewed analyses cannot be recomputed.
- ✓ Counts and thresholds must be non-negative; exceeded count  $\leq$  total count.

### 8. Good practices

- ★ Trend by area as well as by point: a room can drift while each point stays within limits.
- ★ Keep the same period and threshold between analyses for comparability.

### 9. Pitfalls to avoid

- ✗ Trending only exceedances instead of all results.

## 10 Records and evidence

- Reviewed trend analyses with conclusions

## 11 Performance indicators

- Increasing trends without action

## 12 References

- EU GMP Annex 1 (2022) — Manufacture of Sterile Medicinal Products
- ICH Q10 — Pharmaceutical Quality System

### 13. Demonstration

**Case.** Q3 analysis for the grade C area: point C-12 settle plates *Increasing* (mean 8 → 17 CFU), still under the alert limit except once. Conclusion: adverse trend linked to the new material airlock routine; CAPA-2026-044 opened to review the airlock cleaning frequency.

# Recalls

ls\_recall

**ls\_recall** manages recall plans (coordinator and deputy, decision team, authorities, scope, default effectiveness level, target reconciliation rate, initiation target in hours, mock recall interval), and field actions of five types (recall, market withdrawal, stock recovery, field safety corrective action, mock recall): health hazard classification, depth, public warning, affected lots, distribution traced from stock moves, consignee lines with reconciliation, communications with content checklist, effectiveness checks by level A–E, reports, and gated closure. Lots under an open recall cannot be delivered.

| Menu                                                                               | Content       |
|------------------------------------------------------------------------------------|---------------|
| Recalls ► Configuration ► Recall Plans                                             | Plans         |
| Recalls ► Field Actions ► Initiate Recall / Recalls / All Field Actions            | Field actions |
| Recalls ► Execution ► Consignees / Communications / Effectiveness Checks / Reports | Execution     |

| Group                | Main rights                                                          |
|----------------------|----------------------------------------------------------------------|
| Viewer · Coordinator | Read; run recalls                                                    |
| Manager              | Approves plans, closes recalls (including with a justified override) |

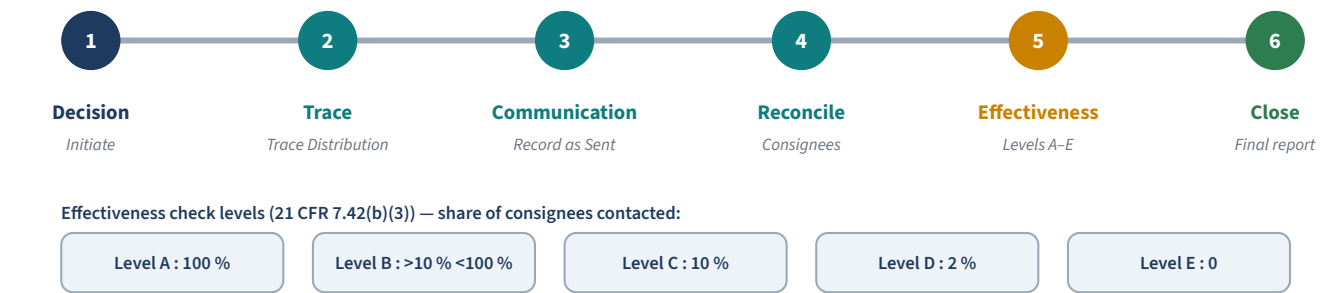
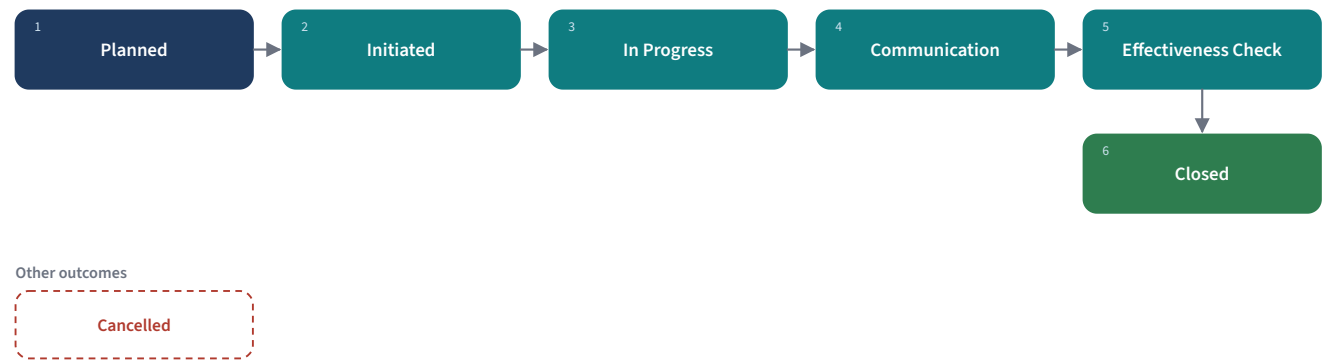


Figure 116 — Recall phases and effectiveness levels.



Model: ls\_recall.execution  
Figure 117 — Lifecycle of a field action.

# Recall plan, with coordinator and deputy

|                   |                                        |                |       |
|-------------------|----------------------------------------|----------------|-------|
| Module            | ls_recall                              | Version        | _____ |
| Menu              | Recalls ► Configuration ► Recall Plans | Effective date | _____ |
| Owner (suggested) | Head of Quality                        | Approved by    | _____ |
| Related SOPs      | SOP-QA-081, SOP-QA-083                 |                |       |

## 1 Objective

Have an approved plan that allows a recall to be decided and started at any time, including outside office hours, by named people.

## 2 Scope

All products marketed by the company, or selected products or categories when several plans exist.

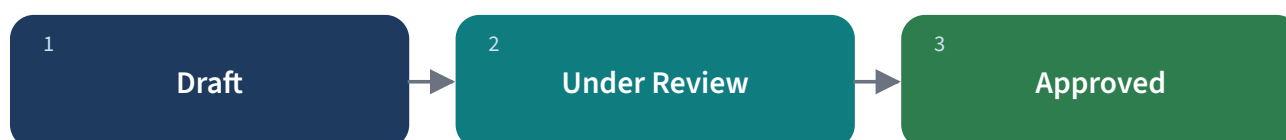
## 3 Responsibilities

| Role (software group)         | Responsibility                                                                               |
|-------------------------------|----------------------------------------------------------------------------------------------|
| Head of Quality               | Approves the plan.                                                                           |
| Recall coordinator and deputy | Initiate and run recalls; available 24/7 by rota.                                            |
| Decision team                 | Decides on recalls with the coordinator (QA, regulatory, medical, supply chain, management). |

## 4 Definitions

| Term                       | Definition                                                        |
|----------------------------|-------------------------------------------------------------------|
| Initiation target          | Hours between the decision and the first communication.           |
| Target reconciliation rate | Share of distributed quantity to be accounted for before closure. |

## 5 Workflow in the software



Other outcomes



Model: ls\_recall.plan

Figure 118 — Lifecycle of a recall plan.

## Procedure



- 1 **Draft.** In **Recall Plans** click **New** : title, coordinator, deputy, decision team, competent authorities, scope (all products, selected products or categories), procedure (the recall procedure text or reference), default effectiveness level, target reconciliation rate, initiation target (hours), mock recall interval (months).
- 2 **Review and approve.** Click **Submit for Review** ; the approver clicks **Approve** (or **Reject** ).
- 3 **Publish contacts.** Give the coordinator/deputy contact list to the security desk and switchboard.
- 4 **Revise.** To change an approved plan click **New Revision** ; the old plan is made obsolete when the new one is approved ( **Set Obsolete** if withdrawn).
- 5 **Test.** Test the plan through mock recalls (SOP-QA-083).

### 7. Controls enforced by the software

- ✓ A plan cannot be approved without a documented procedure or without a deputy coordinator.
- ✓ A plan limited to selected products or categories must list them.
- ✓ Approved plans are frozen for key fields — create a new revision; only draft plans can be deleted.

### 8. Good practices

- ★ Name the deputy from another department than the coordinator's to avoid simultaneous absence.
- ★ Include authority contact details and after-hours numbers in the procedure.

### 9. Pitfalls to avoid

- ✗ Plans naming people who left the company.

## 10 Records and evidence

- Approved plan with revisions

## 11 Performance indicators

- Plan reviewed in the last 12 months
- Coordinator/deputy reachable in mock recall test

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- MDR Articles 87 to 92 — Vigilance

### 13. Demonstration

**Case.** Plan *Recall* — *all products*, coordinator: QA manager; deputy: supply chain manager; initiation target 24 h; default level B at 20 %; target reconciliation 95 %; mock recall every 12 months. Approved by the Head of Quality.

Recalls
Field Actions
Execution
Configuration
7
14
My Company (San Francisco)

New
Recall Plans
[RCP/2026/0001] Recall plan - finished medicin...
Recalls
0
1 / 1

Submit for Review
Draft
Under Review
Approved
Send message
Log note
Activity

Title ?

# Recall plan - finished medicinal products

RCP/2026/0001 version 1

| RESPONSIBILITIES                                                                          | SCOPE                                                                                                                            |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Recall Coordinator ?<br>Deputy ?<br>Decision Team ?<br>Competent Authorities ?<br>Company | <div> OdooBot Scope Type All Products </div> <div>National Medicines Agency (demo) X</div> <div>My Company (San Francisco)</div> |

Procedure
Default Strategy
Approval

Decision route: the recall coordinator convenes the decision team within four hours of a quality defect being classified as potentially recall-relevant.

Out-of-hours: the deputy coordinator holds the duty telephone and may initiate a class I recall alone.

Disposition: returned product is quarantined in the blocked-stock location pending a disposition decision by the quality unit.

Today

OdooBot
10:40 AM
Recall Plan created

Figure 119 — Recall plan with coordinator and deputy. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Executing a recall or withdrawal: distribution tracing, notification, reconciliation

|                   |                                                |                |       |
|-------------------|------------------------------------------------|----------------|-------|
| Module            | ls_recall                                      | Version        | _____ |
| Menu              | Recalls ► Field Actions                        | Effective date | _____ |
| Owner (suggested) | Recall coordinator                             | Approved by    | _____ |
| Related SOPs      | SOP-QA-080, SOP-QA-082, SOP-QA-040, SOP-QA-021 |                |       |

## 1 Objective

Remove defective product from the market quickly and completely: decide, trace distribution, notify consignees and authorities, and reconcile quantities.

## 2 Scope

Recalls, market withdrawals, stock recoveries and field safety corrective actions. Mock recalls follow SOP-QA-083.

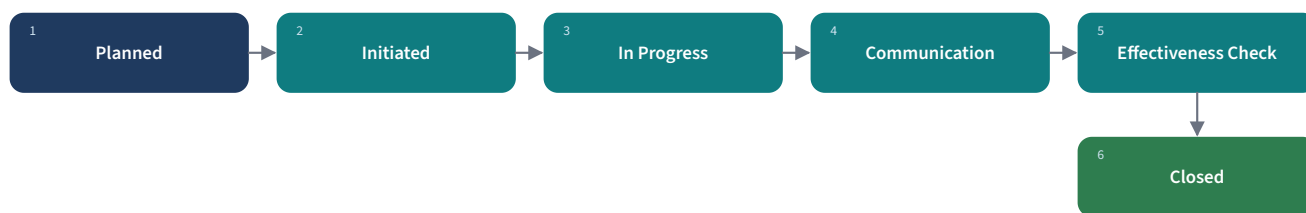
## 3 Responsibilities

| Role (software group)                  | Responsibility                                                  |
|----------------------------------------|-----------------------------------------------------------------|
| <b>Recall coordinator (or deputy)</b>  | Opens and drives the field action.                              |
| <b>Decision team</b>                   | Decides, classifies the health hazard, approves communications. |
| <b>Regulatory affairs</b>              | Notifies the competent authorities.                             |
| <b>Supply chain / customer service</b> | Contacts consignees, receives returns.                          |

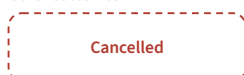
## 4 Definitions

| Term                  | Definition                                                                                                                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Classification</b> | Health hazard class I, II or III (structure of 21 CFR 7.3(m)); the authority may assign it.                                                                                     |
| <b>Depth</b>          | Wholesale, Retail or Consumer/User level.                                                                                                                                       |
| <b>Consignee line</b> | Consignee × lot with quantities shipped, returned, destroyed on site, not recoverable; status Pending Notification → Notified → Response Received → Reconciled, or Discrepancy. |

## 5 Workflow in the software



Other outcomes



Model: ls.recall.execution

Figure 120 — Lifecycle of a field action.

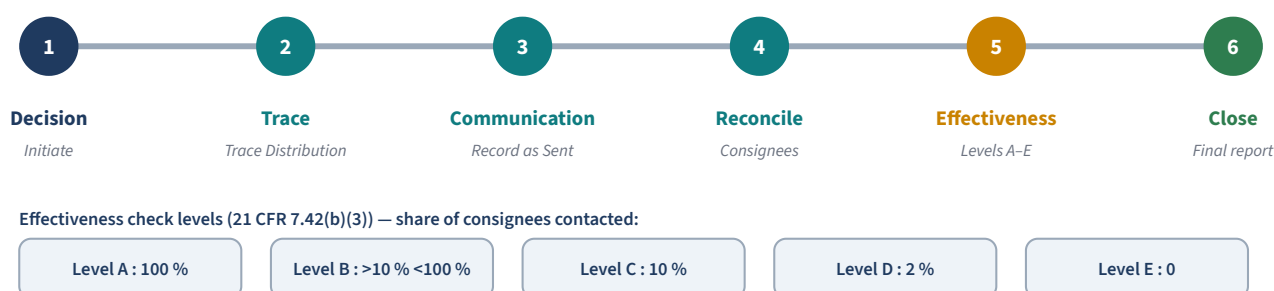


Figure 121 — Phases of a recall.

## 6 Procedure

- 1 **Decide.** The decision team meets; record the decision. Open **Field Actions ► Initiate Recall**: action type, plan, product, lots, reason, classification, depth; click **Open Recall**.
- 2 **Complete the strategy.** Record the health hazard evaluation (who, when), public warning, effectiveness level (Level B percentage), target completion date, quality defect and CAPA references.
- 3 **Initiate.** Click **Initiate** (reason and affected lots required; for a real action, classification, depth and health hazard evaluation required).
- 4 **Trace and start.** Click **Trace Distribution and Start**: consignee lines are created from the delivered stock moves of the lots. Add consignees manually for distribution outside the system. Lots under an open recall are blocked for delivery.
- 5 **Communicate.** Create communications (Recall Notice, Field Safety Notice, Authority Notification, Public Warning...): recipients, subject, body; confirm the content checklist (product identified, reason and hazard, instructions, response requested, contact point); click **Approve Text**, send, then **Record as Sent**; record acknowledgements.
- 6 **Move to communication.** Click **Move to Communication** (at least one communication recorded as sent).
- 7 **Reconcile.** Update each consignee line with quantities returned, destroyed, not recoverable and responses; mark notified; investigate discrepancies.
- 8 **Notify the authority.** Record authority notification date and reference; issue status reports (**Execution ► Reports**) as required.
- 9 **Continue.** Go to effectiveness checks (SOP-QA-082).

## 7. Controls enforced by the software

- ✓ Once initiated, a recall cannot be deleted (cancel it); key fields are under change control.
- ✓ Initiation requires reason and affected lots; a real field action also requires classification, depth and health hazard evaluation.
- ✓ Lots must belong to the product; target completion date  $\geq$  decision date; Level B requires a percentage between 10 % and 100 % exclusive.
- ✓ If no distribution is found, the system says so and consignees must be added manually.
- ✓ Communications: content checklist confirmed, recipients and text required, approval before sending; sent communications are frozen.
- ✓ Deliveries of lots under an open recall are refused.
- ✓ A finalised recall cannot be modified or re-traced.

## 8. Good practices

- ★ Start communication within the plan's initiation target; record the first communication date.
- ★ Call critical consignees (hospitals) in addition to written notices.
- ★ Keep all correspondence attached to the communication records.

## 9. Pitfalls to avoid

- ✗ Tracing only in the ERP when part of the distribution was handled by a third-party logistician — reconcile with their data.
- ✗ Forgetting samples, returns in transit and internal stock.

## 10 Records and evidence

- Field action with consignees, communications, reports
- Authority notifications

## 11 Performance indicators

- Initiation delay (hours) vs target
- Response rate
- Reconciliation rate

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- MDR Articles 87 to 92 — Vigilance
- Regulation (EC) No 1223/2009, Article 23 — Serious undesirable effects

## 13. Demonstration

**Case.** Complaint investigation shows a labelling mix-up on lot 26C044. The decision team decides a Class II recall at pharmacy level. The coordinator opens RCL-2026-002, records the health hazard evaluation, level B 25 %, and initiates. **Trace Distribution and Start** creates 42 consignee lines from deliveries. The recall notice is approved, sent by e-mail and letter within 18 hours; the authority is notified the same day. After 10 days 40 of 42 consignees responded; returned + destroyed = 96 % of shipped quantity.

# Recall effectiveness checks

|                   |                                                                           |                |       |
|-------------------|---------------------------------------------------------------------------|----------------|-------|
| Module            | ls_recall                                                                 | Version        | _____ |
| Menu              | Recalls ► Execution ► Effectiveness Checks · Field Actions ► Close Recall | Effective date | _____ |
| Owner (suggested) | Recall coordinator                                                        | Approved by    | _____ |
| Related SOPs      | SOP-QA-081, SOP-QA-083                                                    |                |       |

## 1 Objective

Verify that consignees received the recall communication and took the requested action, at the level defined in the strategy, and close the recall only when the closure gates are met.

## 2 Scope

All real field actions; mock recalls use the same mechanism (SOP-QA-083).

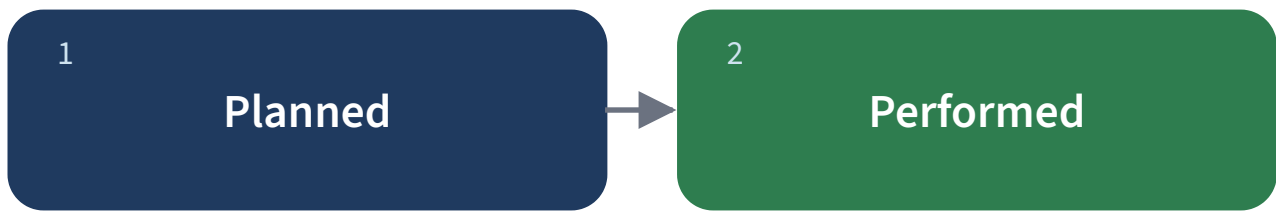
## 3 Responsibilities

| Role (software group) | Responsibility                                                       |
|-----------------------|----------------------------------------------------------------------|
| Recall coordinator    | Plans and performs checks, prepares closure.                         |
| Recall Manager        | Closes; may override failed gates only with a written justification. |

## 4 Definitions

| Term                | Definition                                                                                                    |
|---------------------|---------------------------------------------------------------------------------------------------------------|
| Effectiveness level | A 100 %, B >10 % and <100 %, C 10 %, D 2 %, E no checks (21 CFR 7.42(b)(3)).                                  |
| Outcome             | Contacted – Action Taken, Contacted – No Action Taken, Contacted – Product Not Held, Consignee Not Reachable. |

## 5 Workflow in the software



### Other outcomes



*Model: ls.recall.effectiveness*

Figure 122 — Lifecycle of an effectiveness check.

## 6 Procedure

- 1 **Plan.** Click **Plan Effectiveness Checks** : checks are generated for the number of consignees required by the level.
- 2 **Move to the phase.** Click **Move to Effectiveness Check** .
- 3 **Perform.** Contact each selected consignee (telephone, e-mail, letter, visit, portal); record the outcome and date; click **Record as Performed** .
- 4 **Escalate.** For an unreachable consignee or no action taken, click **Escalate and Retry** (a new attempt is planned).
- 5 **Report.** Prepare the final report: situation summary, actions, product disposition, conclusion; **Review** , **Approve** by another person, **Record Submission** to the authority.
- 6 **Close.** Click **Close Recall** : the wizard shows each closure gate as PASS/FAIL (all consignees notified; no quantity discrepancy; required checks performed; reconciliation rate  $\geq$  plan target for real actions). Enter the closure justification and attestation and confirm.

### 7. Controls enforced by the software

- ✓ Level C/D/E set the number of checks from the consignee count; Level B uses the percentage entered.
- ✓ A performed check needs an outcome and a performed date; only planned checks can be performed; finalised recalls freeze their checks.
- ✓ Closure with failed gates is refused unless the override is ticked with a statement, and only a Recall Manager may do it.
- ✓ Reports: reviewed before approval; approver different from reviewer; approved before submission; the recipient must be recorded; one final report per recall.

## 8. Good practices

- ★ Choose consignees for checks at random, but always include the largest customers.
- ★ Treat "product not held" answers carefully: check the shipment was received.

## 9. Pitfalls to avoid

- ✗ Closing with an override "to meet the deadline".

## 10 Records and evidence

- Effectiveness checks with outcomes
- Final report and submission
- Closure justification and attestation

## 11 Performance indicators

- Effectiveness coverage (%)
- Success rate of checks
- Recalls closed with override (target 0)

## 12 References

- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)
- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall

## 13. Demonstration

**Case (continued).** Level B 25 % of 42 consignees → 11 checks planned (the number is rounded up by the system). 10 answer *Action Taken*; 1 not reachable → escalated, reached at second attempt. Closure gates: 42/42 notified, no discrepancy, 11/11 checks, reconciliation 96 % ≥ 95 %: all PASS. The final report is approved and submitted; the recall is closed.



# Mock recall (periodic exercise)

|                   |                                                         |                |       |
|-------------------|---------------------------------------------------------|----------------|-------|
| Module            | ls_recall                                               | Version        | _____ |
| Menu              | Recalls ► Field Actions ► Initiate Recall (Mock Recall) | Effective date | _____ |
| Owner (suggested) | Head of Quality                                         | Approved by    | _____ |
| Related SOPs      | SOP-QA-080, SOP-QA-081, SOP-QA-082                      |                |       |

## 1 Objective

Test periodically that the recall arrangements work — people reachable, distribution traceable, communication fast — without contacting customers about a real defect.

## 2 Scope

At the interval set in the recall plan (e.g. yearly), for a representative product and lot.

## 3 Responsibilities

| Role (software group)       | Responsibility                                                       |
|-----------------------------|----------------------------------------------------------------------|
| Head of Quality             | Triggers the exercise unannounced to the coordinator where possible. |
| Recall coordinator / deputy | Run the exercise as a real recall, up to the agreed limit.           |
| QA                          | Evaluates and reports the results.                                   |

## 4 Definitions

| Term        | Definition                                                                                                                              |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Mock recall | Action type <i>Mock Recall</i> ; it is excluded from regulatory KPI reporting and its closure does not check the reconciliation target. |

## 5 Procedure

- 1 Prepare.** Choose a distributed lot and success criteria: time to reach the coordinator, time to trace 100 % of distribution, time to draft and approve the notice.
- 2 Trigger.** Out of hours if possible, call the coordinator; start the timer.
- 3 Run.** Open **Initiate Recall** with action type *Mock Recall*; trace distribution; prepare the communication (not sent to real customers unless agreed); perform effectiveness checks on internal or cooperating consignees.
- 4 Close.** Close the mock recall through the closure wizard.
- 5 Evaluate.** Compare times with the plan targets; record gaps; open CAPA for weaknesses; update the plan's last mock recall date.
- 6 Report.** Report the result in management review.

### 6. Controls enforced by the software

- ✓ Mock recalls are identified by their action type and excluded from regulatory KPI reporting; the reconciliation gate does not apply to them.

## 7. Good practices

- ★ Vary the scenario (different product, deputy instead of coordinator, weekend).
- ★ Include the third-party logistician in the exercise.

## 8. Pitfalls to avoid

- ✗ Announced exercises that test nothing.
- ✗ Sending a mock notice to real customers without warning them.

## 9 Records and evidence

- Mock recall record
- Evaluation report
- CAPA

## 10 Performance indicators

- Time to 100 % traceability
- Plan targets met in exercise

## 11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 8 — Complaints and Product Recall
- 21 CFR Part 7, Subpart C — Recalls (7.40 to 7.59)

## 12. Demonstration

**Case.** Saturday 10:00, the Head of Quality calls the deputy: the coordinator is abroad. The deputy opens a mock recall for lot 26A011 at 10:40; distribution traced to 31 consignees by 11:25; notice drafted and approved by 14:10 (target 24 h: met). Gap: the switchboard list still showed the former deputy's number — CAPA opened.

# III

## Industry modules

Procedures specific to each industry: pharmaceutical manufacturing, cosmetics, medical devices, medical plastics and the quality control laboratory.

|                                     |                                        |
|-------------------------------------|----------------------------------------|
| Pharmaceutical — ls_pharma          | Medical plastics — ls_medical_plastics |
| Cosmetics — ls_cosmetics            | Laboratory — ls_lab                    |
| Medical devices — ls_medical_device |                                        |

# Pharmaceutical

ls\_pharma

**ls\_pharma** covers pharmaceutical manufacturing records and their quality decisions: batches (bulk, packaging, finished product) with yield limits and expiry; batch production and control records with master record check, steps, in-process and laboratory controls, area inspections (line clearance), labelling reconciliation, samples and discrepancies; an append-only batch release decision taken by the quality unit; qualified materials (APIs and excipients); stability studies; GS1 serialisation and aggregation; and CTD dossiers. Its controls cite 21 CFR Part 211; deliveries of lots whose batch is not released are refused.

| Menu                                                                                                  | Content       |
|-------------------------------------------------------------------------------------------------------|---------------|
| Pharmaceutical ► Batches ► Manufacturing Batches / Batch Records / Batch Release / Open Discrepancies | Batches       |
| Pharmaceutical ► Materials ► APIs / Excipients / Pharmaceutical Products                              | Materials     |
| Pharmaceutical ► Stability ► Stability Studies / Time Points / Stability Samples                      | Stability     |
| Pharmaceutical ► Serialization ► Serial Numbers / Aggregation / Generate Serial Numbers               | Serialisation |
| Pharmaceutical ► Regulatory ► CTD Dossiers                                                            | Regulatory    |

| Group                                         | Main rights                      |
|-----------------------------------------------|----------------------------------|
| Pharmaceutical / Viewer · Production Operator | Read; execute batch record steps |
| Pharmaceutical / Production Manager           | Manages batches and records      |
| Pharmaceutical / Quality Assurance            | Reviews records, decides release |
| Pharmaceutical / Regulatory Affairs           | CTD dossiers                     |
| Pharmaceutical / Manager                      | Configuration                    |

## EU Qualified Person certification

The batch release decision of ls\_pharma is the quality unit decision described by 21 CFR 211.22 and 211.192. In the EU, certification by a Qualified Person (EU GMP Annex 16) is an additional legal act: state in SOP-QA-090 how and where the QP certifies (register, signature), since the module does not model it.

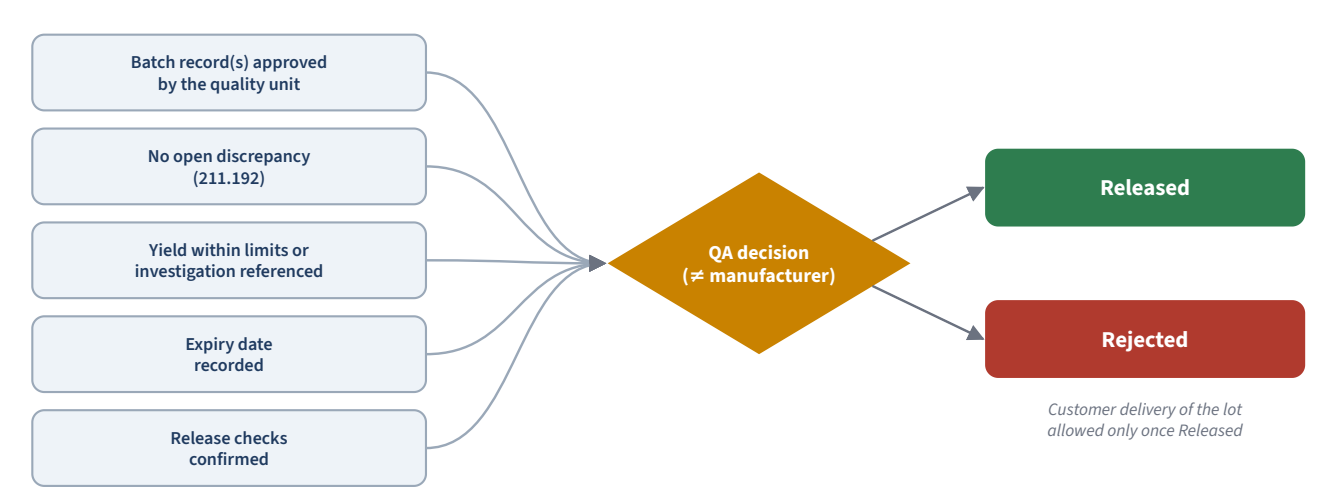
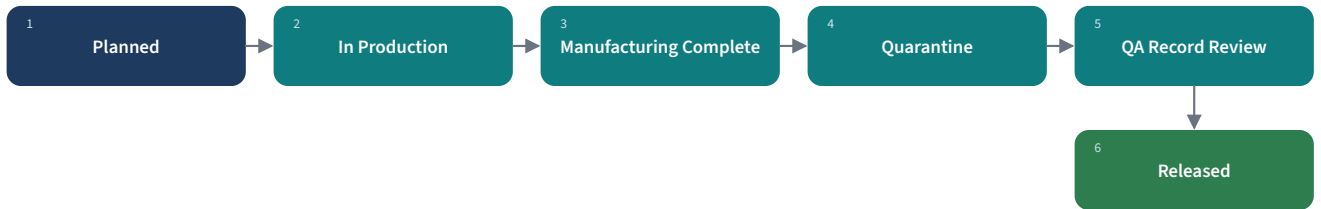


Figure 123 — Release gate of a batch.



Other outcomes



Model: ls.pharma.batch

Figure 124 — Lifecycle of a batch.

# Issue and execution of the batch record

|                   |                                                                  |                |       |
|-------------------|------------------------------------------------------------------|----------------|-------|
| Module            | ls_pharma                                                        | Version        | _____ |
| Menu              | Pharmaceutical ► Batches ► Manufacturing Batches / Batch Records | Effective date | _____ |
| Owner (suggested) | Production Manager                                               | Approved by    | _____ |
| Related SOPs      | SOP-PRD-002, SOP-QA-090, SOP-QA-020                              |                |       |

## 1 Objective

Issue for each batch a production and control record reproduced from the approved master record and checked, and execute it contemporaneously so that every significant step is dated, attributed and, where required, checked by a second person.

## 2 Scope

Manufacturing and packaging/labelling batch records of all pharmaceutical batches (bulk, packaging, finished product).

## 3 Responsibilities

| Role (software group)          | Responsibility                                                                                |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Production Manager</b>      | Creates the batch and issues the record.                                                      |
| <b>Checker (second person)</b> | Checks the reproduction of the master record; checks significant steps and component charges. |
| <b>Production Operators</b>    | Execute steps and record values.                                                              |
| <b>QA</b>                      | Is informed of discrepancies; reviews the record (SOP-QA-090).                                |

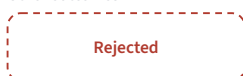
## 4 Definitions

| Term                    | Definition                                                                                            |
|-------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Master record</b>    | Approved master production and control record; referenced by number and version on each batch record. |
| <b>Significant step</b> | Step requiring identification of the person performing it and of the person checking it.              |

## 5 Workflow in the software



Other outcomes



Model: ls.pharma.batch\_record

Figure 125 — Lifecycle of a batch production and control record.

## Procedure

- 1 **Create the batch.** In **Manufacturing Batches** click **New** : reference, batch type, product, lot, manufacturing order, BOM, site, planned quantity, theoretical yield, yield limits, shelf life.
- 2 **Issue the record.** Add a batch record: record type (Manufacturing or Packaging and Labelling), master record reference and version; add steps, controls, planned samples and labelling items as defined in the master record.
- 3 **Check the reproduction.** A second person compares the issued record with the master and records *Master Record Checked By / On*.
- 4 **Start.** Click **Start Production** on the batch and **Start Execution** on the record.
- 5 **Execute steps.** For each step line of the record, perform the step, record the value, the date performed and the performer, then click **Done** on the line; for significant or automated steps, record the checker (*Checked or Supervised By*). A step that does not apply is set to *Not Applicable* with a remark.
- 6 **Components.** Record each component charge (lot, quantity, assay) and have it verified by a second person.
- 7 **Controls.** Record in-process and laboratory control results against their limits.
- 8 **Discrepancies.** Record any discrepancy immediately (classification, description) and click **Start Investigation** ; major events also go through SOP-QA-020.
- 9 **Complete.** Click **Complete Execution** when all steps are done or not applicable; record actual yield and click **Complete Manufacturing** on the batch.
- 10 **Submit.** Click **Submit for Review** on the record; place the batch in quarantine ( **Place in Quarantine** ) pending QA review.

### 7. Controls enforced by the software

- ✓ Execution cannot start until the reproduction of the master record has been checked, dated and attributed.
- ✓ A step cannot be completed without a date and the person who performed it; automated steps need the checker; *not applicable* needs a remark.
- ✓ A component charged and verified by the same user is refused (21 CFR 211.101(d)).
- ✓ The record cannot complete while steps remain open, and cannot be submitted/approved with open discrepancies; closing a discrepancy needs investigation, conclusion and follow-up.
- ✓ Manufacturing cannot be declared complete without actual yield; a batch without batch record cannot proceed.
- ✓ Expiry after manufacturing date; production end after start; minimum yield  $\leq$  maximum; non-draft batches and records cannot be deleted.

### 8. Good practices

- ★ Record at the time of the operation; never pre-fill or back-date.
- ★ Keep the printed master record page references in the step instructions.
- ★ Record discrepancies even when "corrected immediately".

### 9. Pitfalls to avoid

- ✗ The same person charging and verifying components to save time.
- ✗ Steps marked done at the end of the shift.

## 10 Records and evidence

- Batch and batch production and control record with steps, controls, samples, discrepancies

## 11 Performance indicators

- Right-first-time records (no discrepancy)
- Records submitted within 24 h of completion

## 12 References

- 21 CFR 211.186 and 211.188 — Master and batch production and control records
- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 4 — Documentation
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

### 13. Demonstration

**Case.** Batch 26T018 of *Tablets 10 mg*: manufacturing record issued from master MR-TAB10 v7, checked by the shift supervisor. The operator performs step 5 *Dry granulation, 12 min* (value 12 min, done 08:42); the component *API lot A2210, 5.012 kg* is charged by the operator and verified by a second operator. An IPC LOD result is within limits. At the end, actual yield 98.6 % (limits 96–101 %); record submitted for review and batch placed in quarantine.



# Line clearance and label reconciliation

|                   |                                                                                |                |       |
|-------------------|--------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_pharma                                                                      | Version        | _____ |
| Menu              | Pharmaceutical ► Batches ► Batch Records (Area Inspections, Labelling Control) | Effective date | _____ |
| Owner (suggested) | Packaging Manager                                                              | Approved by    | _____ |
| Related SOPs      | SOP-PRD-001, SOP-QA-090                                                        |                |       |

## 1 Objective

Prevent mix-ups by verifying that lines and areas are free of previous materials before and after use, and reconcile the quantities of printed labelling materials issued, used, returned and destroyed.

## 2 Scope

All packaging and labelling operations, and manufacturing areas at product changeover.

## 3 Responsibilities

| Role (software group)       | Responsibility                                 |
|-----------------------------|------------------------------------------------|
| <b>Operator</b>             | Performs the area inspection and label counts. |
| <b>Verifier</b>             | Verifies the inspection (not the same person). |
| <b>Packaging supervisor</b> | Investigates reconciliation differences.       |

## 4 Definitions

| Term                                    | Definition                                                                                                  |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Area inspection (line clearance)</b> | Documented check, before or after use, that no material, document or label of the previous product remains. |
| <b>Reconciliation</b>                   | Issued – used – returned – destroyed; the difference must be within the tolerance.                          |

## 5 Procedure

- Before use.** Add an area inspection: moment *Before Use*, area or line, previous product, performed on/by; record *Pass* or *Fail* with findings; a second person records *Verified By*.
- Fail.** If the inspection fails, describe the findings, clean again, repeat the inspection; do not start.
- Issue labelling.** Add each labelling item (label, leaflet, carton) with its version and code, and the quantity issued.
- Reconcile.** At the end, record used, returned and destroyed; the difference is computed; tick *Reconciled* when within tolerance and record who reconciled.
- Out of tolerance.** Record a discrepancy on the batch record and investigate (missing labels may be on another product).
- After use.** Perform and verify an inspection *After Use*.

## 6. Controls enforced by the software

- ✓ A failed area inspection requires recorded findings.
- ✓ The inspection cannot be verified by the person who performed it.
- ✓ Labelling quantities cannot be negative; unreconciled labels are counted on the record and shown to QA at review.

## 7. Good practices

- ★ Use a checklist of places to inspect (under conveyors, reject bins, printers).
- ★ Destroy excess coded labels and record their destruction.

## 8. Pitfalls to avoid

- ✗ Signing the clearance of a line you did not inspect.
- ✗ Rounding reconciliation differences to zero.

## 9 Records and evidence

- Area inspections with verification
- Labelling reconciliation

## 10 Performance indicators

- Failed clearances
- Batches with unreconciled labels at review

## 11 References

- 21 CFR 211.122 to 211.137 — Packaging and labelling control
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.186 and 211.188 — Master and batch production and control records

## 12. Demonstration

**Case.** Before packaging batch 26T018, the operator inspects line PK-2 (previous product *Capsules 20 mg*): one leaflet of the previous product found under the feeder → *Fail*, findings recorded, line re-cleaned, second inspection *Pass*, verified by the supervisor. Labels: issued 12 500, used 12 180, returned 300, destroyed 20 → difference 0, reconciled.

# Batch record review and batch release by the quality unit

|                   |                                                              |                |       |
|-------------------|--------------------------------------------------------------|----------------|-------|
| Module            | ls_pharma                                                    | Version        | _____ |
| Menu              | Pharmaceutical ► Batches ► Batch Records / Batch Release     | Effective date | _____ |
| Owner (suggested) | Head of Quality Assurance                                    | Approved by    | _____ |
| Related SOPs      | SOP-PRD-001, SOP-PRD-002, SOP-QC-012, SOP-QA-021, SOP-DI-002 |                |       |

## 1 Objective

Release a batch for distribution only after an independent quality unit has reviewed and approved its production and control records, confirmed that every discrepancy is investigated and closed, and verified the release checklist.

## 2 Scope

All pharmaceutical batches made or packaged on site, before distribution.

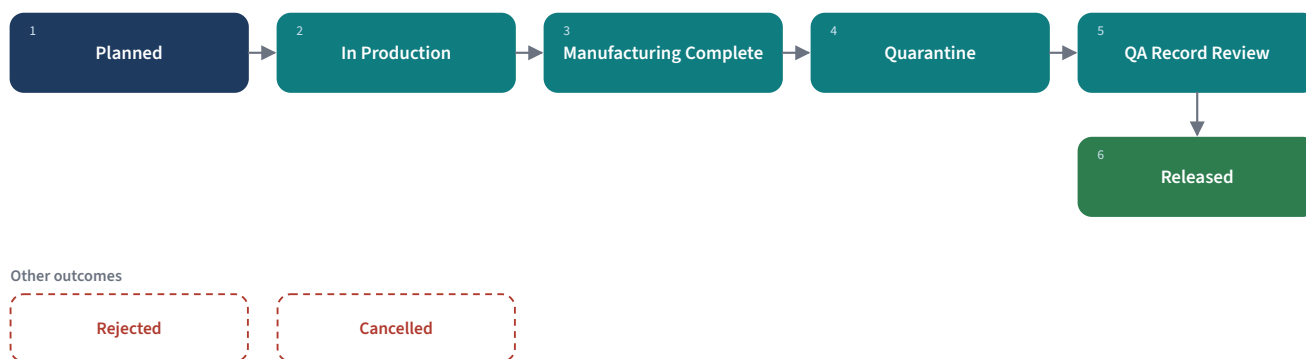
## 3 Responsibilities

| Role (software group)                       | Responsibility                                                                   |
|---------------------------------------------|----------------------------------------------------------------------------------|
| <b>QA reviewer</b>                          | Reviews and approves or rejects each batch record.                               |
| <b>Release authority (QA; QP in the EU)</b> | Takes and records the release decision; cannot be the manufacturer of the batch. |
| <b>Production</b>                           | Answers review questions and completes records returned for completion.          |

## 4 Definitions

| Term                     | Definition                                                                                                                                                                                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Release checklist</b> | Batch records reviewed and approved; discrepancies investigated and closed; yield within limits; component charge-in verified by a second person; laboratory results conform; labelling reconciled; reserve samples retained; covered by the stability programme. |
| <b>Release decision</b>  | Append-only record: Released for Distribution or Rejected, with statement, date, decider and integrity digest.                                                                                                                                                    |

## 5 Workflow in the software



Model: ls.pharma.batch

Figure 126 — Batch lifecycle up to release.

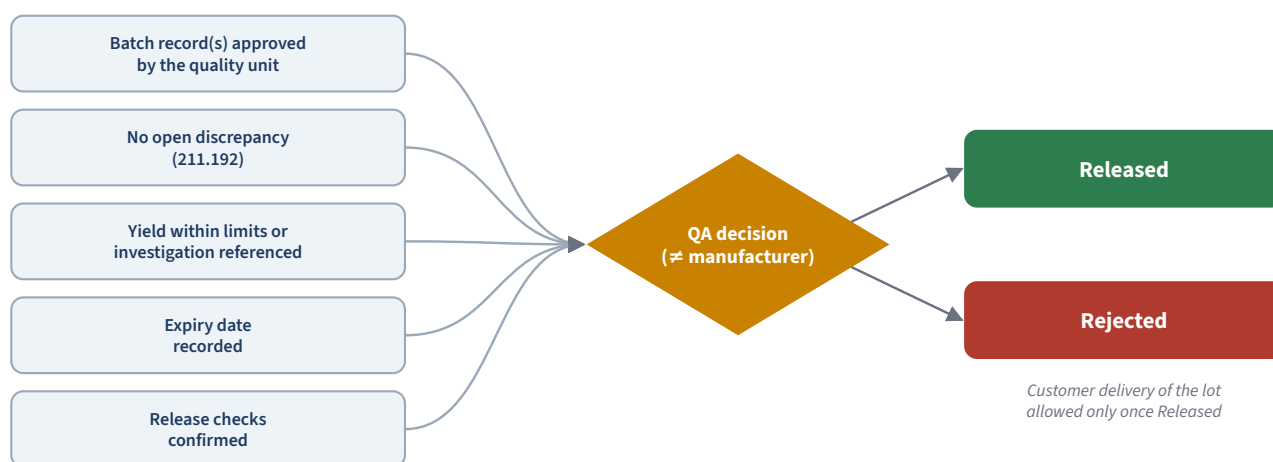


Figure 127 — What the release decision checks.

## 6 Procedure

- 1 **Review the record.** Open the submitted record: steps, controls (non-conforming controls counter), area inspections, labelling reconciliation (unreconciled counter), samples, discrepancies (open counter), audit trail extract where available (SOP-DI-002).
- 2 **Decide on the record.** Click **Approve** with the review conclusion, **Reject** with a written conclusion, or **Return for Completion** if entries are missing.
- 3 **Check laboratory results.** Confirm that the QC results of the batch are approved (SOP-QC-012) and that no OOS investigation is open.
- 4 **Submit the batch.** Click **Submit for QA Review** on the batch.
- 5 **Decide.** Click **Record Release Decision**: decision, statement, and tick each checklist item after verifying it. The decision is recorded as an append-only release decision with an integrity digest.
- 6 **EU QP certification.** Where applicable, the QP certifies the batch in the QP register according to Annex 16, referencing the release decision.
- 7 **Distribution.** Once released, deliveries of the lot are allowed; a rejected batch is handled as rejected material.
- 8 **Verify integrity.** Use **Verify Integrity** on a release decision during audits.

## 7. Controls enforced by the software

- ✓ A batch record cannot be reviewed by the user who executed it; rejection needs a written conclusion; approval is refused with open discrepancies; closed records are frozen.
- ✓ A release decision can only be recorded while the batch is under QA review, once per batch, and not by the person who manufactured it.
- ✓ Release is refused when checklist items are not confirmed, when a batch record is not approved, when yield is outside limits without an investigation reference, or without an expiry date.
- ✓ Release decisions are append-only: cannot be modified, duplicated or deleted.
- ✓ Deliveries of lots whose batch is not released are refused (stock move line control).

## 8. Good practices

- ★ Review by exception using the counters (open discrepancies, non-conforming controls, unreconciled labels) but read the whole record for critical steps.
- ★ Record the reasoning in the decision statement, especially for released batches with closed deviations.

## 9. Pitfalls to avoid

- ✗ Ticking the checklist without evidence.
- ✗ Releasing before the laboratory results are approved.

## 10 Records and evidence

- Approved batch records
- Release decision with checklist and digest
- QP register (EU)

## 11 Performance indicators

- Release lead time
- Batch records returned for completion
- Batches rejected

## 12 References

- 21 CFR 211.22 — Responsibilities of the quality control unit
- 21 CFR 211.192 — Production record review
- EU GMP Annex 16 — Certification by a Qualified Person and Batch Release
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

## 13. Demonstration

**Case.** QA reviews batch 26T018: two records, 0 open discrepancies, 0 non-conforming controls, labels reconciled; yield 98.6 % within 96–101 %; QC results approved. The QA officer (not involved in manufacturing) records *Released for Distribution* with all checklist items ticked. The QP certifies it in the register the same day; the lot can be shipped.

# Qualification of active substances and excipients

|                   |                                                |                |       |
|-------------------|------------------------------------------------|----------------|-------|
| Module            | ls_pharma                                      | Version        | _____ |
| Menu              | Pharmaceutical ► Materials ► APIs / Excipients | Effective date | _____ |
| Owner (suggested) | QC Manager with QA                             | Approved by    | _____ |
| Related SOPs      | SOP-QA-070, SOP-QC-010                         |                |       |

## 1 Objective

Maintain master data of active substances and excipients that are qualified before use: specification basis, approved manufacturers, retest period, storage, and specific requirements (animal origin, controlled substances).

## 2 Scope

All APIs and excipients used in manufacturing.

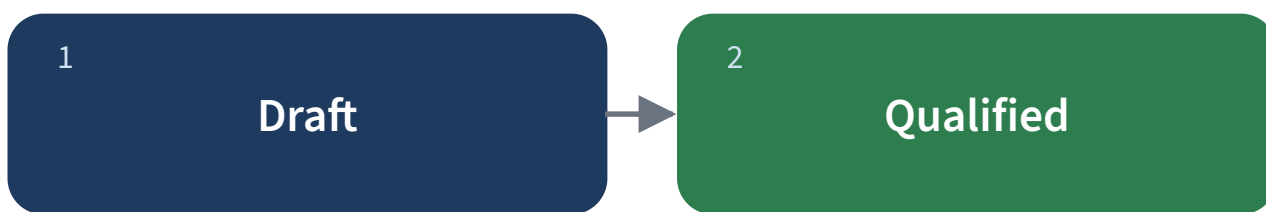
## 3 Responsibilities

| Role (software group) | Responsibility                                                |
|-----------------------|---------------------------------------------------------------|
| QC / QA               | Qualify, restrict or make obsolete materials.                 |
| Purchasing            | Buys only from approved manufacturers (see SOP-QA-070 / 073). |

## 4 Definitions

| Term              | Definition                                                                    |
|-------------------|-------------------------------------------------------------------------------|
| Potency basis     | As Is, Anhydrous, Dried or Active Moiety basis; label assay must be positive. |
| TSE/BSE statement | Required reference for materials of animal origin.                            |

## 5 Workflow in the software



### Other outcomes



*Model: ls.pharma.api*

Figure 128 — States of a pharmaceutical material.

## 6 Procedure

- 1 **Create.** In **Materials** ► **APIs** or **Excipients** click **New**: name, internal code, inventory item, pharmacopoeia and monograph, CAS number, approved manufacturers, retest period, storage condition; for APIs: INN, active moiety, potency basis, label assay, controlled substance and schedule, sterile; for excipients: function, novel excipient, maximum daily intake, label declaration.
- 2 **Animal origin.** Tick *Of Animal Origin* and record the TSE/BSE statement reference.
- 3 **Qualify.** When the supplier is approved and the specification is in place, click **Qualify**.
- 4 **Restrict.** Click **Restrict** to limit use (e.g. pending an investigation); **Declare Obsolete** when no longer used.

### 7. Controls enforced by the software

- ✓ A controlled substance cannot be qualified until its schedule is recorded; a material of animal origin cannot be qualified without its TSE/BSE statement reference.
- ✓ Label assay > 0; retest period and maximum daily intake non-negative.
- ✓ An obsolete material cannot return to draft; a qualified material cannot be qualified twice.

### 8. Good practices

- ★ Keep the approved manufacturer list identical to the supplier qualification scope.

### 9. Pitfalls to avoid

- ✗ Using a new manufacturer's material under an old qualification.

## 10 Records and evidence

- Material master data with state history

## 11 Performance indicators

- Materials used while not qualified (target 0)

## 12 References

- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production
- 21 CFR 211.84 — Testing and approval or rejection of components

### 13. Demonstration

**Case.** Gelatin (excipient, function *Coating Agent*) is of bovine origin: the TSE statement reference *EDQM CEP R1-CEP 2000-xxx* is recorded before **Qualify** is accepted.



# Stability programme

|                   |                            |                |       |
|-------------------|----------------------------|----------------|-------|
| Module            | ls_pharma                  | Version        | _____ |
| Menu              | Pharmaceutical ► Stability | Effective date | _____ |
| Owner (suggested) | QC Manager                 | Approved by    | _____ |
| Related SOPs      | SOP-QC-014, SOP-QA-090     |                |       |

## 1 Objective

Run registration, ongoing, follow-up and post-change stability studies according to approved protocols, pull and test samples on time, and use the results to support shelf life and storage conditions.

## 2 Scope

Stability studies of finished products and intermediates managed in ls\_pharma. Laboratory-side sample testing may be managed in ls\_lab (SOP-QC-014).

## 3 Responsibilities

| Role (software group)        | Responsibility                                       |
|------------------------------|------------------------------------------------------|
| <b>Stability coordinator</b> | Creates studies, generates schedules, pulls samples. |
| <b>Analysts</b>              | Test and record results.                             |
| <b>QC Manager</b>            | Concludes studies and proposes shelf life.           |

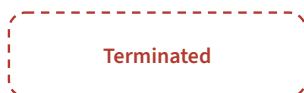
## 4 Definitions

| Term                     | Definition                                                                           |
|--------------------------|--------------------------------------------------------------------------------------|
| <b>Storage condition</b> | Long term, intermediate, accelerated or other (temperature, humidity and tolerance). |
| <b>Time point</b>        | Month of a study at a condition; scheduled date derived from study start.            |

## 5 Workflow in the software



Other outcomes



Model: ls.pharma.stability\_study

Figure 129 — Lifecycle of a stability study.

## 6 Procedure

- 1 Create.** In **Stability Studies** click **New**: title, study type, product, batch, protocol reference, packaging configuration, start, planned duration, storage conditions.

- 2 **Schedule.** Click **Generate Schedule** to create time points per condition; review scheduled dates.
- 3 **Start.** Click **Start Study**; place samples in storage (stability samples).
- 4 **Pull and test.** At each time point click **Pull Samples**, test, record results, then **Mark Tested** and **Complete**. A time point not pulled is recorded with **Mark Missed** and a remark.
- 5 **Out of specification.** Handle non-conforming results through an OOS investigation (SOP-QC-013); significant change is recorded on the study.
- 6 **Conclude.** When all time points are final, write the conclusion (proposed shelf life, storage) and click **Complete Study**; **Terminate Study** with justification if stopped early.

## 7. Controls enforced by the software

- ✓ Time points can only be generated in draft or scheduled studies, and require at least one storage condition; a study needs time points to start.
- ✓ A time point cannot be tested before its samples are pulled; tested requires results; missed requires a remark.
- ✓ Completion requires every time point in a final state and a conclusion (21 CFR 211.166); termination requires a justification.
- ✓ Samples cannot be allocated to another study's time point or stored under a different condition; discarding requires a remark.

## 8. Good practices

- ★ Enrol at least one batch per year per product and strength in the ongoing programme.
- ★ Watch overdue time points weekly.

## 9. Pitfalls to avoid

- ✗ Pulling samples late without recording the actual date.

## 10 Records and evidence

- Study, time points, samples, results, conclusion

## 11 Performance indicators

- Time points pulled on time
- Studies with significant change

## 12 References

- ICH Q1A(R2) — Stability Testing of New Drug Substances and Products
- 21 CFR 211.166 — Stability testing
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control

## 13. Demonstration

**Case.** Ongoing stability of batch 26T018: long term 25 °C/60 % RH and accelerated 40 °C/75 % RH, planned 36 months; time points generated at 0, 3, 6, 9, 12, 18, 24, 36 months. At 6 months accelerated, dissolution is out of specification → OOS investigation opened; the study continues.

# GS1 serialisation and aggregation

|                   |                                |                |       |
|-------------------|--------------------------------|----------------|-------|
| Module            | ls_pharma                      | Version        | _____ |
| Menu              | Pharmaceutical ► Serialization | Effective date | _____ |
| Owner (suggested) | Packaging Manager with IT      | Approved by    | _____ |
| Related SOPs      | SOP-PRD-001, SOP-QA-090        |                |       |

## 1 Objective

Assign a unique identifier to each saleable unit, commission it only for released packaging, aggregate units into containers with SSCC codes, and keep the status of each identifier accurate until it leaves the company.

## 2 Scope

Products flagged as requiring serialisation. Connection to national verification systems (e.g. EU hub) is outside the module and must be described separately.

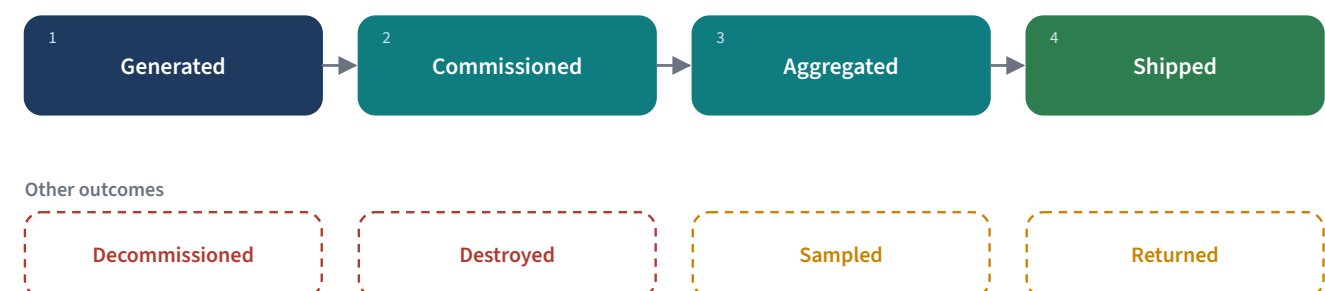
## 3 Responsibilities

| Role (software group)             | Responsibility                             |
|-----------------------------------|--------------------------------------------|
| <b>Packaging operator</b>         | Generates, commissions and aggregates.     |
| <b>Packaging Manager</b>          | Decommissions with reason; supervises.     |
| <b>IT / serialisation partner</b> | Exchanges data with the repository system. |

## 4 Definitions

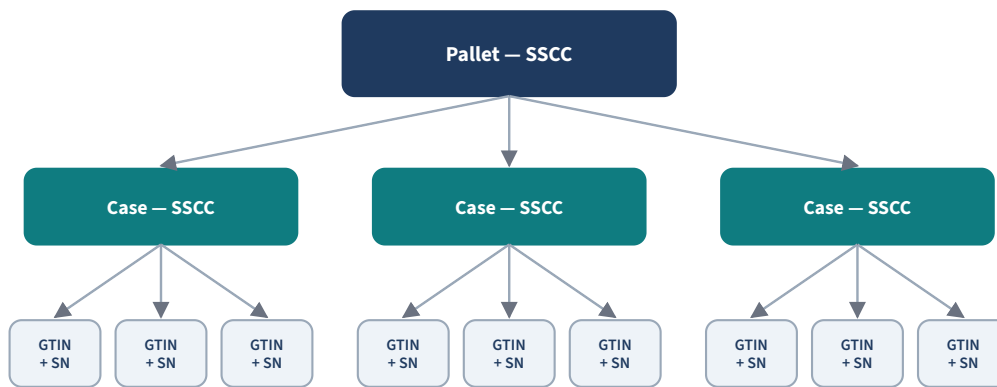
| Term                                 | Definition                                                                                                                                         |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unique identifier</b>             | GTIN-14 product code + serial number + batch + expiry (+ national reimbursement number when required).                                             |
| <b>SSCC</b>                          | 18-digit Serial Shipping Container Code with modulo-10 check digit; built from the company GS1 prefix and extension digit set in company settings. |
| <b>Commissioned / Decommissioned</b> | Active in the supply chain / withdrawn (with reason); other terminal states: destroyed, sampled, returned.                                         |

## 5 Workflow in the software



Model: ls.pharma.serialization

Figure 130 — States of a serialised unit.



Saleable unit: GTIN-14 + serial number (Commissioned before shipping). Container: 18-digit SSCC with modulo-10 check digit.

Figure 131 — Serialisation and aggregation hierarchy.

## 6 Procedure

- 1 **Configure.** Set the GS1 company prefix, SSCC extension digit and serial length in [Configuration ► Company Settings](#); enter the GTIN-14 on each serialised product.
- 2 **Generate.** Use [Serialization ► Generate Serial Numbers](#) for the batch.
- 3 **Commission.** Click [Commission](#) on units actually printed and verified on the line.
- 4 **Aggregate.** Create containers (bundle, case, pallet) with their SSCC; add units or child containers; click [Pack](#).
- 5 **Ship.** Click [Mark Shipped](#) on containers and units shipped.
- 6 **Correct.** Use [Disaggregate](#) on a packed (not shipped) container to rework; [Decommission](#) units damaged, sampled or destroyed with a reason.

### 7. Controls enforced by the software

- ✓ GTIN-14 and SSCC formats and check digits are validated; serialised products must carry a product code.
- ✓ Commissioned units cannot be deleted (decommission instead) and their identifier can no longer change; terminal states are final.
- ✓ Only generated units can be commissioned; units must be commissioned before shipping; containers cannot be packed empty or with uncommissioned units, cannot contain themselves, and cannot be disaggregated once shipped.

### 8. Good practices

- ★ Reconcile generated, commissioned and decommissioned counts per batch at batch record review.
- ★ Keep the rework of aggregated cases under supervision.

### 9. Pitfalls to avoid

- ✗ Commissioning the full range generated instead of the units actually printed.

## 10 Records and evidence

- Serial numbers with status history
- Aggregation hierarchy

## 11 Performance indicators

- Decommissioned units per batch
- Aggregation errors

## 12 References

- Commission Delegated Regulation (EU) 2016/161 — Safety features on medicinal product packaging
- GS1 General Specifications (GTIN, SSCC, check digit)

### 13. Demonstration

**Case.** Batch 26T018 of 12 000 boxes: 12 100 serials generated, 12 000 commissioned; 20 boxes per case (600 cases) and 40 cases per pallet (15 pallets), each with an SSCC. Two damaged boxes are decommissioned with the reason *damaged on line*; their case is disaggregated and repacked.

# Preparation of the CTD dossier (Common Technical Document)

|                   |                                            |                |       |
|-------------------|--------------------------------------------|----------------|-------|
| Module            | ls_pharma                                  | Version        | _____ |
| Menu              | Pharmaceutical ► Regulatory ► CTD Dossiers | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                 | Approved by    | _____ |
| Related SOPs      | SOP-QA-030, SOP-RA-010                     |                |       |

## 1 Objective

Plan and track the preparation of marketing authorisation dossiers section by section, record submission, deficiencies and decision, and keep evidence of each section's source document.

## 2 Scope

New applications, variations, renewals and transfers prepared in CTD format. The module tracks status and references; it does not compile or publish eCTD sequences.

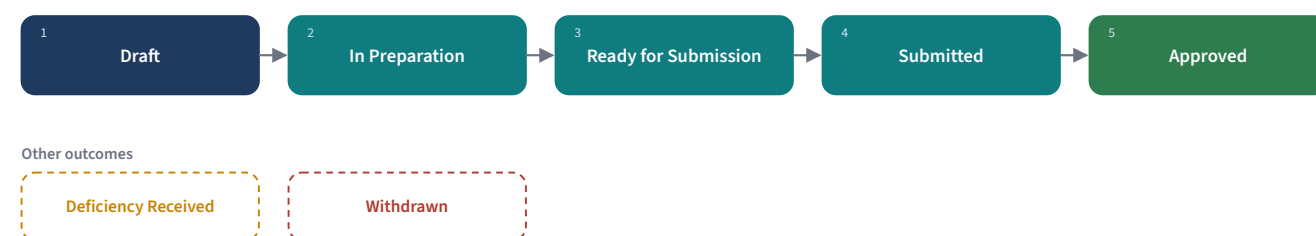
## 3 Responsibilities

| Role (software group)             | Responsibility                                    |
|-----------------------------------|---------------------------------------------------|
| <b>Regulatory Affairs Manager</b> | Owens the dossier and its planning.               |
| <b>Section owners</b>             | Prepare sections and provide document references. |
| <b>Management</b>                 | Approves submission.                              |

## 4 Definitions

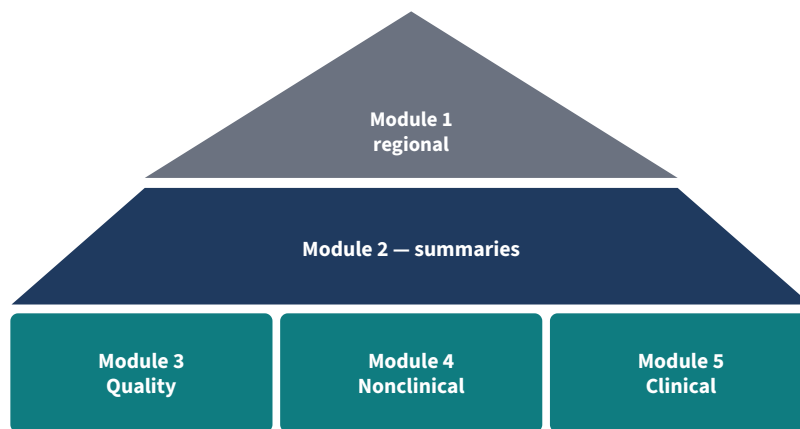
| Term              | Definition                                                                          |
|-------------------|-------------------------------------------------------------------------------------|
| <b>Section</b>    | CTD module (1 to 5), number and title, owner, due date, document reference, status. |
| <b>Deficiency</b> | Question or objection received from the authority on a submitted dossier.           |

## 5 Workflow in the software



Model: ls.pharma.ctd\_dossier

Figure 132 — Lifecycle of a CTD dossier.



ICH M4 structure; ls\_pharma tracks the status of each section (Load CTD Structure).

Figure 133 — CTD structure.

## 6 Procedure

- 1 **Create.** In **CTD Dossiers** click **New** : title, product, dossier type, authority, contact, MA holder, version.
- 2 **Load the structure.** Click **Start Preparation** then **Load CTD Structure** to copy the template sections (configured in **Configuration ► CTD Section Template**).
- 3 **Prepare sections.** Assign owners and due dates; each owner clicks **Start Preparation** on the section, then **Mark Ready** with the document reference.
- 4 **Ready.** When all sections are ready click **Mark Ready** on the dossier.
- 5 **Submit.** After internal approval and submission through the authority's channel, click **Record Submission** with the date.
- 6 **Deficiencies.** On receipt of questions click **Record Deficiency** ; set the affected sections to deficiency with a description; answer and mark them ready again.
- 7 **Decision.** Click **Record Approval** with the authorisation number and decision date, or **Withdraw** .

### 7. Controls enforced by the software

- ✓ A section cannot be declared ready without a document reference; a section in deficiency must describe it; template sections cannot belong to a dossier.
- ✓ The template can only be loaded in preparation; the dossier can only be marked ready when all sections are ready; submission requires *ready*.
- ✓ A deficiency can only be recorded on a submitted dossier; approval requires the authorisation number; decision date  $\geq$  submission date; submitted dossiers cannot be deleted.

### 8. Good practices

- ★ Link each variation dossier to its change request (SOP-QA-030).
- ★ Use the document reference of the controlled document (ls\_document\_management number and version).

### 9. Pitfalls to avoid

- ✗ Sections marked ready with drafts as document reference.

## 10 Records and evidence

- Dossier with sections, submission, deficiencies and decision

## 11 Performance indicators

- Sections late vs due date
- Deficiency cycles per dossier

## 12 References

- ICH M4 — Organisation of the Common Technical Document

### 13. Demonstration

**Case.** Variation for a new API manufacturer (CC-2026-021 follow-up): dossier type *Variation*; only sections 1.x, 2.3, 3.2.S.2 and 3.2.S.4 apply; owners assigned; all ready in 6 weeks; submission recorded; one deficiency on 3.2.S.4 answered; approval recorded with the authority's decision number.



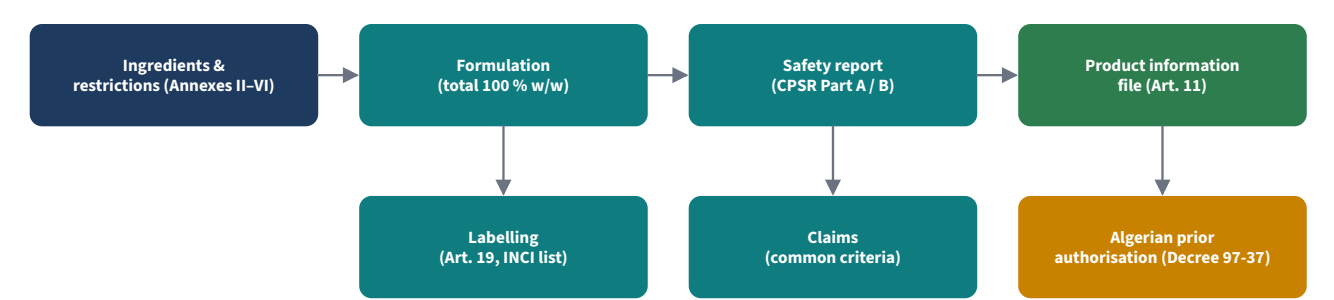
# Cosmetics

## ls\_cosmetics

**ls\_cosmetics** follows the structure of Regulation (EC) No 1223/2009: an ingredient register (INCI, CAS/EC, regulatory category, nanomaterial, CMR, perfume/aroma), the register of Annex II–VI restrictions, versioned formulations totalling 100 % w/w checked against the annexes, the cosmetic product safety report (CPSR, Part A and Part B, approved by the named safety assessor), the product information file (PIF, Article 11) with its ten-year retention, claims assessed against the six common criteria, labelling particulars of Article 19 with an INCI list generated from the formulation, and the Algerian prior authorisation dossier.

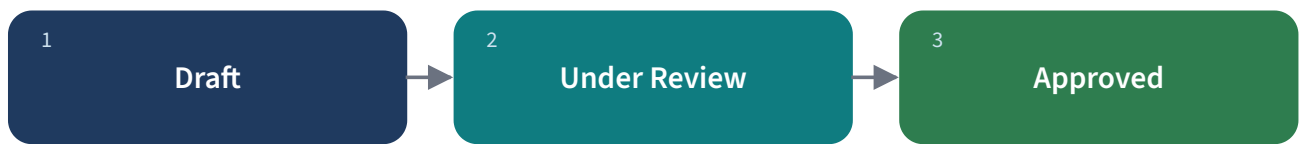
| Menu                                                                                                            | Content                      |
|-----------------------------------------------------------------------------------------------------------------|------------------------------|
| Cosmetics ► Ingredients ► Ingredient Register / Annex Restrictions                                              | Ingredients and restrictions |
| Cosmetics ► Products ► Product Formulations / Finished Products                                                 | Formulations                 |
| Cosmetics ► Safety ► Safety Assessments                                                                         | CPSR                         |
| Cosmetics ► Regulatory ► Product Information Files / Product Claims / Labelling / Algerian Prior Authorisations | Regulatory                   |
| Cosmetics ► Reports ► Retention Status / Safety Reports Due for Review                                          | Monitoring                   |

| Group                         | Main rights                                            |
|-------------------------------|--------------------------------------------------------|
| Cosmetics: Read Only          | Read                                                   |
| Cosmetics: Formulator         | Formulations, labels, claims drafts                    |
| Cosmetics: Safety Assessor    | Safety reports (Part B approval by the named assessor) |
| Cosmetics: Regulatory Manager | Approvals, PIF, restrictions, Algerian dossiers        |



*Each step is approved by a user other than the one who prepared it (four eyes).*

Figure 134 — From ingredients to the PIF and the authorisations.



Other outcomes



*Model: ls.cosmetic.formulation*

*Figure 135 — Lifecycle of a formulation.*

# Development and approval of formulations

|                   |                                                   |                |       |
|-------------------|---------------------------------------------------|----------------|-------|
| Module            | ls_cosmetics                                      | Version        | _____ |
| Menu              | Cosmetics ► Products ► Product Formulations       | Effective date | _____ |
| Owner (suggested) | Formulation Manager with the Regulatory Manager   | Approved by    | _____ |
| Related SOPs      | SOP-COS-002, SOP-COS-005, SOP-COS-006, SOP-QA-030 |                |       |

## 1 Objective

Record each cosmetic formulation as a controlled, versioned composition in % w/w, check it against the annex restrictions, and approve it by a person other than the one who submitted it.

## 2 Scope

All cosmetic formulations placed on the market or used for safety assessment.

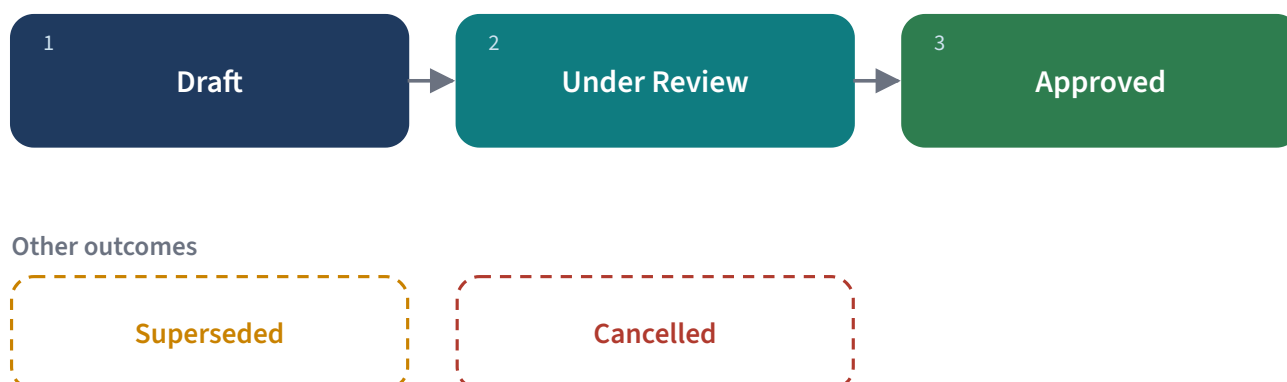
## 3 Responsibilities

| Role (software group)     | Responsibility                                           |
|---------------------------|----------------------------------------------------------|
| <b>Formulator</b>         | Creates the formulation and its composition, submits it. |
| <b>Regulatory Manager</b> | Reviews annex findings and approves (not the submitter). |

## 4 Definitions

| Term                          | Definition                                                                                                          |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Composition</b>            | Lines of ingredients with concentration (% w/w, six decimals); total must equal 100 % within a tolerance of 0.05 %. |
| <b>Blocking annex finding</b> | Presence of an Annex II substance, or concentration above an Annex III–VI limit; approval is refused.               |

## 5 Workflow in the software



Model: ls.cosmetic.formulation

Figure 136 — Lifecycle of a formulation.

## 6 Procedure

- 1 **Check ingredients.** Make sure every ingredient exists in **Ingredient Register** with INCI name, category and annex entries (SOP-COS-006).
- 2 **Create.** In **Product Formulations** click **New** : code, name, version, finished product, BOM, normal and reasonably foreseeable use, targeted population, flags (children under three, external intimate hygiene, nanomaterials, CMR).
- 3 **Compose.** Add the lines with concentrations; check *Total (% w/w) = 100* and the counters *Blocking Annex Findings* and *Lines Without Annex Data*.
- 4 **Submit.** Click **Submit for Review** .
- 5 **Approve.** The Regulatory Manager (not the submitter) clicks **Approve** , or **Return to Draft** with comments.
- 6 **Change.** Any change of an approved formulation is a new version: **Create New Version** (under change control, SOP-QA-030); the previous version becomes superseded.

### 7. Controls enforced by the software

- ✓ A formulation must have a composition to leave draft, and must total 100 % w/w within 0.05 % when submitted.
- ✓ An ingredient linked to an Annex II entry cannot be present in a formulation under review or approved (Article 14(1)(a)); blocking annex findings prevent approval.
- ✓ The submitter cannot approve; approved formulations are frozen; only approved formulations can be revised; non-draft formulations cannot be deleted (retention under Article 11(1)).
- ✓ A concentration rounding to zero at six decimals is refused.

### 8. Good practices

- ★ Record raw material trade names and their INCI breakdown so the label list is exact.
- ★ Flag early the special populations: they require specific assessment in the CPSR.

### 9. Pitfalls to avoid

- ✗ Editing the formulation in a spreadsheet and copying only the final list.

## 10 Records and evidence

- Formulation versions with approval

## 11 Performance indicators

- Formulations returned to draft
- Lines without annex data at submission

## 12 References

- Regulation (EC) No 1223/2009 on cosmetic products
- Regulation (EC) No 1223/2009, Article 14 and Annexes II to VI — Restrictions
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

### 13. Demonstration

**Case.** Face cream *FC-120 v1*: 18 ingredients totalling 100.000 %; phenoxyethanol 0.9 % (Annex V limit 1 %) → within limit; no Annex II substance. Submitted by the formulator, approved by the Regulatory Manager.

# Safety assessment (CPSR, Cosmetic Product Safety Report)

|                   |                                         |                |       |
|-------------------|-----------------------------------------|----------------|-------|
| Module            | ls_cosmetics                            | Version        | _____ |
| Menu              | Cosmetics ► Safety ► Safety Assessments | Effective date | _____ |
| Owner (suggested) | Safety Assessor                         | Approved by    | _____ |
| Related SOPs      | SOP-COS-001, SOP-COS-003                |                |       |

## 1 Objective

Produce for each product a cosmetic product safety report whose Part A information is complete and whose Part B conclusion is written and approved by the qualified safety assessor named in the report.

## 2 Scope

All cosmetic products before placing on the market, and each new formulation version.

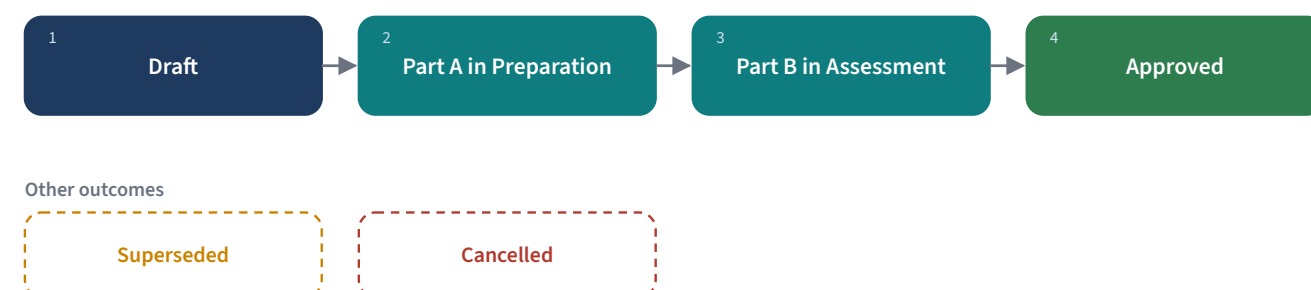
## 3 Responsibilities

| Role (software group)                | Responsibility                                                           |
|--------------------------------------|--------------------------------------------------------------------------|
| <b>Formulator / Regulatory</b>       | Compile Part A data.                                                     |
| <b>Safety Assessor (named in B4)</b> | Performs the assessment and approves Part B with their own user account. |

## 4 Definitions

| Term          | Definition                                                                                                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Part A</b> | A1 composition, A2 physico-chemical characteristics and stability, A3 microbiological quality, A4 impurities/packaging, A5 use, A6 exposure to product, A7 exposure to substances, A8 toxicological profiles, A9 undesirable effects, A10 other information. |
| <b>Part B</b> | B1 conclusion, B2 warnings and instructions, B3 reasoning (including specific assessments), B4 assessor identity, address and qualification.                                                                                                                 |

## 5 Workflow in the software



Model: ls.cosmetic.safety\_assessment

Figure 137 — Lifecycle of a safety report.

## 6 Procedure

**Create.** In **Safety Assessments** click **New** : reference, title, version, formulation (approved).

- 2 **Part A.** Click **Start Part A** and complete sections A1 to A10; record the lowest margin of safety and its basis.
- 3 **Part B.** Click **Start Part B** once all Part A sections are filled; the assessor completes B1–B4, including the specific assessments for children under three or intimate hygiene products when the formulation is flagged.
- 4 **Approve.** The named assessor, logged in with their own account, clicks **Approve as Safety Assessor**.
- 5 **Review.** Monitor **Reports ▶ Safety Reports Due for Review**; update the report when new data, complaints or formulation changes arise.

### 7. Controls enforced by the software

- ✓ Part B requires an approved formulation and all Part A sections filled.
- ✓ Specific assessments are required when the formulation is intended for children under three or exclusively for external intimate hygiene.
- ✓ Approval requires every mandatory section, an assessor user account, and must be performed by that assessor.
- ✓ Approved reports are frozen (issue a new version); non-draft reports cannot be deleted.

### 8. Good practices

- ★ Attach the assessor's qualification evidence (diploma, CV) to the report.
- ★ Use recognised toxicological references and document the source of each profile.

### 9. Pitfalls to avoid

- ✗ Approving on behalf of an external assessor — the approval must come from the assessor's own account.

## 10 Records and evidence

- Approved CPSR with version history

## 11 Performance indicators

- Products marketed without approved CPSR (target 0)
- Reports overdue for review

## 12 References

- Regulation (EC) No 1223/2009, Article 10 and Annex I — Safety assessment (CPSR)
- Regulation (EC) No 1223/2009 on cosmetic products

### 13. Demonstration

**Case.** CPSR for FC-120 v1: Part A completed with stability (3 months accelerated), challenge test, exposure 1.54 g/day; lowest MoS 212 (phenoxyethanol). The external toxicologist named in B4 logs in and clicks **Approve as Safety Assessor**.

# Product information file (PIF) and its retention period

|                   |                                                    |                |       |
|-------------------|----------------------------------------------------|----------------|-------|
| Module            | ls_cosmetics                                       | Version        | _____ |
| Menu              | Cosmetics ► Regulatory ► Product Information Files | Effective date | _____ |
| Owner (suggested) | Responsible Person (Regulatory Manager)            | Approved by    | _____ |
| Related SOPs      | SOP-COS-002, SOP-COS-004, SOP-COS-005, SOP-COS-007 |                |       |

## 1 Objective

Keep for each product a complete product information file, available at the stated address, and retain it for ten years after the last batch was placed on the market.

## 2 Scope

All cosmetic products for which the company is the responsible person.

## 3 Responsibilities

| Role (software group)     | Responsibility                                    |
|---------------------------|---------------------------------------------------|
| <b>Responsible Person</b> | Owns the PIF and its availability to authorities. |
| <b>Regulatory Manager</b> | Compiles, activates, archives.                    |

## 4 Definitions

| Term                         | Definition                                                                                                                             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Article 11(2) content</b> | (a) description, (b) safety report, (c) manufacturing method and GMP statement, (d) proof of claimed effects, (e) animal testing data. |
| <b>Retention period</b>      | Ten years from the date the last batch was placed on the market (Article 11(1)).                                                       |

## 5 Workflow in the software



Model: ls.cosmetic.pif

Figure 138 — Lifecycle of a product information file.

## 6 Procedure

- Create.** In **Product Information Files** click **New**: file reference and name, product, responsible person and basis of designation (written mandate reference if applicable), address where the file is kept, languages.
- Complete.** Fill items (a) to (e): description; approved safety report and formulation; approved label; manufacturing method, GMP statement, standard, site and evidence; claims and proof (or justification of absence); animal testing data or the statement that none was performed; notification reference and date.
- Activate.** Click **Activate** when the product is placed on the market (record *First Placed on the Market*).
- Maintain.** Update the file for each formulation, label or claim version; record reviews.

5 **Enter retention.** When the product is discontinued, record the date of the last batch placed on the market and click **Enter Retention Period**; the retention end date is computed.

6 **Archive.** After the retention end date click **Archive**.

### 7. Controls enforced by the software

- ✓ Activation requires all Article 11(2) items, an approved safety report that concludes the product is safe, and the animal testing statement.
- ✓ The last batch date cannot precede the first placing; retention requires the last batch date; archiving is refused before the ten-year retention end.
- ✓ A file that left draft cannot be deleted; only an active file can return to draft.

### 8. Good practices

- ★ Keep the PIF in the language(s) accepted by the authorities of each market.
- ★ Check the PIF at each product change, not only at launch.

### 9. Pitfalls to avoid

- ✗ PIFs pointing to superseded safety reports or labels.

## 10 Records and evidence

- PIF with contents, activation and retention dates

## 11 Performance indicators

- Products marketed without active PIF (target 0)
- PIFs not reviewed in 12 months

## 12 References

- Regulation (EC) No 1223/2009, Article 11 — Product information file
- ISO 22716:2007 — Cosmetics — Good Manufacturing Practices

### 13. Demonstration

**Case.** PIF for FC-120: all items completed, CPSR approved (*safe*), label v2 approved, two claims approved, statement *no animal testing performed*. Activated on 1 March 2027. The product is discontinued; last batch placed on the market on 30 June 2031 → retention until 30 June 2041.



# Substantiation of claims

|                   |                                         |                |       |
|-------------------|-----------------------------------------|----------------|-------|
| Module            | ls_cosmetics                            | Version        | _____ |
| Menu              | Cosmetics ► Regulatory ► Product Claims | Effective date | _____ |
| Owner (suggested) | Regulatory Manager                      | Approved by    | _____ |
| Related SOPs      | SOP-COS-003, SOP-COS-005                |                |       |

## 1 Objective

Allow only claims that meet the six common criteria (legal compliance, truthfulness, evidential support, honesty, fairness, informed decision-making) and are supported by adequate evidence.

## 2 Scope

All claims made on labels, advertising, websites and social media for cosmetic products.

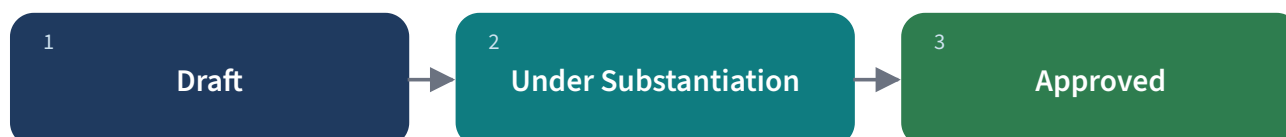
## 3 Responsibilities

| Role (software group)                                  | Responsibility                              |
|--------------------------------------------------------|---------------------------------------------|
| <b>Marketing / formulator</b>                          | Proposes claims and evidence.               |
| <b>Assessor</b>                                        | Assesses each criterion with justification. |
| <b>Regulatory Manager (approver, not the assessor)</b> | Approves or rejects.                        |

## 4 Definitions

| Term                   | Definition                                                                                                 |
|------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Common criteria</b> | The six criteria of Regulation (EU) No 655/2013.                                                           |
| <b>Evidence</b>        | Studies (with number of subjects), consumer tests, literature, expert opinions — dated, not in the future. |

## 5 Workflow in the software



Other outcomes



Model: ls.cosmetic.claim

Figure 139 — Lifecycle of a claim.

## 6 Procedure

- 1 **Create.** In **Product Claims** click **New** : reference, wording, product, formulation, medium; tick *No Animal Testing Claim* where relevant.
- 2 **Substantiate.** Click **Start Substantiation** ; add supporting evidence records; for each of the six criteria tick *Met* and write the justification.
- 3 **Animal testing claims.** Record the Article 20(3) declaration when the claim refers to the absence of animal testing.
- 4 **Decide.** A user other than the assessor clicks **Approve** or **Reject** (with decision reason).
- 5 **Withdraw.** Click **Withdraw** when a claim is no longer used; the substantiation record remains.

### 7. Controls enforced by the software

- ✓ All six criteria must be marked met with a written justification before approval; an evidentially supported claim must carry evidence.
- ✓ A claim on absence of animal testing needs the Article 20(3) declaration.
- ✓ The assessor cannot approve; only draft claims can be deleted (withdraw instead); state transitions are enforced.
- ✓ Study evidence must record the number of subjects and cannot be dated in the future.

### 8. Good practices

- ★ Keep the exact wording used on pack and in advertising; a small change is a new claim.
- ★ Prefer studies with adequate design (sample size, controls) for efficacy claims.

### 9. Pitfalls to avoid

- ✗ Using a claim approved for one formulation on another.

## 10 Records and evidence

- Claims with evidence, criteria and decision

## 11 Performance indicators

- Claims used without approval (target 0)
- Rejected claims

## 12 References

- Regulation (EC) No 1223/2009, Article 20, and Regulation (EU) No 655/2013 — Claims and common criteria

### 13. Demonstration

**Case.** Claim "*Hydration +32 % after 4 weeks*" for FC-120: evidence = corneometry study on 30 subjects; all six criteria met with justification; the assessor is the regulatory officer; the Regulatory Manager approves.

# Validation of labelling and of the INCI list

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_cosmetics                          | Version        | _____ |
| Menu              | Cosmetics ► Regulatory ► Labelling    | Effective date | _____ |
| Owner (suggested) | Regulatory Manager                    | Approved by    | _____ |
| Related SOPs      | SOP-COS-001, SOP-COS-003, SOP-COS-004 |                |       |

## 1 Objective

Ensure that each label carries all Article 19(1) particulars and an ingredient list generated from the approved formulation, and approve the label before artwork release.

## 2 Scope

All labels, outer and inner packaging, leaflets and adjacent notices of cosmetic products.

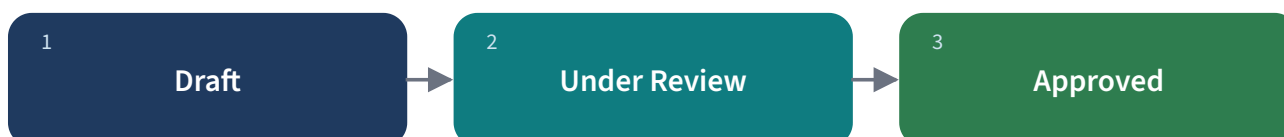
## 3 Responsibilities

| Role (software group)     | Responsibility                                                    |
|---------------------------|-------------------------------------------------------------------|
| <b>Regulatory officer</b> | Prepares labelling particulars and generates the ingredient list. |
| <b>Regulatory Manager</b> | Approves the label.                                               |

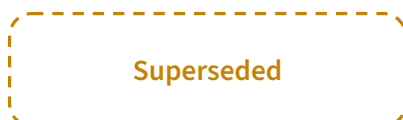
## 4 Definitions

| Term                         | Definition                                                                                                                                                                                                                                                                             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INCI list</b>             | Ingredients in descending order of weight at the time of addition; ingredients below 1 % in any order after those above 1 %; nanomaterials followed by ( <i>nano</i> ); perfume/aroma compositions as <i>parfum/aroma</i> ; colorants may be listed last; +/- marker for shade ranges. |
| <b>Durability indication</b> | Date of minimum durability, or period after opening when durability exceeds 30 months, per Article 19(1)(c).                                                                                                                                                                           |

## 5 Workflow in the software



Other outcomes



Model: *ls.cosmetic.label*

Figure 140 — Lifecycle of labelling particulars.

## 6 Procedure

- 1 **Create.** In **Labelling** click **New** : label reference, name, version, approved formulation, product, responsible person and address, imported product and country of origin, nominal content (or exemption with reason), durability indication, storage conditions, precautions, professional use, batch number rule, function of the product (or *clear from presentation*), languages, artwork file.
- 2 **Generate the list.** Click **Generate Ingredient List** , choose *List Colorants Last* and the *may contain* marker if applicable, review and **Apply to Label** . Use **Regenerate from Formulation** after a formulation change while the label is draft.
- 3 **Submit.** Click **Submit for Review** .
- 4 **Approve.** The Regulatory Manager checks the artwork against the particulars and clicks **Approve** .
- 5 **Release artwork.** Only approved label versions are sent to the printer (see SOP-PRD-002 for reconciliation of printed labels).

### 7. Controls enforced by the software

- ✓ Approval requires all Article 19(1) particulars and an approved formulation.
- ✓ An imported product needs its country of origin; the function must be stated or declared clear from presentation; nominal content or a justified exemption is required; period after opening needs a number of months; durability rules of Article 19(1)(c) are checked.
- ✓ Approved labels are frozen (new version) and their ingredient list can no longer be regenerated; only draft labels can be deleted.

### 8. Good practices

- ★ Compare the printed proof with the approved label word by word, including the INCI list.
- ★ Keep the list prefix ("Ingredients") in the languages of the markets.

### 9. Pitfalls to avoid

- ✗ Typing the INCI list by hand instead of generating it from the formulation.

## 10 Records and evidence

- Approved label versions with artwork

## 11 Performance indicators

- Labels approved first time
- Artwork released without approved label (target 0)

## 12 References

- Regulation (EC) No 1223/2009, Article 19 — Labelling

### 13. Demonstration

**Case.** Label FC-120 v2: 50 ml, PAO 12 months (durability > 30 months), function *face cream*, list generated: *Ingredients: Aqua, Glycerin, ... , Parfum, CI 77891*; approved by the Regulatory Manager before the artwork order.

# Maintenance of the register of regulatory restrictions on ingredients

|                   |                                                                    |                |       |
|-------------------|--------------------------------------------------------------------|----------------|-------|
| Module            | ls_cosmetics                                                       | Version        | _____ |
| Menu              | Cosmetics ► Ingredients ► Annex Restrictions / Ingredient Register | Effective date | _____ |
| Owner (suggested) | Regulatory Manager                                                 | Approved by    | _____ |
| Related SOPs      | SOP-COS-001, SOP-RA-010                                            |                |       |

## 1 Objective

Keep the register of Annex II to VI entries and the ingredient register current with the consolidated regulation, so that formulation checks rely on up-to-date limits.

## 2 Scope

All annex entries relevant to the ingredients used, and all ingredients of the register.

## 3 Responsibilities

| Role (software group)     | Responsibility                                                     |
|---------------------------|--------------------------------------------------------------------|
| <b>Regulatory Manager</b> | Monitors amendments, updates entries, assesses impact on products. |

## 4 Definitions

| Term                 | Definition                                                                                                                                                                                                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Annex entry</b>   | Annex (II prohibited, III restricted, IV colorants, V preservatives, VI UV-filters), reference number, substance, CAS/EC, product type/body part, maximum concentration, conditions and warnings, source reference, consolidation date, applicability dates. |
| <b>CMR substance</b> | Carcinogenic, mutagenic or toxic for reproduction; flag and category must agree.                                                                                                                                                                             |

## 5 Procedure

- 1 Monitor.** Check monthly for amendments of the annexes (official journal, consolidated version); record them as regulatory watch (SOP-RA-010 principles).
- 2 Update.** Create or update the entries in **Annex Restrictions**: source reference, consolidation date, *Applicable From/Until* for transitional periods.
- 3 Link ingredients.** In **Ingredient Register** link each ingredient to its applicable annex entries; set CMR, nanomaterial and perfume flags.
- 4 Assess impact.** List approved formulations containing the affected ingredients; open a change request for reformulation or relabelling (SOP-QA-030) within the transitional period.
- 5 Record.** Keep the impact assessment with the source reference.

## 6. Controls enforced by the software

- ✓ *Applicable Until* cannot precede *Applicable From*; a numeric limit must be > 0; Annex II entries cannot carry a permitted concentration.
- ✓ Ingredients: CMR flag and category must agree; an impurity cannot also be a subsidiary technical material; a perfume composition must be labelled *parfum* or *aroma*.
- ✓ Annex II ingredients are refused in formulations under review or approved.

## 7. Good practices

- ★ Record the consolidation date of the source used, to prove which version was applied.
- ★ Watch the transitional dates: placing on the market and making available often differ.

## 8. Pitfalls to avoid

- ✗ Entries created from a supplier datasheet instead of the official text.

## 9 Records and evidence

- Annex entries with sources
- Impact assessments

## 10 Performance indicators

- Amendments not assessed within 30 days

## 11 References

- Regulation (EC) No 1223/2009, Article 14 and Annexes II to VI — Restrictions
- Regulation (EC) No 1223/2009 on cosmetic products

## 12. Demonstration

**Case.** An amendment restricts a fragrance allergen to 0.01 % in leave-on products from a future date. The Regulatory Manager updates the Annex III entry with *Applicable From*, finds 3 formulations above the limit and opens CC-2027-008 for reformulation before the date.

Cosmetics
Products
Ingredients
Safety
Regulatory
Reports
7
14
My Company (San Francisco)

New
Annex Restrictions

Substance Name

Substance name as published

IDENTIFICATION

Annex

Reference Number ?

Inci Name

Cas Number

Ec Number

Company ?

My Company (San Francisco)

LIMIT

Numeric Limit Applies ?

☐

Product Type / Body Part ?

PROVENANCE

Source Reference ?

Consolidation Date ?

Applicable From

Applicable Until

Conditions of Use

Label Wording

Other Column

Internal Note

Conditions as published in the annex

Figure 141 — Annex restriction record. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Prior authorisation dossier in Algeria

|                   |                                                        |                |       |
|-------------------|--------------------------------------------------------|----------------|-------|
| Module            | ls_cosmetics                                           | Version        | _____ |
| Menu              | Cosmetics ► Regulatory ► Algerian Prior Authorisations | Effective date | _____ |
| Owner (suggested) | Regulatory Manager (Algeria)                           | Approved by    | _____ |
| Related SOPs      | SOP-COS-003, SOP-RA-010, SOP-RA-011                    |                |       |

## 1 Objective

Prepare, submit and follow the prior authorisation dossier required in Algeria for cosmetic products, track the decision deadline and formal notices, and keep the authorisation reference.

## 2 Scope

Cosmetic products manufactured or imported for the Algerian market. The legal texts are those cited by the module (Executive Decree 97-37 as amended by 10-114); their current wording must be verified by Regulatory Affairs before use of this SOP.

## 3 Responsibilities

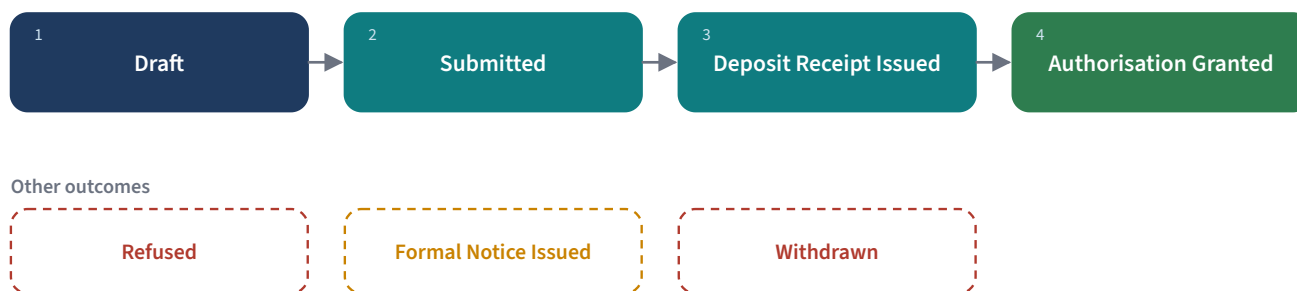
| Role (software group)           | Responsibility                                                                              |
|---------------------------------|---------------------------------------------------------------------------------------------|
| <b>Regulatory Manager</b>       | Compiles the dossier, submits it to the Direction de Wilaya du Commerce, records decisions. |
| <b>Operator's legal/finance</b> | Provide the administrative documents.                                                       |

## 4 Definitions

| Term                   | Definition                                                                                                                                                                                                                                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deposit receipt</b> | Receipt issued at submission; the 45-day decision period coded in the module runs from its date.                                                                                                                                                                                                                       |
| <b>Formal notice</b>   | Notice that an element of the authorisation fails; the module computes a one-month (30-day) compliance deadline.                                                                                                                                                                                                       |
| <b>Dossier items</b>   | 16 items: commercial register extract, tax identification number, statutes, CNRC filing certificate, tax roll extract, CNAS/CASNOS certificate, denomination, use and directions, qualitative composition, analyses, toxicity testing, batch identification, precautions, label model, responsible persons, trademark. |



## 5 Workflow in the software



Model: *ls.cosmetic.dz\_authorization*

Figure 142 — Lifecycle of an Algerian prior authorisation dossier.

## 6 Procedure

- 1 **Create.** In **Algerian Prior Authorisations** click **New** : denomination, authorisation type, product, PIF, formulation, operator, wilaya direction, submission mode.
- 2 **Complete.** Tick each of the 16 items when provided and attach documents; *Dossier Complete* becomes true.
- 3 **Submit.** Click **Submit Dossier** with the submission date.
- 4 **Receipt.** Click **Register Deposit Receipt** with its reference and date; the decision due date is computed (45 days).
- 5 **Decision.** Click **Record Authorisation** with the authorisation reference, or **Record Refusal** with the reasoned refusal.
- 6 **Formal notice.** If notified, click **Register Formal Notice** with date and subject; act within the compliance deadline; click **Resolve Formal Notice** .
- 7 **Withdrawal.** Record the withdrawal of an authorisation with **Record Withdrawal** .

### 7. Controls enforced by the software

- ✓ Submission requires a complete dossier (missing items are listed).
- ✓ Receipt date  $\geq$  submission; decision date  $\geq$  receipt; a granted dossier needs its authorisation reference; a refusal needs its reason.
- ✓ Formal notices need a date and a subject; state transitions are enforced; submitted dossiers cannot be deleted.

### 8. Good practices

- ★ Keep scans of the stamped deposit receipt and of the decision in the dossier.
- ★ Record open questions about the procedure as regulatory questions (SOP-RA-011) rather than guessing.

### 9. Pitfalls to avoid

- ✗ Relying on the coded deadlines without checking the text in force.

## 10 Records and evidence

- Dossier with items, receipt, decision and notices

## 11 Performance indicators

- Decisions overdue vs computed due date
- Dossiers refused

## 12 References

- Algeria — Executive Decree No. 97-37 of 14 January 1997, as amended by Executive Decree No. 10-114 of 18 April 2010 (as cited in ls\_cosmetics; to be verified by Regulatory Affairs)

### 13. Demonstration

**Case.** Dossier DZ-2027-004 for FC-120: 16/16 items provided; submitted 3 February; deposit receipt 5 February → decision due 22 March; authorisation recorded on 15 March with its reference, linked to the PIF.

# Medical devices

ls\_medical\_device

**ls\_medical\_device** follows the structure of Regulation (EU) 2017/745 (MDR): device register with risk class and classification rationale, Basic UDI-DI and UDI-DI assignments, risk management files (ISO 14971) with risks, controls and residual acceptability, clinical evaluations and PMCF evaluation reports, technical documentation by section (Annex II/III, or Annex XIII for custom-made devices), CE marking and certificate records with notified bodies, post-market surveillance plans and periodic reports. A device cannot be placed on the market without approved technical documentation and, when its class requires it, a valid certificate.

| Menu                                                                                                       | Content         |
|------------------------------------------------------------------------------------------------------------|-----------------|
| Medical Devices ▶ Devices ▶ Device Register / UDI Assignments                                              | Devices and UDI |
| Medical Devices ▶ Risk Management ▶ Risk Management Files / Risks                                          | ISO 14971       |
| Medical Devices ▶ Clinical ▶ Clinical Evaluations / PMCF Evaluation Reports                                | Clinical        |
| Medical Devices ▶ Regulatory ▶ Technical Documentation / Documentation Sections / CE Marking               | Regulatory      |
| Medical Devices ▶ Post-Market Surveillance ▶ Surveillance Plans / Periodic Reports / Open Reporting Period | PMS             |
| Medical Devices ▶ Monitoring ▶ Expired Certificates / Overdue Post-Market Reports                          | Monitoring      |

| Group                        | Main rights                                              |
|------------------------------|----------------------------------------------------------|
| Viewer · User                | Read; prepare records                                    |
| Regulatory Affairs · Manager | Approve files, change market status, manage certificates |

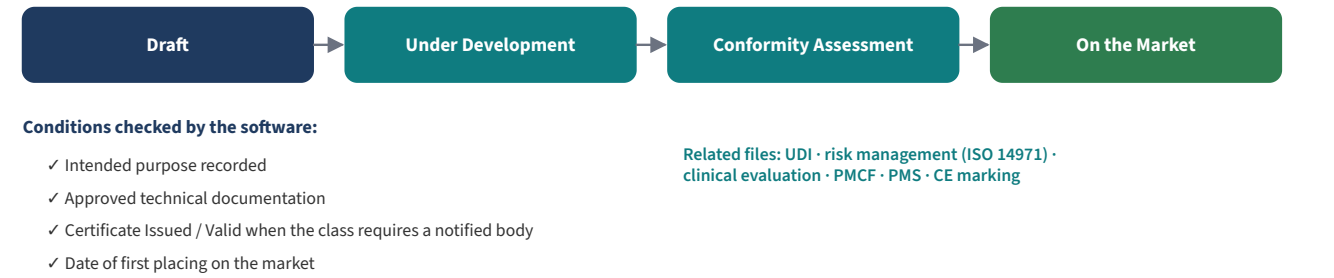


Figure 143 — Device lifecycle and the gates checked by the software.

# Device registration and justification of its risk class

|                   |                                             |                |       |
|-------------------|---------------------------------------------|----------------|-------|
| Module            | ls_medical_device                           | Version        | _____ |
| Menu              | Medical Devices ► Devices ► Device Register | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                  | Approved by    | _____ |
| Related SOPs      | SOP-MD-002, SOP-MD-005, SOP-MD-006          |                |       |

## 1 Objective

Register every medical device with its intended purpose and characteristics, justify its risk class by the applicable classification rule, and control its market status through documented gates.

## 2 Scope

All medical devices designed, manufactured or placed on the market by the company, including custom-made devices.

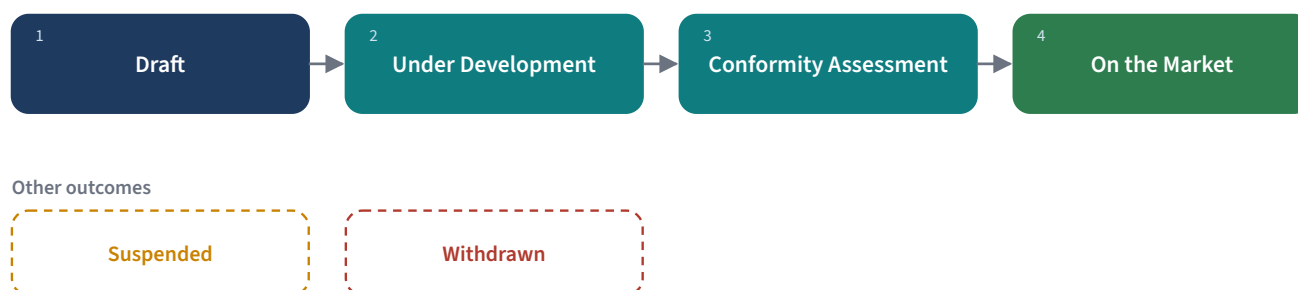
## 3 Responsibilities

| Role (software group)     | Responsibility                                                      |
|---------------------------|---------------------------------------------------------------------|
| <b>Regulatory Affairs</b> | Registers devices, classifies them, changes market status.          |
| <b>R&amp;D / quality</b>  | Provide intended purpose, characteristics and design documentation. |

## 4 Definitions

| Term                 | Definition                                                                                                                                                    |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk class</b>    | Class I, IIa, IIb, III (with sub-cases) according to the classification rules of MDR Annex VIII; configured in <a href="#">Configuration ► Risk Classes</a> . |
| <b>Market status</b> | Draft → Under Development → Conformity Assessment → On the Market; Suspended; Withdrawn.                                                                      |

## 5 Workflow in the software



Model: ls.md.device

Figure 144 — Lifecycle of a device.



#### Conditions checked by the software:

- ✓ Intended purpose recorded
- ✓ Approved technical documentation
- ✓ Certificate Issued / Valid when the class requires a notified body
- ✓ Date of first placing on the market

Related files: UDI · risk management (ISO 14971) · clinical evaluation · PMCF · PMS · CE marking

Figure 145 — Gates before placing on the market.

## 6 Procedure

- 1 **Register.** In **Device Register** click **New** : reference, name, model/catalogue reference, nomenclature code, product, legal manufacturer, authorised representative, risk class, classification rule and rationale, intended purpose, characteristics (implantable, sterile, measuring function, reusable surgical instrument, software, custom-made, direct marking), documentation retention years.
- 2 **Develop.** Click **Start Development** .
- 3 **Conformity assessment.** Click **Start Conformity Assessment** once the intended purpose is recorded; prepare the technical documentation, risk file, clinical evaluation and CE marking (SOP-MD-003 to 006).
- 4 **Place on the market.** Click **Place on the Market** with the date of first placing on the market.
- 5 **Suspend / resume / withdraw.** Use **Suspend** , **Resume** , **Withdraw** , or **Change Status with Justification** to record the reason.

### 7. Controls enforced by the software

- ✓ Changing the market status requires the Regulatory Affairs or Manager access level.
- ✓ Conformity assessment requires the intended purpose; placing on the market requires the date, completed conformity assessment, approved technical documentation and, when the class requires a notified body, an issued or valid certificate.
- ✓ A custom-made implantable device requires a classification rationale; withdrawal date  $\geq$  first placing date.

### 8. Good practices

- ★ Write the classification rationale citing the rule and the reasoning for borderline cases.
- ★ Keep the intended purpose identical in the IFU, technical documentation and clinical evaluation.

### 9. Pitfalls to avoid

- ✗ Changing the intended purpose without reclassification.

## 10 Records and evidence

- Device record with status history and reasons

## 11 Performance indicators

- Devices on market without valid certificate (target 0)

## 12 References

- Regulation (EU) 2017/745 on medical devices (MDR)
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** Sterile single-use catheter CAT-14: class IIa by rule 7 (surgically invasive, short term), rationale recorded; development started; after approved technical documentation and a certificate issued by the notified body, placed on the market on 1 September.

The screenshot displays the 'Device Register' form in the Odoo 19.0 Medical Devices application. The interface includes a top navigation bar with various modules like 'Medical Devices', 'Devices', 'Risk Management', etc. The main form is titled 'Device Register' and shows details for a device with reference 'MD/2026/00002' and name 'Demo Implantable Pulse Generator'. The form is organized into several sections: 'IDENTIFICATION', 'ECONOMIC OPERATORS', 'CLASSIFICATION', and 'CHARACTERISTICS'. The 'IDENTIFICATION' section includes fields for 'Model or Catalogue Reference' (DEMO-IPG-200), 'Basic UDI-DI', 'Nomenclature Code', 'Product', and 'Company' (My Company (San Francisco)). The 'ECONOMIC OPERATORS' section includes 'Legal Manufacturer' and 'Authorised Representative'. The 'CLASSIFICATION' section shows 'Risk Class' as 'Class III', 'Classification Rule' as 'Annex VIII, rule to be confirmed by the', and 'Notified Body Involvement Required' as checked. The 'CHARACTERISTICS' section includes 'Implantable' (checked), 'Placed on the Market Sterile' (checked), 'Has a Measuring Function' (unchecked), and 'Reusable Surgical Instrument' (unchecked). A right-hand sidebar shows a chat window with 'OdooBot' and a message 'Medical Device created'.

Reference ?  
**MD/2026/00002**

Device Name ?  
**Demo Implantable Pulse Generator**

**IDENTIFICATION**

Model or Catalogue Reference ? DEMO-IPG-200

Basic UDI-DI ?

Nomenclature Code ?

Product ?

Company ? My Company (San Francisco)

**ECONOMIC OPERATORS**

Legal Manufacturer ?

Authorised Representative ?

**CLASSIFICATION**

Risk Class ? Class III

Classification Rule ? Annex VIII, rule to be confirmed by the

Notified Body Involvement Required ? ☒

**CHARACTERISTICS**

Implantable ? ☒

Placed on the Market Sterile ? ☒

Has a Measuring Function ? ☐

Reusable Surgical Instrument ? ☐

Send message Log note Activity

Today

OdooBot 10:40 AM  
Medical Device created

Figure 146 — Device register form. (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Assignment of the UDI (Unique Device Identification)

|                   |                                             |                |       |
|-------------------|---------------------------------------------|----------------|-------|
| Module            | ls_medical_device                           | Version        | _____ |
| Menu              | Medical Devices ► Devices ► UDI Assignments | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                  | Approved by    | _____ |
| Related SOPs      | SOP-MD-001                                  |                |       |

## 1 Objective

Assign one Basic UDI-DI per device and UDI-DIs per packaging level from an accredited issuing entity, define the production identifier, register them in the UDI database and retire them in a controlled way.

## 2 Scope

All devices subject to UDI obligations.

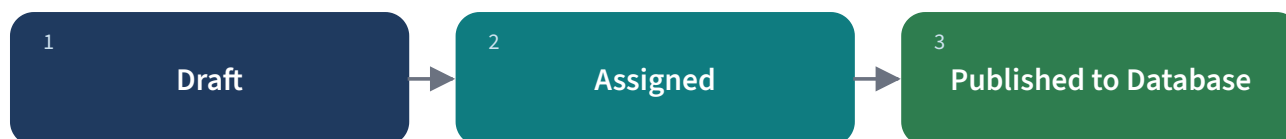
## 3 Responsibilities

| Role (software group) | Responsibility                              |
|-----------------------|---------------------------------------------|
| Regulatory Affairs    | Assigns and publishes identifiers.          |
| Packaging / labelling | Applies the UDI carrier and direct marking. |

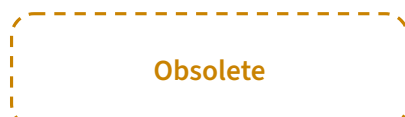
## 4 Definitions

| Term            | Definition                                                                                                                 |
|-----------------|----------------------------------------------------------------------------------------------------------------------------|
| Basic UDI-DI    | Primary identifier of a device model, key in the UDI database; one per device.                                             |
| UDI-DI / UDI-PI | Device identifier per packaging level / production identifier (lot, serial, manufacturing date, expiry, software version). |

## 5 Workflow in the software



Other outcomes



Model: ls.md.udi

Figure 147 — Lifecycle of a UDI assignment.

## 6 Procedure

- 1 **Create.** In **UDI Assignments** click **New**: device, identifier kind (Basic UDI-DI or UDI-DI), identifier, issuing entity (note when *Other*), packaging level and quantity, carrier type, direct marking, production identifier components.
- 2 **Assign.** Record the assignment date and click **Assign**.
- 3 **Publish.** After registration in the UDI database, record the database name, submission date and reference, and click **Publish to Database**.
- 4 **Retire.** Click **Mark Obsolete** with the reason when the identifier is no longer used; a change requiring a new UDI-DI is a new assignment.

### 7. Controls enforced by the software

- ✓ A device carries a single Basic UDI-DI.
- ✓ Leaving draft requires the assignment date; publication requires the database name and submission date; retirement requires a reason.
- ✓ Assigned identifiers cannot be deleted; published identifiers cannot be modified (create a new assignment).

### 8. Good practices

- ★ Define in this SOP which changes trigger a new UDI-DI (e.g. change of packaging quantity, sterilisation state).

### 9. Pitfalls to avoid

- ✗ Reusing a UDI-DI after a change that requires a new one.

## 10 Records and evidence

- UDI assignments with publication data

## 11 Performance indicators

- Devices on market with unpublished UDI (target 0)

## 12 References

- MDR Article 27 and Annex VI — Unique Device Identification
- GS1 General Specifications (GTIN, SSCC, check digit)

### 13. Demonstration

**Case.** CAT-14: Basic UDI-DI assigned (GS1), UDI-DI for the unit (box of 1) and the shelf box (10), PI = lot + expiry; published in the database with its reference.



# Risk management according to ISO 14971

|                   |                                         |                |       |
|-------------------|-----------------------------------------|----------------|-------|
| Module            | ls_medical_device                       | Version        | _____ |
| Menu              | Medical Devices ► Risk Management       | Effective date | _____ |
| Owner (suggested) | Risk management lead (Regulatory / R&D) | Approved by    | _____ |
| Related SOPs      | SOP-QA-060, SOP-MD-004, SOP-MD-007      |                |       |

## 1 Objective

Maintain for each device a risk management file identifying hazards, hazardous situations and harms, estimating and controlling risks, evaluating residual risks and the overall benefit-risk, and updating it with production and post-production information.

## 2 Scope

All devices of the register, throughout their lifecycle.

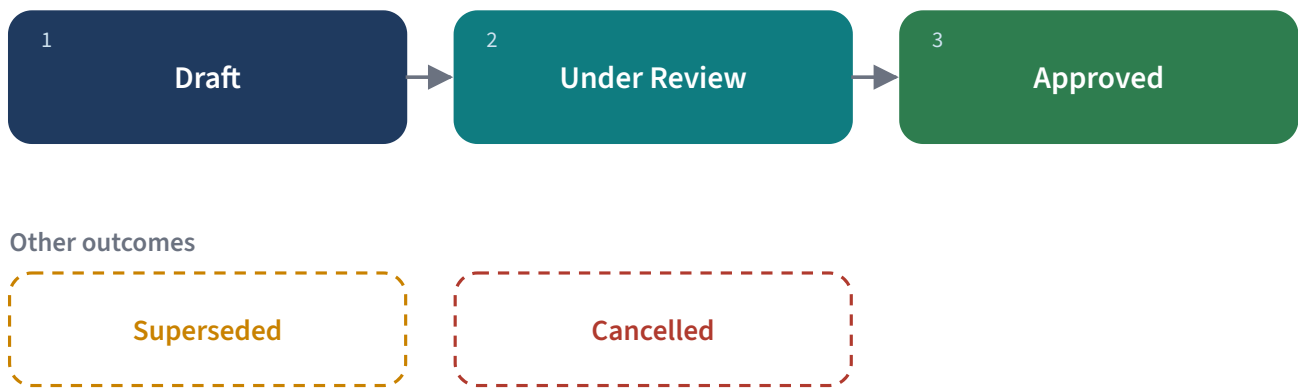
## 3 Responsibilities

| Role (software group)                 | Responsibility                                      |
|---------------------------------------|-----------------------------------------------------|
| <b>Responsible person of the file</b> | Leads the analysis; cannot approve the file.        |
| <b>Cross-functional team</b>          | R&D, clinical, quality, manufacturing, post-market. |
| <b>Regulatory Affairs / Manager</b>   | Approves the file.                                  |

## 4 Definitions

| Term                          | Definition                                                                                             |
|-------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Risk index</b>             | Combination of severity and probability; the residual index cannot exceed the initial index.           |
| <b>Residual acceptability</b> | Acceptability after control; a difference from the value proposed by the matrix needs a justification. |

## 5 Workflow in the software



Model: ls.md.technical\_file

Figure 148 — Lifecycle of the regulated files (risk file, clinical, technical documentation, PMS).

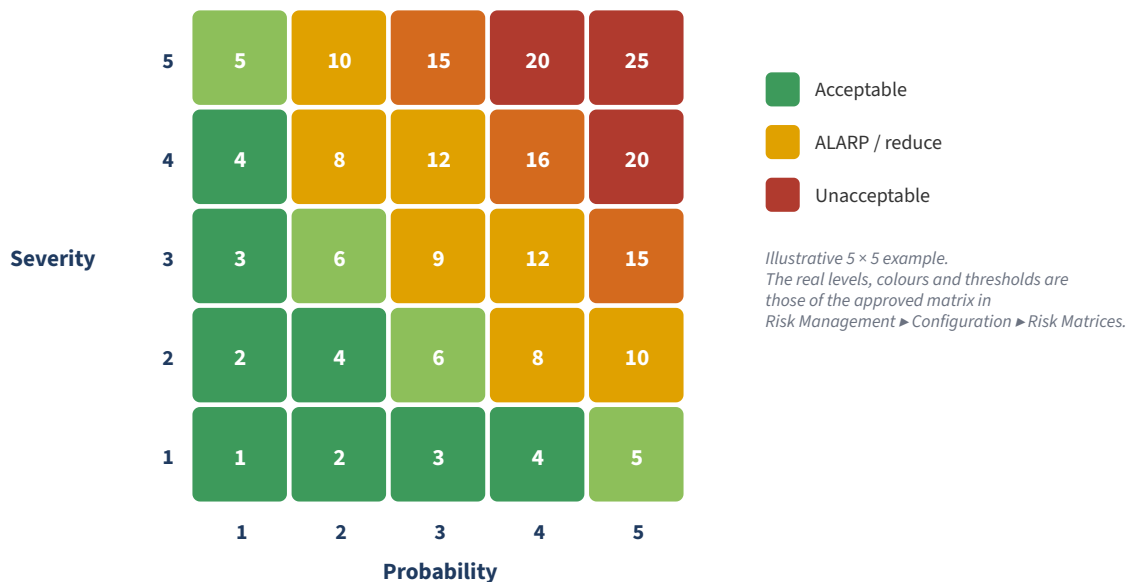


Figure 149 — Illustrative risk matrix.

## 6 Procedure

- Create the file.** In **Risk Management Files** click **New** : title, device, version, scope, risk acceptability policy, standard reference, responsible.
- Identify risks.** Add risks: identifier, hazard and category, sequence of events, hazardous situation, harm, affected party, initial severity and probability.
- Control.** For each risk choose the control option, describe the measure, record implementation and effectiveness evidence, residual severity and probability, residual acceptability (justify deviations from the proposed value), disclosure of residual risk in the IFU.
- New hazards.** If a control introduces a new hazard, tick it, describe it and add it as a new risk.
- Conclude.** Write the overall benefit-risk conclusion and tick overall residual risk acceptable when justified.
- Review and approve.** Click **Submit for Review** ; the approver (Regulatory Affairs/Manager, not the responsible person) clicks **Approve** .
- Update.** Feed production and post-production information (complaints, PMS, PMCF); create a new version when risks change ( **Supersede** the old one).

## 7. Controls enforced by the software

- ✓ A file cannot be submitted without at least one risk; approval requires the overall benefit-risk conclusion, no unacceptable residual risk and no risk without a control measure.
- ✓ Residual index  $\leq$  initial index; a reduced residual risk needs a control measure; a new hazard needs a description; an acceptability different from the matrix proposal needs a justification.
- ✓ Approved files are closed for editing — create a new version.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

## 8. Good practices

- ★ Link device risks to the general register (ls\_risk\_management) only if you need its workflow; keep one reference file per device.
- ★ Review the file at each PMS report.

## 9. Pitfalls to avoid

- ✗ Using "information for safety" as the main control for high risks.

## 10 Records and evidence

- Approved risk management file with versions

## 11 Performance indicators

- Files not updated after a PMS report
- Risks without control measure (target 0)

## 12 References

- ISO 14971:2019 — Medical devices — Application of risk management
- Regulation (EU) 2017/745 on medical devices (MDR)

## 13. Demonstration

**Case.** CAT-14 risk *R-07 tip detachment* → *vessel embolism*: initial S5/P3; controls: bonding process validated (protective), pull test on each lot (protective), IFU warning (information). Residual S5/P1 acceptable with justification. The file is approved by Regulatory Affairs.

# Clinical evaluation and PMCF (Post-Market Clinical Follow-up)

|                   |                            |                |       |
|-------------------|----------------------------|----------------|-------|
| Module            | ls_medical_device          | Version        | _____ |
| Menu              | Medical Devices ► Clinical | Effective date | _____ |
| Owner (suggested) | Clinical evaluation lead   | Approved by    | _____ |
| Related SOPs      | SOP-MD-003, SOP-MD-007     |                |       |

## 1 Objective

Demonstrate conformity with the relevant safety and performance requirements through a planned clinical evaluation, and keep it updated with post-market clinical follow-up.

## 2 Scope

All devices; the depth depends on class, novelty and available evidence.

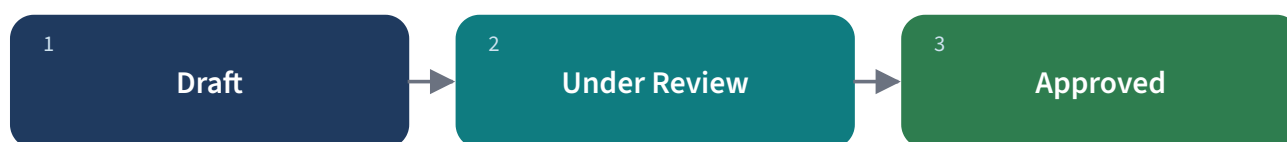
## 3 Responsibilities

| Role (software group)               | Responsibility                                               |
|-------------------------------------|--------------------------------------------------------------|
| <b>Evaluator (qualified)</b>        | Plans and writes the clinical evaluation; cannot approve it. |
| <b>Regulatory Affairs / Manager</b> | Approves.                                                    |
| <b>PMCF author</b>                  | Writes PMCF evaluation reports.                              |

## 4 Definitions

| Term               | Definition                                                                                                             |
|--------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Equivalence</b> | Claim that clinical data of another device apply; requires a demonstration.                                            |
| <b>PMCF</b>        | Continuous process to collect clinical data after placing on the market; a justification is required when not applied. |

## 5 Workflow in the software



Other outcomes



Model: ls.md.technical\_file

Figure 150 — Lifecycle of a clinical evaluation / PMCF report.

## 6 Procedure

- 1 **Plan.** In **Clinical Evaluations** click **New**: device, version, evaluator and qualification, plan scope, requirements supported by clinical data, intended benefits, target population, acceptance criteria, literature search protocol.
- 2 **Collect and appraise.** Record evidence sources, equivalence (and its demonstration), clinical investigation (reference) or the justification for none; appraisal and analysis of data.
- 3 **Conclude.** Write the benefit-risk conclusion, confirm conformity, record residual gaps; state whether PMCF applies (plan summary) or not (justification).
- 4 **Approve.** Submit for review; Regulatory Affairs/Manager (not the evaluator) approves.
- 5 **PMCF reports.** Periodically create PMCF evaluation reports: period, activities, data, findings, new or emerging risks, off-label use, benefit-risk conclusion, actions; approve (not the author).
- 6 **Update.** When a PMCF report requires it, update the clinical evaluation (new version) and the risk file.

### 7. Controls enforced by the software

- ✓ Equivalence claimed requires a demonstration; a declared clinical investigation needs its reference, otherwise a justification is required; PMCF applicable needs a plan summary, not applicable needs a justification.
- ✓ Approval requires the benefit-risk conclusion; the evaluator/author cannot approve.
- ✓ PMCF report: new risks and required actions must be described; the linked clinical evaluation must belong to the same device.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

### 8. Good practices

- ★ Keep the literature search reproducible (databases, dates, search strings).
- ★ Align acceptance criteria with the state of the art.

### 9. Pitfalls to avoid

- ✗ Equivalence claimed on superficial similarity.

## 10 Records and evidence

- Clinical evaluations and PMCF reports with versions

## 11 Performance indicators

- Clinical evaluations past their review date
- PMCF reports overdue

## 12 References

- MDR Article 61 and Annex XIV — Clinical evaluation and PMCF
- Regulation (EU) 2017/745 on medical devices (MDR)

### 13. Demonstration

**Case.** Clinical evaluation of CAT-14 v2: literature search on 3 databases, 41 relevant publications, no equivalence claimed, own clinical data from a 60-patient study; PMCF applicable (registry follow-up). The annual PMCF report identifies no new risk; benefit-risk remains acceptable.

# Technical documentation

|                   |                                                        |                |       |
|-------------------|--------------------------------------------------------|----------------|-------|
| Module            | ls_medical_device                                      | Version        | _____ |
| Menu              | Medical Devices ► Regulatory ► Technical Documentation | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                             | Approved by    | _____ |
| Related SOPs      | SOP-MD-001, SOP-MD-006                                 |                |       |

## 1 Objective

Compile, for each device, technical documentation structured by section according to the applicable annex, with evidence for each section, approved and retained for the required period.

## 2 Scope

Technical documentation (Annex II and III) of all devices; Annex XIII documentation for custom-made devices.

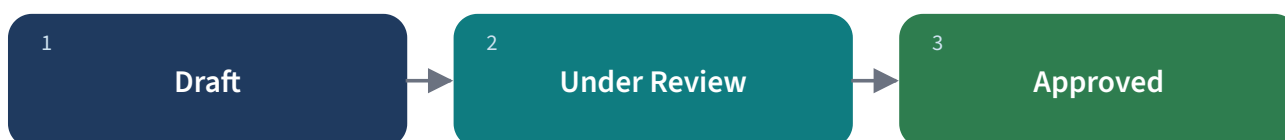
## 3 Responsibilities

| Role (software group)               | Responsibility                   |
|-------------------------------------|----------------------------------|
| <b>Section owners</b>               | Provide the evidence references. |
| <b>Responsible person</b>           | Compiles; cannot approve.        |
| <b>Regulatory Affairs / Manager</b> | Approves.                        |

## 4 Definitions

| Term           | Definition                                                                                                                                                                                         |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Section</b> | Part of the documentation from a template (Configuration ► Documentation Section Templates), mandatory or not; marked complete with an evidence reference, or not applicable with a justification. |

## 5 Workflow in the software



Other outcomes



Model: ls.md.technical\_file

Figure 151 — Lifecycle of technical documentation.

## 6 Procedure

- 1 **Create.** In **Technical Documentation** click **New**: title, device, annex (Annex II/III, or Annex XIII for custom-made), version, responsible, storage location; load the sections from the templates.
- 2 **Complete sections.** For each section record the evidence reference (controlled document number and version) and mark it complete; for a non-applicable section record the justification (Annex II point 4(a) for GSPR).
- 3 **Check completeness.** *Missing Mandatory Sections* must be zero.
- 4 **Approve.** Submit for review; Regulatory Affairs/Manager (not the responsible person) approves. This is a gate for placing on the market (SOP-MD-001).
- 5 **Maintain.** Changes to the device create a new version of the documentation; retain until the date computed from the device retention years.

### 7. Controls enforced by the software

- ✓ Custom-made devices use Annex XIII and other devices cannot; the file must contain sections; approval requires all mandatory sections complete.
- ✓ A section complete requires an evidence reference; a non-applicable section requires a justification; approved documentation is closed for editing.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

### 8. Good practices

- ★ Use `Is_document_management` references so that each evidence is a controlled version.

### 9. Pitfalls to avoid

- ✗ Sections marked complete with a folder name instead of a document reference.

## 10 Records and evidence

- Approved technical documentation with sections and versions

## 11 Performance indicators

- Devices with incomplete documentation

## 12 References

- MDR Annexes II and III — Technical documentation
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** TD of CAT-14 (Annex II/III): 26 sections; section *Biocompatibility* complete with evidence *REP-BIO-2027-03 v1*; section *Software* not applicable with justification; 0 mandatory section missing; approved.

# CE marking and relationship with the notified body

|                   |                                                                             |                |       |
|-------------------|-----------------------------------------------------------------------------|----------------|-------|
| Module            | ls_medical_device                                                           | Version        | _____ |
| Menu              | Medical Devices ► Regulatory ► CE Marking · Configuration ► Notified Bodies | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                                                  | Approved by    | _____ |
| Related SOPs      | SOP-MD-001, SOP-MD-005, SOP-MD-007                                          |                |       |

## 1 Objective

Record the conformity assessment route, the notified body and its certificates, the EU declaration of conformity and the affixing of CE marking, and keep certificate status (valid, suspended, withdrawn, expired) under control.

## 2 Scope

All devices placed on the EU market.

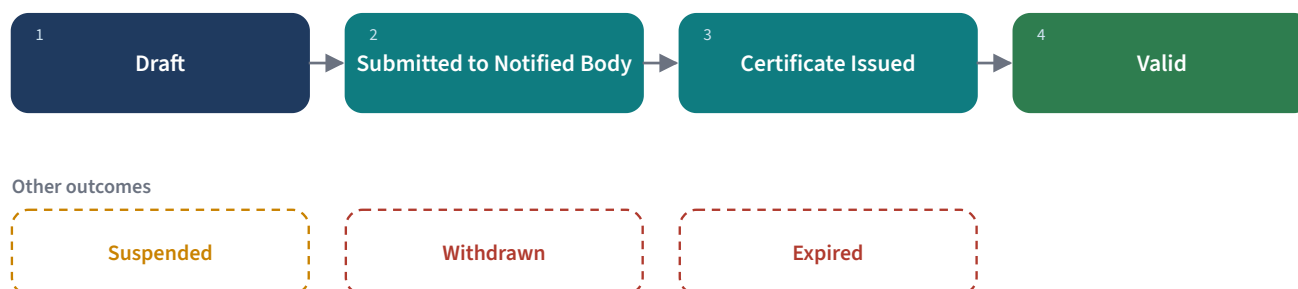
## 3 Responsibilities

| Role (software group)     | Responsibility                                                    |
|---------------------------|-------------------------------------------------------------------|
| <b>Regulatory Affairs</b> | Manages certificates and the relationship with the notified body. |
| <b>Management</b>         | Signs the declaration of conformity.                              |

## 4 Definitions

| Term                               | Definition                                                                      |
|------------------------------------|---------------------------------------------------------------------------------|
| <b>Conformity assessment route</b> | Annex(es) applied; routes involving a notified body require one to be selected. |
| <b>Notified body</b>               | Registered with its designation verification date (not in the future).          |

## 5 Workflow in the software



Model: ls.md.ce\_marking

Figure 152 — Lifecycle of a CE marking / certificate record.

## 6 Procedure

- 1 Register the notified body.** In **Configuration ► Notified Bodies** record name, number and designation verification date.



- 2 **Create the record.** In **CE Marking** click **New** : device, route, notified body, scope, conditions and restrictions, declaration of conformity number and date, Basic UDI-DI.
- 3 **Submit.** Click **Submit to Notified Body** when the application is filed.
- 4 **Certificate issued.** Record the certificate number, issue and expiry dates, and click **Record Certificate Issued** ; then **Activate** (status Valid).
- 5 **Affix CE.** Record that CE marking is affixed and its date.
- 6 **Surveillance.** Record the next surveillance audit due; monitor **Monitoring ► Expired Certificates**.
- 7 **Suspension / withdrawal.** Use **Suspend** or **Withdraw** with reason and date; assess the device market status (SOP-MD-001).

### 7. Controls enforced by the software

- ✓ Managing certificate status requires the Regulatory Affairs or Manager access level.
- ✓ A route involving a notified body requires one; CE affixed requires its date; issued requires certificate number and issue date; an expired certificate cannot be activated.
- ✓ Only valid certificates can be suspended; suspension and withdrawal need a reason; issued records are frozen for their identifying data and cannot be deleted.

### 8. Good practices

- ★ Plan recertification at least 12 months before expiry.
- ★ Keep the notified body correspondence attached to the record.

### 9. Pitfalls to avoid

- ✗ Placing a new variant on the market outside the certificate scope.

## 10 Records and evidence

- CE marking and certificate records
- Declarations of conformity

## 11 Performance indicators

- Certificates expiring within 12 months
- Devices on market with expired certificate (target 0)

## 12 References

- MDR Articles 20 and 52, Annexes IX to XI — CE marking and conformity assessment
- Regulation (EU) 2017/745 on medical devices (MDR)

### 13. Demonstration

**Case.** CAT-14: route Annex IX (QMS + technical documentation assessment) with notified body NB 0000 (placeholder number); certificate issued 15 August, expiry 5 years later; activated; CE affixed 1 September.

# Post-market surveillance and periodic reports

|                   |                                                |                |       |
|-------------------|------------------------------------------------|----------------|-------|
| Module            | ls_medical_device                              | Version        | _____ |
| Menu              | Medical Devices ► Post-Market Surveillance     | Effective date | _____ |
| Owner (suggested) | PMS lead (Regulatory Affairs)                  | Approved by    | _____ |
| Related SOPs      | SOP-QA-040, SOP-QA-041, SOP-MD-003, SOP-MD-004 |                |       |

## 1 Objective

Collect and analyse post-market data according to an approved plan, report periodically (PMS report or periodic safety update report depending on class), and feed the risk file, clinical evaluation and labelling.

## 2 Scope

All devices on the market.

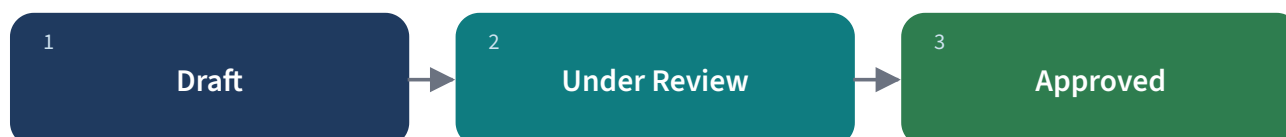
## 3 Responsibilities

| Role (software group)                    | Responsibility                                             |
|------------------------------------------|------------------------------------------------------------|
| <b>PMS lead / responsible person</b>     | Writes the plan and reports; cannot approve own documents. |
| <b>Regulatory Affairs / Manager</b>      | Approves; records notified body submissions.               |
| <b>Complaint and vigilance functions</b> | Provide data (SOP-QA-040, SOP-QA-041).                     |

## 4 Definitions

| Term                     | Definition                                                                                                                                                                                                                               |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PMS plan elements</b> | Information sources, proactive collection, indicators and thresholds, analysis methods, communication methods, referenced procedures, corrective action process, traceability tools, PMCF plan or justification (Annex III section 1.1). |
| <b>Periodic report</b>   | Type and interval set on the device/plan; created per period with <i>Open Reporting Period</i> .                                                                                                                                         |

## 5 Workflow in the software



Other outcomes



Model: ls.md.technical\_file

Figure 153 — Lifecycle of a PMS plan / periodic report.

## 6 Procedure

- 1 **Plan.** In **Surveillance Plans** create the plan with all Annex III elements, PMCF plan reference or justification, report type and interval, plan review frequency; submit and approve.
- 2 **Open the period.** Use **Open Reporting Period** to create the periodic reports due.
- 3 **Report.** Fill the report: sales volume, user population, complaints, serious and non-serious incidents, FSCA, trend signal (with description), analysis summary, actions, updates required (risk file, clinical evaluation, labelling/IFU), benefit-risk determination and conclusion.
- 4 **Approve.** Submit for review (analysis summary required); approve (not the author).
- 5 **Submit.** When required, send it to the notified body and click **Record Notified Body Submission**.
- 6 **Act.** Implement the updates flagged in the report (SOP-MD-003, SOP-MD-004, SOP-QA-030); monitor **Monitoring**  
► **Overdue Post-Market Reports.**

### 7. Controls enforced by the software

- ✓ Approving a plan or report requires the Regulatory Affairs or Manager access level; the responsible person/author cannot approve.
- ✓ The plan must contain every Annex III section 1.1 element, and a PMCF plan reference or a justification for its absence.
- ✓ Report periods of the same device cannot overlap; submission requires the analysis summary; approval requires the benefit-risk determination and conclusion; a trend signal requires its description; only approved reports can be recorded as submitted to the notified body.
- ✓ Approval requires the Regulatory Affairs or Manager access level; the author/responsible person cannot approve (segregation of duties).
- ✓ Only a draft can be submitted; only a record under review can be approved or returned to draft; approved records are frozen — create a new version; non-draft records cannot be deleted (cancel instead).

### 8. Good practices

- ★ Define trend thresholds numerically in the plan (e.g. complaint rate per 10 000 units).
- ★ Reconcile incident counts with the vigilance records before approval.

### 9. Pitfalls to avoid

- ✗ Reports written from memory rather than from complaint and incident data.

## 10 Records and evidence

- Approved PMS plan and periodic reports
- Notified body submission references

## 11 Performance indicators

- Overdue periodic reports (target 0)
- Trend signals and their follow-up

## 12 References

- MDR Articles 83 to 86 and Annex III — Post-market surveillance, PMS report and PSUR
- MDR Articles 87 to 92 — Vigilance
- ISO 13485:2016 — Medical devices — Quality management systems

### 13. Demonstration

**Case.** CAT-14 (class IIa): PSUR every 2 years. The 2027–2028 report: 1.2 million units, 38 complaints, 2 serious incidents reported, no FSCA, no trend signal; risk file update not required; benefit-risk favourable. Approved and made available to the notified body.

# Medical plastics

ls\_medical\_plastics

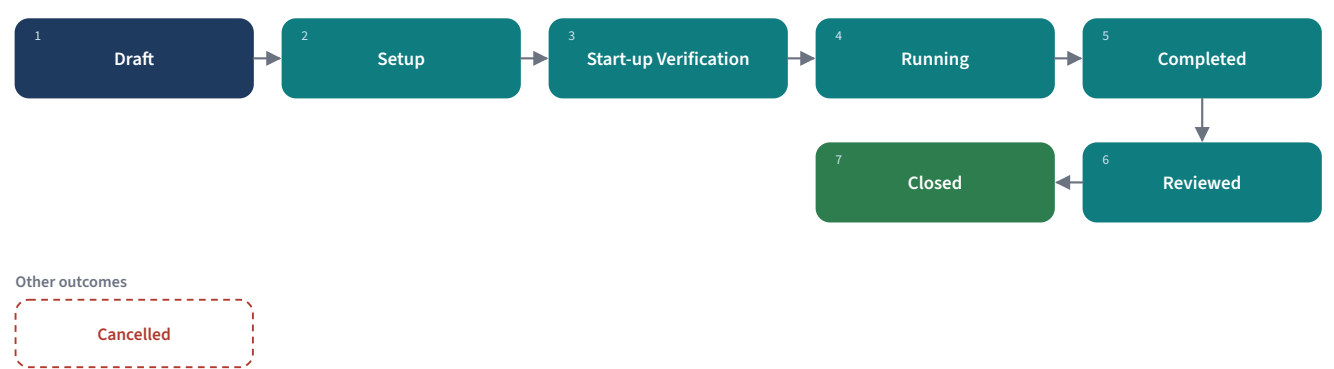
**ls\_medical\_plastics** supports injection moulding of primary packaging and device components: components with approved material grades, qualified material grades, moulds (tools) with a cavity register, shot counters and preventive maintenance by shots and by time, versioned moulding parameter specifications (process windows) approved by people other than the author, moulding runs with setup, start-up verification, immutable parameter readings, material consumption, scrap by cavity and independent review, and a traceability enquiry by lot, material lot, component or tool.

| Menu                                                                                          | Content       |
|-----------------------------------------------------------------------------------------------|---------------|
| Medical Plastics ▶ Products ▶ Components / Packaging                                          | Components    |
| Medical Plastics ▶ Tools ▶ Tool Register / Maintenance Records / Due For Maintenance          | Tools         |
| Medical Plastics ▶ Production ▶ Injection Moulding / Moulding Parameters / Parameter Readings | Production    |
| Medical Plastics ▶ Reports ▶ Moulding Analysis / Traceability Enquiry                         | Reports       |
| Medical Plastics ▶ Configuration ▶ Material Grades / Scrap Reasons                            | Configuration |

| Group                                | Main rights                           |
|--------------------------------------|---------------------------------------|
| Medical Plastics / Viewer · Operator | Read; run production, record readings |
| Medical Plastics / Tool Technician   | Tool maintenance                      |
| Medical Plastics / Process Engineer  | Parameter specifications, run review  |
| Medical Plastics / Manager           | Approvals and configuration           |



Figure 154 — Maintenance triggered by shots or time (illustrative).



Model: ls.mp.injection\_molding

Figure 155 — Lifecycle of a moulding run.

# Qualification and commissioning of moulds

|                   |                                          |                |       |
|-------------------|------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                      | Version        | _____ |
| Menu              | Medical Plastics ► Tools ► Tool Register | Effective date | _____ |
| Owner (suggested) | Tooling / Process Engineering Manager    | Approved by    | _____ |
| Related SOPs      | SOP-MP-002, SOP-MP-003, SOP-VAL-002      |                |       |

## 1 Objective

Register each mould with its cavities and the components it produces, qualify it before first production, and place it in service only when qualified with at least one active cavity.

## 2 Scope

Injection moulds, family moulds, inserts, hot runners, core pin sets and fixtures used for GMP components.

## 3 Responsibilities

| Role (software group)   | Responsibility                                               |
|-------------------------|--------------------------------------------------------------|
| <b>Tool Technician</b>  | Registers the tool and cavity register, records maintenance. |
| <b>Process Engineer</b> | Qualifies the tool (IQ/OQ/PQ as defined by validation).      |
| <b>Manager</b>          | Places in service, quarantines, decommissions.               |

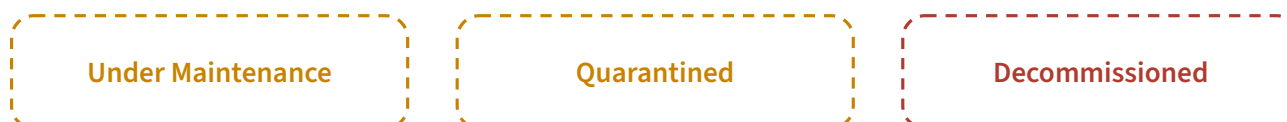
## 4 Definitions

| Term                            | Definition                                                                       |
|---------------------------------|----------------------------------------------------------------------------------|
| <b>Cavity register</b>          | One record per cavity number with marking and status (Active, Blocked, Removed). |
| <b>Requalification interval</b> | Months after which the tool must be requalified.                                 |

## 5 Workflow in the software



### Other outcomes



Model: ls.mp.tool

Figure 156 — Lifecycle of a mould.

## 6 Procedure

- 1 **Register.** In **Tool Register** click **New** : name, code, type, tool maker, serial number, acquisition date, steel, storage location, default work centre, components produced, cavity count, opening shot count, maintenance intervals (shots, months), requalification interval.
- 2 **Cavities.** Check the cavity register (number and marking of each cavity).
- 3 **Qualify.** Perform the tool qualification (dimensional study per cavity, process capability) under SOP-VAL-002; record the qualification date and click **Qualify** .
- 4 **Place in service.** Click **Place In Service** (at least one active cavity required).
- 5 **Quarantine / decommission.** Use **Quarantine** when the tool is suspect and **Decommission** at end of life (no run in progress).
- 6 **Requalify.** Monitor *Requalification Due*; requalify after major repair or modification.

### 7. Controls enforced by the software

- ✓ Qualification and placing in service require a qualification date; transitions are allowed only from the proper states; a tool without active cavity cannot be placed in service.
- ✓ The cavity count cannot be reduced below a registered cavity number (parts remain traceable to that cavity).
- ✓ A tool with runs in progress cannot be sent to maintenance, quarantined or decommissioned.

### 8. Good practices

- ★ Engrave cavity numbers on the parts so that scrap and complaints can be traced to the cavity.
- ★ Store tools with their last-shot samples.

### 9. Pitfalls to avoid

- ✗ Moving a tool to another press without checking the specification work centre restriction.

## 10 Records and evidence

- Tool record, cavity register, qualification evidence

## 11 Performance indicators

- Tools in service past requalification (target 0)

## 12 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance
- EU GMP Annex 15 (2015) — Qualification and Validation

### 13. Demonstration

**Case.** Mould T-032 (8 cavities) for syringe plunger SP-5: registered with maintenance interval 250 000 shots or 6 months; OQ/PQ done with a dimensional study of all 8 cavities; qualified on 12 May; placed in service.

# Preventive maintenance of moulds (by shot count and by time) and cavity blocking

|                   |                                                                      |                |       |
|-------------------|----------------------------------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                                                  | Version        | _____ |
| Menu              | Medical Plastics ► Tools ► Due For Maintenance / Maintenance Records | Effective date | _____ |
| Owner (suggested) | Tooling Manager                                                      | Approved by    | _____ |
| Related SOPs      | SOP-MP-001, SOP-GEN-003                                              |                |       |

## 1 Objective

Maintain moulds before wear affects parts, by whichever of the shot-count and time intervals comes first, and block faulty cavities with a recorded reason so that production continues on good cavities only.

## 2 Scope

All moulds in service.

## 3 Responsibilities

| Role (software group)   | Responsibility                                         |
|-------------------------|--------------------------------------------------------|
| <b>Tool Technician</b>  | Performs maintenance and records findings and actions. |
| <b>Process Engineer</b> | Decides on cavity blocking and requalification.        |
| <b>Operator</b>         | Reports defects by cavity.                             |

## 4 Definitions

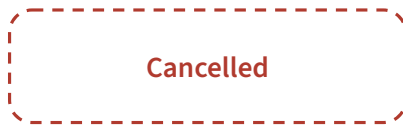
| Term                               | Definition                                                                                  |
|------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Maintenance status</b>          | OK, Due Soon, Overdue, from shots since last maintenance and months since last maintenance. |
| <b>Resets maintenance baseline</b> | Flag on a maintenance event that resets the shot counter baseline.                          |



## 5 Workflow in the software



Other outcomes



Model: *ls.mp.tool.maintenance*

Figure 157 — Lifecycle of a maintenance event.



Preventive maintenance is triggered by shot count or elapsed time, whichever comes first.

Figure 158 — Shots since last maintenance against the interval.

## 6 Procedure

- 1 **Monitor.** Daily, open **Tools ► Due For Maintenance**.
- 2 **Plan.** Create a maintenance record: tool, type (preventive, corrective, cleaning, repair, modification, requalification), planned date.
- 3 **Stop the tool.** When runs are finished, click **Send To Maintenance** on the tool.
- 4 **Perform.** Click **Start** on the maintenance; record shot count at intervention, performer or provider, findings, actions taken, cavities affected, *Requalification Required*, *Resets Maintenance Baseline*; click **Complete**.
- 5 **Return to service.** Click **Return To Service** on the tool (wizard); requalify first when required.
- 6 **Block a cavity.** When a cavity produces defects, click **Block** on the cavity with a reason; **Unblock** after repair and verification.

### 7. Controls enforced by the software

- ✓ A maintenance cannot be completed without recording the actions taken; only draft maintenance can start; completed maintenance cannot be cancelled or deleted (tool history).
- ✓ A cavity cannot be blocked or removed without a recorded reason; only active cavities can be blocked.
- ✓ A tool with runs in progress cannot be sent to maintenance.

### 8. Good practices

- ★ Keep the last parts of the run before maintenance and the first parts after it for comparison.
- ★ Trend corrective maintenance per tool to adjust the preventive interval.

## 9. Pitfalls to avoid

- ✗ Blocking a cavity physically without recording it — parts counts and traceability become wrong.

## 10 Records and evidence

- Maintenance records
- Cavity status history

## 11 Performance indicators

- Tools overdue for maintenance (target 0)
- Corrective / preventive ratio

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- ISO 13485:2016 — Medical devices — Quality management systems

## 13. Demonstration

**Case.** T-032 reaches 236 000 shots (Due Soon). After the current run the technician sends it to maintenance, cleans vents and replaces ejector pins, records the shot count, ticks *Resets Maintenance Baseline*, completes. Cavity 6 showed flash: the engineer blocks it with reason *flash on parting line*; production continues on 7 cavities until repair.

# Process windows: drafting, review and approval

|                   |                                                     |                |       |
|-------------------|-----------------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                                 | Version        | _____ |
| Menu              | Medical Plastics ► Production ► Moulding Parameters | Effective date | _____ |
| Owner (suggested) | Process Engineering Manager                         | Approved by    | _____ |
| Related SOPs      | SOP-MP-004, SOP-QA-030, SOP-VAL-006                 |                |       |

## 1 Objective

Define for each component/tool combination the validated process window (parameters with target, minimum and maximum, critical parameters and monitoring frequency), approved through author–reviewer–approver separation.

## 2 Scope

All moulding parameter specifications used in production.

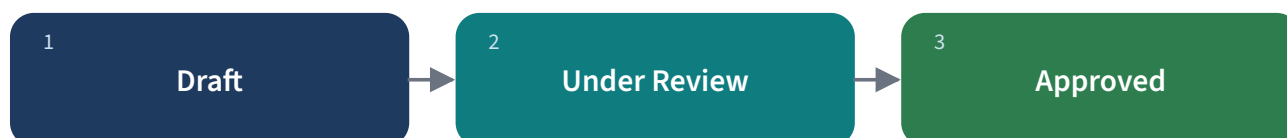
## 3 Responsibilities

| Role (software group)            | Responsibility                                         |
|----------------------------------|--------------------------------------------------------|
| <b>Author (Process Engineer)</b> | Writes the specification from process validation data. |
| <b>Reviewer</b>                  | Reviews (not the author).                              |
| <b>Approver</b>                  | Approves (neither author nor reviewer).                |

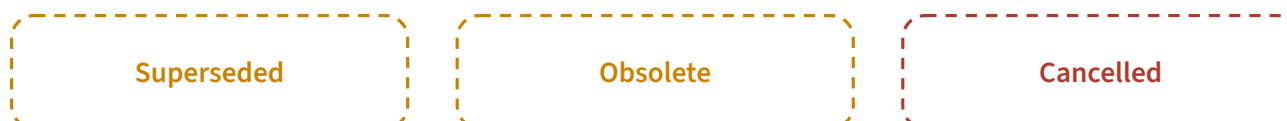
## 4 Definitions

| Term                        | Definition                                                                                                            |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Critical parameter</b>   | Parameter whose deviation may affect quality; recording is mandatory and out-of-tolerance at start-up blocks the run. |
| <b>Monitoring frequency</b> | At setup only, at start-up verification, hourly, once per shift, once per lot, continuous.                            |

## 5 Workflow in the software



### Other outcomes



Model: ls.mp.molding\_parameter

Figure 159 — Lifecycle of a parameter specification.

## Procedure

- 1 **Create.** In **Moulding Parameters** click **New** : component, tool, work centre (optional restriction), version, effective from, periodic review interval; for versions after the first, the reason for change.
- 2 **Add parameters.** For each parameter: name, code, unit, value type (numeric or qualitative), target, minimum, maximum (or expected value), critical, monitoring frequency, recording required.
- 3 **Submit.** Click **Submit For Review**.
- 4 **Review.** The reviewer clicks **Record Review** (or **Return To Draft**).
- 5 **Approve.** The approver clicks **Approve**; the previous approved version is superseded.
- 6 **Change.** Use **New Version** under change control; **Set Obsolete** when the combination is no longer produced (no run in progress).

## 7. Controls enforced by the software

- ✓ The tool must be declared as producing the component; one approved specification per component and tool — supersede before approving another.
- ✓ A specification needs at least one parameter; later versions need a reason for change.
- ✓ Author ≠ reviewer ≠ approver; approval requires review.
- ✓  $\text{Min} \leq \text{target} \leq \text{max}$ ; qualitative parameters need an expected value; critical parameters must have recording enabled.
- ✓ Approved specifications cannot be cancelled (make obsolete); a specification used by runs in progress cannot be made obsolete; non-draft or referenced specifications cannot be deleted.

## 8. Good practices

- ★ Derive windows from the process validation (DoE, OQ/PQ), not from habits.
- ★ Keep the number of critical parameters small and meaningful.

## 9. Pitfalls to avoid

- ✗ Widening a window after a failed run without change control.

## 10 Records and evidence

- Approved specifications with versions

## 11 Performance indicators

- Specifications past periodic review

## 12 References

- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance
- ISO 13485:2016 — Medical devices — Quality management systems
- EU GMP Annex 15 (2015) — Qualification and Validation

## 13. Demonstration

**Case.** SP-5 on T-032, v2: melt temperature 230 °C (220–240, critical, hourly), injection pressure 850 bar (800–900, critical, continuous), cooling time 8 s (7–9), visual aspect *no flash* (qualitative, once per lot). Author and reviewer from process engineering, approver from QA; v1 superseded.

# Start-up, operation and closure of a moulding run

|                   |                                                    |                |       |
|-------------------|----------------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                                | Version        | _____ |
| Menu              | Medical Plastics ► Production ► Injection Moulding | Effective date | _____ |
| Owner (suggested) | Moulding Production Manager                        | Approved by    | _____ |
| Related SOPs      | SOP-MP-003, SOP-MP-005, SOP-MP-006, SOP-QA-020     |                |       |

## 1 Objective

Run production only with a released component, a tool in service and an approved specification; verify start-up parameters before running; record readings, material and scrap; have the run reviewed by someone other than its operator or setter before closure.

## 2 Scope

All injection moulding runs of GMP components.

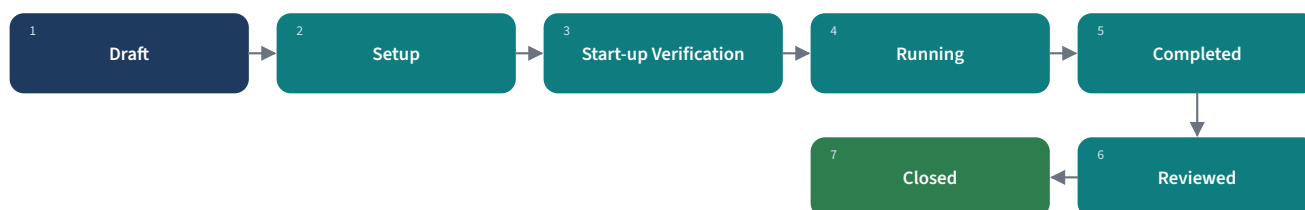
## 3 Responsibilities

| Role (software group)                   | Responsibility                                     |
|-----------------------------------------|----------------------------------------------------|
| <b>Setter</b>                           | Sets up the tool and press.                        |
| <b>Operator</b>                         | Runs production, records readings and scrap.       |
| <b>Reviewer (Process Engineer / QA)</b> | Reviews the run; cannot be the operator or setter. |

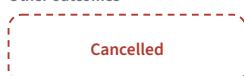
## 4 Definitions

| Term                         | Definition                                                                                                      |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Start-up verification</b> | Readings of every parameter required at start-up, within tolerance for critical ones, before the run can start. |
| <b>Shot counter</b>          | Tool counter at start and end; shots = end – start.                                                             |

## 5 Workflow in the software



Other outcomes



Model: ls.mp.injection\_molding

Figure 160 — Lifecycle of a moulding run.

## 6 Procedure

- 1 **Create the run.** In **Injection Moulding** click **New** : component, tool, work centre, manufacturing order, produced lot, operator, shift, shot counter at start. The approved specification is proposed.
- 2 **Setup.** Click **Start Setup** ; the setter records setup readings ( **Record Readings** ); click **Setup Complete** .
- 3 **Start-up verification.** Record start-up verification readings for the required parameters and the material consumption (grade, lot, quantity); click **Confirm Start-up and Run** .
- 4 **Run.** Record in-process readings at the specified frequency; record scrap by reason and cavity; block cavities if needed (SOP-MP-002).
- 5 **Complete.** Record the end shot counter, quantities produced and rejected; click **Complete Run** .
- 6 **Deviations.** If readings were out of tolerance, open a deviation (SOP-QA-020) and record its reference.
- 7 **Review.** The reviewer (not operator nor setter) clicks **Review** , or **Return For Correction** with a comment.
- 8 **Close.** Click **Close Run** .

### 7. Controls enforced by the software

- ✓ Production requires a released component, a tool in service declared for the component, and an approved specification for this component/tool (and work centre if restricted).
- ✓ Start-up verification must be complete; critical parameters out of tolerance at start-up block the run; material consumption must be recorded before running.
- ✓ Completion requires the end shot counter ( $\geq$  start); rejected  $\leq$  produced.
- ✓ Review requires completion, a reviewer who is neither operator nor setter, and a deviation reference if readings were out of tolerance; closure requires review.
- ✓ Readings cannot be captured outside setup/start-up/running; closed runs are frozen; runs past draft cannot be deleted; cancellation requires a reason.

### 8. Good practices

- ★ Record readings at the time they are taken — the timestamp is part of the evidence.
- ★ Quarantine parts produced between the last good reading and an out-of-tolerance one.

### 9. Pitfalls to avoid

- ✗ Starting production "while waiting" for the start-up verification.

## 10 Records and evidence

- Run with readings, material, scrap, review

## 11 Performance indicators

- Reject rate per run
- Runs with out-of-tolerance readings
- Runs returned for correction

## 12 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- GHTF/SG3/N99-10:2004 — Quality Management Systems — Process Validation Guidance

### 13. Demonstration

**Case.** Run MR-2027-118: SP-5 on T-032 (7 active cavities), lot SP5-2711. Setup and start-up readings within limits; material PP grade lot R-5531 recorded; 7 in-process readings; injection pressure once at 905 bar (max 900) for 10 min → parts of that period quarantined, deviation DEV/2027/00012 recorded. Completed: 96 600 shots, 676 200 parts, 1 900 rejects. Reviewed by the process engineer and closed.

# Correction of parameter readings (without altering the original value)

|                   |                                                    |                |       |
|-------------------|----------------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                                | Version        | _____ |
| Menu              | Medical Plastics ► Production ► Parameter Readings | Effective date | _____ |
| Owner (suggested) | Moulding Production Manager with QA                | Approved by    | _____ |
| Related SOPs      | SOP-MP-004, SOP-DI-002                             |                |       |

## 1 Objective

Correct an erroneous parameter reading by a new, justified reading that supersedes the original, so that the original value, the correction and its reason all remain visible.

## 2 Scope

All parameter readings of moulding runs.

## 3 Responsibilities

| Role (software group)    | Responsibility                                  |
|--------------------------|-------------------------------------------------|
| <b>Operator / setter</b> | Records the correcting reading with its reason. |
| <b>Reviewer</b>          | Checks corrections at run review.               |

## 4 Definitions

| Term                       | Definition                                                                                                                            |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Superseding reading</b> | New reading linked to the reading it corrects (same run, same parameter), with a mandatory reason; the original is marked superseded. |

## 5 Procedure

- 1 Detect.** An erroneous reading is noticed (typing error, wrong instrument, wrong parameter).
- 2 Record the correction.** Use **Record Readings** on the run, select the parameter, enter the correct value, select the reading it supersedes and write the reason for correction.
- 3 Never edit.** Do not try to modify or delete the original reading — the system refuses it.
- 4 Review.** At run review, check each correction: reason plausible, time close to the original, same person or supervisor.

### 6. Controls enforced by the software

- ✓ Readings cannot be modified or deleted.
- ✓ A correcting reading must record a reason, belong to the same run and address the same parameter as the reading it supersedes.
- ✓ Qualitative parameters require an observed value; no reading on a run that is not open for readings.



## 7. Good practices

- ★ Explain the correction factually ("typed 285 instead of 258, see press display photo").

## 8. Pitfalls to avoid

- ✗ Using corrections to bring out-of-tolerance values back into range without evidence — a data integrity breach.

## 9 Records and evidence

- Original and superseding readings with reasons

## 10 Performance indicators

- Corrections per 100 readings
- Corrections without evidence found at review

## 11 References

- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- MHRA — GxP Data Integrity Guidance and Definitions (2018)
- EU GMP Annex 11 — Computerised Systems (check the revision in force)

## 12. Demonstration

**Case.** The operator types a cooling time of 80 s instead of 8.0 s. Two minutes later she records a new reading 8.0 s superseding the first, reason "*decimal point omitted — press display shows 8.0 s*". At review the engineer sees both readings and accepts the correction.

# Material traceability

|                   |                                                                                     |                |       |
|-------------------|-------------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_medical_plastics                                                                 | Version        | _____ |
| Menu              | Medical Plastics ► Configuration ► Material Grades · Reports ► Traceability Enquiry | Effective date | _____ |
| Owner (suggested) | Moulding Quality Manager                                                            | Approved by    | _____ |
| Related SOPs      | SOP-MP-004, SOP-QA-070, SOP-QA-081                                                  |                |       |

## 1 Objective

Use only qualified material grades approved for each component, record the material lots consumed by each run, and be able to trace forward (material lot → parts) and backward (part lot → material, tool, cavity, run) within minutes.

## 2 Scope

All resins, masterbatches and colorants used for GMP components.

## 3 Responsibilities

| Role (software group)      | Responsibility                                              |
|----------------------------|-------------------------------------------------------------|
| Quality / Process Engineer | Qualify material grades and approve them for components.    |
| Operators                  | Record material lots at start-up and at each change.        |
| Quality                    | Performs traceability enquiries for complaints and recalls. |

## 4 Definitions

| Term                 | Definition                                                                                             |
|----------------------|--------------------------------------------------------------------------------------------------------|
| Qualification status | Draft, Under Evaluation, Qualified, Conditionally Qualified (conditions recorded), Rejected, Obsolete. |
| Direct drug contact  | Grade used for parts in contact with the drug; a lot reference is mandatory at consumption.            |

## 5 Procedure

- 1 Qualify the grade.** In **Material Grades**: polymer type, resin manufacturer, approved suppliers, compendial references, master file, change notification agreement, properties; **Start Evaluation**, then **Qualify** (or **Qualify Conditionally** with conditions) with the qualification date and requalification due date.
- 2 Approve for components.** On the component list the approved material grades and the primary grade; release the component (**Release**).
- 3 Record consumption.** On each run, record material grade, lot and quantity before start-up and at every lot change.
- 4 Trace.** Use **Reports ► Traceability Enquiry** by produced lot, material lot, component and period, or tool and period; **Open Results** or **Print Records**.
- 5 Change of material.** A resin change (grade, supplier, site) goes through change control and requalification.

## 6. Controls enforced by the software

- ✓ A grade cannot be qualified without a qualification date; conditional qualification requires recorded conditions; requalification date after qualification.
- ✓ Only qualified grades approved for the component can be consumed; a direct-drug-contact grade requires a lot reference; the lot must belong to the product.
- ✓ A component cannot be released without at least one approved grade; the primary grade must be among the approved grades; sterile components need a sterilisation method.

## 7. Good practices

- ★ Obtain a change notification agreement from resin suppliers.
- ★ Test the traceability enquiry during mock recalls (SOP-QA-083).

## 8. Pitfalls to avoid

- ✗ Mixing two resin lots in the dryer without recording both.

## 9 Records and evidence

- Material grades with qualification
- Material consumption per run
- Traceability enquiry outputs

## 10 Performance indicators

- Time to answer a traceability enquiry
- Runs without material record (target 0)

## 11 References

- ISO 13485:2016 — Medical devices — Quality management systems
- ISO 15378:2017 — Primary packaging materials for medicinal products
- ISO 10993-1 — Biological evaluation of medical devices

## 12. Demonstration

**Case.** A resin supplier notifies a contamination risk on lot R-5531. The quality manager runs the enquiry *By Material Lot*: 3 runs, lots SP5-2711 to SP5-2713, all on T-032. The lots are blocked and a deviation opened within 1 hour.

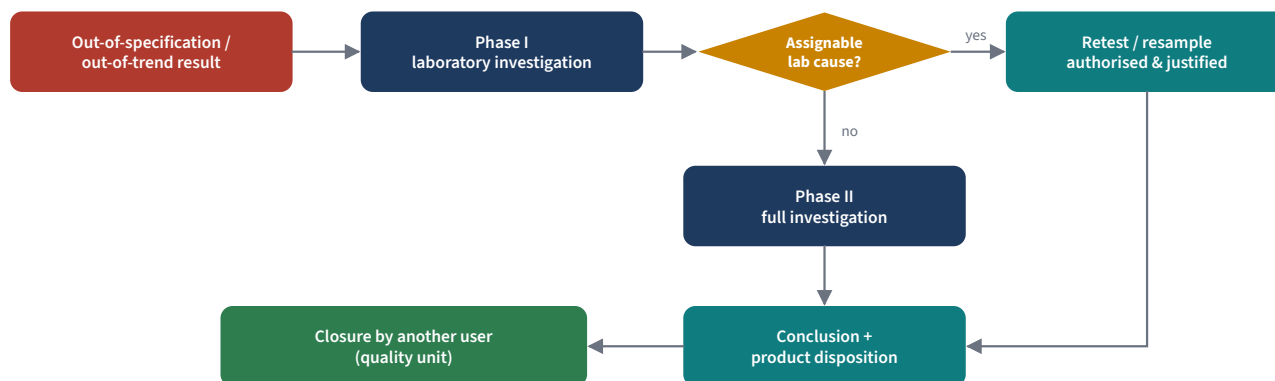
# Laboratory

ls\_lab

**ls\_lab** supports the quality control laboratory: controlled test methods and product specifications (draft, review, approval, versions), samples from receipt to reporting, results with automatic evaluation and second-person review, OOS/OOT investigations in two phases with retest and resample authorisation, stability studies with time points and pull samples, and certificates of analysis. Approvals and certificate issue record a *signature intent* (user, time, meaning), which the module states is **not** a 21 CFR Part 11 electronic signature. Lots held by an open investigation cannot be delivered.

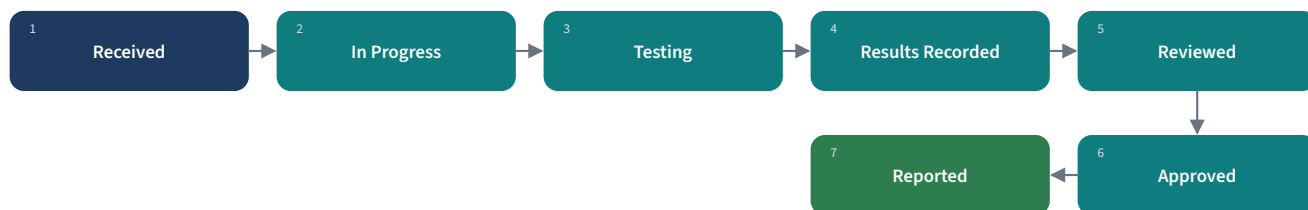
| Menu                                                                         | Content                           |
|------------------------------------------------------------------------------|-----------------------------------|
| Laboratory ► Tests ► Test Methods / Test Results                             | Methods and results               |
| Laboratory ► Specifications ► Product Specifications / OOS / OOT             | Specifications and investigations |
| Laboratory ► Samples ► Sample Receipt / Sample Register                      | Samples                           |
| Laboratory ► Stability ► Stability Studies / Time Points / Stability Samples | Stability                         |
| Laboratory ► Reports ► Certificates of Analysis / Test Result Analysis       | Reporting                         |

| Group                       | Main rights                                                    |
|-----------------------------|----------------------------------------------------------------|
| Laboratory Viewer · Analyst | Read; perform tests and record results                         |
| Laboratory Reviewer         | Review results and samples (not their own)                     |
| Laboratory Manager          | Approve methods, specifications, samples; close investigations |

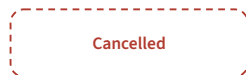


*The lot stays blocked for delivery while the investigation is open (control on stock.move.line).*

Figure 161 — OOS investigation phases.



Other outcomes



Model: *ls.lab.sample*

Figure 162 — Lifecycle of a laboratory sample.

# Approval of test methods and specifications

|                   |                                                                             |                |       |
|-------------------|-----------------------------------------------------------------------------|----------------|-------|
| Module            | ls_lab                                                                      | Version        | _____ |
| Menu              | Laboratory ► Tests ► Test Methods · Specifications ► Product Specifications | Effective date | _____ |
| Owner (suggested) | QC Manager                                                                  | Approved by    | _____ |
| Related SOPs      | SOP-VAL-006, SOP-QC-012, SOP-QA-030                                         |                |       |

## 1 Objective

Use only approved, validated test methods and approved specifications with explicit acceptance criteria, and control their versions.

## 2 Scope

Test methods and specifications for raw materials, packaging, in-process, finished product, stability and water.

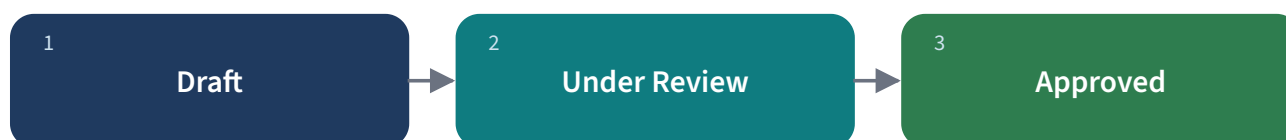
## 3 Responsibilities

| Role (software group)          | Responsibility                          |
|--------------------------------|-----------------------------------------|
| <b>Analyst / QC specialist</b> | Drafts methods and specifications.      |
| <b>Laboratory Manager</b>      | Approves; makes obsolete with a reason. |

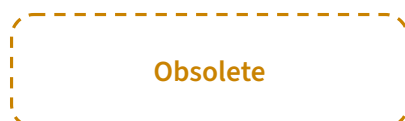
## 4 Definitions

| Term                            | Definition                                                                                                                                 |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Method validation status</b> | Not Required, Planned, In Progress, Validated, Revalidation Due (with validation reference).                                               |
| <b>Criterion type</b>           | Between Min and Max, Not Less Than, Not More Than, Text Match, Pass/Fail, Informative Only; <i>Mandatory</i> and <i>Show on CoA</i> flags. |

## 5 Workflow in the software



Other outcomes



Model: *ls.lab.test\_method*

Figure 163 — Lifecycle of a test method and of a specification.

## 6 Procedure

- 1 **Method.** In **Test Methods** click **New**: name, code, version, technique, result type, unit, decimals, reference document, procedure summary, instrument category, validation status and reference. **Submit for Review** then **Approve**.
- 2 **Specification.** In **Product Specifications** click **New**: name, code, version, product, type, effective date; add criteria lines (approved method, criterion type and limits, unit, mandatory, show on CoA). **Submit for Review** then **Approve**.
- 3 **Revise.** Use **Create Revision** for any change; the new version supersedes the old when approved.
- 4 **Obsolete.** Click **Set Obsolete** with the obsolescence reason; only one approved specification per product and type.

### 7. Controls enforced by the software

- ✓ A specification without criteria cannot be approved; only approved methods can be used in a specification;  $\text{min} \leq \text{max}$ ; a text criterion needs the expected text.
- ✓ Only one approved specification per product and type — make the old one obsolete first.
- ✓ Approved records are frozen (blocked fields listed); criteria of approved specifications cannot change; obsolescence requires a reason; only draft records can be deleted.

### 8. Good practices

- ★ Link each method to its validation report (SOP-VAL-006) before approval.
- ★ Align criteria with the registered specification (marketing authorisation, pharmacopoeia).

### 9. Pitfalls to avoid

- ✗ Approving a method whose validation is still *In Progress* without a documented justification.

## 10 Records and evidence

- Approved methods and specifications with versions

## 11 Performance indicators

- Methods with revalidation due
- Specifications revised per year

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ICH Q2(R2) — Validation of Analytical Procedures

### 13. Demonstration

**Case.** Specification *Tablets 10 mg — finished product v4*: assay 95.0–105.0 % (HPLC method AM-012 v3, validated), dissolution NLT 80 % (Q) at 30 min, related substances NMT 0.5 %, appearance text match. Approved; v3 made obsolete with reason "*new impurity limit per variation 2027-02*".

# Receipt and registration of samples

|                   |                                       |                |       |
|-------------------|---------------------------------------|----------------|-------|
| Module            | ls_lab                                | Version        | _____ |
| Menu              | Laboratory ► Samples ► Sample Receipt | Effective date | _____ |
| Owner (suggested) | QC Manager                            | Approved by    | _____ |
| Related SOPs      | SOP-QC-012, SOP-QC-010                |                |       |

## 1 Objective

Register every sample received by the laboratory with its identity, specification and storage, so that it can be tested against the right criteria and traced.

## 2 Scope

Raw material, packaging, in-process, finished product, stability, retain, water and cleaning verification samples.

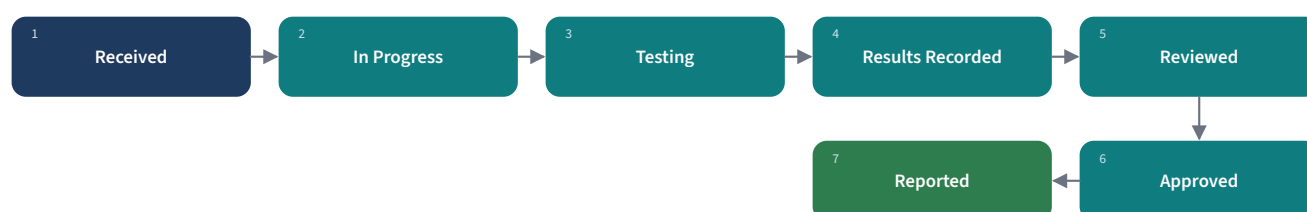
## 3 Responsibilities

| Role (software group)     | Responsibility                                        |
|---------------------------|-------------------------------------------------------|
| <b>Sample receiver</b>    | Checks labelling and condition, registers the sample. |
| <b>Laboratory Manager</b> | Sets priorities and due dates.                        |

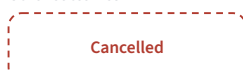
## 4 Definitions

| Term               | Definition                                                                                                       |
|--------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Sample type</b> | Raw Material, Packaging Material, In Process, Finished Product, Stability, Retain, Water, Cleaning Verification. |

## 5 Workflow in the software



Other outcomes



Model: ls.lab.sample

Figure 164 — Sample lifecycle.

## 6 Procedure

- Check.** Check label, container integrity, quantity and storage condition on arrival; refuse or annotate non-conforming samples.
- Register.** In **Sample Receipt** click **New**: sample type, product, lot, batch reference, approved specification, quantity, sampling date, receipt date, due date, priority, sampled by, sampling point, storage condition.



- 3 **Start.** Click **Start** to accept the sample into the laboratory; test result lines are prepared from the specification.
- 4 **Store.** Store the sample in the recorded condition until testing.
- 5 **Cancel.** If the sample cannot be tested, use **Cancel** with a reason — never delete.

#### 7. Controls enforced by the software

- ✓ The specification must be approved; the lot must belong to the product.
- ✓ State transitions are enforced; cancelled samples keep their reason; records that left draft cannot be deleted.

#### 8. Good practices

- ★ Label the container with the system sample reference.
- ★ Record the actual receipt time for time-sensitive samples (microbiology, water).

#### 9. Pitfalls to avoid

- ✗ Testing a sample against the specification of another product version.

### 10 Records and evidence

- Sample register

### 11 Performance indicators

- Samples received vs tested on time
- Samples cancelled

### 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories

#### 13. Demonstration

**Case.** Finished product sample of batch 26T018 received: 60 tablets, specification *Tablets 10 mg — FP v4* (approved), priority normal, due in 5 working days; registered and started.

# Entry, review and approval of results

|                   |                                                               |                |       |
|-------------------|---------------------------------------------------------------|----------------|-------|
| Module            | ls_lab                                                        | Version        | _____ |
| Menu              | Laboratory ► Tests ► Test Results · Samples ► Sample Register | Effective date | _____ |
| Owner (suggested) | QC Manager                                                    | Approved by    | _____ |
| Related SOPs      | SOP-QC-011, SOP-QC-013, SOP-QC-015, SOP-DI-002                |                |       |

## 1 Objective

Record each result contemporaneously with its analyst, date and instrument, have it reviewed by a second analyst, and approve the sample only when all mandatory results are reviewed and no investigation is open.

## 2 Scope

All results of laboratory samples.

## 3 Responsibilities

| Role (software group)                | Responsibility                                  |
|--------------------------------------|-------------------------------------------------|
| <b>Analyst</b>                       | Performs the test and records the result.       |
| <b>Laboratory Reviewer</b>           | Reviews results (not their own) and the sample. |
| <b>Laboratory Manager (approver)</b> | Approves the sample (not the reviewer).         |

## 4 Definitions

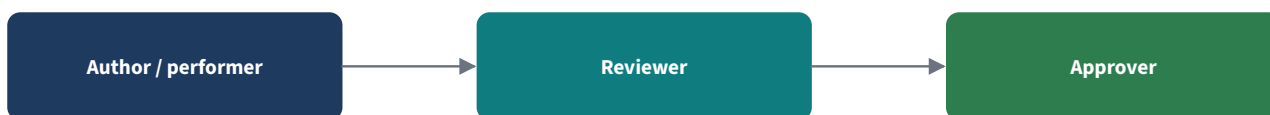
| Term                | Definition                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------|
| <b>Evaluation</b>   | Pending, Conforms, Out Of Specification, Informative — computed against the specification line. |
| <b>Out of trend</b> | Flag set by the analyst/reviewer with a mandatory justification.                                |

## 5 Workflow in the software



Model: *ls.lab.test\_result*

Figure 165 — Lifecycle of a test result.



The software refuses the same person in two roles on one record wherever the rule exists  
(e.g. result review, SOP approval, batch release, excursion closure, protocol approval).

Figure 166 — Analyst, reviewer and approver are different people.

## 6 Procedure

- 1 **Test.** Click **Start Testing** on the sample.
- 2 **Record.** For each result line enter the numeric, text or pass/fail result, analyst, test date, instrument reference, remarks; click **Record Result**. Attach raw data or reference it.
- 3 **OOS / OOT.** If the evaluation is *Out Of Specification* or the result is out of trend, open an investigation (SOP-QC-013) — do not retest on your own.
- 4 **Confirm.** When all mandatory results are recorded, click **Confirm Results Recorded**.
- 5 **Review.** A reviewer who did not enter results reviews each result (**Review** on the result) and then the sample (**Review**).
- 6 **Approve.** The approver (not the reviewer) clicks **Approve** and confirms the signature intent (meaning *Approved*).
- 7 **Report.** Click **Mark Reported** once the results are communicated or the CoA issued (SOP-QC-015).

### 7. Controls enforced by the software

- ✓ A result cannot be recorded on a sample not open for entry, nor entered twice; a criterion must belong to the sample's specification.
- ✓ The analyst cannot review their own result; the result entrant cannot review the sample; the reviewer cannot approve it.
- ✓ Approval requires every mandatory test recorded and reviewed and no open investigation.
- ✓ Reviewed results cannot be modified (record a retest under an investigation); results with data cannot be deleted; results linked to an investigation cannot be reset.
- ✓ An out-of-trend flag requires a justification; a retest requires a documented retest authorisation.

### 8. Good practices

- ★ Review raw data and the audit trail, not only the value typed (SOP-DI-002).
- ★ Record instrument references to link results with calibration status.

### 9. Pitfalls to avoid

- ✗ Resetting a result to draft to change an inconvenient value.
- ✗ Averaging an OOS result with retest results.

## 10 Records and evidence

- Results with analyst, reviewer, dates
- Sample approval with signature intent

## 11 Performance indicators

- Right-first-time results
- Review turnaround
- OOS rate

## 12 References

- 21 CFR 211.160 to 211.176 — Laboratory controls
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)
- ISO/IEC 17025:2017 — Competence of testing and calibration laboratories

### 13. Demonstration

**Case.** Sample of 26T018: 6 results entered by analyst A; assay 99.2 % (Conforms). Reviewer B reviews each result and the sample; manager C approves (signature intent *Approved*, timestamp recorded); sample marked reported after the CoA is issued.

# Investigation of OOS (out-of-specification) and OOT (out-of-trend) results

|                   |                                                |                |       |
|-------------------|------------------------------------------------|----------------|-------|
| Module            | ls_lab                                         | Version        | _____ |
| Menu              | Laboratory ► Specifications ► OOS / OOT        | Effective date | _____ |
| Owner (suggested) | QC Manager with QA                             | Approved by    | _____ |
| Related SOPs      | SOP-QC-012, SOP-QA-020, SOP-QA-021, SOP-QA-090 |                |       |

## 1 Objective

Investigate every OOS or OOT result in a structured way — laboratory phase first, full investigation when no laboratory cause is found — authorise retests or resamples only with written scientific justification, and decide on the product with QA.

## 2 Scope

All OOS and OOT results of release, stability, raw material and in-process testing.

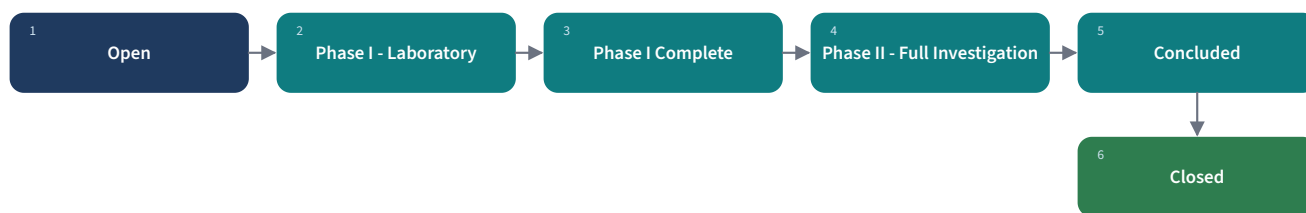
## 3 Responsibilities

| Role (software group)                      | Responsibility                                       |
|--------------------------------------------|------------------------------------------------------|
| <b>Investigator (analyst's supervisor)</b> | Conducts Phase I and Phase II.                       |
| <b>QA / quality unit approver</b>          | Approves closure; must not be the investigator.      |
| <b>Production</b>                          | Contributes to the manufacturing review in Phase II. |

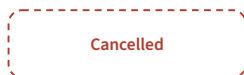
## 4 Definitions

| Term                    | Definition                                                                                                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase I</b>          | Laboratory investigation: analyst interview, calculations, instrument performance, standards and reagents, sample integrity, method adherence.                                  |
| <b>Phase II</b>         | Full investigation including manufacturing review, when Phase I finds no assignable laboratory cause.                                                                           |
| <b>Final conclusion</b> | Assignable Laboratory Cause, Assignable Manufacturing Cause, No Assignable Cause Identified; disposition Reject, Release, Remain In Quarantine, Further Investigation Required. |

## 5 Workflow in the software

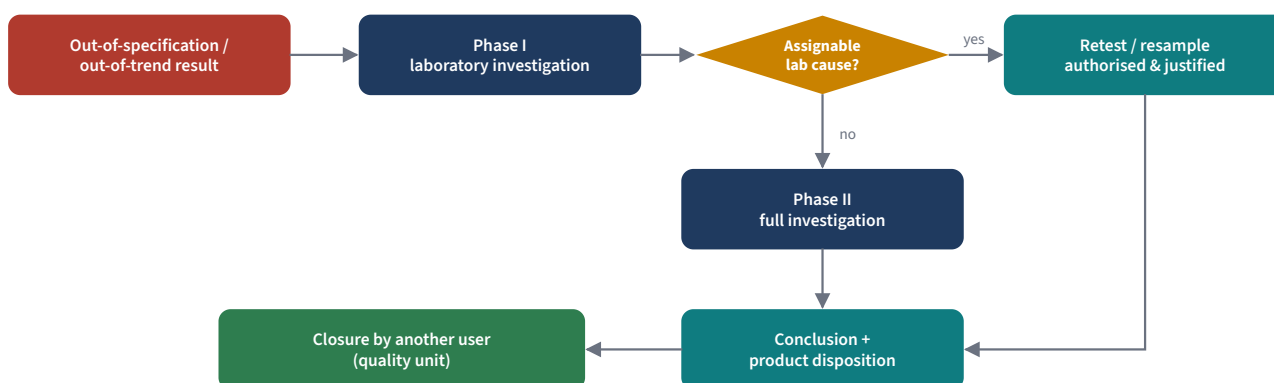


Other outcomes



Model: ls.lab.oos

Figure 167 — Lifecycle of an OOS/OOT investigation.



The lot stays blocked for delivery while the investigation is open (control on stock.move.line).

Figure 168 — Phase I, Phase II, conclusion and closure.

## 6 Procedure

- 1 **Open.** In **Specifications ► OOS / OOT** click **New** : type (OOS or OOT), originating result, investigator, description. The lot is held: deliveries are refused while it is open.
- 2 **Phase I.** Click **Start Phase I** ; tick each check performed (analyst interviewed, calculation verified, instrument performance, standards and reagents, sample integrity, method adherence); record Phase I findings and conclusion; click **Complete Phase I** .
- 3 **Laboratory cause.** If an assignable laboratory cause is identified, invalidate the result with justification and, when justified in writing, click **Authorise Retest** ; the retest is recorded as a retest linked to the investigation.
- 4 **Phase II.** If no laboratory cause, click **Start Phase II** : review manufacturing (batch record, deviations, equipment), determine root cause; resample only with **Authorise Resample** and justification; record Phase II findings; **Complete Phase II** .
- 5 **Conclude.** Click **Conclude** with the final conclusion, product disposition and justification, CAPA reference if any.
- 6 **Close.** A quality unit member who did not conduct the investigation clicks **Close** .

## 7. Controls enforced by the software

- ✓ Investigations cannot be deleted — cancel with a reason.
- ✓ Phase I needs findings and a conclusion; Phase II applies only when Phase I found no laboratory cause, and is mandatory in that case before concluding.
- ✓ Retest and resample require a written scientific justification; a retest result cannot be recorded without authorisation.
- ✓ Closure requires a final conclusion and a product disposition, by a user other than the investigator.
- ✓ Lots held by an open investigation cannot be delivered; a sample cannot be approved while an investigation is open.

## 8. Good practices

- ★ Never test into compliance: the number of retests and the decision rule are defined before retesting.
- ★ Report OOS trends in management review.

## 9. Pitfalls to avoid

- ✗ Invalidating an OOS result without evidence of a laboratory error.
- ✗ Closing the OOS with *Release* while a manufacturing cause is under investigation.

## 10 Records and evidence

- Investigation with phases, checks, authorisations, conclusion, disposition

## 11 Performance indicators

- OOS invalidated in Phase I (%)
- Median time to closure
- OOS per product

## 12 References

- FDA Guidance — Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production (Revision 1, 2022)
- 21 CFR 211.192 — Production record review
- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control

## 13. Demonstration

**Case.** Dissolution of 26T018 at 30 min: 76 % (NLT 80 %). Phase I: vessel 4 paddle height found out of tolerance at post-test check → assignable laboratory cause; the manager authorises a retest with justification. Retest on 6 new units: 88–93 %. Conclusion *Assignable Laboratory Cause*, disposition *Release*; closed by QA (not the investigator); CAPA opened for apparatus verification frequency.

# Stability studies in the laboratory

|                   |                                    |                |       |
|-------------------|------------------------------------|----------------|-------|
| Module            | ls_lab                             | Version        | _____ |
| Menu              | Laboratory ► Stability             | Effective date | _____ |
| Owner (suggested) | QC Manager                         | Approved by    | _____ |
| Related SOPs      | SOP-QC-002, SOP-QC-012, SOP-QC-013 |                |       |

## 1 Objective

Run stability studies with an approved protocol, generate pull samples at each time point, test them against the stability specification and record missed time points with reasons.

## 2 Scope

Stability studies managed on the laboratory side (long term, intermediate, accelerated, in-use, photostability, ongoing/annual). Programme-level stability of pharmaceutical batches may be managed in ls\_pharma (SOP-QC-002); choose one module per study to avoid double records.

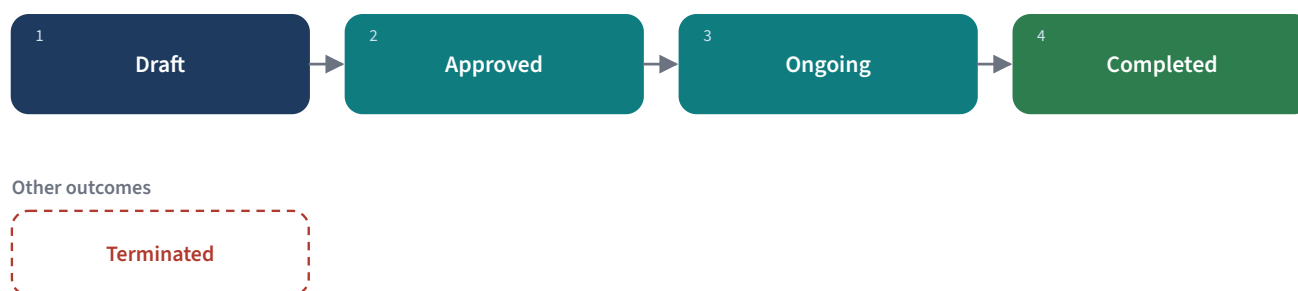
## 3 Responsibilities

| Role (software group)     | Responsibility                                              |
|---------------------------|-------------------------------------------------------------|
| <b>Stability analyst</b>  | Pulls and tests samples.                                    |
| <b>Laboratory Manager</b> | Approves protocol, starts, completes or terminates studies. |

## 4 Definitions

| Term              | Definition                                              |
|-------------------|---------------------------------------------------------|
| <b>Time point</b> | Planned, Sampled, Tested, Completed, Missed, Cancelled. |

## 5 Workflow in the software



Model: ls.lab.stability\_study

Figure 169 — Lifecycle of a laboratory stability study.

## 6 Procedure

- Create.** In **Stability Studies** click **New** : product, lot, batch, study type, storage condition, container closure, specification, protocol reference, planned end; add time points.
- Approve and start.** Click **Approve Protocol** ; set the start date; click **Start Study** .



- 3 **Pull.** At each time point click **Create Pull Sample** on the time point: a laboratory sample is created and tested (SOP-QC-012).
- 4 **Progress.** Click **Mark Tested** then **Complete** on the time point; if not pulled, **Mark Missed** with notes.
- 5 **Close.** Click **Complete** on the study when all time points are final, or **Terminate** with a reason.

### 7. Controls enforced by the software

- ✓ A study cannot be approved without time points; a start date is required to begin (due dates are derived from it).
- ✓ Pull samples only for ongoing studies and planned time points; only sampled time points can be marked tested, only tested ones completed, only planned ones missed (with notes).
- ✓ Completion requires every time point completed, missed or cancelled; termination requires a reason; the lot must belong to the product.

### 8. Good practices

- ★ Pull within the time window defined in the protocol and record the actual date.

### 9. Pitfalls to avoid

- ✗ Managing the same study in both ls\_pharma and ls\_lab.

## 10 Records and evidence

- Study, time points, samples, results

## 11 Performance indicators

- Time points missed
- Studies completed on time

## 12 References

- ICH Q1A(R2) — Stability Testing of New Drug Substances and Products
- 21 CFR 211.166 — Stability testing

### 13. Demonstration

**Case.** Photostability study on 26T018 per ICH Q1B conditions: 2 time points (exposed, dark control); pull samples created and tested; both completed; study completed with conclusion *photostable in the proposed pack*.

# Issue of certificates of analysis

|                   |                                                 |                |       |
|-------------------|-------------------------------------------------|----------------|-------|
| Module            | ls_lab                                          | Version        | _____ |
| Menu              | Laboratory ► Reports ► Certificates of Analysis | Effective date | _____ |
| Owner (suggested) | QC Manager                                      | Approved by    | _____ |
| Related SOPs      | SOP-QC-012, SOP-QA-090                          |                |       |

## 1 Objective

Issue certificates of analysis only from approved samples, showing the results and conclusion against the specification, and revise them in a traceable way.

## 2 Scope

Certificates of analysis for customers and for internal release.

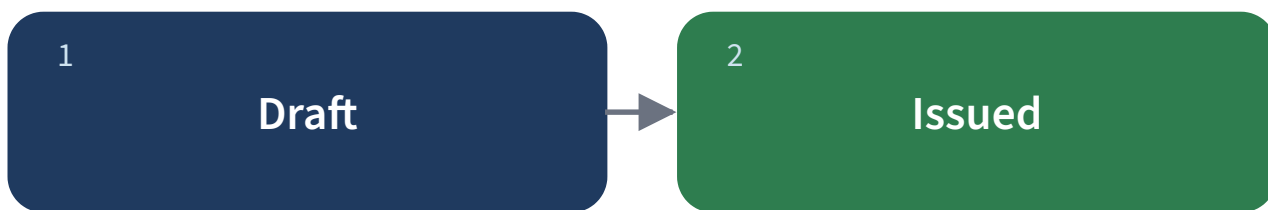
## 3 Responsibilities

| Role (software group) | Responsibility                               |
|-----------------------|----------------------------------------------|
| QC officer            | Prepares the certificate.                    |
| Laboratory Manager    | Issues it (signature intent <i>Issued</i> ). |

## 4 Definitions

| Term       | Definition                                                                            |
|------------|---------------------------------------------------------------------------------------|
| Conclusion | Conforms To Specification / Does Not Conform To Specification (Pending until issued). |
| Revision   | New version superseding an issued certificate, with a revision reason.                |

## 5 Workflow in the software



### Other outcomes



Model: *ls.lab.coa*

Figure 170 — Lifecycle of a certificate of analysis.

## 6 Procedure

- 1 **Create.** In **Reports ► Certificates of Analysis** click **New** and select the approved sample; enter version, customer, remarks; results flagged *Show on CoA* in the specification are included.
- 2 **Issue.** Click **Issue Certificate** and confirm the signature intent.
- 3 **Send.** Send the PDF to the customer or attach it to the batch file.
- 4 **Revise.** If a correction is needed, click **Create Revision** with the revision reason; the previous certificate becomes superseded.
- 5 **Cancel.** Only a draft certificate can be cancelled, with a reason.

### 7. Controls enforced by the software

- ✓ A certificate can only be issued from an approved sample; issued certificates cannot be modified or deleted.
- ✓ Only an issued certificate can be revised, with a revision reason; only drafts can be cancelled, with a reason.

### 8. Good practices

- ★ Include the specification version and the method references on the certificate.
- ★ Check the customer-specific requirements (units, additional tests) before issue.

### 9. Pitfalls to avoid

- ✗ Editing the PDF manually after issue.

## 10 Records and evidence

- Issued and superseded certificates

## 11 Performance indicators

- Certificates revised after issue

## 12 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 6 — Quality Control
- 21 CFR 211.160 to 211.176 — Laboratory controls
- ICH Q7 — Good Manufacturing Practice for Active Pharmaceutical Ingredients

### 13. Demonstration

**Case.** CoA for batch 26T018: 6 results, conclusion *Conforms To Specification*; issued by the Laboratory Manager. A customer asks for the dissolution profile: a revision v2 is created with reason "*customer request — add dissolution at 15/20/30 min*"; v1 is superseded.

# IV

## Regulatory affairs

Regulatory intelligence for import and export. Only phase 1 of ls\_import\_export (the cited provision registry and regulatory questions) exists; import-export operations (authorisations, customs, banking) are later phases and are not covered.

Import-export (phase 1) — ls\_import\_export

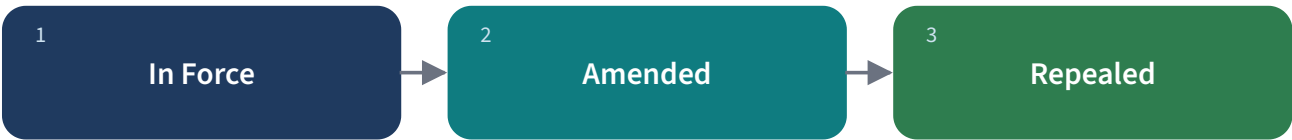
# Import-export compliance (phase 1)

ls\_import\_export

**ls\_import\_export** is built on a *Truth Protocol*: the module asserts no regulatory requirement that is not first recorded as a cited provision with a source, an effective date and an in-force status. The registry ships **empty**; every provision is entered by a person who read the source. Expected but unverified rules are recorded as **regulatory questions** for a qualified adviser. A compliance requirement cannot exist without at least one cited provision. Phase 1 is non-behavioural: it does not yet block purchases, sales or shipments.

| Menu                                                                                                | Content  |
|-----------------------------------------------------------------------------------------------------|----------|
| Import & Export Compliance ► Registry ► Provisions / Compliance Requirements / Regulatory Questions | Registry |

| Group                             | Main rights                                         |
|-----------------------------------|-----------------------------------------------------|
| Viewer · User · Officer · Manager | From read-only to registry maintenance and approval |



Other outcomes



Model: *ls.import\_export.provision*  
Figure 171 — *Statuses of a provision.*

# Regulatory watch: register of cited texts

|                   |                                                                              |                |       |
|-------------------|------------------------------------------------------------------------------|----------------|-------|
| Module            | ls_import_export                                                             | Version        | _____ |
| Menu              | Import & Export Compliance ► Registry ► Provisions / Compliance Requirements | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager (import-export)                                   | Approved by    | _____ |
| Related SOPs      | SOP-RA-011, SOP-COS-006, SOP-COS-007                                         |                |       |

## 1 Objective

Maintain a registry of the legal and regulatory provisions that govern regulated imports and exports, each cited from its official source with effective date and status, and derive compliance requirements only from cited provisions.

## 2 Scope

Algerian and international texts applicable to the import and export of medicinal products, medical devices and cosmetics (e.g. pharmaceutical agency, ministry, customs, central bank, foreign trade rules), as selected by Regulatory Affairs.

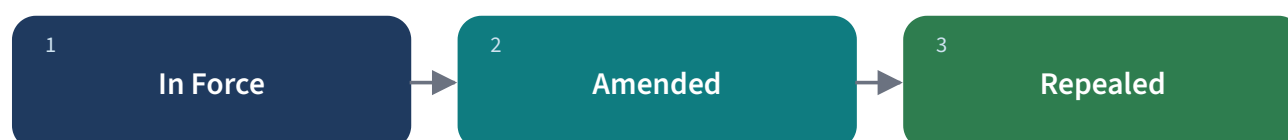
## 3 Responsibilities

| Role (software group)             | Responsibility                                                 |
|-----------------------------------|----------------------------------------------------------------|
| <b>Regulatory Officer</b>         | Reads the source, records provisions and citations.            |
| <b>Regulatory Affairs Manager</b> | Validates entries and requirements; decides on status changes. |

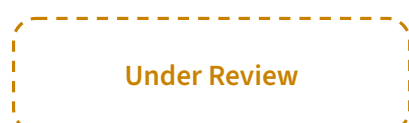
## 4 Definitions

| Term                          | Definition                                                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Provision</b>              | Legal text or article recorded with source, effective date and status: In Force, Amended, Repealed, Under Review. |
| <b>Compliance requirement</b> | Operational obligation derived from one or more cited provisions.                                                 |

## 5 Workflow in the software



Other outcomes



Model: ls.import\_export.provision

Figure 172 — Statuses of a provision.

## 6 Procedure

- 1 **Monitor sources.** Define the official sources to monitor (official journal, agency websites, customs notes) and the frequency (e.g. weekly).
- 2 **Record a provision.** In **Registry ► Provisions** create the provision with its official reference, title, source, effective date and status; add citations (article, exact wording or summary with reference).
- 3 **Derive requirements.** In **Compliance Requirements** create the requirement and cite at least one provision.
- 4 **Update.** When a text is amended or repealed, change the status and create the new provision; review the requirements citing it.
- 5 **Open questions.** When a rule is expected but not verified, record a regulatory question (SOP-RA-011) instead of a requirement.

### 7. Controls enforced by the software

- ✓ A compliance requirement must cite at least one provision; the last cited provision cannot be removed.
- ✓ The registry contains nothing that has not been entered by a person — there is no pre-loaded law.

### 8. Good practices

- ★ Record the date on which the source was consulted.
- ★ Never paraphrase a legal obligation without the citation next to it.

### 9. Pitfalls to avoid

- ✗ Entering requirements from memory or from a consultant's slide without the text.

## 10 Records and evidence

- Provisions with citations
- Requirements with cited provisions

## 11 Performance indicators

- Provisions under review for more than 30 days
- Requirements whose provisions were amended

### 12. Demonstration

**Case.** The officer reads a newly published executive decree on the import of medical devices, records it as a provision *In Force* from its effective date with citations of the articles on prior import authorisation, and creates the requirement "*Import programme approval before first import of the year*" citing those articles.



# Handling open regulatory questions

|                   |                                                              |                |       |
|-------------------|--------------------------------------------------------------|----------------|-------|
| Module            | ls_import_export                                             | Version        | _____ |
| Menu              | Import & Export Compliance ► Registry ► Regulatory Questions | Effective date | _____ |
| Owner (suggested) | Regulatory Affairs Manager                                   | Approved by    | _____ |
| Related SOPs      | SOP-RA-010, SOP-COS-007                                      |                |       |

## 1 Objective

Record, assign and answer regulatory questions whose answer is not verified, so that no rule is assumed or guessed and each answer is traceable to a source or a qualified adviser.

## 2 Scope

Questions about import-export rules, and more generally any regulatory point where the applicable requirement is uncertain.

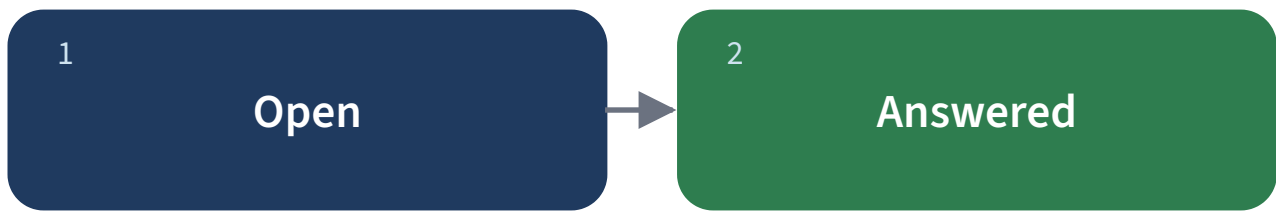
## 3 Responsibilities

| Role (software group)             | Responsibility                                                           |
|-----------------------------------|--------------------------------------------------------------------------|
| <b>Any user</b>                   | Raises a question with context.                                          |
| <b>Regulatory Affairs Manager</b> | Assigns, obtains the answer (authority, lawyer, consultant), records it. |
| <b>Management</b>                 | Decides how to operate while the question is open.                       |

## 4 Definitions

| Term          | Definition                       |
|---------------|----------------------------------|
| <b>Status</b> | Open, Answered, Will Not Answer. |

## 5 Workflow in the software



### Other outcomes



*Model: `ls.import_export.regulatory.question`*

Figure 173 — Statuses of a regulatory question.

## 6 Procedure

- 1 **Raise.** In **Regulatory Questions** create the question: precise wording, context, products concerned, urgency.
- 2 **Assign.** Assign it to a responsible person and a due date.
- 3 **Answer.** Obtain the answer from an authoritative source; record the answer, its source (letter, meeting minutes, legal opinion) and date; set *Answered*. If a provision results, record it (SOP-RA-010).
- 4 **Decline.** If the question will not be pursued, set *Will Not Answer* with the reason.
- 5 **Interim decision.** While open, record the interim operating decision taken by management (e.g. the most conservative interpretation).

### 7. Controls enforced by the software

- ✓ Statuses are limited to Open, Answered and Will Not Answer.

### 8. Good practices

- ★ Prefer written answers from authorities; keep minutes of oral answers.

### 9. Pitfalls to avoid

- ✗ Treating an informal answer as a legal requirement without recording its source.

## 10 Records and evidence

- Questions with answers and sources

## 11 Performance indicators

- Open questions older than 60 days

## 12. Demonstration

**Case.** Question: *"Does a change of the importer's storage site require a new import authorisation for cosmetics?"* — assigned to the regulatory manager; answer received by letter from the competent directorate after 3 weeks; recorded with the letter reference; the provision cited in the letter is added to the registry.



## Procedures supported by OCA modules

Procedures supported by community (OCA) modules present in the repository. Before relying on them, note the installation findings reported in the annex *Known limitations*: on a fresh Odoo 19.0 database, `auditlog`, `base_user_role`, `mgmtsystem_kpi` and `mgmtsystem_review` (through its dependency `base_external_dbsource`) failed to install because of view references renamed from *tree* to *list*. Screenshots are therefore given only for `maintenance_plan`, `maintenance_equipment_certification` and `quality_control_oca`, which installed.

|                                                                              |                                                        |
|------------------------------------------------------------------------------|--------------------------------------------------------|
| Management review — <code>mgmtsystem_review</code>                           | Roles and access review — <code>base_user_role</code>  |
| Quality indicators — <code>mgmtsystem_kpi</code>                             | Technical audit log — <code>auditlog</code>            |
| Preventive maintenance — <code>maintenance_plan</code>                       | Generic inspections — <code>quality_control_oca</code> |
| Equipment certificates —<br><code>maintenance_equipment_certification</code> |                                                        |

# Management review

|                   |                                                  |                |       |
|-------------------|--------------------------------------------------|----------------|-------|
| Module            | mgmtsystem_review                                | Version        | _____ |
| Menu              | Management Systems ► Management System ► Reviews | Effective date | _____ |
| Owner (suggested) | General Manager with the Head of Quality         | Approved by    | _____ |
| Related SOPs      | SOP-QA-005, SOP-GEN-002, SOP-QA-062, SOP-EM-005  |                |       |

## 1 Objective

Review at planned intervals the suitability, adequacy and effectiveness of the quality system and decide on improvements and resources.

## 2 Scope

The whole quality management system, at least once a year (more often in the first years or after major changes).

## 3 Responsibilities

| Role (software group)  | Responsibility                        |
|------------------------|---------------------------------------|
| <b>Top management</b>  | Chairs the review, takes decisions.   |
| <b>Head of Quality</b> | Prepares inputs, records the review.  |
| <b>Process owners</b>  | Present their indicators and actions. |

## 4 Definitions

| Term           | Definition                                                                                                                                                                                                                                                                         |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs</b>  | Status of previous actions; changes; customer feedback and complaints; process performance and product conformity; nonconformities and CAPA; audit results; monitoring results; supplier performance; risks and opportunities; adequacy of resources; new regulatory requirements. |
| <b>Outputs</b> | Decisions on improvement, changes to the QMS, and resource needs.                                                                                                                                                                                                                  |

## 5 Workflow in the software

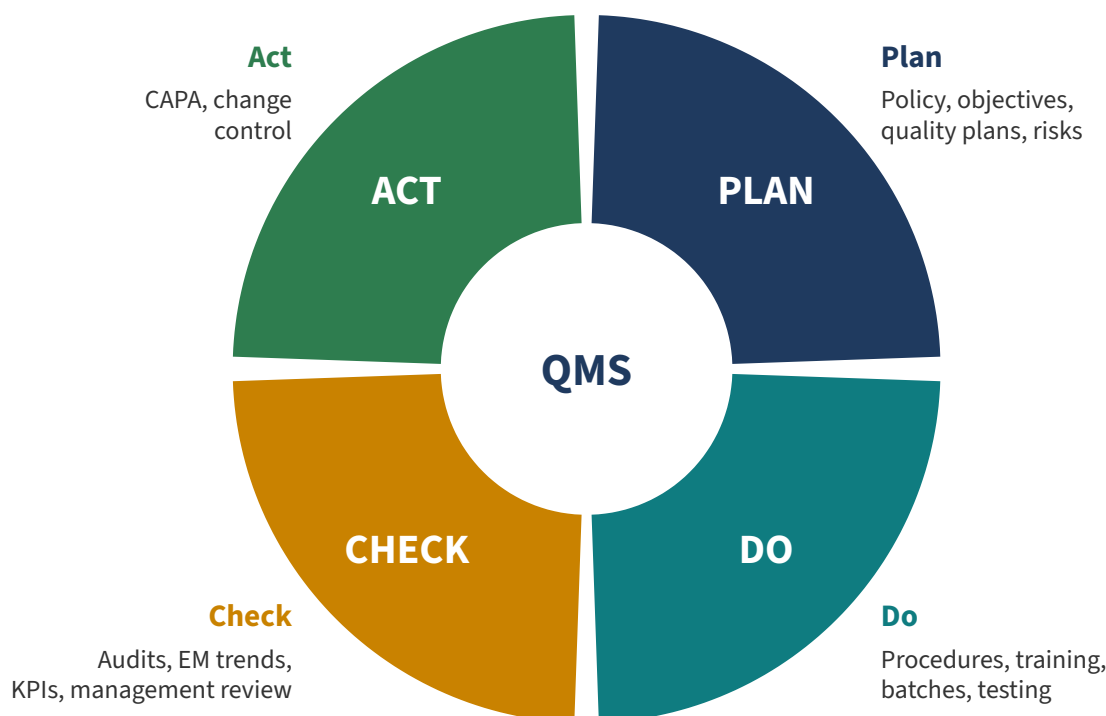


Figure 174 — Management review closes the quality cycle.

## 6 Procedure

- 1 **Prepare inputs.** Collect from the suite: objectives (SOP-QA-005), CAPA status (ls\_capa), deviations and complaints trends, audit results (ls\_audit), supplier ratings, EM trends, risk profile, training compliance, recalls, validation status.
- 2 **Create the review.** In the management review record (mgmtsystem\_review): name, date, participants, policy, changes; update the KPI history with the dedicated button (KPIs from mgmtsystem\_kpi).
- 3 **Hold the meeting.** Discuss each input; record decisions and actions as review lines.
- 4 **Conclude and close.** Write the conclusion; close the review.
- 5 **Follow up.** Turn decisions into CAPA, changes or objectives with owners and dates; report their status at the next review.
- 6 **Alternative.** If mgmtsystem\_review is not installed (see limitations), record the review as a quality record of type *Management Review* in ls\_qms (SOP-QA-004), with the minutes attached.

### 7. Controls enforced by the software

- ✓ The review record has states Open and Closed; closed reviews keep their KPI history snapshot.

### 8. Good practices

- ★ Send the input pack one week before the meeting.
- ★ Record decisions as verifiable actions, not as intentions.

## 9. Pitfalls to avoid

- ✗ A review reduced to a slide presentation without decisions.

## 10 Records and evidence

- Management review record or quality record with minutes
- Actions arising

## 11 Performance indicators

- Actions from the last review completed on time

## 12 References

- ISO 9001:2015 — Quality management systems — Requirements
- ISO 13485:2016 — Medical devices — Quality management systems
- ICH Q10 — Pharmaceutical Quality System
- EudraLex Vol. 4, EU GMP Part I, Chapter 1 — Pharmaceutical Quality System

## 13. Demonstration

**Case.** Annual review 2027: inputs from 14 audits, 62 deviations, 11 CAPA overdue, EM trends stable except corridor C. Decisions: add one QA reviewer (resource), revise the deviation classification SOP (change CC-2028-002), new objective on CAPA timeliness. Recorded, closed, actions assigned.

# Quality indicators (KPI)

|                   |                          |                |       |
|-------------------|--------------------------|----------------|-------|
| Module            | mgmtsystem_kpi           | Version        | _____ |
| Menu              | Management Systems ► KPI | Effective date | _____ |
| Owner (suggested) | Head of Quality          | Approved by    | _____ |
| Related SOPs      | SOP-QA-005, SOP-GEN-001  |                |       |

## 1 Objective

Define a small set of quality indicators with clear formulas, sources, frequencies and thresholds, and review them regularly.

## 2 Scope

Indicators of the quality system and GMP processes (e.g. deviation closure time, CAPA on time, complaint rate, right-first-time batches, OOS rate, EM excursions, training compliance, supplier rating).

## 3 Responsibilities

| Role (software group) | Responsibility                     |
|-----------------------|------------------------------------|
| Head of Quality       | Owns the KPI list.                 |
| Process owners        | Provide values and comment trends. |

## 4 Definitions

| Term        | Definition                                                                                |
|-------------|-------------------------------------------------------------------------------------------|
| KPI         | Indicator computed periodically from a defined source, with thresholds (green/amber/red). |
| KPI history | Stored values over time, snapshotted into management reviews.                             |

## 5 Procedure

- 1 Define.** For each indicator write: name, formula, data source (menu or report of the suite), frequency, owner, thresholds, direction.
- 2 Configure.** Create the KPI in mgmtsystem\_kpi (based on the OCA *kpi* module) with its computation and thresholds; or, if the module is not available, follow it as a quality objective with measurements in ls\_qms (SOP-QA-005).
- 3 Collect.** Compute or enter values at the defined frequency; check the source data.
- 4 Review.** Monthly in the quality meeting; quarterly and annually in management review (SOP-GEN-001).

## 6. Controls enforced by the software

- ✓ Indicator computation and history are managed by the kpi / mgmtsystem\_kpi modules; check the installation limits before relying on them.



## 7. Good practices

- ★ Prefer indicators computed from system data rather than typed by hand.
- ★ Pair each lagging indicator (e.g. complaints) with a leading one (e.g. CAPA on time).

## 8. Pitfalls to avoid

- ✗ Too many indicators, none of them acted upon.

## 9 Records and evidence

- KPI definitions and history

## 10 Performance indicators

- KPIs out of threshold without action

## 11 References

- ISO 9001:2015 — Quality management systems — Requirements
- ICH Q10 — Pharmaceutical Quality System

## 12. Demonstration

**Case.** KPI *CAPA closed on time (%)* =  $\text{CAPA closed} \leq \text{due date} / \text{CAPA closed}$ , monthly, source [Quality ► CAPA ► CAPA Records](#), green  $\geq 90\%$ , red  $< 75\%$ . In March: 71 % → red; discussed in the quality meeting; action: weekly CAPA review.

# Preventive maintenance of equipment

|                   |                                               |                |       |
|-------------------|-----------------------------------------------|----------------|-------|
| Module            | maintenance_plan                              | Version        | _____ |
| Menu              | Maintenance ► Maintenance ► Maintenance Plans | Effective date | _____ |
| Owner (suggested) | Maintenance Manager                           | Approved by    | _____ |
| Related SOPs      | SOP-MET-001, SOP-MP-002, SOP-VAL-005          |                |       |

## 1 Objective

Maintain GMP equipment at planned intervals so that it remains in its qualified state, and record every intervention.

## 2 Scope

Production, utility and laboratory equipment registered in Odoo Maintenance. Moulds follow SOP-MP-002; measuring instruments follow SOP-MET-002.

## 3 Responsibilities

| Role (software group)      | Responsibility                                                    |
|----------------------------|-------------------------------------------------------------------|
| <b>Maintenance Manager</b> | Defines plans and kinds of maintenance.                           |
| <b>Technicians</b>         | Perform maintenance requests and record work done.                |
| <b>QA</b>                  | Is informed of interventions that may affect the qualified state. |

## 4 Definitions

| Term                    | Definition                                                                                                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Maintenance plan</b> | Per equipment: kind, interval and interval step, duration, start date, planning horizon, instructions (PDF, slide or text); generates maintenance requests; next maintenance date computed. |

## 5 Procedure

- Create plans.** On each equipment add maintenance plans: kind, interval (e.g. 3 months), duration, start date, horizon, instructions.
- Execute.** Perform the generated maintenance requests; record the work done and parts replaced; close the request.
- Assess impact.** When an intervention may affect the qualified state (e.g. replacement of a critical part), raise a change request (SOP-QA-030) or a revalidation (SOP-VAL-005) before release to production.
- Review.** Review overdue requests weekly; adjust intervals with failure history.

### 6. Controls enforced by the software

- ✓ Next maintenance dates are computed from the plan interval; requests are generated within the planning horizon.

### 7. Good practices

- ★ Attach the GMP-relevant instructions (cleaning after maintenance, line clearance) to the plan.

## 8. Pitfalls to avoid

- ✗ Equipment returned to production after intervention without QA being informed.

## 9 Records and evidence

- Maintenance plans and requests with work done

## 10 Performance indicators

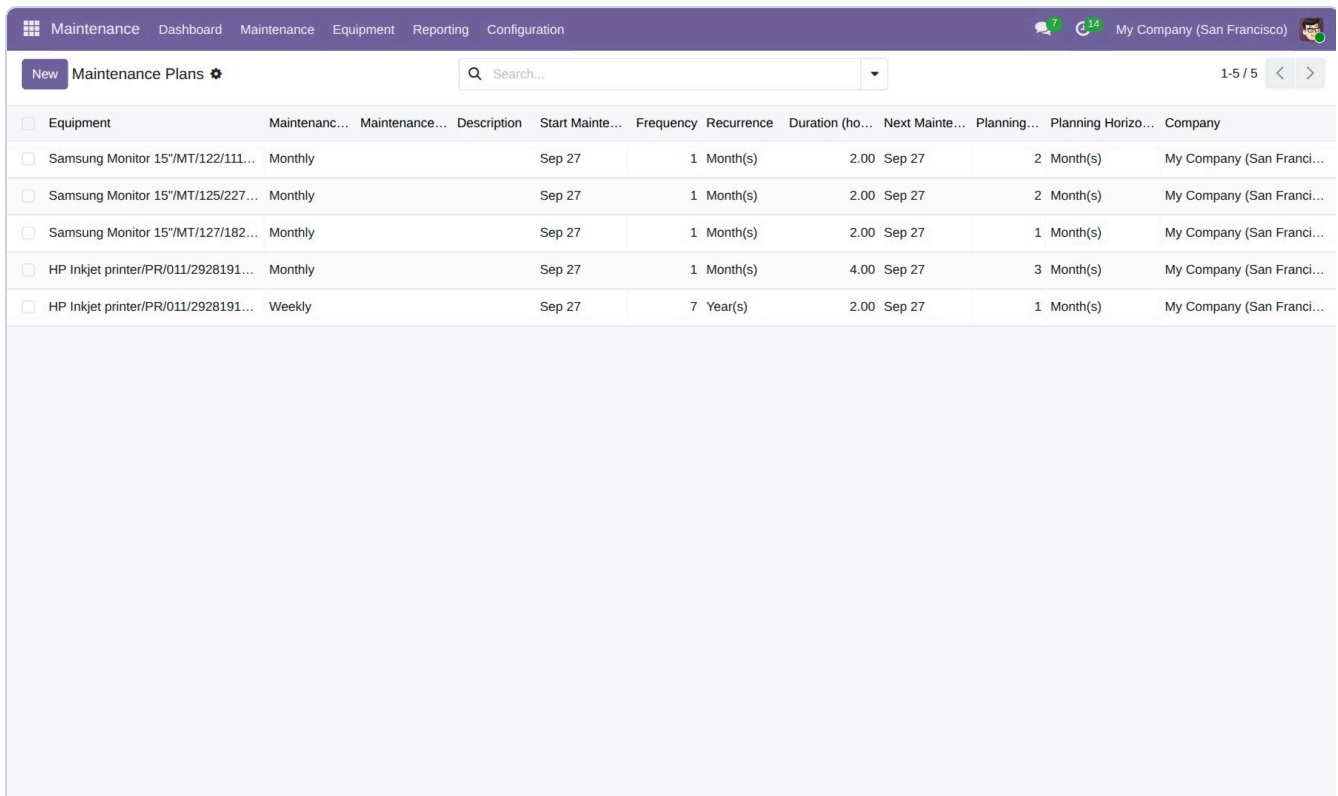
- Preventive maintenance done on time (%)
- Corrective interventions per equipment

## 11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment
- 21 CFR 211.68 — Automatic, mechanical and electronic equipment
- ISO 13485:2016 — Medical devices — Quality management systems

## 12. Demonstration

**Case.** Tablet press TP-02: plan *Preventive* — 3 months, 4 h, instructions PDF; the request of June is done: turret cleaned, punches inspected; no part replaced → no revalidation needed; request closed.



| <input type="checkbox"/> Equipment                           | Maintenanc... | Maintenance... | Description | Start Mainte... | Frequency | Recurrence | Duration (ho... | Next Mainte... | Planning... | Planning Horizo... | Company                   |
|--------------------------------------------------------------|---------------|----------------|-------------|-----------------|-----------|------------|-----------------|----------------|-------------|--------------------|---------------------------|
| <input type="checkbox"/> Samsung Monitor 15"/MT/122/111...   | Monthly       |                |             | Sep 27          |           | 1 Month(s) | 2.00            | Sep 27         | 2           | Month(s)           | My Company (San Franci... |
| <input type="checkbox"/> Samsung Monitor 15"/MT/125/227...   | Monthly       |                |             | Sep 27          |           | 1 Month(s) | 2.00            | Sep 27         | 2           | Month(s)           | My Company (San Franci... |
| <input type="checkbox"/> Samsung Monitor 15"/MT/127/182...   | Monthly       |                |             | Sep 27          |           | 1 Month(s) | 2.00            | Sep 27         | 1           | Month(s)           | My Company (San Franci... |
| <input type="checkbox"/> HP Inkjet printer/PR/011/2928191... | Monthly       |                |             | Sep 27          |           | 1 Month(s) | 4.00            | Sep 27         | 3           | Month(s)           | My Company (San Franci... |
| <input type="checkbox"/> HP Inkjet printer/PR/011/2928191... | Weekly        |                |             | Sep 27          |           | 7 Year(s)  | 2.00            | Sep 27         | 1           | Month(s)           | My Company (San Franci... |

Figure 175 — Preventive maintenance plans (maintenance\_plan). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

# Regulatory certificates of equipment

|                   |                                                                         |                |       |
|-------------------|-------------------------------------------------------------------------|----------------|-------|
| Module            | maintenance_equipment_certification                                     | Version        | _____ |
| Menu              | Maintenance ► Configuration ► Certificates / Certificate Reminder Rules | Effective date | _____ |
| Owner (suggested) | Maintenance Manager / EHS                                               | Approved by    | _____ |
| Related SOPs      | SOP-GEN-003, SOP-MET-004                                                |                |       |

## 1 Objective

Keep the regulatory certificates of equipment (pressure vessels, lifting equipment, electrical installations, etc.) current, with reminders before renewal dates.

## 2 Scope

Equipment subject to statutory inspections or certifications.

## 3 Responsibilities

| Role (software group) | Responsibility                                   |
|-----------------------|--------------------------------------------------|
| Maintenance Manager   | Records certificates and renewals.               |
| EHS                   | Defines which equipment needs which certificate. |

## 4 Definitions

| Term          | Definition                                                                         |
|---------------|------------------------------------------------------------------------------------|
| Certificate   | Name, certificate number, date, renewal date, file, notes, linked equipment.       |
| Reminder rule | Number of days before renewal at which a reminder is sent (activated in settings). |

## 5 Procedure

- 1 Configure reminders.** Enable certificate expiration notifications in the settings and create reminder rules (e.g. 60 and 15 days before).
- 2 Record.** On the equipment add each certificate with number, date, renewal date and file.
- 3 Renew.** Schedule the inspection before the renewal date; record the new certificate; archive the old one.
- 4 Stop use.** If a certificate expires, stop using the equipment until renewed.

### 6. Controls enforced by the software

- ✓ Reminders are sent according to the rules when notifications are enabled.

### 7. Good practices

- ★ Keep inspection reports with their certificate.

### 8. Pitfalls to avoid

- ✗ Relying on the inspection body to remind you.

## 9 Records and evidence

- Equipment certificates with files

## 10 Performance indicators

- Expired certificates of equipment in use (target 0)

## 11 References

- EudraLex Vol. 4, EU GMP Part I, Chapter 3 — Premises and Equipment

### 12. Demonstration

**Case.** Autoclave AC-02 pressure vessel certificate renewed every 24 months; reminders at 60 and 15 days; the inspection is performed 3 weeks before the renewal date and the new certificate attached.

# Role-based access management and periodic review of access rights

|                   |                                      |                |       |
|-------------------|--------------------------------------|----------------|-------|
| Module            | base_user_role                       | Version        | _____ |
| Menu              | Settings ► Users & Companies ► Roles | Effective date | _____ |
| Owner (suggested) | System Owner with IT                 | Approved by    | _____ |
| Related SOPs      | SOP-DI-011, SOP-QA-013               |                |       |

## 1 Objective

Grant access rights through documented roles matching job functions and the segregation of duties of the SOPs, and review them periodically.

## 2 Scope

All users of the Odoo installation, all LS module groups and standard Odoo groups.

## 3 Responsibilities

| Role (software group)   | Responsibility                                                         |
|-------------------------|------------------------------------------------------------------------|
| <b>System Owner</b>     | Defines roles, approves access requests, performs the periodic review. |
| <b>IT administrator</b> | Implements approved changes; never grants rights without approval.     |
| <b>Line managers</b>    | Request access and removal for their staff.                            |

## 4 Definitions

| Term                 | Definition                                                                                               |
|----------------------|----------------------------------------------------------------------------------------------------------|
| <b>Role</b>          | Named set of groups (base_user_role) assigned to users, optionally with <i>From</i> and <i>To</i> dates. |
| <b>Access matrix</b> | Document listing, per job function, the roles and the LS groups granted.                                 |

## 5 Procedure

- Design roles.** Build roles from the responsibilities tables of the SOPs (e.g. *QA officer* = Deviation Investigator + CAPA Coordinator + Document Approver + Laboratory Reviewer...). Check that no role combines incompatible duties (e.g. batch manufacturer and release authority).
- Request and approve.** Access requests are made by the line manager and approved by the System Owner; record them (ticket or quality record).
- Assign.** Assign roles to users with dates when temporary (holidays, projects). Without base\_user\_role (see limitations), assign groups directly and keep the access matrix as the reference.
- Link to training.** Grant critical roles only to users with valid qualifications (SOP-QA-013).
- Remove.** Remove access the day a person leaves or changes job.
- Review.** Every 6 months (critical systems) export users and groups, compare with the matrix and HR list, correct differences, record the review.

## 6. Controls enforced by the software

- ✓ Role lines can be limited in time (From/To); each LS module enforces its own group checks and segregation-of-duties rules regardless of the role mechanism.

## 7. Good practices

- ★ Give administrators separate personal accounts for administration and for business use.
- ★ Keep at least two people per critical approval role.

## 8. Pitfalls to avoid

- ✗ Granting *Manager* groups to everyone to avoid access errors — this defeats segregation of duties.

## 9 Records and evidence

- Access matrix
- Access requests and approvals
- Periodic review records

## 10 Performance indicators

- Differences found at review
- Leavers with active accounts (target 0)

## 11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- 21 CFR Part 11 — Electronic Records; Electronic Signatures
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

## 12. Demonstration

**Case.** Semi-annual review: 142 active users; 3 leavers still active (disabled the same day), 2 users with Deviation Manager who changed jobs (removed), 1 missing Laboratory Reviewer backup (granted after training). Review recorded as a quality record.

# Technical audit log

|                   |                                                                          |                |       |
|-------------------|--------------------------------------------------------------------------|----------------|-------|
| Module            | auditlog                                                                 | Version        | _____ |
| Menu              | Settings ► Technical ► Audit ► Rules / Logs (menu defined by the module) | Effective date | _____ |
| Owner (suggested) | IT administrator with the System Owner                                   | Approved by    | _____ |
| Related SOPs      | SOP-DI-001, SOP-DI-002                                                   |                |       |

## 1 Objective

Use the OCA auditlog module, where installed, to log technical operations (including reads, sessions and HTTP requests) useful for security investigations, and keep it distinct from the GxP audit trail.

## 2 Scope

Technical and security logging of selected models. GxP record changes are covered by `ls_audit_trail` (SOP-DI-001 to 004).

## 3 Responsibilities

| Role (software group) | Responsibility                                     |
|-----------------------|----------------------------------------------------|
| IT administrator      | Configures rules and investigates security events. |
| System Owner / QA     | Approves the scope and reviews findings.           |

## 4 Definitions

| Term                                               | Definition                                                                                                                                                             |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>auditlog rule</b>                               | Model with <code>log_read</code> , <code>log_write</code> , <code>log_create</code> , <code>log_unlink</code> options, log type (full or fast), capture of the record. |
| <b>Difference with <code>ls_audit_trail</code></b> | auditlog entries are not hash-chained; <code>ls_audit_trail</code> provides tamper-evidence, verification and evidence packs.                                          |

## 5 Procedure

- 1 Decide the need.** Define which technical events must be logged (e.g. reads of personal data, access to configuration) and why `ls_audit_trail` does not cover them.
- 2 Configure.** Create rules for the selected models and subscribe them; keep reads logging limited (volume).
- 3 Review.** Review the logs during security investigations and periodically for administrator activity.
- 4 Retain and purge.** Define retention and purge the logs by the module's scheduled action according to the IT policy.

## 6. Controls enforced by the software

- ✓ At the time of writing, auditlog failed to install on a fresh Odoo 19.0 database (view reference `view_auditlog_http_request_tree` not found). Fix and validate the module before relying on this SOP.



## 7. Good practices

- ★ Do not duplicate GxP auditing in two tools with different rules — use ls\_audit\_trail for GxP.

## 8. Pitfalls to avoid

- ✗ Logging reads on large models, degrading performance.

## 9 Records and evidence

- auditlog rules
- Investigation extracts

## 10 Performance indicators

- Security investigations using the log

## 11 References

- EU GMP Annex 11 — Computerised Systems (check the revision in force)
- PIC/S PI 041-1 — Good Practices for Data Management and Integrity in Regulated GMP/GDP Environments (2021)

## 12. Demonstration

**Case.** Reads of the HR employee model are logged to support investigations of personal data access; changes to GxP records remain in ls\_audit\_trail.

# Generic inspection controls, for example at reception

|                   |                                                             |                |       |
|-------------------|-------------------------------------------------------------|----------------|-------|
| Module            | quality_control_oqa                                         | Version        | _____ |
| Menu              | Quality Control ► Inspections ► Inspections · Tests ► Tests | Effective date | _____ |
| Owner (suggested) | QC Manager                                                  | Approved by    | _____ |
| Related SOPs      | SOP-QC-011, SOP-QA-070                                      |                |       |

## 1 Objective

Perform simple, questionnaire-based inspections (e.g. visual and documentary checks at reception) with recorded answers and a pass/fail outcome, where a full laboratory sample is not needed.

## 2 Scope

Incoming visual/identity checks, packaging inspections and other attribute inspections. Analytical testing is managed in *ls\_lab*.

## 3 Responsibilities

| Role (software group) | Responsibility                               |
|-----------------------|----------------------------------------------|
| <b>Inspector</b>      | Performs the inspection and records answers. |
| <b>Supervisor</b>     | Approves inspections waiting for approval.   |

## 4 Definitions

| Term                     | Definition                                                                                  |
|--------------------------|---------------------------------------------------------------------------------------------|
| <b>Test</b>              | Set of questions (qualitative with allowed values, or quantitative with min/max).           |
| <b>Inspection states</b> | Plan, Draft, Ready, Waiting supervisor approval, Quality success, Quality failed, Canceled. |

## 5 Procedure

- 1 Define tests.** In [Quality Control ► Tests ► Tests](#) create the test and its questions with acceptable values or ranges; group tests by category.
- 2 Create the inspection.** Create an inspection for the object (e.g. the product/lot received) and select the test. Note: automatic creation on receipts requires a stock integration module that is not present in the repository — inspections are created manually.
- 3 Record answers.** Answer each question; the outcome is computed; confirm the inspection.
- 4 Approve.** Inspections requiring approval go to *Waiting supervisor approval*; the supervisor approves.
- 5 Act on failure.** A failed inspection leads to quarantine and a deviation or supplier complaint.

### 6. Controls enforced by the software

- ✓ The inspection outcome (success) is computed from the answers against the defined values or ranges.

## 7. Good practices

- ★ Keep tests short and specific to the material category.

## 8. Pitfalls to avoid

- ✗ Using generic inspections for analytical tests that need method, specification and review (use ls\_lab).

## 9 Records and evidence

- Inspections with answers and outcome

## 10 Performance indicators

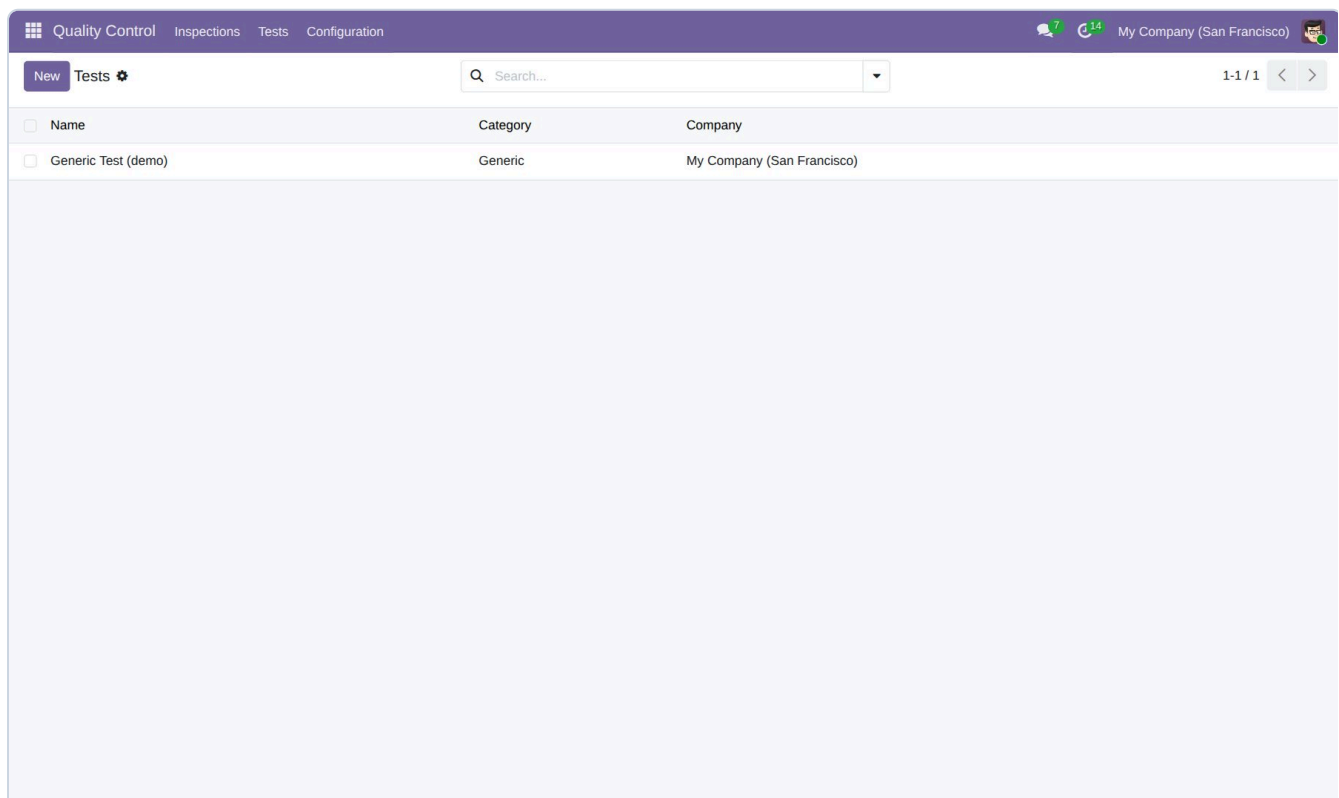
- Failed inspections per supplier

## 11 References

- ISO 13485:2016 — Medical devices — Quality management systems
- 21 CFR 211.84 — Testing and approval or rejection of components
- EudraLex Vol. 4, EU GMP Part I, Chapter 5 — Production

## 12. Demonstration

**Case.** Reception of printed cartons: test *Carton — reception* with 6 questions (supplier label present, quantity matches, artwork version, colour vs standard, no damage, CoC present). One carton lot has artwork version 3 instead of 4 → *Quality failed*, lot quarantined, supplier complaint opened.



The screenshot shows the Odoo 19.0 Quality Control interface. The top navigation bar includes 'Quality Control', 'Inspections', 'Tests', and 'Configuration'. The user is logged in as 'My Company (San Francisco)'. The main view is 'Tests', showing a table with the following data:

| Name                | Category | Company                    |
|---------------------|----------|----------------------------|
| Generic Test (demo) | Generic  | My Company (San Francisco) |

Figure 176 — Inspection test definitions (*quality\_control\_oca*). (Screenshot of the running software (Odoo 19.0 Community, LS modules, demonstration data).)

## Part VI — Procedures needed but not covered by the suite

These procedures depend on infrastructure and organisation, not on the modules. They are nevertheless required by GMP and data integrity guidance (EU GMP Annex 11, PIC/S PI 041-1) for the suite to be used in a regulated way. Each outline below lists what the SOP must contain; the content must be written with IT and the hosting provider.

| Proposed code | Procedure                                                 | What it must contain                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SOP-IT-010    | Backup and restore                                        | Scope (database, filestore, configuration), frequency and retention, encryption, off-site copy, restore test at defined intervals with evidence (restore to a test environment and verification of records, audit trail chain and signature chain — SOP-DI-002, SOP-DI-012), responsibilities.                                                                                                 |
| SOP-IT-011    | Business continuity and disaster recovery                 | Recovery time and recovery point objectives per process, fallback on paper (forms, later transcription and verification), crisis roles, periodic exercise, return to normal.                                                                                                                                                                                                                   |
| SOP-IT-012    | IT incident management                                    | Detection, classification (including data integrity impact), escalation to QA, investigation, link to deviation/CAPA, communication, closure.                                                                                                                                                                                                                                                  |
| SOP-IT-013    | Physical security of servers                              | Location, access control and logs, environment (power, temperature, fire), visitor rules, cloud provider controls when hosted.                                                                                                                                                                                                                                                                 |
| SOP-IT-014    | Qualification of the hosting                              | Supplier assessment of the hosting/cloud provider (SOP-QA-070), service agreement, infrastructure qualification (IQ), security certifications, change notification.                                                                                                                                                                                                                            |
| SOP-VAL-010   | Validation of the suite itself (IQ, OQ, PQ of the system) | User requirements, supplier assessment of the modules, risk-based test strategy (SOP-VAL-006), installation qualification of Odoo and modules (versions, installation log without errors), operational qualification of the controls listed in this handbook, performance qualification with real data and users, traceability matrix, report, periodic review and change control of upgrades. |

### Why these matter here

Several controls of the suite depend on the database (append-only triggers, hash chains, scheduled integrity checks). A restore, a migration or a server change can therefore affect regulated evidence: backup/restore and validation procedures must include the verification of the audit trail and signature chains after any restore.

# Annexes

## Annex A — SOP register

Register of the SOPs of this handbook, to steer their drafting. Copy it into your document system and complete owners, dates and status.

| Code       | Title                                                             | Module                          | Owner (suggested)                     | Drafting status  | Target date |
|------------|-------------------------------------------------------------------|---------------------------------|---------------------------------------|------------------|-------------|
| SOP-QA-001 | Drafting, review, approval and publication of quality documents   | ls_qms · ls_document_management | Head of Quality Assurance             | To draft / adapt |             |
| SOP-QA-002 | Revision, withdrawal and archiving of documents                   | ls_qms · ls_document_management | QMS Manager                           | To draft / adapt |             |
| SOP-QA-003 | Periodic review of documents                                      | ls_qms                          | QMS Manager                           | To draft / adapt |             |
| SOP-QA-004 | Management and retention of quality records                       | ls_qms · ls_document_management | QMS Manager                           | To draft / adapt |             |
| SOP-QA-005 | Quality objectives and measurement of their achievement           | ls_qms                          | Head of Quality / Management          | To draft / adapt |             |
| SOP-QA-010 | Creation and approval of training courses                         | ls_training                     | Training Manager                      | To draft / adapt |             |
| SOP-QA-011 | Session planning and issue of certifications                      | ls_training                     | Training Manager                      | To draft / adapt |             |
| SOP-QA-012 | Training linked to the publication of a new document              | ls_qms · ls_training            | QMS Manager with the Training Manager | To draft / adapt |             |
| SOP-QA-013 | Follow-up of qualifications and requalification                   | ls_training                     | Training Manager                      | To draft / adapt |             |
| SOP-QA-020 | Reporting, assessment and classification of deviations            | ls_deviation                    | Head of Quality Assurance             | To draft / adapt |             |
| SOP-QA-021 | Investigation, extension to other batches and product disposition | ls_deviation                    | Head of Quality Assurance             | To draft / adapt |             |
| SOP-QA-022 | CAPA management from identification to closure                    | ls_capa                         | CAPA Manager (QA)                     | To draft / adapt |             |
| SOP-QA-023 | Verification of CAPA effectiveness                                | ls_capa                         | CAPA Manager (QA)                     | To draft / adapt |             |
| SOP-QA-030 | Change request and impact assessment                              | ls_change_control               | Change Control Manager                | To draft / adapt |             |
| SOP-QA-031 | Approval, implementation and verification of changes              | ls_change_control               | Change Control Manager                | To draft / adapt |             |
| SOP-QA-040 | Reception, assessment and investigation of complaints             | ls_complaint                    | Complaint Manager (QA)                | To draft / adapt |             |
| SOP-QA-041 | Adverse events: reportability assessment and reporting deadlines  | ls_complaint                    | Vigilance officer / Complaint Manager | To draft / adapt |             |
| SOP-QA-050 | Internal audit programme                                          | ls_audit                        | Audit Manager                         | To draft / adapt |             |
| SOP-QA-051 | Auditor qualification and impartiality                            | ls_audit                        | Audit Manager                         | To draft / adapt |             |
| SOP-QA-052 | Conducting an audit: preparation, fieldwork, findings             | ls_audit                        | Lead auditor                          | To draft / adapt |             |

| Code        | Title                                                                                | Module                               | Owner (suggested)                            | Drafting status  | Target date |
|-------------|--------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------|------------------|-------------|
| SOP-QA-053  | Audit report and follow-up of findings                                               | ls_audit                             | Audit Manager                                | To draft / adapt |             |
| SOP-QA-060  | Quality risk management: matrix, assessment, control and acceptance of residual risk | ls_risk_management                   | Risk Manager                                 | To draft / adapt |             |
| SOP-QA-061  | FMEA (failure modes and effects analysis)                                            | ls_risk_management                   | Risk Manager                                 | To draft / adapt |             |
| SOP-QA-062  | Periodic review of risks                                                             | ls_risk_management                   | Risk Manager                                 | To draft / adapt |             |
| SOP-QA-070  | Qualification and approval of suppliers                                              | ls_supplier_qualification            | Supplier Manager (QA)                        | To draft / adapt |             |
| SOP-QA-071  | Supplier audits                                                                      | ls_supplier_qualification            | Supplier Manager (QA)                        | To draft / adapt |             |
| SOP-QA-072  | Supplier performance evaluation and requalification                                  | ls_supplier_qualification            | Supplier Manager (QA)                        | To draft / adapt |             |
| SOP-QA-073  | Control of purchases from non-approved suppliers                                     | ls_supplier_qualification · purchase | Supplier Manager with the Purchasing Manager | To draft / adapt |             |
| SOP-DI-001  | Definition of the audit trail scope (rules per model)                                | ls_audit_trail                       | Data Integrity lead / System owner           | To draft / adapt |             |
| SOP-DI-002  | Audit trail review and integrity verification                                        | ls_audit_trail                       | QA / Data Integrity lead                     | To draft / adapt |             |
| SOP-DI-003  | Evidence packs for inspections                                                       | ls_audit_trail                       | Audit Trail Auditor (QA)                     | To draft / adapt |             |
| SOP-DI-004  | Retention and controlled purge of audit data                                         | ls_audit_trail                       | QA with the Audit Trail Administrator        | To draft / adapt |             |
| SOP-DI-010  | Electronic signature policy: meanings, policies, signer commitment                   | ls_electronic_signature              | QA with the System Owner                     | To draft / adapt |             |
| SOP-DI-011  | Management of credentials, account lockout and security alerts                       | ls_electronic_signature              | IT security with QA                          | To draft / adapt |             |
| SOP-DI-012  | Verification of the signature chain                                                  | ls_electronic_signature              | Signature Auditor (QA)                       | To draft / adapt |             |
| SOP-VAL-001 | Validation master plan (VMP)                                                         | ls_validation                        | Validation Manager                           | To draft / adapt |             |
| SOP-VAL-002 | Drafting and approval of qualification protocols: DQ, IQ, OQ, PQ                     | ls_validation                        | Validation Manager                           | To draft / adapt |             |
| SOP-VAL-003 | Protocol execution and management of discrepancies                                   | ls_validation                        | Validation Manager                           | To draft / adapt |             |
| SOP-VAL-004 | Summary report and validation status                                                 | ls_validation                        | Validation Manager                           | To draft / adapt |             |
| SOP-VAL-005 | Periodic revalidation                                                                | ls_validation                        | Validation Manager                           | To draft / adapt |             |

| Code        | Title                                                                                | Module                      | Owner (suggested)               | Drafting status  | Target date |
|-------------|--------------------------------------------------------------------------------------|-----------------------------|---------------------------------|------------------|-------------|
| SOP-VAL-006 | Validation of processes, cleaning, analytical methods and computerised systems (CSV) | ls_validation               | Validation Manager              | To draft / adapt |             |
| SOP-MET-001 | Management of the instrument inventory                                               | ls_calibration              | Calibration Manager (metrology) | To draft / adapt |             |
| SOP-MET-002 | Planning and performing calibrations (as-found and as-left values)                   | ls_calibration              | Calibration Manager             | To draft / adapt |             |
| SOP-MET-003 | Impact assessment of an out-of-tolerance result                                      | ls_calibration              | Calibration Manager with QA     | To draft / adapt |             |
| SOP-MET-004 | Calibration certificates                                                             | ls_calibration              | Calibration Manager             | To draft / adapt |             |
| SOP-EM-001  | Monitoring programme: areas, sampling points and rationale                           | ls_environmental_monitoring | EM Manager (QA / microbiology)  | To draft / adapt |             |
| SOP-EM-002  | Setting and approval of alert and action limits                                      | ls_environmental_monitoring | EM Manager                      | To draft / adapt |             |
| SOP-EM-003  | Sampling, analysis and review of results                                             | ls_environmental_monitoring | EM Manager                      | To draft / adapt |             |
| SOP-EM-004  | Management of excursions                                                             | ls_environmental_monitoring | EM Manager with QA              | To draft / adapt |             |
| SOP-EM-005  | Trend analysis                                                                       | ls_environmental_monitoring | EM Manager                      | To draft / adapt |             |
| SOP-QA-080  | Recall plan, with coordinator and deputy                                             | ls_recall                   | Head of Quality                 | To draft / adapt |             |
| SOP-QA-081  | Executing a recall or withdrawal: distribution tracing, notification, reconciliation | ls_recall                   | Recall coordinator              | To draft / adapt |             |
| SOP-QA-082  | Recall effectiveness checks                                                          | ls_recall                   | Recall coordinator              | To draft / adapt |             |
| SOP-QA-083  | Mock recall (periodic exercise)                                                      | ls_recall                   | Head of Quality                 | To draft / adapt |             |
| SOP-PRD-001 | Issue and execution of the batch record                                              | ls_pharma                   | Production Manager              | To draft / adapt |             |
| SOP-PRD-002 | Line clearance and label reconciliation                                              | ls_pharma                   | Packaging Manager               | To draft / adapt |             |
| SOP-QA-090  | Batch record review and batch release by the quality unit                            | ls_pharma                   | Head of Quality Assurance       | To draft / adapt |             |
| SOP-QC-001  | Qualification of active substances and excipients                                    | ls_pharma                   | QC Manager with QA              | To draft / adapt |             |
| SOP-QC-002  | Stability programme                                                                  | ls_pharma                   | QC Manager                      | To draft / adapt |             |

| Code        | Title                                                                            | Module              | Owner (suggested)                               | Drafting status  | Target date |
|-------------|----------------------------------------------------------------------------------|---------------------|-------------------------------------------------|------------------|-------------|
| SOP-PRD-003 | GS1 serialisation and aggregation                                                | ls_pharma           | Packaging Manager with IT                       | To draft / adapt |             |
| SOP-RA-001  | Preparation of the CTD dossier (Common Technical Document)                       | ls_pharma           | Regulatory Affairs Manager                      | To draft / adapt |             |
| SOP-COS-001 | Development and approval of formulations                                         | ls_cosmetics        | Formulation Manager with the Regulatory Manager | To draft / adapt |             |
| SOP-COS-002 | Safety assessment (CPSR, Cosmetic Product Safety Report)                         | ls_cosmetics        | Safety Assessor                                 | To draft / adapt |             |
| SOP-COS-003 | Product information file (PIF) and its retention period                          | ls_cosmetics        | Responsible Person (Regulatory Manager)         | To draft / adapt |             |
| SOP-COS-004 | Substantiation of claims                                                         | ls_cosmetics        | Regulatory Manager                              | To draft / adapt |             |
| SOP-COS-005 | Validation of labelling and of the INCI list                                     | ls_cosmetics        | Regulatory Manager                              | To draft / adapt |             |
| SOP-COS-006 | Maintenance of the register of regulatory restrictions on ingredients            | ls_cosmetics        | Regulatory Manager                              | To draft / adapt |             |
| SOP-COS-007 | Prior authorisation dossier in Algeria                                           | ls_cosmetics        | Regulatory Manager (Algeria)                    | To draft / adapt |             |
| SOP-MD-001  | Device registration and justification of its risk class                          | ls_medical_device   | Regulatory Affairs Manager                      | To draft / adapt |             |
| SOP-MD-002  | Assignment of the UDI (Unique Device Identification)                             | ls_medical_device   | Regulatory Affairs Manager                      | To draft / adapt |             |
| SOP-MD-003  | Risk management according to ISO 14971                                           | ls_medical_device   | Risk management lead (Regulatory / R&D)         | To draft / adapt |             |
| SOP-MD-004  | Clinical evaluation and PMCF (Post-Market Clinical Follow-up)                    | ls_medical_device   | Clinical evaluation lead                        | To draft / adapt |             |
| SOP-MD-005  | Technical documentation                                                          | ls_medical_device   | Regulatory Affairs Manager                      | To draft / adapt |             |
| SOP-MD-006  | CE marking and relationship with the notified body                               | ls_medical_device   | Regulatory Affairs Manager                      | To draft / adapt |             |
| SOP-MD-007  | Post-market surveillance and periodic reports                                    | ls_medical_device   | PMS lead (Regulatory Affairs)                   | To draft / adapt |             |
| SOP-MP-001  | Qualification and commissioning of moulds                                        | ls_medical_plastics | Tooling / Process Engineering Manager           | To draft / adapt |             |
| SOP-MP-002  | Preventive maintenance of moulds (by shot count and by time) and cavity blocking | ls_medical_plastics | Tooling Manager                                 | To draft / adapt |             |
| SOP-MP-003  | Process windows: drafting, review and approval                                   | ls_medical_plastics | Process Engineering Manager                     | To draft / adapt |             |



| Code        | Title                                                                      | Module                              | Owner (suggested)                          | Drafting status  | Target date |
|-------------|----------------------------------------------------------------------------|-------------------------------------|--------------------------------------------|------------------|-------------|
| SOP-MP-004  | Start-up, operation and closure of a moulding run                          | ls_medical_plastics                 | Moulding Production Manager                | To draft / adapt |             |
| SOP-MP-005  | Correction of parameter readings (without altering the original value)     | ls_medical_plastics                 | Moulding Production Manager with QA        | To draft / adapt |             |
| SOP-MP-006  | Material traceability                                                      | ls_medical_plastics                 | Moulding Quality Manager                   | To draft / adapt |             |
| SOP-QC-010  | Approval of test methods and specifications                                | ls_lab                              | QC Manager                                 | To draft / adapt |             |
| SOP-QC-011  | Receipt and registration of samples                                        | ls_lab                              | QC Manager                                 | To draft / adapt |             |
| SOP-QC-012  | Entry, review and approval of results                                      | ls_lab                              | QC Manager                                 | To draft / adapt |             |
| SOP-QC-013  | Investigation of OOS (out-of-specification) and OOT (out-of-trend) results | ls_lab                              | QC Manager with QA                         | To draft / adapt |             |
| SOP-QC-014  | Stability studies in the laboratory                                        | ls_lab                              | QC Manager                                 | To draft / adapt |             |
| SOP-QC-015  | Issue of certificates of analysis                                          | ls_lab                              | QC Manager                                 | To draft / adapt |             |
| SOP-RA-010  | Regulatory watch: register of cited texts                                  | ls_import_export                    | Regulatory Affairs Manager (import-export) | To draft / adapt |             |
| SOP-RA-011  | Handling open regulatory questions                                         | ls_import_export                    | Regulatory Affairs Manager                 | To draft / adapt |             |
| SOP-GEN-001 | Management review                                                          | mgmtsystem_review                   | General Manager with the Head of Quality   | To draft / adapt |             |
| SOP-GEN-002 | Quality indicators (KPI)                                                   | mgmtsystem_kpi                      | Head of Quality                            | To draft / adapt |             |
| SOP-GEN-003 | Preventive maintenance of equipment                                        | maintenance_plan                    | Maintenance Manager                        | To draft / adapt |             |
| SOP-GEN-004 | Regulatory certificates of equipment                                       | maintenance_equipment_certification | Maintenance Manager / EHS                  | To draft / adapt |             |
| SOP-IT-001  | Role-based access management and periodic review of access rights          | base_user_role                      | System Owner with IT                       | To draft / adapt |             |
| SOP-IT-002  | Technical audit log                                                        | auditlog                            | IT administrator with the System Owner     | To draft / adapt |             |
| SOP-QC-020  | Generic inspection controls, for example at reception                      | quality_control_oca                 | QC Manager                                 | To draft / adapt |             |
| SOP-IT-010  | Backup and restore                                                         | — (not covered)                     | IT / QA                                    | To draft / adapt |             |
| SOP-IT-011  | Business continuity and disaster recovery                                  | — (not covered)                     | IT / QA                                    | To draft / adapt |             |
| SOP-IT-012  | IT incident management                                                     | — (not covered)                     | IT / QA                                    | To draft / adapt |             |

| Code        | Title                                | Module          | Owner (suggested) | Drafting status  | Target date |
|-------------|--------------------------------------|-----------------|-------------------|------------------|-------------|
| SOP-IT-013  | Physical security of servers         | — (not covered) | IT / QA           | To draft / adapt |             |
| SOP-IT-014  | Qualification of the hosting         | — (not covered) | IT / QA           | To draft / adapt |             |
| SOP-VAL-010 | Validation of the suite (IQ, OQ, PQ) | — (not covered) | IT / QA           | To draft / adapt |             |

## Annex B — Glossary

| Term                               | Meaning                                                                                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ALCOA+                             | Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available — expected properties of GxP data. |
| Audit trail                        | Secure, computer-generated, time-stamped record of creation, modification and deletion of records.                                            |
| CAPA                               | Corrective and Preventive Action.                                                                                                             |
| CoA                                | Certificate of Analysis.                                                                                                                      |
| CPSR                               | Cosmetic Product Safety Report (Regulation (EC) No 1223/2009, Annex I).                                                                       |
| CSV                                | Computerised System Validation.                                                                                                               |
| CTD / eCTD                         | Common Technical Document (ICH M4) / its electronic format.                                                                                   |
| DQ / IQ / OQ / PQ                  | Design / Installation / Operational / Performance Qualification.                                                                              |
| EM                                 | Environmental Monitoring.                                                                                                                     |
| FMEA / RPN                         | Failure Modes and Effects Analysis / Risk Priority Number (S × O × D).                                                                        |
| FSCA                               | Field Safety Corrective Action (medical devices).                                                                                             |
| GTIN / SSCC                        | Global Trade Item Number / Serial Shipping Container Code (GS1).                                                                              |
| GxP                                | Good practices (manufacturing, laboratory, distribution, clinical...).                                                                        |
| INCI                               | International Nomenclature of Cosmetic Ingredients.                                                                                           |
| MDR                                | Regulation (EU) 2017/745 on medical devices.                                                                                                  |
| OOS / OOT                          | Out of specification / out of trend result.                                                                                                   |
| PIF                                | Product Information File (cosmetics, Article 11).                                                                                             |
| PMCF / PMS / PSUR                  | Post-Market Clinical Follow-up / Post-Market Surveillance / Periodic Safety Update Report.                                                    |
| QP                                 | Qualified Person (EU GMP Annex 16).                                                                                                           |
| Segregation of duties              | Rule that the same person cannot perform and approve (or review and approve) the same record.                                                 |
| SOP                                | Standard Operating Procedure.                                                                                                                 |
| UDI (Basic UDI-DI, UDI-DI, UDI-PI) | Unique Device Identification and its components.                                                                                              |
| VMP                                | Validation Master Plan.                                                                                                                       |

## Annex C — Known limitations observed while preparing this handbook

- OCA modules that failed to install** on a fresh Odoo 19.0 Community database (source build of the 19.0 branch dated 26 September 2026) during the preparation of this handbook: *auditlog* (external ID `auditlog.view_auditlog_http_request_tree` not found), *base\_user\_role* (`base_user_role.view_res_users_role_tree` not found), and *mgmtsystem\_kpi* and *mgmtsystem\_review* through their dependency *base\_external\_dbsource* (`base_external_dbsource.view_dbsource_tree` not found). In each case the view was renamed from *tree* to *list* but a `ref=` still uses the old identifier. The project's own

container image (19.0-20260723) and database were not available for comparison; the production database may already contain these modules from an earlier installation. SOP-GEN-001, SOP-GEN-002, SOP-IT-001 and SOP-IT-002 describe fall-back procedures.

- **Electronic signatures:** no production model of the 22 LS modules inherits *ls.signature.mixin*; signature policies can therefore not yet be applied to LS records (see the Electronic signatures chapter). *ls\_validation* re-authenticates signers; *ls\_lab* records a signature intent that it states is not a Part 11 signature; *ls\_supplier\_qualification* keeps a chained decision log without re-authentication.
- **Hand-offs between modules** (deviation → CAPA, complaint → CAPA, audit finding → CAPA, excursion → deviation...) are made by typed references, not by relational links. The procedures must require the references to be written and checked.
- **Translations:** the LS modules ship no French translation; the interface and the screenshots are in English.
- **Training:** *ls\_training* states that it does not implement 21 CFR Part 11 (no electronic signature, no field-level audit trail of its own). There is no automatic link between a published document and a training course (SOP-QA-012).
- **Supplier purchase control** is installed at level *warn* by default; SOP-QA-073 requires *block* for GMP companies.
- **quality\_control\_oca:** no stock integration module is present in the repository; inspections are created manually.
- **Demonstration data:** many LS models have no demo records; screenshots of those models show the empty form of a new record with its real fields and status bar.

## Annex D — How this handbook was produced

Workflows, buttons, menus, groups and controls were extracted from the module source code (repository *youneskh/LSS*, branch *master*) and from a database where the 22 LS modules were installed with demonstration data on Odoo 19.0 Community. State lists come from that database (file *data/states\_from\_db.txt*). Screenshots were captured from that running instance. The book is generated by *docs/sop\_handbook/build.py*; regenerate it after module changes so that the procedures stay aligned with the software.